Meeting of the Board
17 – 20 February 2025
Songdo, Incheon, Republic of Korea
Provisional agenda item 11

GCF/B.41/02/Add.04

27 January 2025

Consideration of funding proposals – Addendum IV

Funding proposal package for FP256

Summary

This addendum contains the following six parts:

a) A funding proposal titled "Intensification of Agriculture and Agroforestry Techniques (IAAT) for Climate Resilient Food and Nutrition Security: Tombouctou, Gao, Mopti, Koulikoro and Segou regions of Mali";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Secretariat’s assessment;
d) Independent Technical Advisory Panel’s assessment;
e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
f) Gender documentation.

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Funding Proposal

Project/Programme title:	*Intensification of Agriculture and Agroforestry Techniques (IAAT) for climate resilient food and nutrition security: Tombouctou, Gao, Mopti, Koulikoro and Segou regions of Mali*
Country(ies):	*Mali*
Accredited Entity:	*Save the Children Australia*
Date of first submission:	*[2023/12/21]*
Date of current submission	*[2025/01/21]*
Version number	*[vB.41]*

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that the annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-SCA-MALI-2023-12-15"

LIST OF ACRONYMS	
AE	Accredited Entity
AEDD	Agence de l'Environnement et du Developpement Durable (Agency for Environment and Sustainable Development)
AOM	Agricultural Office Manager
AMA	Approved Master Agreement
AR6	Sixth Assessment Report (IPCC)
CACs	Community Action Cycles
CAG	Community Action Group
CNCCM	National Climate Change Committee (in English)
COP	Conference of the Parties
CRR	Rural Resource Center
CSA	Climate Smart Agriculture
CSOs	Civil Society Organizations
EE	Executing Entity
ENPV	Environmental Net Present Value
ESMS	Environmental and Social Management System
ESS	Environmental and Social Safeguard
ESAP	Environmental and Social Action Plan
EWE	Extreme Weather Event
EWS	Early Warning System
FMCA	Financial Management and Capacity Assessment
GAP-RUs	Groupe d’Alerte Précoce et de Réponse aux Urgence -Early Warning and Emergency Response Group
GCF	Green Climate Fund
GDP	Gross Domestic Product
GESI	Gender Equality and Social Inclusion
GHG	Greenhouse Gas
GoM	Government of Mali
GSAN	Groupe de Soutien aux Actions de Nutrition – Support Group for Nutrition Action
IAAT	Intensification of Agriculture and Agroforestry Techniques for climate resilient food and nutrition security
IFAD	International Fund for Agricultural Development
IRR	International Rate of Return
IPCC	Intergovernmental Panel on Climate Change
IRMF	Integrated Results Management Framework
LCC	Local Committee of Coordination
LDC	Least Developed Country
LLA	Locally Led Adaptation
ME&S	Ministry of Environment and Sanitation
MEAL	Monitoring, Evaluation, Accountability and Learning
MFIs	Micro Finance Institutions
MoALF	Ministry of Agriculture, Livestock, and Fisheries
MAWCF	Ministry for the Advancement of Women, Children and Families
MCF	Mali Climate Fund
NAPA	National Adaptation Plan of Action

NCC	National Committee of Coordination
NDA	National Designated Authority
NDC	Nationally Determined Contribution
NGO	Non-governmental Organization
NPV	Net Present Value
PAYG	Pay-As-You-Go
PDESC	Programme de Développement Economique, Social et Culturel (Economic, Social, and Cultural Development Program)
PIU	Project Implementation Unit
PMU	Project Management Unit
PSC	Project Steering Committee
RCC	Regional Committee of Coordination
RCP	Representative Concentration Pathway
RDA	Regional Directorate of Agriculture
SAP	Simplified Approval Process
SBC	Social and Behavior Change
SC	Save the Children
SCA	SC Australia
SCI	Save the Children International
SDG	Sustainable Development Goals
SCUS	Save the ChildrenFederation, Inc.
SEAH	Sexual Exploitation, Abuse and Harassment
SME	Small and Medium Enterprises
SNCC	Stratégie nationale sur le changement climatique (National Climate Change Strategy)
SOP	Standard Operating Procedure
SPEI	Standardized Precipitation Evapotranspiration Index
SSP	Shared Socioeconomic Pathway
ToC	Theory of Change
UNDP	United Nations Development Program
UNFCCC	United Nations Framework Convention on Climate Change
USAID	U.S. Agency for International Development
USD	United States Dollar
VC	Value Chain
VSLA	Village Savings and Loan Associations

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	**A.2. Public or private sector**	Public	
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF [Request for Proposals](), indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable			
A.4. Result area(s)	*Check the applicable [GCF result area(s)]() that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*			
			GCF contribution	**Co-financers' contribution[1]**
	Mitigation total		**25%**	**15%**
	☒ Energy generation and access		8.5%	5%
	☐ Low-emission transport			
	☐ Buildings, cities, industries and appliances			
	☒ Forestry and land use		16.5%	10%
	Adaptation total		**75%**	**85%**
	☒ Most vulnerable people and communities		37.5%	42.5%
	☒ Health and well-being, and food and water security		37.5%	42.5%
	☐ Infrastructure and built environment			
	☐ Ecosystems and ecosystem services			
A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	2,446,543 tCO2eq	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	1,132,680	
			Direct: 431,700	Indirect: 700,980
			2.4%	3.8%
A.7. Total financing (GCF + co-finance[2])	43,717,918 USD	**A.9. Project size**	Small (Up to USD 50 million)	
A.8. Total GCF funding requested	33,717,918 USD *For multi-country proposals, please fill out annex 17.*			

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] Refer to the Policy of Co-financing of the GCF.

<table>
<tr><td rowspan="2">A.10. Financial instrument(s) requested for the GCF funding</td><td colspan="3">Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</td></tr>
<tr><td colspan="2">☒ Grant: 33,717,918 USD
☐ Loan Enter number
☐ Guarantee Enter number</td><td>☐ Equity Enter number
☐ Results-based payment Enter number</td></tr>
<tr><td>A.11. Implementation period</td><td>5 years (60 months)</td><td>A.12. Total lifespan</td><td>15 years (180 months)</td></tr>
<tr><td>A.13. Expected date of AE internal approval</td><td>This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.
2024/11/22</td><td>A.14. ESS category</td><td>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.
C</td></tr>
<tr><td>A.15. Has this FP been submitted as a CN before?</td><td>Yes ☒ No ☐</td><td>A.16. Has Readiness or PPF support been used to prepare this FP?</td><td>Yes ☐ No ☒</td></tr>
<tr><td>A.17. Is this FP included in the entity work programme?</td><td>Yes ☒ No ☐</td><td>A.18. Is this FP included in the country programme?</td><td>Yes ☒ No ☐</td></tr>
<tr><td>A.19. Complementarity and coherence</td><td colspan="3">Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.
Yes ☒ No ☐</td></tr>
<tr><td>A.20. Executing Entity information</td><td colspan="3">Entity: Save the Children International (SCI)
Country of registration: Mali
Type of organization: Non-for-profit

Entity: Save the Children Federation, Inc. (SCUS)
Country of registration: Unted States
Type of Organization: Non-for-profit

A SCI Mali and SCUS will co-Execute the project and implement in collaboration with an implementing entity - the Environment and Sustainable Development Agency (ESDA – known nationally in French as AEDD).</td></tr>
<tr><td colspan="4">A.21. Executive summary (max. 750 words, approximately 1.5 pages)</td></tr>
</table>

Climate change evidence and problem:

1. Mali is ranked 11th most vulnerable to climate change and 23rd least ready for climate change adaptation and mitigation out of 192 countries in the Notre Dame Global Adaptation Initiative (ND-GAIN index.[3] Over the last 60 years, Mali has experienced the following key climate trends: a mean annual temperature increase of 0.15°C per decade, seasonal variability and decreased precipitation with increasing frequency and severity of climate hazards, particularly droughts and floods across the regions, and related to this is increased heat stress and emerging pests and diseases. Mali is ranked among the countries most vulnerable to drought according to the Intergovernmental Panel on Climate Change (IPCC),[4] with a drought severity index of -5, which is the highest level of drought, based on the Palmer Severity Index.[5] Floods are Mali's most frequent climate hazard, making up 50% of all climate hazard occurrences from 1980 to 2020.[6] It is estimated that 400,000 people are currently affected by droughts each year in Mali with drought risk impacting those in the Mopti and Gao regions in particular[7] and that 500,000 people are currently affected by floods each year in Mali.
2. Under both Representative Concentration Pathway (RCP) 4.5 and RCP 8.5 emissions scenarios, Mali is forecast to experience temperatures increasing up to 3°C in RCP 8.5 in the 2041-2070 period. Projections show that against a baseline historical period of 1981–2010, Mali is likely to experience temperature increases of ~1°C across the project regions in both RCP 4.5 and RCP 8.5 in the 2011-2040 period (2030 projection), rising to ~2°C in the 2041-2070 period (2050 projection) under emissions scenarios RCP 4.5, and nearly 3°C under emissions scenario RCP 8.5.[8] In addition, rainfall events will be more erratic and intense and, within the 5 regions of the project, precipitation is forecast to decrease overall.[9] Increased temperatures are projected to lead to more frequent and intense droughts and extreme heat with "very hot" (daily maximum temperature above 35 °C) days forecast to increase. The erratic rainfall and reduced absorption capacity of the soil from extreme heat are forecast to lead to increased frequency and intensity of floods with shorter, heavier events with the Standardized Precipitation Evapotranspiration Index (SPEI) drought index forecasting a decrease in SPEI score in all Shared Socioeconomic Pathways (SSP) scenarios.[10] Mali has 5 distinct agro-climatic zones that are in large part defined by their variation in rainfall, with a variation of up to ~800mm on an annual basis, resulting in regional variation in the forecast impacts of climate change.[11] For example, the northern region of Tombouctou in the Saharan zone is forecast to experience greater exposure to drought than the southern region of Sikasso in the North Guinea zone (including a worse SPEI score by 0.58 and over 120 more days over 40°C /year under the 2040-2059 projection for SSP 5-8.5).[12]
3. Mali's socio-economic context increases its vulnerability to climate hazards which are present against a backdrop of a Least Developed Country (LDC) with high levels of poverty and food insecurity, dependence on subsistence agriculture, and a fast-growing young population. Just under half of the population lives below the national poverty line and most of the population lives rurally. [13] The highly climate-sensitive agriculture sector employs 80% of the country's workforce and accounts for ~36% of Mali's Gross Domestic Product (GDP) with the dominant products including food crops and livestock which represent ~45% and ~30% of Mali's agricultural GDP respectively.[14,15] [16] The agricultural sector is primarily rainfed subsistence farming and agropastoralism (less than 10% of arable land is under irrigation) with smallholders representing an estimated 86% of total farmers (holding less than 10ha)[17]. Due to the country's reliance on subsistence farming, the impacts of climate change on crop and livestock production have a major impact on food security. Limited income and cash reserves mean that smallholder farmers have limited capacity to respond to changes in food prices or an availability shock.[18] Firewood and charcoal account for ~ 78% of energy use in Mali's households largely collected from natural forests and agroforestry systems which are also highly sensitive to climate change[19].
4. Erratic rainfall already affects the flood patterns of 4.2 million hectares of wetlands in the country and generates a negative impact on soil-stabilizing vegetation growth[20] in 2021, drought induced a 10.5% decrease in cereal production, threatening the livelihoods of over three million people.[21] The impacts of water scarcity on livelihoods have been felt particularly acutely in the North of Mali: droughts led to the drying up of Lake Faguibine, resulting in the disappearance of the irrigated farming lands and the lack of water for livestock.[22] In livestock production, rising temperatures can directly cause heat stress which is a major concern for animals, as it affects their feed intake, milk production, and reproductive efficiency, and it can increase the risk of diseases and parasitic infections in livestock.[23] Together, these impacts compound to result in growing food insecurity: the share of the population estimated to be either facing severe food insecurity (1.3 million) or at risk of doing so (4 million) is projected to hit to 24% in August 2023, up from 15% in early 2023.[24] The levels of acutely malnourished children aged 6-59 months also increased by ~53% from 2020/2021 to 2021/2022 and reached 1.2 million children.[25]
5. Climate change and related hazards are projected to continue and increase the negative impacts on agriculture and natural sources, leading to increasing food and water insecurity and impacting the livelihoods and well-being of agriculture-dependent households in Mali. For example, the land suitability for many crops (e.g. Maize, Millets, Wheat, Banana, and Cotton) is projected to decrease by 2050 due to changes in

temperature and rainfall patterns across Mali[26] In addition, the yield losses are estimated at 51%-57% for maize and 7-12% for millet by 2050 under without adaptation conditions.[27] Temperature rises will also negatively impact livestock production through a decrease in fodder availability for livestock. The fodder availability is predicted to decline between 5%-36% by 2050 depending on the location and type of livestock.[28,29]

6. Mali's total green greenhouse gas (GHG) emissions are rising but remain very low relative to global averages. Emissions have been rising steadily with total emissions estimated at 44 million tons of Carbon Dioxide equivalent CO2eq in 2022.[30] However, Mali's emissions represent only 0.01% of total CO2 emissions worldwide and Mali's per capita emissions (0.19 tons per capita) are approximately only 4% of the global average (4.69 tons per capita).[31] Agriculture is the dominant source of GHG emissions, with an estimated between 70%[32] and 80%[33] of total GHG emissions. Without mitigation efforts, these emissions are projected to increase due to population growth, the expansion of electricity access, deforestation for agriculture and mining, and an increase in agricultural production and mining. In light of this, the government of Mali has submitted revised Nationally Determined Contributions (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC) in 2021 which aims to reduce emissions by 25% in the agriculture sector.[34]

Purpose, activities, and climate results:

7. Intensification of Agriculture and Agroforestry Techniques (IAAT) will support the Government of Mali (GoM) to advance progress on its national climate change adaptation and mitigation priorities and commitments by increasing the climate resilience in agriculture systems and reducing GHG emissions. IAAT will achieve this by working with government, civil society, private sector, and communities to increase capacity and support climate risk-informed decisions, including scaling Locally Led Adaptation (LLA) actions to address vulnerability linked to food and water insecurity. The IAAT will focus on i) improving extension services for Climate Smart Agriculture (CSA) adoption, ii) supporting the development of CSA and agroforestry value chains, iii) reducing GHG emissions from agricultural systems, and iv) increasing institutional capacity and knowledge for climate action. The promotion of CSA and agroforestry technologies will be the entry points for broader agroecological outcomes. The project outcomes will cover climate change resilience, environmental health, gender equity, social inclusion, soil health, biodiversity conservation, and resource use efficiency which provide common points of reference for CSA and agroforestry.

8. The IAAT will be implemented in 5 regions (Gao, Koulikoro, Mopti, Segou, and Tombouctou) and highly vulnerable 12 circles[35] covering 48 communes. Five regions and 12 circles were selected based on national and regional workshops, stakeholder consultations, and climate change vulnerability assessment. IAAT will

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[3] NDC Gain index (2021). Avalable here
[4] Makougoum C. T. P., (2015) changement climatique au Mali : Impact de lasécheresse sur l'agriculture et stratégies d'adaptation
[5] Makougoum C. T. P., (2015) changement climatique au Mali : Impact de lasécheresse sur l'agriculture et stratégies d'adaptation
[6] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here
[7] International Committee of the Red Cross (2021) Country overview Mali - Red Cross Red Crescent Climate Centre. Available here
[8] Using the Climate Information Platform, this analysis has taken the projections for the capital cities for each of the regions (cities of the same name), acknowledging that there will be variation within the region itself.
[9] Potsdam Institute for Climate Impact Research (PIK) (2020), Climate risk profile: Mali - Agrica. Available here
[10] World Bank, (2023), Climate Knowledge Portal, Mali Mean Projections. Available here
[11] Hummel, D., Doevenspeck, M. and Samimi, C. (2012), Climate Change, Environment and Migration in the Sahel, Selected Issues with a Focus on Senegal and Mali (pp.44-49), Available here
[12] World Bank, (2023), Climate Knowledge Portal Mali profile.Multi-Model Ensemble. Available here
[13] https://data.worldbank.org/indicator/SI.POV.NAHC?locations=ML
[14] N'Diaye, I., Aune, J.B., Synnevåg, G., Yossi, H., & Hamadoun, A. (Eds.). (2020) Adaptation de l'Agriculture et de l'Élevage au Changement Climatique au Mali: Résultats et leçons apprises au Sahel. Available here
[15] Douyon A, Worou ON, Diama A, Badolo F, Denou RK, Touré S, Sidibé A, Nebie B and Tabo R (2022) Impact of Crop Diversification on Household Food and Nutrition Security in Southern and Central Mali, Available here
[16] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here
[17] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here
[18] World Bank, (2020), Poverty and Inequality Platform, accessible here
[19] African development bank (2015), Renewable energy use in Africa: Mali country profile. Available here
[20] USAID, (2023), Climate links: Mali, Available here
[21] International Committee of the Red Cross, (2022) Climate change in Mali: "We drilled deep but found nothing", Link
[22] International Committee of the Red Cross, (2021) Climate change turns Mali's Lake Faguibine into desert and forces people to move, Link
[23] Rahimi, J., Mutua, J.Y., Notenbaert, A.M.O. et al. (2020), Will dairy cattle production in West Africa be challenged by heat stress in the future?. Climatic Change 161, 665–685. Available here
[24] IMF (2023) Mali: Selected Issues, Available here
[25] Relief Web, (2022), Breaking the Spiral of the Food and Nutrition Crisis in Mali, Available here
[26] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[27] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[28] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[29] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[30] World Bank (2023) Total greenhouse gas emissions. Available here.
[31] Mali: CO2 profile (2021) Our World in Data using Gloab Carbon Project 2023 data. Land use is not included in the estimation. Available here
[32] CGIAR, Climate Smart Agriculture in Mali, Link
[33] Estimated FAO total emissions from agriculture (Available here) as a percentage of the World Bank (2023) Total greenhouse gas emissions. (Available here). NB that there will be some discrepancy due to differing methodologies employed.
[34] Revised Nationally Determined Contributions (2021) Available here
[35] A circle is the second-level administrative unit in Mali. Mali is divided into eight regions and regions are sub-divided into 49 cercles. Circles are subdivided into 703 communes.

select 48 communes within 12 circles before the start of project implementation based on circle-level workshops and stakeholder consultations. This project will establish circle-level LLA plan development committees composed of officials from the line ministries (e.g., agriculture, forestry, livestock, water, etc.) and representatives from the local government. The committees will provide input into LLA plans, approve activities, and monitor progress as the plans are implemented. The committees will also facilitate the integration of LLA plans into the government's local development programs.

9. IAAT's conflict-sensitive activities are related to training and capacity building, technology transfer (e.g. solar pump and biodigester), and involvement of private sector value chain actors to promote the CSA value chain in the conflict-affected areas. The entry points which are also connection points identified in the conflict analysis are community and social institutions and community structures such as CACs and VSLAs. IAAT will use Albarka Resilience Food Security Activity (hereafter Albarka) approach and lessons to address conflict situations that may hinder the implementation of IAAT's activities. Albarka used CACs and VSLAs as entry points to make it possible to strengthen awareness-raising, information-sharing, and behavior-change activities. Village chiefs and community leaders play key roles in designing and implementing CACs, including conflict resolution. In addition, IAAT will develop and consult the conflict-sensitive IAAT guide for implementing and monitoring activities in the fields.

10. IAAT will cover 431,700 direct beneficiaries, primarily 172,680 smallholder farmers including youth and women agripreneurs, to increase their resilience to current and future risks of droughts, floods, and heatwaves by building on-farm capacity and the supporting market and institutional environment for sustained adoption of climate-smart agriculture farming. In addition, the IAAT will indirectly support a further 700,980 beneficiaries, by working with the public and private extension services to increase technical capacity amongst smallholder farmers across the project regions. Strengthening the capacity of public extension agents will create public goods and foster scaling out CSA and agroforestry in Mali. Results details are presented in Section B, Table B 3.3. The land use impacts associated with the adoption of CSA practices across smallholder farms and public degraded lands, solar-based water pumps, and biodigesters result in a total carbon emissions reduction of 2,446,543 tCO_2eq, equivalent to ~163,103 tCO_2eq per year over 15 years (Annex 22).

11. The two Executing Entities (EEs) are Save the Children International (SCI) Mali, and Save the Children Federation, Inc. (SCUS). Mali regional and local government authorities will also engage in the project implementation. The AEDD (implementing entity) coordinates climate adaptation and mitigation efforts and acts as the secretariat for the National Climate Change Committee (CNCCM). The CNCCM plays a consultative role and facilitates the mobilization of resources and stakeholders for climate-related projects in Mali. Given Save the Children is a child-responsive organization with a Malian presence in remote communities, it's important to acknowledge the differentiated climate impacts (see paragraph 19) and adaptation needs of children in Mali and how the IAAT will benefit children within the small-scale farming households and communities through increased food and water security, reduced open cook fires and potentially local economy prosperity, which positively contribute to health and well-being and safeguarding children's rights, including the right to education.

12. In addition to delivering the 3 Components, the IAAT will also achieve the following co-benefits as listed below. Engagement in national and sub-national levels of stakeholders for adaptation and mitigation planning, enhanced enabling environment, and adoption of a range of CSA and agroforestry technologies will help ensure these co-benefits are sustained and improved over time.

- Co-benefit 1Gender: Increased gender equality and economic empowerment
- Co-benefit 2 Environment: Improved soil health and water resources from land use improvements

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Socio-economic Context

13. Mali has a young and growing population facing significant challenges to improve their livelihoods in the face of climate change. The median age in Mali is 16.3 years in 2023,[36] compared to the World average of 30 years.[37] More than 47% of the population is less than 15 years old,[38] and 40% are aged between 15 and 40.[39] The rapid population growth (3.2% from 2020 to 2021)[40] has outpaced job creation, leading to high unemployment rates, particularly among youth aged 15 to 24.[41] Less than half of this age group's workforce is employed,[42] with an estimated overall youth unemployment rate of 15%, potentially rising to 50% when underemployment is considered.[43] This pervasive unemployment is a significant driver of poverty and poses a threat to social cohesion and food security within communities[44] This situation is exacerbated by various stressors, including security issues and increasing risk of droughts, floods, heatwaves, and pests and diseases in Mali.[45] The population faces significant challenges in coping with an unpredictable environment, largely due to limited access to resources (inputs, equipment, finance, etc.) and limited awareness of and skills related to existing technologies and practices, particularly driven by widespread poverty[46]. Mali's extreme poverty has been on the rise, reaching 19.1% in 2022 from 15.9% in 2021, further emphasized by declining purchasing power and leading to limited capacity to adapt to climate change, especially in rural areas.[47]

14. Vulnerability to climate change is highly linked with the rural populations, which account for 90% of the country's poverty.[48] About 56% of Malians live in villages, surrounded by agriculture, forestry, and grazing lands,[49] with low population density, reaching just one person per km^2 in remote eastern and northeastern regions[50]. The country's arid and flat landscape, comprised of plains and plateaus, limits agricultural productivity.[51] While the Senegal and Niger rivers provide water resources, lands outside the Niger valley are often barren or desert, making agriculture difficult.[52] Recurring droughts and the resulting increase in migration, especially from herders,[53] have led to a concentration of people along the Niger River, where more agricultural opportunities and livelihood prospects exist,[54]. The seasonal floods in some sections of the Niger River and the rich alluvial soil in the central delta provide fertile agricultural soil and pasture for livestock.[55]

15. The combination of insecurity and a highly variable climate context threatens the livelihood resilience of these vulnerable populations in Mali, particularly for its young and rural populations[56]. Mali has been plagued by conflict and instability for several decades, leading to deteriorating security and socio-economic conditions. The fragile state of the country has weakened governance, increased inequality, and caused heightened insecurity, prompting communities to organize for self-defense. The crisis has resulted in 3,500 battle-related deaths across 2012-2020, 412,000 internally displaced people as of December 2022, and an estimated 3.9 million people in need of protection assistance with many at risk of human rights violations.[57,58,59] Climate change exacerbates the conflict, as competition for scarce resources fuels tensions and weakens social bonds, especially between livestock herders and sedentary farmers.[60] The conflict has disrupted public services, destroyed homes and infrastructure, and severely affected the allocation and deployment of resources and technologies in climate-sensitive locations,

[36] Worldometers, (2020) Mali Population, Link
[37] Wisevoter, "Median age by country", Link
[38] Statista, (2021), Mali: Age structure from 2011 to 2021, Link
[39] FAO, (2017), Youth employment in Mali, Link
[40] World Bank, (2021) Population growth (Annual percentage – Mali), Link
[41] ICCO Cooperation, (2018) Youth employment in Mali, Link
[42] Ibid.
[43] ARCO, (2020) Market analysis to foster employment of young people in Mali, Link
[44] FAO, (2017), Youth employment in Mali, Link
[45] USAID, (2017), Decentralized Governance And Climate Change Adaptation, Link
[46] Ibid.
[47] World Bank, (2023), "Mali Presentation", Available here: Link
[48] Ibid.
[49] Britannica (2023), Country Profile: Mali, Link
[50] Ibid.
[51] Ibid.
[52] Ibid.
[53] Ibid.
[54] Ibid.
[55] Ibid.
[56] USAID, (2017), Decentralized Governance And Climate Change Adaptation, Link
[57] European Commission, (2023), Mali Fact Sheet, Available here
[58] Care International, (2021), Mali sees highest levels of displacement in its recent history due to a dangerous combination of conflict and climate change, Available here
[59] World Bank, (1990-2020), Battle-related deaths (number of people) – Mali, Available here
[60] Africa Center for Strategic Studies, (2021), The Growing Complexity of Farmer-Herder Conflict in West and Central Africa, Available here

significantly impacting vulnerable groups, including women and youth.[61] Recent developments indicate a potential escalation in the conflict, despite counter-terrorism efforts by Malian forces.[62]

16. Poor access to land for agriculture and agroforestry exacerbates conflicts between rural communities. Although the 2017 Agricultural Land Law aimed to provide farmers with increased access to land ownership, its implementation at the local level has been inconsistent, resulting in uncertainty regarding long-term land ownership.[63,64,65] Some studies indicate that in some locations, the lack of law enforcement and ambiguity regarding land ownership contributes to disputes and heightens tensions between communities, particularly between sedentary farmers and transhumant pastoralists, who compete for access to water and pasture.[66] But currently, land ownership rights are well executed by the Territorial Collectives (sub-national government structures in Mali) and local governments. Insecurity regarding land tenure affects one-third of Mali's population, stemming from factors such as low literacy rates, limited knowledge of land registration processes, and financial constraints.[67] Women face additional challenges in accessing land rights despite the presence of equitable policies and laws,[68] as usufructuary rights are typically allocated by male heads of settler families, disadvantaging women.[69]

17. Access to finance is also a major challenge hampering the capacity to mitigate and adapt to climate change. While 83% of households have access to formal financial institutions,[70] the expansion of financial services has mostly been driven by microfinance institutions and mobile network operators.[71] People in rural areas, including farmers, are highly underserved. More than 70% of loan requests submitted by smallholder farmers are rejected in Mali.[72] Banks are concentrated in urban areas and cater to large firms, leaving rural areas, low-income earners, and Small and Medium Enterprises (SMEs) underserved.[73] Micro-finance institutions (MFIs) were established to address this gap but have encountered difficulties, including limited financial resources and exposure to insecurity. The prioritization of a small customer segment by the banking system, combined with the challenging conditions faced by microfinance institutions, has led to the development of alternative solutions to improve financial inclusion in Mali, such as cooperative networks and Village Savings and Loans Associations (VSLAs). Cooperative networks emphasize collective ownership and local management[74], while VSLAs promote community-based microfinance and empowerment of women.[75] To address climate change challenges, the Mali Climate Fund (MCF) supports policy implementation and resilience capacities, mobilizing financing, promoting clean technologies, and fostering partnerships.[76]

18. Gender inequalities in Mali are deeply entrenched and pervasive, as evidenced by the country's low ranking of 155th out of 170 countries on the UNDP Gender Inequality Index.[77] Despite several reforms, significant socio-economic and political barriers persist.[78] Women and girls face structural inequalities and societal norms that hinder their health, education, participation in governance, and social and economic independence.[79] Their representation in decision-making processes is disproportionately low, including fewer high-profile public positions and under-representation in government roles.[80] Women's labor force participation is also significantly lower than men, with a notable gap in skilled jobs. In Mali, the labor force participation rate among females is 54.2%, and among males is 80.5% for 2022.[81] Educational attainment for women is among the lowest globally, with limited access to literacy and numeracy skills.[82] Only 26% of adult women were literate in 2018.[83] In rural areas, women are predominantly viewed as farmers responsible for feeding their households, lacking decision-making power and facing challenges in accessing advisory services and acquiring knowledge about production technologies and markets.[84] Unpaid domestic work, combined with inadequate infrastructure and services, further restricts their time for business activities. Access to finance is also a critical constraint for women entrepreneurs, with limited

[61] Weathering Risk, (2022), Mali: How climate and environmental change compound conflict and inequality, Available here
[62] Security Council Report, June 2023 Monthly Forecast : Africa – Mali, Link
[63] World Agroforestry, (2023), Agroforestry: A pathway to harnessing nature-based solutions for ecosystems and livelihoods in Sub-Saharan Africa, Available here
[64] Land portal (2022), Mali – context and land governance, Available here
[65] Land links (2020), Country profile: Mali, Available here
[66] Land links (2020), Country profile: Mali, Available here
[67] Land portal (2022), Mali – context and land governance, Available here
[68] Land links (2020), Country profile: Mali, Available here
[69] Land portal (2022), Mali – context and land governance, Available here
[70] BCEAO, Financial Inclusion in Mali
[71] Fowowe B., Financial Inclusion, Gender Gaps and Agricultural Productivity in Mali, Link
[72] World Bank, (2015), Improved Access to Credit Helps Boost Agricultural Production in Mali, Link
[73] Fowowe B., Financial Inclusion, Gender Gaps and Agricultural Productivity in Mali, Link
[74] Government of Canada, Inclusive finance reaching financially excluded persons in Mali, Link
[75] Care International, The resilience champions, Link
[76] Mali Climate Fund, Presentation, Link
[77] IMF, (2023), Fragility, Demographics, Gender Inequality, Available here
[78] AGRA, (2021), Value4Her Synthesis report, Available here
[79] UN Women, Women in Mali, Available here
[80] IMF, (2023), Fragility, Demographics, Gender Inequality, Available here
[81] World Bank, Gender Data Portal: Mali, Available here
[82] AGRA, (2021), Value4Her Synthesis report, Available here
[83] AGRA, (2021), Value4Her Synthesis report, Available here
[84] AGRA, (2021), Value4Her Synthesis report, Available here

collateral, business management knowledge, and confidence in financial negotiations.[85] Financial institutions often lack gender-responsive products and services, contributing to insufficient capital support.[86] Furthermore, despite land tenure policies granting women the right to own land, women still encounter difficulties in controlling agricultural land and acquiring necessary resources for agribusiness. These multifaceted challenges hinder women's capacity to engage in revenue-generating activities, exacerbating gender disparities in entrepreneurship and business ownership.[87]

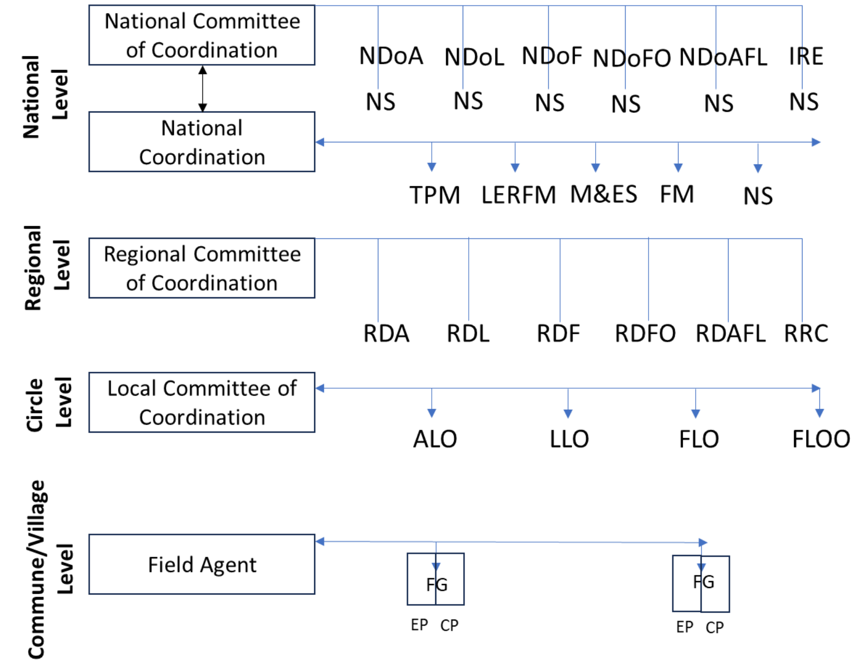
19. In Mali, nearly five million children urgently require assistance to address their food security, health, nutrition, and education needs[88]. Newborns face the prospect of 9.9 times more crop failures than their elders[89]. The escalating impacts of climate change have significantly heightened the risk of moderate to severe food insecurity, posing grave challenges for children in particular. Decreasing agricultural production and rising food prices due to climate change further threaten farming families and vulnerable households, exacerbating food insecurity. For children, diminished food supplies have profound consequences, hindering their mental and physical development and educational progress. Farming families and households facing crop losses, seed/food inflation and debt often resort to desperate measures such as forced early child marriages for girls or withdrawing children from formal education to support the household by participating in farming/income generating activities. For children in remote and rural communities, vulnerability is heightened further due to limited access to resources, services and support. This critical situation underscores the urgent need to enhance climate adaptation capacity in agriculture and related sectors. Strengthening these systems is vital to bolstering the local food supply and ensuring the food and nutrition security of Mali's children.

Mali's Agricultural Extension System

20. The organizational chart for the management of agriculture extension in Mali is presented below (Figure B 1.1). Mali has three levels of administrative offices- national, regional and circle level. There are 8 regions, 49 circles, and 703 communes. Communes are further divided into villages and circle-level administrative offices oversee the communes and villages. The agriculture extension in Mali operates at the national, regional, and local (circle) levels. At the national level, the National Committee of Coordination (NCC) and National Coordination (NC) liaise with various national directors (agriculture, fisheries, livestock, functional literacy, and farmer organization) and the Institute of Rural Economy to promulgate the activities of agriculture extension. The Regional Committee of Coordination (RCC) and the Regional Director of Agriculture (RDA) work with various desk officers of agriculture, fisheries, livestock, functional literacy, farmer organization, and research to manage and supervise agriculture extension activities at the regional level. At the circle level, the agricultural office manager (AOM) and the Local Committee of Coordination (LCC) work with the local officers for livestock, forestry, and farmer organizations to ensure the effective delivery of extension to the farmers in the villages. In addition, various private sector institutions also offer agriculture extension and advisory service-related training, with content and methods determined by the National Directorate for Technical and Vocational Education under the Ministry of Education[90]. For example, The Agro-Pastoral Training Center of Segou, the Professional Training Center for the Promotion of Agriculture in the Sahel, and The Rural Polytechnic Training Center, offer training programs for agropastoral technicians. Agriculture universities and colleges also offer training on extension and advisory services to young professionals and farmers. Other private sector actors such as agriculture input suppliers, equipment sellers, contractors, transporters, aggregators, traders, importers, exporters, and others also offer extension and advisory services.

Figure B 1.1: Structure of agriculture extension system in Mali

[85] AGRA, (2021), Value4Her Synthesis report, Available here
[86] AGRA, (2021), Value4Her Synthesis report, Available here
[87] AGRA, (2021), Value4Her Synthesis report, Available here
[88] WFP. 2023. Food Security in Mali. Available here.
[89] Save the Children, 2021, Born into the climate crisis. Available here.
[90] Feed the Future (USAID). 2018. Mali: In-depth assessment of extension and advisory services. Available here.



NDoA: National Directorate of Agriculture,
NDoL: National Directorate of Livestock,
NDoF: National Directorate of Fisheries,
NDoFO: National Directorate of Farmers Organization,
NDoAFL: National Directorate of Literacy and Functional Languages,
IRE: Institute of Rural Economy,
NS: National Supervisor,
TPM: Training Program Manager,
LERFM: Link Extension-Research-Farmer Manager,
M&ES: Monitoring and Evaluation Specialist,
FM: Financial Manager,
RDA: Regional Directorate of Agriculture,
RDL: Regional Directorate of Livestock,
RDF: Regional Directorate of Forestry,
RDFO: Regional Directorate of Farmers Organization,
RDAFL: Regional Directorate of Literacy and Functional Languages,
RRC: Regional Research Center,
ALO: Agricultural Local Office,
LLO: Livestock Local Office,
FLO: Forestry Local Office,
LFOO: Local Farmers Organization Office,
FG: Farmer Group,
EP: Experimental Plot,
CP: Control Plot

21. The field agent and farmer groups are key elements of the agriculture extension system at the village level. The field agents, therefore, need to be well-trained in diverse areas of agriculture/agroforestry and constantly assisted by technical specialists according to activities to be implemented to ensure extension delivery at the village level. The field agents set up experimental and control plots to demonstrate proven technologies to farmer groups. The performance of any extension system must be flexible to constantly reflect the socioeconomic and politico-cultural contexts and the environmental practices of producers. The agriculture extension, therefore, focused on organizing an extension system, building partnerships, practical technology transfer to field staff members and producers, the linkage between extension, research, and farmers, program planning, involvement of women in programs, and increasing production and productivity of farmers.

22. Training and knowledge transfer under the agriculture extension are planned, structured, organized, and implemented using the “training and visits" model of extension delivery in Mali. There are annual program review workshops, monthly technology review workshops, fortnight training, specific training, and experience exchange visits. These training schemes are implemented after situational analysis with beneficiaries to identify, select and prioritize issues of concern to producers. The field agent is the essential link in this system because he/she is the implementer of programs who comes in direct contact with farmers. The field agents should be multifaceted to meet the diverse demands of producers in areas such as business development, production, storage, processing, and marketing. field agents are therefore trained in many areas and supervised to transfer knowledge in diverse areas of agriculture to producers.

Climate Context

23. At the national level, Mali has high exposure to climate hazards and is ill-equipped to adapt to climate change. Mali is ranked 175 in vulnerability to climate change and 157th in readiness to adapt to the impacts of climate change, out of 197 countries in the ND-GAIN index.[91] Mali has increased its overall ND-GAIN ranking since 1995 slipping from 172 to 176. The impacts of climate change arise both directly through climate drivers and hazards (key drivers include temperature and precipitation, and hazards include droughts and floods), and as secondary impacts e.g. resource-driven conflict. The following sections outline Mali’s climate trends including current and future climate risks, their impacts on livelihood and food security, and the country’s current GHG profile.

Climate context – historical and current trends

Climate Drivers

24. ***Temperature:*** Mali is one of the hottest countries in the world with an average mean temperature of ~28°C. Temperature is seasonal with the coldest average temperatures of around 22°C in December / January in the dry season and the hottest average temperatures in May / June at 34°C, as precipitation increases entering into the wet season.[92] This results in widespread extreme heat events with the majority of Mali (in terms of geography)

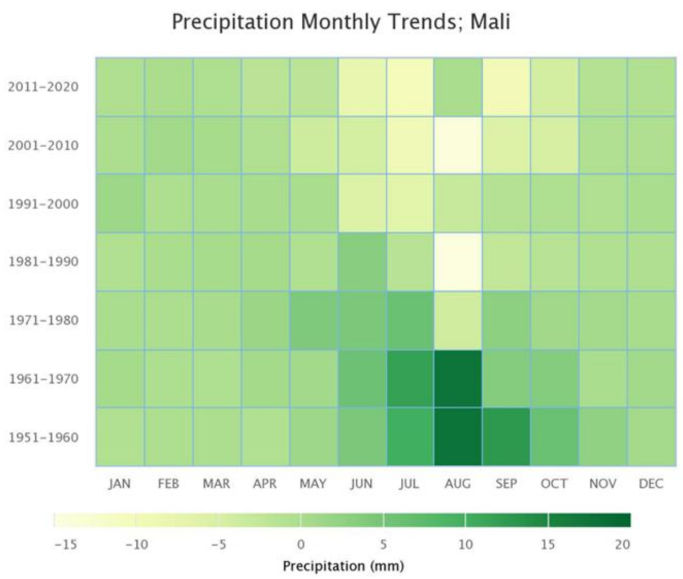
[91] NDC Gain index. Available here.1.2
[92] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here

experiencing >200 'very hot days' per year.[93] Since the 1960s two trends have been evident in temperature change. Firstly, the mean annual temperature in Mali has increased by 0.7°C, a rate of approximately 0.15°C per decade, with higher increases experienced in the northern parts of the country. Secondly, these temperature increases are not linear and display multi-decadal variability, as evident in **Error! Reference source not found.**

Figure B 1.2: Observed average annual mean temperature of Mali (1901-2021) [94]

25. ***Precipitation****:* There is a common annual rainfall cycle with little or no rainfall in November-March, which then increases until the peak of the rainy season in August and then decreases sharply. However, precipitation levels vary significantly across Mali with the north of the country experiencing low rainfall of <100mm annually, compared to the southern-most parts of the country which experience over ten times that amount.[95] Trends in precipitation are hard to characterise as rainfall in the Sahel is highly variable on both annual and inter-decadal time scales.[96] However, examining historic precipitation data shows that the seasonal distribution of precipitation has changed over time with the wet months (June, July, August, and September) getting drier since the 1950s making the seasons less pronounced. Secondly, in spite of interdecadal variation, there has been a general trend of decreased precipitation in all regions in Mali since the 1950s.[97]

Figure B.1.3: Monthly precipitation trends in Mali by decade (1951-2020)[98]



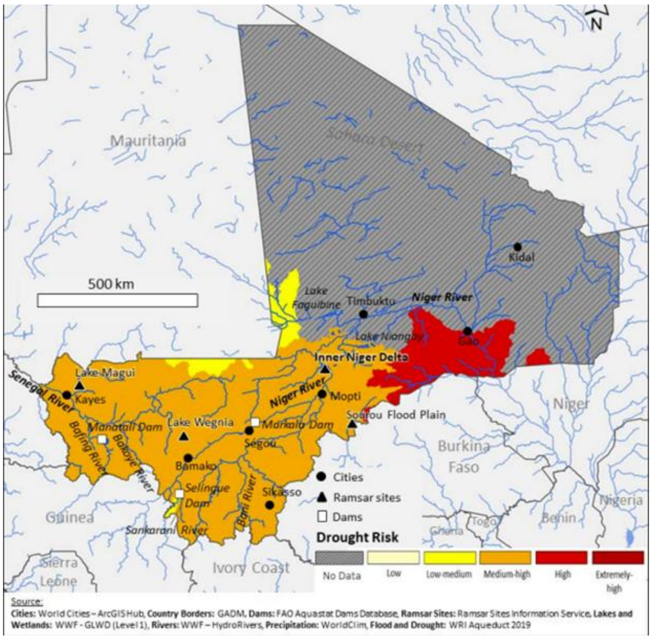
Climate Hazards

26. ***Droughts:*** Droughts are increasing in occurrence and intensity in Mali, this is especially evident during recent decades. Mali is ranked among the countries that are most vulnerable to drought according to the Intergovernmental Panel on Climate Change,[99] with a drought severity index of -5, which is the highest level of drought, based on the Palmer severity index.[100] At present, it is estimated that four hundred thousand people are

[93] Defined as days with a daily maximum temperature of >35°C by the Potsdam Institute for Climate Impact Research (PIK) (2020), Climate risk profile: Mali - Agrica. Available here
[94] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here
[95] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here
[96] International Committee of the Red Cross (2021) Country overview Mali - Red Cross Red Crescent Climate Centre. Available here
[97] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here
[98] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here
[99] Makougoum C. T. P., (2015) changement climatique au Mali : Impact de lasécheresse sur l'agriculture et stratégies d'adaptation
[100] Makougoum C. T. P., (2015) changement climatique au Mali : Impact de lasécheresse sur l'agriculture et stratégies d'adaptation

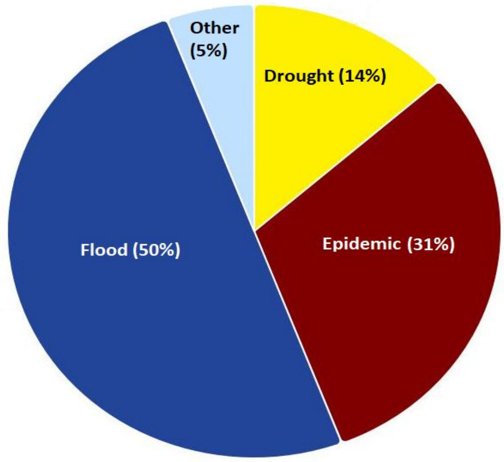
currently affected by droughts each year in Mali with drought risk impacting those in the Mopti and Gao regions in particular.[101]

Figure B.1.4: Figure B.1.4: Drought risk in Mali. USAID 2021[102]



27. ***Floods:*** Floods are the most frequent climate hazards in Mali making up 50% of all climate hazard occurrences from 1980 to 2020.[103] Mali is principally affected by floods during the rainy season each year, most significantly in several localities in the inner delta of the Niger River, generally from August to October or early November. It is estimated that 500,000 people are currently affected by floods each year in Mali.

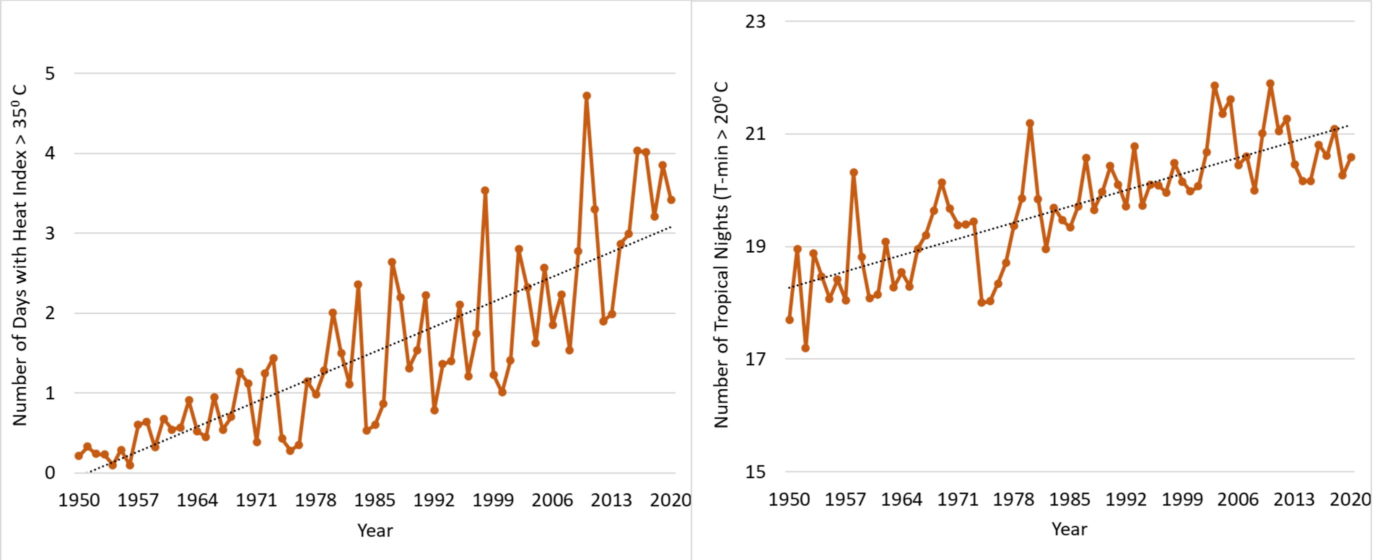
Figure 1.5: Average annual natural hazard occurrence in Mali (1980-2020[89]*)*



28. ***Heat stress:*** Average annual number of days with heat index > 35⁰C between 1950 and 2020 has significantly increased in Mali (Figure 1.6 left). Similarly, the average annual number of tropical nights (T- min > 20⁰ C) has also increased (Figure 1.6 right). These historical trends show that both maximum and minimum temperatures are gradually increasing in Mali. This is gradually changing the suitability of crop and livestock production across Mali with increasing heat stresses.

Figure 1.6: Annual number of days with Heat Index > 35⁰ C (Left) and number of tropical nights, T-min > 20⁰ C (Right)[89]

[101] International Committee of the Red Cross (2021) Country overview Mali - Red Cross Red Crescent Climate Centre. Available here
[102] Winrock International, (2021), Mali Water Resources Profile Overview, Available here
[103] World Bank, (2023), Climate Knowledge Portal Mali profile. Available here



Climate context – projected trends of temperature, precipitation, and corresponding hazards

29. Climate change projections for Mali are disproportionately difficult to generate and subject to challenge. This is in part due to data availability constraints including a lack of accurate domestic data sources, gaps in the historical record, and funding constraints to improve data collection.[104] In addition, given the multi-decadal variability in temperature and precipitation, as well as a limited understanding of the impact of aerosols on projected temperatures and precipitation levels,[105] projections vary significantly between models. In this analysis, the Representative Concentration Pathways (RCPs) RCP 4.5 and RCP 8.5 are predominantly used to analyse climate trends. The projections for 2030 (2011-2040) and 2050 (2041-2070) are used as more distant time horizons are subject to increasing variability and challenge, for example the 2080 projection.

30. ***Temperature:*** Under both RCP 4.5 and RCP 8.5 emissions scenarios in the 2030 projection time frame, Mali is likely to witness temperature increases of ~1°C across the project regions. This rises to ~2°C under emissions scenario RCP 4.5, and nearly 3°C under emissions scenario RCP 8.5 for the 2050 projection time frame as displayed in Table B.1.1.

Table B.1.1: Temperature projections: increased average medium temperature (°C) under different time series and emissions scenarios against baseline historical period of 1981-2010, by region [IAAT project regions only]

Region	Projection for 2030		Projection for 2050	
	RCP 4.5	RCP 8.5	RCP 4.5	RCP 8.5
Gao	1.04	1.11	1.90	2.78
Koulikoro	0.95	1.11	1.83	2.64
Mopti	1.04	1.20	2.02	2.85
Segou	1.01	1.25	2.05	2.91
Tombouctou	1.02	1.21	1.88	2.70

31. Alongside temperatures, extreme heat is also projected to rise with the number of 'very hot days' (days with a daily maximum temperature of >35°C) expected to increase across Mali. The forecast increase is most significant in the south of the country (namely Sikasso, Bamako, and parts of Kayes and Koulikoro) where ~50-75 additional very hot days per year are projected under RCP 6.0 by 2050.[106]

32. ***Precipitation****:* Precipitation projections are inconclusive due to i) variability across different regions and ii) inconsistency in directional trends between under RCP 4.5 versus RCP 8.5 scenarios in the Projection for 2050-time frame. This is primarily due to the high uncertainty and year-to-year variability, which is demonstrated by the range of projections under different climate models. However, the projections of the ensemble model are outlined in table B.1.2. In the 2030 projection, precipitation is expected to increase in all regions of the IAAT project with the percentage increase ranging from 1% to 10% in both RCP 4.5 and RCP 8.5. There is less consistency across

[104] International Committee of the Red Cross (2021) Country overview Mali - Red Cross Red Crescent Climate Centre. Available here
[105] Schewe J., Levermann A., (2022), Advancing Earth and Space Science, Vol 49, Issue 18. Available here
[106] Potsdam Institute for Climate Impact Research (PIK) (2020), Climate risk profile: Mali - Agrica. Available here

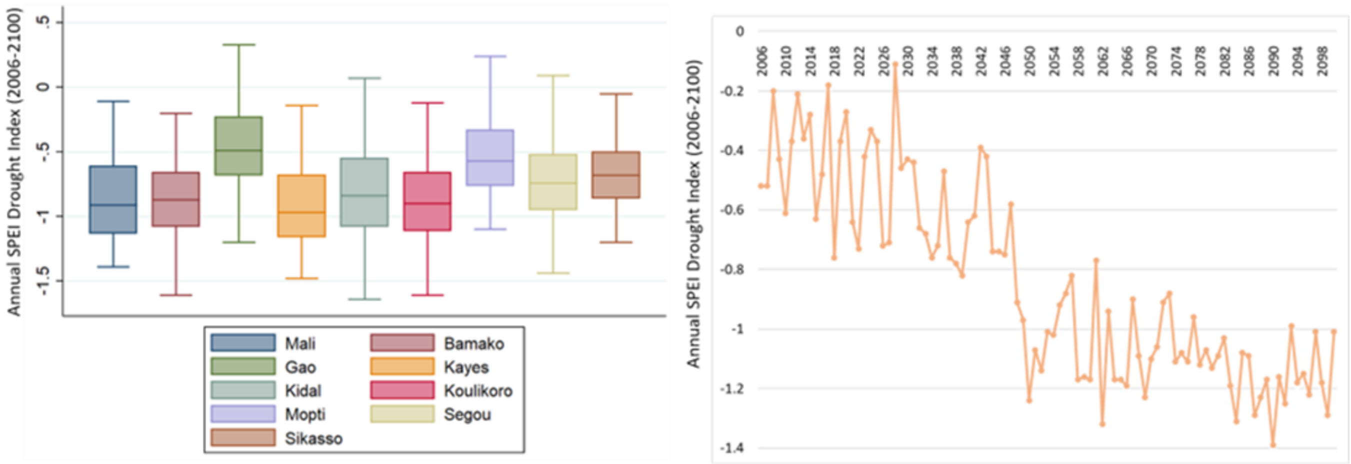
regions in the 2050 projection: in the Koulikoro and Segou regions, precipitation is expected to decrease under the RCP 4.5 scenario but increase or stay the same under the 8.5 scenario. The opposite trend is projected in the Mopti region with precipitation expected to stay the same under the 4.5 scenario but increase under the 8.5 scenario. The Tombouctou and Gao regions display the same trend as in the projection for 2030 with precipitation expected to increase under both emissions' pathway scenarios and to a greater degree under scenario RCP 8.5. Despite limited consensus on projected levels of precipitation across Mali, there is however broad consensus that the rainfall events will be more erratic and more intense.[107]

Table B.1.2: Precipitation projections: change in average precipitation from baseline (1981-2010) under different time series and emissions scenarios by regions.

Region	Projection for 2030		Projection for 2050	
	RCP 4.5	RCP 8.5	RCP 4.5	RCP 8.5
Gao	6%	10%	4%	17%
Koulikoro	3%	2%	-1%	3%
Mopti	2%	3%	0%	-1%
Segou	4%	3%	-2%	0%
Tombouctou	1%	10%	4%	8%

33. ***Droughts***: Drought projections rely on precipitation projections, as well as projections of temperature, evapotranspiration, and soil moisture. The inability to accurately project precipitation means that drought projections are also inconclusive.[108] It should be noted that some other models predict almost no change in the national crop land area exposed to at least one drought per year in response to global warming, others project up to a threefold increase in drought exposure.[109] However, the Standardized Precipitation-Evapotranspiration index (a measure of drought severity by intensity and duration) is summarized below in Figure B.1.7. Panel (a) in the Figure shows the distribution of SPEI from 2006 to 2100 in Mali and its regions. The average SPEI is below 0 for Mali and its regions that indicate more frequent drought across the regions. Only a few years are predicted with access rainfall (above average with SPEI >0) for Gao, Kidal, Mopti, and Segou. Panel (b) in Future B.1.7 presents the predicted average SPEI for Mali from 2006-2100. The average SPEI will be decreasing starting from -0.52 in 2006 to -1.01 in 2100[110]. This information shows increasing severe drought probability across the regions in Mali.

Figure B.1.7: SPEI Drought Index projection from 2006 to 2100- a) distribution of annual SPEI in Mali and its regions, b) average annual SPEI for Mali from 2006-2100[111]



a. Distribution of annual SPEI in Mali and its regions

b. Average annual SPEI for Mali from 2006 to 2100

Note: SPEI value below zero indicates drought conditions and above 0 indicates access rainfall

[107] Potsdam Institute for Climate Impact Research (PIK) (2020), Climate risk profile: Mali - Agrica. Available here
[108] Global Facility for Disaster Reduction and Recovery, Disaster Risk Profile, Mali. 2019, Available here
[109] Potsdam Institute for Climate Impact Research (PIK) (2020), Climate risk profile: Mali - Agrica. Available here
[110] Potsdam Institute for Climate Impact Research (PIK) (2020), Climate risk profile: Mali - Agrica. Available here
[111] World Bank, (2023), Climate Knowledge Portal, Mali Mean Projections. Available here

34. ***Floods:*** High temperatures make soil dry and reduce soil's absorption rate, which increases the likelihood of floods, especially in the face of extreme rainfall events as in Mali's wet season. With temperatures projected to rise, and the trends in precipitation showing increases in certain regions of Mali as well as increased erratic rainfall events, the conditions suggest that floods will either stay the same or get more frequent. This is corroborated by the Think Hazard 'High' rating for both the River Flood and the Urban Flood hazards across all five IAAT regions. [112]

35. ***Pest and diseases:*** Increasing pest and disease infestation in crops and livestock has been observed in recent years across Mali with increasing temperature and changes in rainfall patterns[113]. An assessment of the expected impact of a changed climate on the most common pests and diseases impacting 16 important crops and 7 major livestock types in Mali shows the increasing risk of infestation or outbreak levels for many pests and diseases[114, 115]. These assessments of pest and disease risks in crops and livestock considered the biology and environmental requirements of each pest or disease, the endemic zone, the relative frequency of outbreaks within endemic zones, and infection rates. The assessments indicated that the major effect of climate change on pests and diseases will be through influence on vector, and pathogen transmission, crop/livestock management practices, and land use.

Climate change impacts

Impact on livelihoods and resulting food insecurity.

36. Agriculture in Mali is largely reliant on rainfall and suitable temperatures for the crops and livestock production which is highly vulnerable to the current and forecast changes in temperature, precipitation, and the corresponding climate hazards. This presents a major challenge for Mali as agriculture accounts for 80% of livelihoods[116] and ~36% of GDP[117]. The agricultural sector is predominantly based on rainfed, subsistence agriculture, and agropastoralism, with 86% of farmers being smallholders who work on small parcels of land (generally less than 1.5-3 ha).[118] The dominant sources of agricultural production are food crops (predominantly millet, sorghum, maize and rice) and livestock (predominantly cattle, chickens, goats and sheep) representing ~45% and ~30% of Mali's agricultural GDP respectively.[119] The cotton production contributes a large proportion of export and support about 4 million people in Mali. Due to the significance of agriculture in Mali for employment and food consumption, the impacts of climate change on crop suitability and yield have a significant effect on food security.

37. **Precipitation impacts:** Dependence on rainfall has been a major constraint on crop and livestock production historically, as limited and decreasing precipitation has generated stress and failure.[120] Erratic rainfall already affected the flood patterns of 4.2 million hectares of wetlands in the country and generated a negative impact on soil-stabilizing vegetation growth.[121] In addition, precipitation changes have also impacted the growing patterns of crops, the reduced period of heavy precipitation has resulted in a shortened growing period in Mali and corresponding reduced productivity of cereals across Mali. Water scarcity also negatively impacted the health and productivity of livestock both through reduced direct access to water for livestock consumption, and due to the decreased vegetation and fodder for animal feed. To mitigate against the risk of water scarcity, nomadic pastoralist farmers have been forced to remain near permanent water sources, disrupting their habitual migration patterns. This in turn leads to considerable overgrazing which could further strain livestock production and land degradation.[122] The impacts of water scarcity on livelihoods have been felt acutely in the North of Mali. For example, the prevalence of droughts in Mali led to the drying up of Lake Faguibine in the north of Mali (80 km from Tombouctou), which was the main source of livelihood for the communities.[123] The depletion of Lake Faguibine bears several consequences including the disappearance of the irrigated farming lands and the lack of water for livestock production, leading to disputes between farmers and livestock herders.[124]

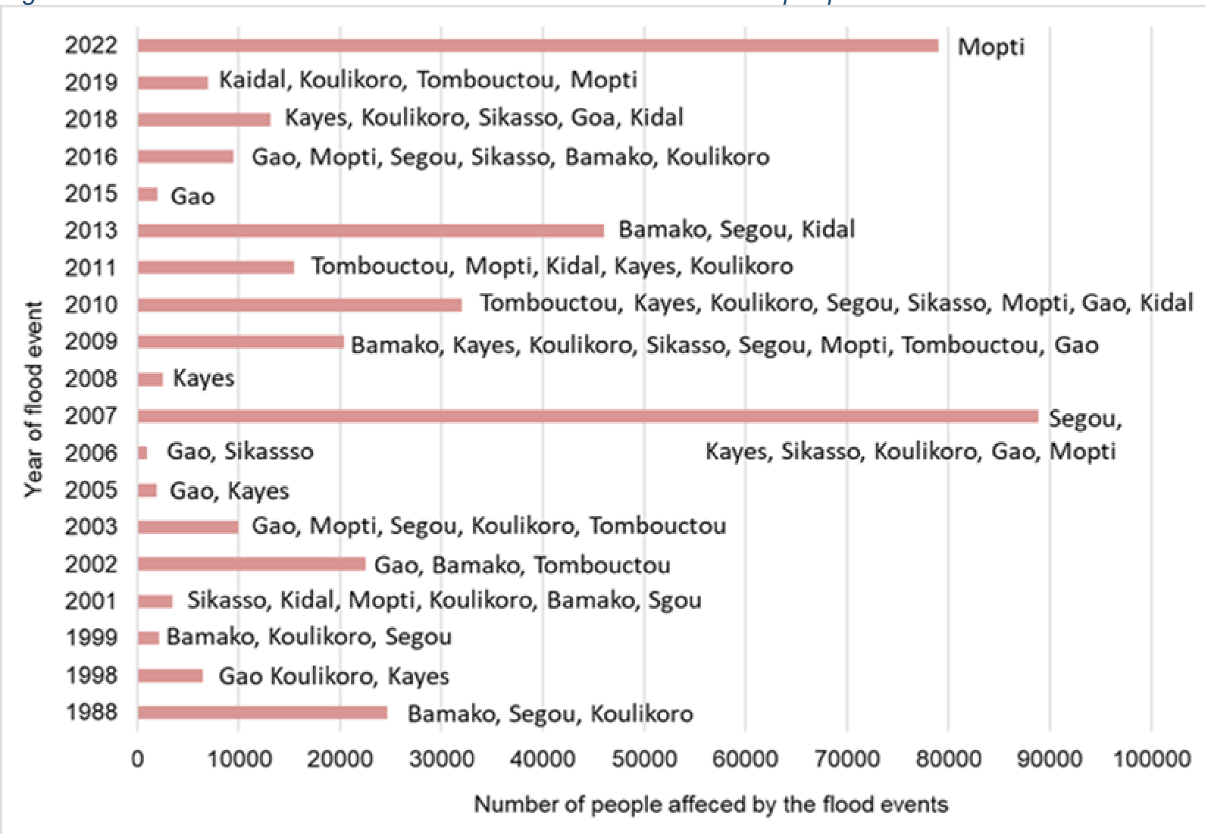
38. **Temperature and drought impacts:** The severity of drought is highly correlated to an increase in temperature in Mali which is associated with an increased evaporation rate, leading to increased water requirements for crop and livestock production. During the last 10 years maize yields across Sub-Saharan Africa have decreased, something attributed to the increased droughts in the 2016-2020 period.[125] In 2021, drought-induced about a 10.5% decrease

[112] Think Hazard Mali profile. Available here
[113] World Bank, (2011). Vulnerability, risk reduction and adaptation to climate change in Mali, Available here
[114] USAID, (2014), Expected impacts on pests and diseases afflicting livestock in Mali. Available here
[115] USAID, (2014), Expected impacts on pests and diseases afflicting selected crops in Mali. Available here.
[116] N'Diaye, I., Aune, J.B., Synnevåg, G., Yossi, H., & Hamadoun, A. (Eds.). (2020) Adaptation de l'Agriculture et de l'Élevage au Changement Climatique au Mali: Résultats et leçons apprises au Sahel. Available here
[117] Douyon A, Worou ON, Diama A, Badolo F, Denou RK, Touré S, Sidibé A, Nebie B and Tabo R (2022) Impact of Crop Diversification on Household Food and Nutrition Security in Southern and Central Mali, Available here
[118] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here
[119] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here
[120] USAID, Climate risks in food for peace geographies: Mali
[121] USAID, (2023), Climate links: Mali, Available here
[122] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here
[123] International Committee of the Red Cross, (2021) Climate change turns Mali's Lake Faguibine into desert and forces people to move, Link
[124] International Committee of the Red Cross, (2021) Climate change turns Mali's Lake Faguibine into desert and forces people to move, Link
[125] Derbile E. K., Bonye, S. Z., Yiridomoh G. Y., (2022), Mapping vulnerability of smallholder agriculture in Africa: Vulnerability assessment of food crop farming and climate change adaptation in Ghana, Available here

in cereal production in Mali, threatening the livelihoods of over three million people.[126] In livestock production, rising temperatures cause heat stress which affects their feed intake, milk production, and reproductive efficiency and increases the risk of diseases and parasitic infections in livestock.[127] Indirectly, temperature rises also negatively impact livestock production through decreased fodder availability which is predicted to decline by 5%-36% by 2040-69.[128] The cereal yield is predicted to decline in the future that will also likely generate major food tensions in Mali, due to their predominance in farms and nutrition representing 68% of all daily caloric intake and 70% of cultivated area.[129] According to the Center of Africa-Europe Relations, the predicted losses in livelihoods due to declining resource availability could generate an overall loss of welfare ranging from USD 70 to 142 million, increasing the share of the population at risk of hunger from 44% to over 70%.[130]

39. **Flood impacts:** Floods and excessive rainfall have created waterlogging and soil erosion in many places of Mali damaging crops and washing away fertile topsoil. Frequent floods are also impacting livestock rearing by decreasing and damaging grazing areas. Figure B.1.8 presents flood incidences and the number of affected people in Mali[131]. The high frequency of severe flood events was observed in Koulikoro, Gao, Mopti, and Segou regions. These events have a direct impact on access to land for agriculture, crop damages, and reduce households' ability to secure inputs e.g. fertilizers, seeds, and machinery, which can further constrain agricultural production. In addition to agricultural damage, the various impacts of floods include the destruction of dwellings, loss of assets, disappearances, and even deaths.[132] The economic damages provoked by flooding can reach USD 600 million per year. In 2020, 6,000 houses and over 7,000 tons of food were damaged by seasonal flooding, affecting 80,000 individuals, mostly women and children.[133]

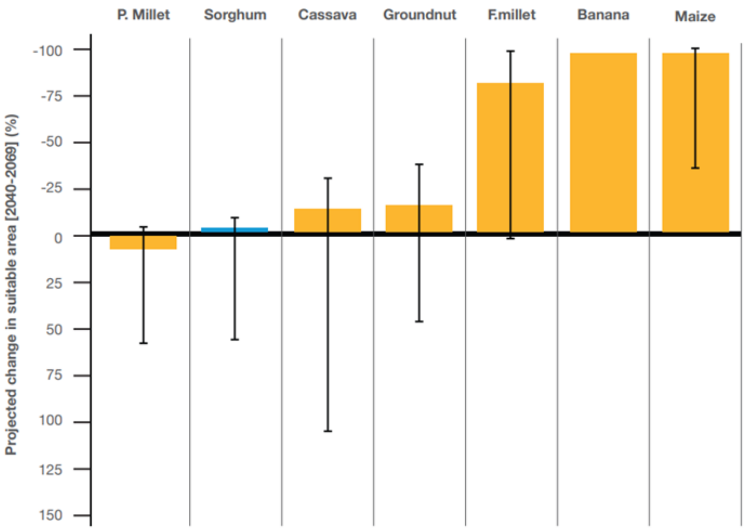
Figure B. 1.8: Severe floods incidence and the number of affected people in Mali[134].



[126] International Committee of the Red Cross, (2022) Climate change in Mali: "We drilled deep but found nothing", Link
[127] Rahimi, J., Mutua, J.Y., Notenbaert, A.M.O. et al. (2020), Will dairy cattle production in West Africa be challenged by heat stress in the future?. Climatic Change 161, 665–685. Available here
[128] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[129] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[130] ECDPM, (2019), The when and how of climate conflict: The case of Mali, (2019), Available here
[131] EM-DAT (2023). Disaster events. Available here
[132] International Committee of the Red Cross, (2021) Climate fact sheets: Mali, Link
[133] SIPRI, (2021) climate-related security risks and peacebuilding in Mali, Link
[134] EM-DAT (2023). Disaster events. Available here

40. It is also forecast that land degradation will accelerate due to the forecast climate changes resulting in an increased negative impact on agricultural production. Figure B.1.8 models the impact of changing climate conditions and land degradation on available agricultural land for different crop types, predicting a decline of suitable land for all key crops in Mali except pearl millet. [135] The extent of land degradation could lead to the incapacity to grow maize, banana, and finger millet by 2040-2069.[136]

Figure B.1.9: Projected percentage change in crop cultivation suitable areas in Mali, 2040-2069 [137]



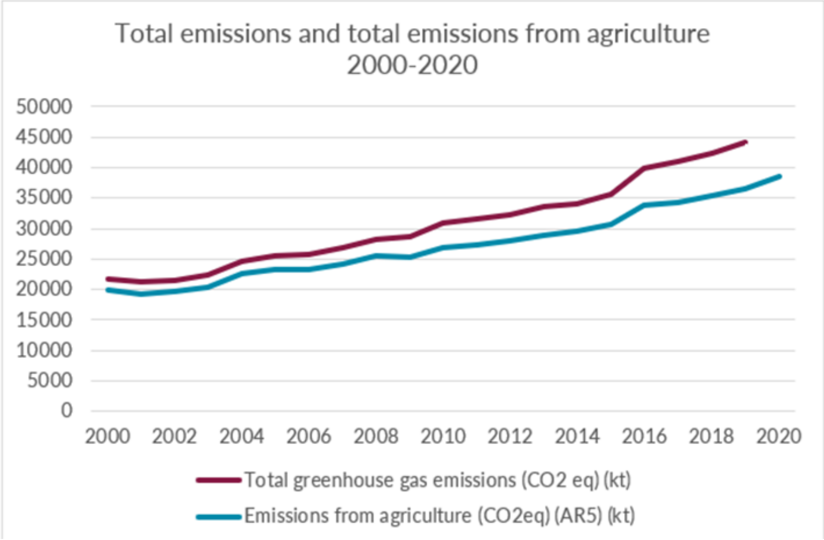
41. **Resulting impact on food security:** To date, the cycle of droughts, floods, and hotter temperatures, as well as the proliferation of pests, has generated severe impacts on crop production, decreasing yields and agricultural land, and reducing the nutritional value of some crops.[138] Droughts have contributed to severe food crises in Mali and the wider Sahel, including those in 1972-1974, 1983-1895, 2002-2003, 2011-2013, and 2015-2018.[139] Reduced production of crops and livestock led to reduced food availability and increased food prices in Mali. For example, due to the 10% decrease in cereal production in 2021, the prices for the main cereals and pulses in 2022 rose by an average of 79% for millet, 95% for sorghum and 55% for maize compared to the same period in 2021.[140] This is particularly significant in Mali as food represents the primary household expenditure. In 2021, food as a share of the household budget reached over 55%,[141] and this figure did not take into account the food produced by a household for personal consumption. Poor food security negatively impacts the health of Malians, particularly children. The share of the population that is undernourished rose from just over 3% in 2017 to almost 10% in 2020[142] whilst the levels of acutely malnourished children aged 6-59 months increased by ~53% from 2020/2021 to 2021/202, reaching 1.2 million children September-2021 to August-2022.[143] Food insecurity further increased in 2023: the share of the population estimated to be either facing severe food insecurity or at risk of doing so is projected to 24% in August 2023, up from 15% in early 2023.[144]

Mali's GHG profile

42. Mali's total greenhouse gas (GHG) emissions are rising but remain very low relative to global averages. Emissions have been rising steadily with total emissions estimated at 44 million tons CO2eq in 2022.[145] However, Mali's emissions represent only 0.01% of total CO2 emissions worldwide and Mali's per capita emissions (0.19 t per capita) are approximately only 4% of the global average (4.69 t per capita).[146] Agriculture is the dominant source of GHG emissions in Mali, with estimates suggesting it accounts for between 70%[147] to 80%[148] of total GHG emissions.

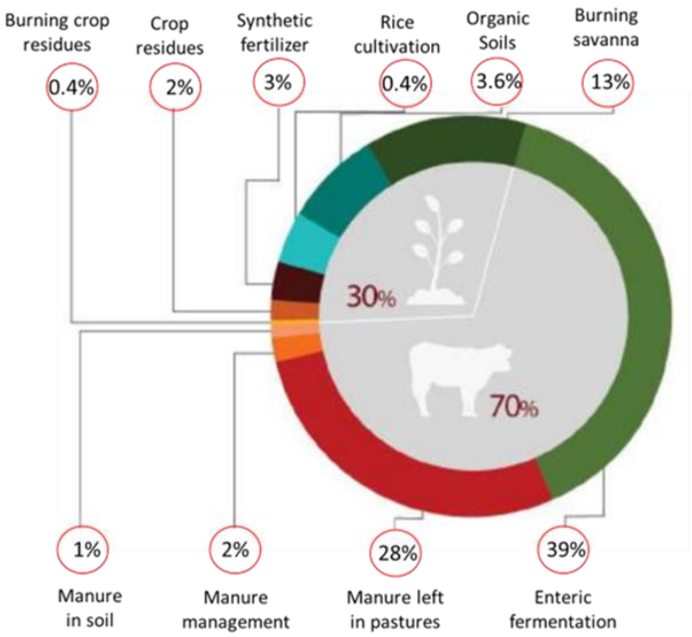
[135] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[136] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[137] World Bank, (2019), Climate Smart Agriculture Investment Plan, Available here
[138] Ibid.
[139] GFDRR, (2019) Disaster Risk Profile Mali, Available here
[140] Relief Web, (2022), Breaking the Spiral of the Food and Nutrition Crisis in Mali, Available here
[141] World Bank Group (2022), Africa's Pulse, No. 25, Available here
[142] IMF (2023) Mali: Selected Issues, Available here
[143] Relief Web, (2022), Breaking the Spiral of the Food and Nutrition Crisis in Mali, Available here
[144] IMF (2023) Mali: Selected Issues, Available here
[145] World Bank (2023) Total greenhouse gas emissions. Available here.
[146] Mali: CO2 profile (2021) Our World in Data using Gloab Carbon Project 2023 data. Land use is not included in the estimation. Available here
[147] CGIAR, Climate Smart Agriculture in Mali, Link
[148] Estimated FAO total emissions from agriculture (Available here) as a percentage of the World Bank (2023) Total greenhouse gas emissions. (Available here). NB that there will be some discrepancy due to differing methodologies employed.

Figure B.1.10: Total GHG emissions and total emissions from agriculture, 2000-2020[149,150]



43. In agriculture, crop production contributes 30% of total agricultural emissions and livestock production contributes 70%[151]. Cultivation of organic soils is a major GHG contributor, with savannah burning attributed to 21% of agricultural emissions. Within the livestock subsector, enteric fermentation is a major emissions source (39% of agricultural emissions), while manure left on pastures follows closely behind (28% of agricultural emissions). Nitrogen emissions are mostly induced by fertilization and the conversion of forest areas. The conversion of forest areas into farms, and the increasing demand of urban centers for charcoal and wood to satisfy household energy needs are speeding the degradation of soils, natural resources, and increasing emissions trends. The GHG emission from fossil fuel use in agriculture is not separately available but included in crop production related emissions estimation. The demand for energy use is continuously increasing with the expansion of underground water pumping based irrigation systems in Mali. Some recent studies present potential of replacing fossil fuel-based irrigation pump by solar pumps as an alternative to increase smallholders' access to energy and reduce carbon footprint[152].

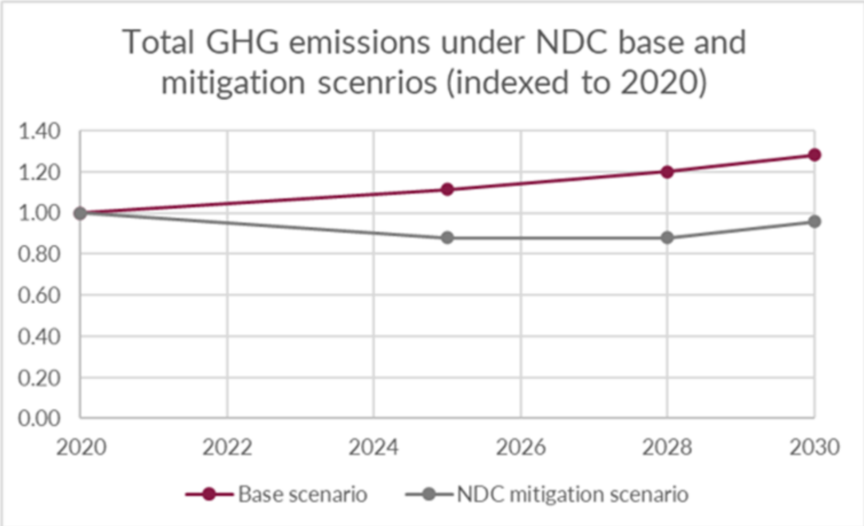
Figure B 1.12: Agricultural greenhouse emissions[145]



[149] FAO, (2023), Total emissions from agriculture, Available here
[150] World Bank, (2023), Total greenhouse gas emissions, Available here
[151] CIAT, ICRISAT, BFS/USAID (2020). Climate-Smart Agriculture in Mali. Available here
[152] IWMI (2019), Suitability for farmer-led solar irrigation development in Mali. Available here

44. Without mitigation efforts, these emissions are projected to increase due to population growth, the expansion of electricity access, and an increase in agricultural production and mining. In light of this, the government of Mali submitted revised Nationally Determined Contributions (NDC) to the UNFCCC in 2021 which include the aim to reduce emissions by 25% in the agriculture sector.[153]

Figure B.1.13: Total GHG emissions under NDC base and mitigation scenarios (indexed to 2020, where 2020 GHG emissions =1)[154]



Limitations in adaptive capacity to adjust to climate change

45. Smallholder farmers have limited capacity to adapt to the current and projected climate change impacts and risks outlined above increasing their vulnerability to climate change. Despite the presence of adaptive technologies and practices in Mali, implementation is limited at the household and community levels with four key themes constraining this ability: the availability and capacity to implement CSA techniques; information availability; availability and accessibility of finance; and institutional capacity.

46. Suitable **CSA techniques** exist in Mali, but their implementation is uneven across the country. For example, suitable techniques include the use of improved and climate-adaptive seeds and breeds; soil and water management practices e.g. Zai pits, irrigation, and field contouring; changing crop/livestock distribution and densities; and farm crop and livestock diversification. However, they have been employed to mixed effect, for example: improved seeds are only used for 23% of rice and 5% of maize cultivation whilst 62% of farmers highlighted the lack of waste and water management as one the main constraints for the implementation of agricultural best practices.[155,156] The main challenges limiting adoption are limited technical capacity amongst smallholder farmers (partially attributed to farmer illiteracy) and limited access to the financial inputs required to invest in these practices.[157]

47. **Access to the market** for climate-smart technologies and other inputs (e.g. climate resilient seeds/breeds, planting materials, livestock feed and health services, solar pumps, micro-irrigation kits, soil amendments, etc) is another challenge for the majority of smallholder farmers in Mali. This indicates farmers' low ability to access other actors of climate-smart technologies and inputs suppliers in the markets. The limitations in promoting climate change adaptation in vulnerable agriculture households and communities in Mali are associated with limited connection to input markets and local service providers, weak private sector involvement to ensure supplies/services and low smallholders' access to output markets.

48. **Information sharing** in the form of early warning systems does exist in Mali, but information is not consistently communicated to the rural farmers in good time, which undermines individuals' ability to deploy resources to act quickly to mitigate climate hazards.[158] In addition, information on which techniques are optimal, and how techniques should be implemented is uneven across Mali. There is limited information available on the costs and benefits of different high-tech and low-tech options and how to best combine them in the context of Mali's

[153] Revised Nationally Determined Contributions (2021) Available here
[154] Ministry of The Environment, Sanitation and Sustainable Development, (2021), Revised Nationally Determined Contributions, Available here
[155] IAAT Field Data collection 2023 – Farmer respondents
[156] INSTAT, (2019), Enquête Agricole de Conjoncture Intégrée aux Conditions de Vie des Ménages 2019-2020, Available here
[157] Ouédraogo et al., (2019), Uptake of Climate-Smart Agricultural Technologies and Practices: Actual and Potential Adoption Rates in the Climate-Smart Village Site of Mali, Available here
[158] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here

climate.[159] This is driven in part by shortcomings of extension services which are under-resourced and ill-equipped to assist farmers with adopting innovative CSA practices.[160] This is partially because extension agents lack up-to-date knowledge, meaning that farmers are trained on technologies and practices that are not 'future fit'.

49. **Access to finance** for smallholder farmers continues to be low in Mali and across West Africa. The use of agricultural credit to purchase inputs is low,[161] with over 70% of farmers in Mali refusing agricultural loans, predominantly due to the perception of high risk.[162] Financial resources are even less accessible to women in large part due to societal norms which mean women rarely own the land required as collateral for financial loans. As a result, adaptive capacity is further reduced for female-headed households.[163] In addition, market infrastructure challenges mean that farmers are constrained both in buying inputs and selling products that make adopting adaptive techniques more attractive.

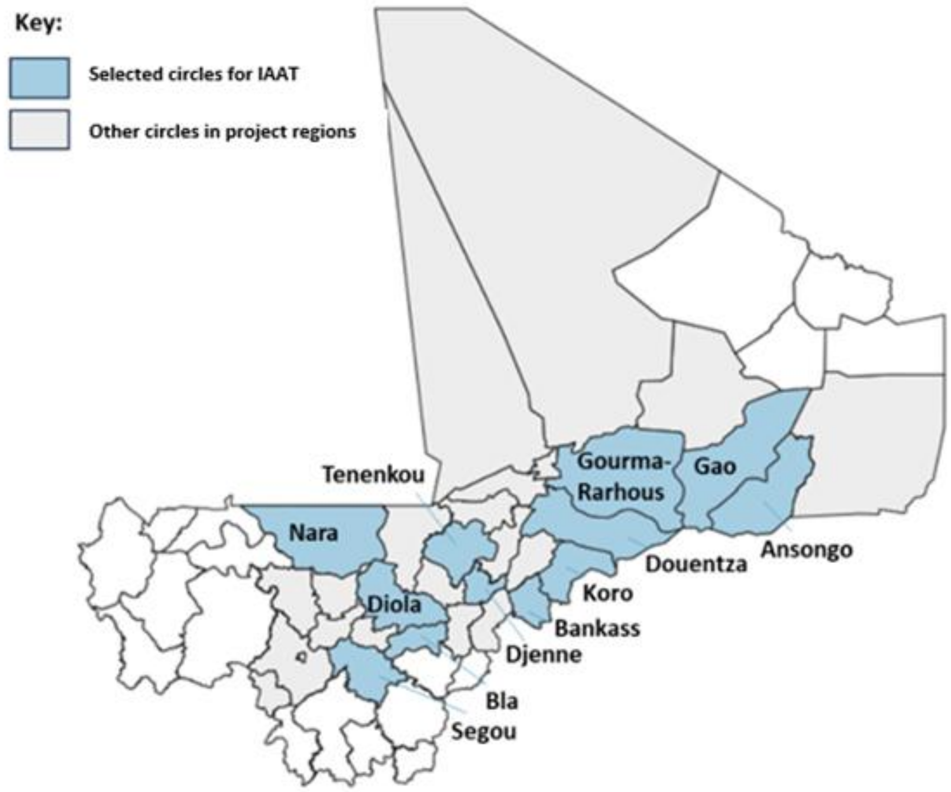
50. **Institutional shortcomings** including poor use of planning mechanisms and a lack of adequate government funding impact smallholder farmers due to the coordination role formal and informal institutions must play to implement adaptive techniques. This is further hindered by the lack of trust in formal institutions, in part derived from backdrop of conflict in Mali. With several adaptive techniques requiring coordination, distrust and inefficiencies of organizations exacerbate adaptation gaps.[164]

Focus areas of IAAT implementation

51. IAAT aims to target the most climate-vulnerable communities within Mali and to subsequently build resilience by supporting 431,700 direct beneficiaries (that include 86,340 farming households and 172,680 farmers) to adapt to climate change and support low-carbon agriculture development pathways. Guided by the IPCC vulnerability framework, the three components of exposure, sensitivity, and adaptive capacity have been assessed to determine vulnerability at the circle level for the five target regions of the project (Gao, Koulikoro, Mopti, Segou, and Tombouctou). [165,166] The full methodology and the output are provided in **Annex 2, Feasibility Study, Section VIII: Vulnerability and Targeting Assessment**. This analysis resulted in the selection of 12 circles across five regions of Mali that have high climate vulnerability and are sufficiently secure to work (at the time of proposal development).[167] IAAT will select 48 communes within the 12 circles based on stakeholders' consultation in the 1st six months of the project start.

Figure B.1.14: Map of selected circles for IAAT

[159] https://www.adaptation-fund.org/wp-content/uploads/2015/01/AFB.PPRC_.6.9%20Proposal%20for%20Mali.pdf
[160] World Bank & FAO, "A Blueprint for Strengthening Food System Resilience in West Africa: Regional Priority Intervention Areas Public", 2021
[161] World Bank & FAO, "A Blueprint for Strengthening Food System Resilience in West Africa: Regional Priority Intervention Areas Public", 2021
[162] CIAT, ICRISAT, BFS/USAID. 2021. Climate-Smart Agriculture in Mali. CSA Country Profiles for Africa Series. International Center for Tropical Agriculture (CIAT); Bureau for Food Security, United States Agency for International Development (BFS/USAID), Washington, D.C. 25 p.
[163] World Bank, 2019, "Mali Climate-Smart Agriculture Investment Plan" https://doi.org/10.1596/32741
[164] https://www.brookings.edu/wp-content/uploads/2022/04/2022_Conflicts-in-Sahel_Case-of-Mali.pdf
[165] Mali is divided into regions, then circles, then communes, then villages. Please see the Mali context chapter for a full explanation of the administrative levels within Mali.
[166] There have been changed to the administrative boundaries in Mali. Due to availability of data, this analysis follows historic boundaries, acknowledging that some of the circles may have changed categorization. In addition, some circles were not selected based on the Save the Children conflict monitoring and suggestions received from the Government of Mali to avoid high conflict locations.
[167] The security situation will be monitored monthly by the Save the Children security expert. Current security assumptions are based on analysis conducted in September 2024.



Related projects/interventions

52. The high vulnerability to climate change and threatened livelihood in Mali have driven development partners to strongly focus on the improvement of mitigation and adaptation capacities, in partnership with the Malian government. Several projects are currently implemented or being designed to enhance the adaptation and mitigation capacities of smallholder farmers, in both agriculture and agroforestry. These projects aim to address the root causes leading to exacerbated exposure and vulnerability to climate change, including limited access to resources (inputs, equipment, finance, technologies), limited development of and access to markets, and limited awareness and adoption of Climate Smart Agriculture (CSA) and agroforestry practices and technologies.

53. Notable examples of ongoing initiatives include the World Bank's West Africa Food System Resilience Program (Phase 2) and the Regional Sahel Pastoralism Support Project II (PRAPS II) and the Africa Hydromet Program (FP012), which aim to increase on-farm adoption of CSA and agroforestry practices among smallholder farmers by strengthening national extension services and developing training and vocational education and training programs for youth and women. Additionally, the International Fund for Agricultural Development (IFAD) is currently implementing the Africa Integrated Climate Risk Management Program (FP162), funded by the Green Climate Fund (GCF), which promotes the adoption of CSA and agroforestry technologies, especially Early Warning Systems (EWS) and related products. The project aims to build institutional capacity for the development of efficient EWS, drive the creation of skilled human capital for the dissemination of EWS, and improve the knowledge of farmers. Financial inclusion and the development of climate-smart value chains for smallholder farmers in agriculture and agroforestry are focal points in projects such as IFAD's Inclusive Finance in Agricultural Value Chain Project (INCLUSIF) and Inclusive Green Financing Initiative (FP183). These projects aim to enhance access to finance while fostering climate-resilient and low-emission agricultural practices. Other projects such as the Multi-Energy Project for Resilience and Integrated Land Management (MERIT) launched by IFAD, focus on the installation of technologies such as biodigesters and/or solar water pumps, to facilitate the productive use of energy (PUE) and enhance the productivity of smallholder farmers. Collectively, these projects exemplify a concerted effort to empower smallholder farmers in embracing climate-smart practices both in agriculture and agroforestry, ensuring financial inclusivity, and providing them with access to sustainable technologies to enhance their overall resilience. Table B.1.3. illustrates potential synergetic areas with other projects in Mali. Further information regarding project complementarity to GCF IAAT is provided in **Annex 2 – Feasibility Study, Section III- Mali Context, Sub-section 3.8. Existing Agriculture and Agroforestry Projects in Mali.**

Table B.1.3: Illustration of potential synergies with existing projects in Mali

Synergetic areas	Illustrative list of projects

Support for dissemination of CSA techniques amongst smallholder farmers	• World Bank – West Africa Food System Resilience Program, Phase 2 • World Bank – Regional Sahel Pastoralism Support Project II (PRAPS II) • FP162: GCF & IFAD – Africa Integrated Climate Risk Management Program (also ref Component 3: Institutional Capacity and Knowledge) • FP012: GCF & World Bank – Africa Hydromet Program
Financial Inclusion and CSA value chain development, with a focus on smallholder farmers	• IFAD – Inclusive Finance in Agricultural Value Chain Project (INCLUSIF) 2019-2024 • FP183: GCF & IFAD – Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries - Phase I
Smallholder farmers access to biodigesters and solar irrigation technologies	• IFAD - Multi-Energy Project for Resilience and Integrated Land Management (MERIT)

54. The GCF IAAT project seeks to bring a meaningful contribution to the ecosystem already created in Mali, by ensuring close alignment with government priorities and collaborating with Non-Governmental Organization (NGOs) and Civil Society Organizations (CSOs) to harness the synergies identified across the respective projects. Conversations with key stakeholders in Mali have driven the iteration of project design to reflect the most pressing needs in the regions targeted by the project and capitalize on existing synergies to create economies of scale and avoid duplication of activities. Despite significant efforts led and improvement achieved by the government and development partners, the adaptation and mitigation gap for smallholder farmers remains significant and requires additional support to progress towards reduced vulnerability to climate change and enhanced livelihood for populations in Mali. The GCF IAAT project aims to create value by adopting an end-to-end approach to the agricultural and agroforestry value chain, providing support to all stakeholders including communities, extension services, private sector, government, etc. A subset of 22 agriculture and agroforestry practices has been prioritized, in accordance with government priorities, and will be promoted through the implementation phase. Furthermore, the project endeavors to foster collaboration, coordination, and experience sharing between development partners, research institutes, government, and all relevant stakeholders, to ensure complementarity between initiatives and create a sustainable market and enabling environment for CSA and agroforestry technologies. This will be achieved through enhanced stakeholder convening and contribution to existing databases.

55. IAAT will bring Save the Children's global experience in Locally Led Adaptation (LLA) to address climate change impacts in agriculture and natural resource management across resilience food security projects including Albarka in Mali. This approach focuses on protecting local rights, integrating scientific and local knowledge, decentralizing and co-creating decision-making opportunities, strengthening capacities, institutions, and leadership, and fostering local ownership to advance climate action. Save the Children used the LLA approach in its resilience food security projects in Vanuatu (FP184), Solomon Islands (SAP027), Sierra Leone (SAP036), Mozambique (SAP042), Niger (Wadata), Mali (Albakra), Nepal (Sabal), Tanzania (Lishe Endelevu), and Ethiopia (Transformation to Food Security). IAAT will also use the experience and expertise of Save the Children, as a child-responsive organization with presence in remote and last mile communities in Mali to advocate and raise awareness on children's health and wellbeing when supporting targeted farmer households and communities through IAAT interventions addressing food and water insecurity.

56. The GCF IAAT project will provide a platform to collaborate with the Albarka project and build upon its results and experience while delivering a cross-cutting project aimed at low-carbon and climate resilient pathways. The Albarka Resilience Food Security Activity project is a USAID-funded project, implemented by a consortium led by Save the Children, aiming to improve food security and resilience of communities in conflict-affected areas in Mali through strengthening local systems and community participation. The project has implemented several activities to improve the resilience of communities, especially farmers, women, and youth, in three of the five regions targeted by GCF IAAT (Gao, Mopti, and Tombouctou). These activities include the establishment and strengthening of strategic organizations for community development and enhanced adaptation and resilience capacities, including Village Savings and Loans Associations (VSLAs) for enhanced access to finance, village-level Community Action Cycles (CACs) to foster ownership of decision-making processes by communities, and Early Warning Groups (GAP-RUs) for the efficient dissemination of Early Warning Systems (EWS) and related products and information. Albarka also aims to enhance the business capacities of women and youth and create sustainable markets for producers. GCF IAAT will leverage the wealth of knowledge developed by Albarka to replicate activities in other communes and regions and emphasize the integration of climate change mitigation and adaptation into existing activities. This will include integrating adaptation and mitigation planning into community action plans and Economic, Social and Cultural Development Plans (PDESC) at the commune level, strengthening the capacity of VSLAs to invest in CSA and agroforestry, strengthening the business capacity and reach of SMEs and entrepreneurs involved in CSA and agroforestry, and creating linkages between multiple stakeholders involved in

the CSA and agroforestry value chains (farmers, cooperatives, private sector, financial institutions, etc.). Albarka will also inform IAAT on the conflict management approach implemented in the conflict-sensitive locations in Gao, Mopti, and Tombouctou regions. Albarka's conflict management approach will be included in the CSA and agroforestry training of the IAAT project. Collaboration with the Albarka project on co-financing is detailed in Section C, and with regards to implementation in Section B.4.

Alignment with national priorities

57. The project is consistent with Mali's climate policies and frameworks specifically targeted toward climate-resilient agriculture and agroforestry, including *National Adaptation Programme of Action* (NAPA-2007), *National Climate Change Policy* (NCCP -2017), *National Climate Change Strategy* (NCCS-2017), *Nationally Determined Contributions* (NDC 2016 and 2021), and *Strategic Axis 4: Protection of the environment and strengthening of resilience to climate change of the strategic framework document for economic recovery and sustainable development* (CREDD2019-2023). The NAPA aims to promote the integration of agroforestry solutions into national climate adaptation measures in response to the need to cope with adverse and irregular climatic conditions like rainfall variability while increasing the productivity of food systems. Similarly, the NCCP and NCCS encourage participatory and decentralized management of natural resources; strengthening national capacities and research on climate change; and promoting climate-smart income-generating livelihood options, sustainable land management, increased access to climate-ready crop varieties, and sustainable irrigation practices. This national ambition for decentralization of natural resource management is aligned with IAAT's LLA approach. The NDC commits to a 25% emissions reduction target for agriculture and stresses the value in conserving soils and nutrient management, promoting climate-smart agriculture and agroforestry, and reducing land degradation as agroecological practices for addressing climate change. Mali's 2021 NDC emphasized creating an enabling environment for agriculture and natural resource management focusing on the following key actions for Climate change adaptation and mitigation,

- Technical support to downscale global and regional climate impact and vulnerability assessments for the national and local levels in a way that can support local and sector planning, including identifying practical adaptation measures.
- Technology and knowledge transfer to support monitoring, forecasting, analysis, early warning, and preparedness for natural hazards in agriculture and natural resource management sectors.

  Investment in capacity building to support planning and management of adaptation and mitigation measures within broader sectoral objectives and plans.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Summary of Proposed Project Approach

58. **This project will contribute to alleviating the key barriers by achieving the following paradigm shift statement**: "**IF** smallholder farmers in highly vulnerable regions in Mali can access and adopt climate resilient technologies, knowledge, and low carbon agricultural practices, **THEN** their food, nutrition and water security will be improved **BECAUSE** their adaptative and mitigation capacities, access to markets and sustainable livelihoods, and that of public and private institutions will be strengthened to respond to and reduce the climate change risk and impacts". Save the Children will use its global evidence-based child sensitive approach to ensure that improvements in income and household resilience will lead to investments that directly improve children health and wellbeing, including nutrition and access to education. **In order to achieve the above paradigm statement, the project will target the following 4 outcomes:**

- Outcome 1: Increased climate-resilient agricultural crop and food production in the targeted regions in Mali
- Outcome 2: Enhanced and more sustainable livelihoods for the most vulnerable in targeted communities
- Outcome 3: Adaptation and mitigation considerations and best practices are embedded in institutional agriculture and agroforestry planning.
- Outcome 4: Reduced GHG emissions from agricultural system.

Barriers Addressed by the Project

59. **The project's outcomes have been designed to alleviate the barriers currently limiting the transition to a low emissions and climate-resilient agricultural system in Mali.** To identify the most pertinent barriers, Save the Children in partnership with the AEDD conducted an analysis (based on a review of existing research and extensive stakeholder consultation) to identify gaps in the current adaptive capacity of smallholder farmers in Mali and the barriers to closing them (**Annex 2: Feasibility Study, Section VI**). In addition, an analysis was conducted to identify the barriers to low-carbon development in Agriculture (**Annex 2: Feasibility Study, Section VII**). This analysis identified the following 7 barriers as the most pressing to overcome when supporting the transition to a low emissions and climate-resilient agricultural system for smallholder farmers in Mali.

60. **Barrier 1: Technical know-how and capacity gaps amongst smallholder farmers to implement and integrate climate-smart agriculture and low emissions land management technologies and practices.** Technical know-how and capacity gaps are one of the largest barriers preventing the adoption of low-emissions and climate resilience agricultural and land management practices, with 60% of farmers surveyed during IAAT field data collection citing "insufficient knowledge" as a barrier to implementing climate-resilient agricultural practices. Although awareness of relevant practices is relatively high amongst smallholder farmers, (various studies and surveys show rates of 60-90%, varying by CSA technique), studies such as Ouédraogo et. al. 2019 found that adoption rates remain low due to a lack of technical capacity to adopt the new practices.[168]

61. **Barrier 2: Limited capacity of public and private extension services at the local level to promote access to inclusive and reliable agro-advisory services on climate resilient practices.** Extension services are the key knowledge dissemination channel for agricultural practices in Mali, but their limited reach and the low quality of education and information provided to smallholder farmers a key factor limiting smallholder adoption of these practices. At present, farmers are underserved by extension services with an estimated ~40% of smallholder farmers reached.[169] For those farmers that can access extension services, the service provided is often ill-equipped to assist farmers with adopting innovative CSA practices, in large part because extension agents lack up-to-date knowledge, meaning that farmers are trained on technologies and practices that are not 'future fit'. For example, there is little to no information available through extension services relating to solar irrigation. In addition, extension services lack an inclusive approach with women and youth systematically excluded. This is driven by a multitude of factors including a lack of representation with women making up only 10-25% of extension agents, and the average age of extension agents being 50 years old.[170, 171]

62. **Barrier 3: Lack of access for smallholder farmers to mature, connected, and transparent local markets and value chains for the sale of CSA compatible crops.** The majority of smallholder farmers in Mali practice subsistence farming due to both lack of access to markets to sell their agricultural products, and value-additive services to support the cultivation of higher-value, and market-ready products. Limited connectivity between value chain actors, and in particular between smallholder farmers and value-added input providers and processing services such as cleaning and grading, means that smallholder farmers generally produce low-value, unprocessed crops. In addition, smallholder farmers have limited connectivity to broader markets for the sale of their crops (either national or export) due to low levels of organization, capacity, and limited road access to urban centers from

[168] Ouédraogo et al, Uptake of Climate-Smart Agricultural Technologies and Practices: Actual and Potential Adoption Rates in the Climate-Smart Village Site of Mali, *Sustainability* 2019, *11*(17),
[169] Alliance for a green revolution in Africa (2021) Agra Mali Report Final. Available here
[170] USAID, "Mali: In-depth Assessment of Extension and Advisory Services", 2018 https://www.digitalgreen.org/wp-content/uploads/2017/09/DLEC-Mali-In-depth-Assessment-Extension-Final.pdf
[171] World Bank & FAO, "A Blueprint for Strengthening Food System Resilience in West Africa: Regional Priority Intervention Areas Public", 2021

rural areas.[172] Farmers also have limited access to information on the fair price of agricultural outputs, for example on how the weight and quality of rice impacts its sales price, in part due to competing routes to market for smallholder farmers (cooperatives, aggregators, wholesalers), many of which lack contracts.[173] Combined, these challenges present a major barrier to cultivating climate-resilient crop varieties and adopting practices because subsistence farming alone does not provide a strong enough incentive to adjust techniques or crop varieties to CSA techniques, without access to the additional income offered from broader markets[174].

63. **Barrier 4: Inability for smallholder and subsistence farmers to invest in inputs for climate change adaptation (seeds, fertilizer, labour, equipment) due to limited financial resources.** CSA and low-emissions farming practices typically require a relatively high upfront investment from the smallholder farmer in comparison to their total income and resources; example investments may include seeds for new climate-resilient varieties and breeds, labour for implementing water management techniques, or purchase of new equipment such as solar irrigation and biodigester systems. This provides a barrier to smallholder farmers in Mali as they typically have limited cash reserves and limited access to finance for loans or alternative funding mechanisms. Just under half of Mali's population live below the national poverty line, with smallholder farmers particularly likely to have low capital.[175] Smallholder subsistence farmers produce crops and rear livestock predominantly for household consumption and generate household income by selling a small proportion of their total production. Seasonal harvesting periods compound this challenge with farmers facing cashflow issues caused by semi-annual harvesting of crops. Few farmers can use credit to purchase inputs with over 70% of farmers in Mali refused agricultural loans, predominantly due to the perception of high risk by the financial institutions. [176,177] In addition, few farmers use risk mitigation techniques that can increase access to finance such as crop insurance primarily due to low awareness.[178] By contrast, research shows that farmers with access to credit are more likely to adopt CSA practices, consistent with the understanding that new practices require investment.[179]

64. **Barrier 5: Limited consideration of the needs of women or youth in the development and management of agricultural and agribusiness systems, leading to significant bottlenecks for their empowerment.** Challenges faced by all smallholder farmers are generally exacerbated for women largely due to the strong patriarchal setting in Mali. Educational imbalances (both formal and informal) and social and cultural norms hinder women in agricultural entrepreneurship. For example, high levels of female illiteracy mean many women lack the tangible skills required to set up businesses or to access services (e.g., extension services or financial service institutions).[180] Social and cultural norms also limit women's ability to engage in agribusiness: cultural decision-making norms have typically excluded women meaning women often lack the confidence to start businesses or expand production; women also often require permission from their husbands to buy or sell outputs and experience reduced decision making power within the family unit.[181] In combination with land-ownership norms (women are less likely to own land than men, and those who do own land often receive smaller and less fertile parcels), women have limited access to the collateral often needed to obtain financing.[182] As a result, few female producers can access agricultural inputs, machinery, and other factors of production required for agribusiness, for example, it is estimated that only 51% of the women engaged in agricultural activities have access to merchant equipment.[183]

65. A similar pattern exists for young farmers in Mali as banks (microfinance institutions included) require collateral, which is often an insurmountable barrier to disadvantaged youths who do not have savings to serve as collateral, nor possess any land or other resources.[184] This is also true when social capital (as opposed to assets) is used to secure a line of credit. Informal credit (e.g., through communities) requires high stocks of social capital (in terms of a broad network and a status of reliability and creditworthiness) which youths generally do not have. Youth lack decision-making power in the household; although youth may be assigned a specific plot to tender, there are many instances of this being "taken back" by the head of the family on account that they spent too much time on it and neglect the family field.[185] In addition, there is insufficient organization of youth into collectives, to support the more profitable selling of their agricultural outputs.[186]

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[172] https://www.worldbank.org/en/results/2022/05/17/afw-creating-markets-for-smallholder-farmers-in-mali
[173] Stakeholder Consultation with Faso Jigi/PACCEM, a union of cooperatives in Ségou
[174] World Bank & FAO, "A Blueprint for Strengthening Food System Resilience in West Africa: Regional Priority Intervention Areas Public", 2021
[175] https://data.worldbank.org/indicator/SI.POV.NAHC?locations=ML
[176] World Bank & FAO, "A Blueprint for Strengthening Food System Resilience in West Africa: Regional Priority Intervention Areas Public", 2021
[177] CIAT, ICRISAT, BFS/USAID. 2021. Climate-Smart Agriculture in Mali. CSA Country Profiles for Africa Series. International Center for Tropical Agriculture (CIAT); Bureau for Food Security, United States Agency for International Development (BFS/USAID), Washington, D.C. 25 p.
[178] https://www.gsma.com/mobilefordevelopment/wp-content/uploads/2020/05/Agricultural_Insurance_for_Smallholder_Farmers_Digital_Innovations_for_Scale.pdf
[179] Ouédraogo et al, Uptake of Climate-Smart Agricultural Technologies and Practices: Actual and Potential Adoption Rates in the Climate-Smart Village Site of Mali, *Sustainability* 2019, *11*(17),
[180] AGRA, (2021), "Value 4 Her" Synthesis Report on Agri-enterprise event held in Bamako Mali, Available here
[181] AGRA, (2021), "Value 4 Her" Synthesis Report on Agri-enterprise event held in Bamako Mali, Available here
[182] World Bank, 2019, "Mali Climate-Smart Agriculture Investment Plan" https://doi.org/10.1596/32741
[183] WFP, (2017), Gender, Access and Use of Credit, Capital and Insurance Services in Mali, Available here
[184] Food & Business Knowledge Platform, (2016), Youth inclusiveness in agricultural transformation: The case of Mali, Available here
[185] Food & Business Knowledge Platform, (2016), Youth inclusiveness in agricultural transformation: The case of Mali, Available here
[186] Food & Business Knowledge Platform, (2016), Youth inclusiveness in agricultural transformation: The case of Mali, Available here

66. **Barrier 6: Market, technology, and cultural barriers to the adoption of low-carbon agriculture technologies/ practices such as agroforestry, solar irrigation, and biodigesters.** High-potential low-carbon technologies that provide reduced emissions in combination with increased productivity and adaptive capacity (such as solar irrigation and biodigester systems, or practices such as agroforestry (see **Annex 2: Feasibility Study, Section VII, Sub-section 7.2** for more information) continue to have low adoption within Mali. In addition to the barriers experienced for adaptation practices, low-carbon pathways face additional market and technology barriers.

67. Solar irrigation and biodigester markets are considered nascent, with the Alliance for Biodigester in West and Central Africa (AB/ WCA) estimating that by 2019 there were only 1,600 biodigesters in the country, and the market foundations are still in the process of being developed for both technologies.[187,188] Across both technologies, limited capital provides a significant barrier: a high upfront investment cost makes the technologies inaccessible to most smallholder farmers within their existing resources.[189,190] Lack of knowledge regarding the technologies' productive use amongst smallholder farmers combined with limited installation, operations or maintenance services accessible to smallholder farmers further reduces the interest in the technologies. For solar irrigation, initial barriers are compounded by limited marketing attempts from solar pump distributors to resource-poor farmers and underdeveloped markets for other items in the irrigation value chain, for example. for inputs (such as seeds and fertilizers) and outputs (irrigated agricultural products).[191,192] For biodigesters, legacy donor projects have also served to compound the challenges by skewing local understanding of the return on investment: by historically focussing solely on the energy production and excluding the other valuable outputs of the system such as fertilizer return on investment has traditionally been under-valued in Mali.[193]

68. For agroforestry, additional barriers are also present that limit its adoption. As agroforestry often requires multiple years to reach a return on investment (trees compatible with agroforestry in Mali such as Acacia Senegal take ~5 years to reach maturity but then deliver stable output for a further 10 years), certainty on land ownership is key to enable adoption at the local level. [88] The connection between land ownership and agroforestry is also compounded by cultural norms that associate the act of tree planting with land ownership. However, this is often incompatible with the often complex and unclear land ownership practices observed in rural areas, with the state being the ultimate owner of most of the rural land.[194] As a result, there is reticence amongst farmers to adopt the practice on the land they farm.

69. **Barrier 7: Institutional barriers due to insufficient capacity, resources and sometimes trust within public sector institutions and local community governance mechanisms limit the adoption of mitigation and adaptation practices.** The presence of de-centralized commune-level governance structures and administration (particularly through the PDESC planning cycle) in Mali provides an opportunity to support locally-led adaptive capacity building intervention and has been associated with the adoption of specific new adaptive techniques such as water harvesting.[195] However, a lack of knowledge of climate change at the local government level limits communes from effectively leveraging the PDESC to foster improved adaptation and mitigation practices. [196] This is compounded by challenges regarding coordination between the national and local levels which also present a barrier to implementing CSA and agroforestry supportive policies that are approved at the national level. Policies relating to agroforestry provide a good example of this, where there is a disconnect in policy implementation in part due to an 'incomplete' transfer of authority from national to local government, with much decision-making, budgetary control, and policy execution remaining central, in spite of the 'decentralized' model. This makes it difficult for local governments and authorities to implement practices like agroforestry on state-owned lands as there is a conception that they do not have the authority to do so despite official policy support.[197]

70. Completing demands for resources also limits the ability of public sector institutions to support CSA and agroforestry adoption. For example, in 2023, due to budgetary pressures, the government was not able to fully finance the National Response Plan (NRP), which is the main state response to support households in the areas most affected by food and nutrition insecurity.[198] Furthermore, the active conflict in Mali provides a competing demand for resources at both the national and local levels, with communes also citing a lack of budget to implement widescale agroforestry adoption. In addition, lower trust in formal institutions than in local informal institutions can be found in some regions of Mali, in part derived from the backdrop of conflict.[199] Informal institutions rely on social capital, kinship, and religious networks for authority with their support required to be able

[187] Alliance for Biodigester in West and central Africa (AB/ WCA) Interim Office Report, 2019. Available here.
[188] Interview with the SNV representative for the ABC and MERIT biodigester projects.
[189] Agrilinks, 2022 Closing the Demand-Supply Gap in Mali's Solar-Powered Irrigation Value Chains, Available here
[190] R. van Veenhuizen and M. Dubbeling (RUAF Foundation), M. Ba (IAGU), S. Niang (UCAD), M. Kamaté (AEDR Teriya-Bugu), J. Lekoto (Songhaï) and J.M. Médoc (CIRAD), Policy brief: Integrated development of biogas in West Africa, 2017, Available here
[191] AICCRA, Pay-as-you-go model makes solar-powered irrigation affordable for farmers in Mali, Available here
[192] Agrilinks, 2022 Closing the Demand-Supply Gap in Mali's Solar-Powered Irrigation Value Chains, Available here
[193] Stakeholder consultation for IAAT project April-July 2023: Interview with the SNV representative for the ABC and MERIT biodigester projects.
[194] Land links (2020), Country profile: Mali, Available here
[195] USAID (2017) *Decentralized governance and climate change adaptation: A case study on Mali*. rep. Available here
[196] Ministère du Développement Rurale, Plan Triennale de Campagne Agricole Consolidé et Harmonisé (2023)
[197] Duan, Faye, (2020) "Policy Landscape for the Scaling-up of Agroforestry in Mali." Oxfam Research Backgrounder series. Available here
[198] Handicap International, (2022), Breaking the Spiral of the Food and Nutrition Crisis in Mali, Link
[199] Ahmadou Aly Mbaye, Nancy Benjamin, "Institutions, informality and conflict in the Sahel: the case for Mali" 2022, Available here

to implement many community-level changes. [200] With several adaptive techniques requiring coordination, distrust, and inefficiencies of organizations exacerbate adaptation gaps.

Outline of Project Components to Address Barriers

71. This project's paradigm shift will therefore be achieved through four interlocking components which will deliver low-emissions and resilient impact pathways to address the barriers outlined above and result in the project's 4 outcomes.

72. **Outcome 1: Increased climate-resilient agricultural crop and food production in the targeted regions in Mali.** To achieve outcome 1, **if** the project i) *improves the technical capacities and inclusivity of extension services in climate-smart agriculture and agroforestry production (Output 1.1 and Activity 1.1.1)* and ii) delivers *increased adoption of climate resilient practices in the production of crops, livestock, and agroforestry by smallholder farmers (Output 1.2 and Activity 1.2.1)* **then** smallholder farmers will increase their productivity and the volume of agricultural outputs produced will increase through yield increases and reduction of losses **because** smallholder farmers will have increased awareness, knowledge, interest and capacity to select and implement appropriate farming practices in their farms that are more adapted and resilient to climate hazards whilst also targeting increased productivity. The practices shared by the extension services in the new curriculum have been carefully selected based on the findings of existing studies to deliver both increased resilience to climate change, increased productivity, and low carbon emissions when implemented on smallholder farms in Mali. This outcome will directly seek to alleviate barrier 1, "*Technical know-how and capacity gaps amongst smallholder farmers to implement and integrate climate-smart agriculture and low emissions land management technologies and practices",* and barrier 2 *"Limited capacity of public and private extension services at the local level to promote access to inclusive and reliable agro-advisory services on climate resilient practices".*

73. **Outcome 2: Improved and inclusive access to finance and markets for vulnerable communities to enhance sustainable livelihoods.** To achieve outcome 2, **if** the project i) supports the creation of more connected value chains for agroforestry and climate-smart agriculture crops and livestock that reach more smallholder farmers (Output 2.1 and Activities 2.1.1 and 2.2.1 and ii) support smallholder farmers, especially youth and women to overcome barriers to entrepreneurship in CSA and agribusiness (Output 2.2 and Activities 2.2.1 and 2.2.2), **then** the most vulnerable people in the targeted communities will be able to access more diverse, reliable and sustainable livelihoods, **because** smallholder farmers and women and youth entrepreneurs will be able to establish and maintain additional income-generating activities, such as the sale of crops in new markets or the provision of processing services for agricultural outputs. As part of this component, actors across the value chain will be supported to better serve smallholder farmers, both through increased connections between smallholder farmers and processing services as well as by supporting private sector businesses to actively target farmers and develop products for the often-underserved women and youth segments, particularly with regard to financial products. This outcome will directly seek to alleviate barrier 3, *"Lack of access for smallholder farmers to mature, connected and transparent local markets and value chains for the sale of CSA compatible crops",* barrier 4, *"Inability for smallholder and subsistence farmers to invest in inputs for climate change adaptation (seeds, fertilizer, labour, equipment) due to limited financial resources"* and barrier 5, *"Limited consideration of the needs of women or youth in the development and management of agricultural and agribusiness systems, leading to significant bottlenecks for their empowerment".* It will also indirectly contribute to the alleviation of barrier 6, in partnership with component 3, *"market, technology and cultural barriers for the adoption of low-carbon agriculture technologies/ practices such as agroforestry, solar irrigation, and biodigesters."*

74. **Outcome 3: Adaptation and mitigation considerations and best practices are embedded in institutional agriculture and agroforestry planning.** To achieve outcome 3, **if** the project i) increases institutional capacity in climate change adaptation and mitigation planning as well as knowledge of best practices to address agriculture and NRM-related climate risks (Output 3.1 and Activities 3.3.1 and 3.1.2) and ii) enables increased knowledge sharing and coordination of best practices in CSA and agroforestry across stakeholders (Output 3.2 and Activity 3.2.1) **then** mitigation and adaptation considerations and best practices will be embedded in the public sector and local governance planning for agriculture and agroforestry and consequently government actions **because** knowledge and capacity barriers will be removed for institutional stakeholders. This will directly address barrier 7, the *Institutional barrier due to insufficient capacity, resources and sometimes trust within public sector institutions and local community governance mechanisms limit the adoption of mitigation and adaptation practices."* Output 3.1 and Output 3.2 also address barrier 2 ("*limited capacity of public and private extension services at the local level to promote access to inclusive and reliable agro-advisory services on climate resilient practices*") and barrier 1

[200] Ahmadou Aly Mbaye, Nancy Benjamin, "Institutions, informality and conflict in the Sahel: the case for Mali" 2022, Available here

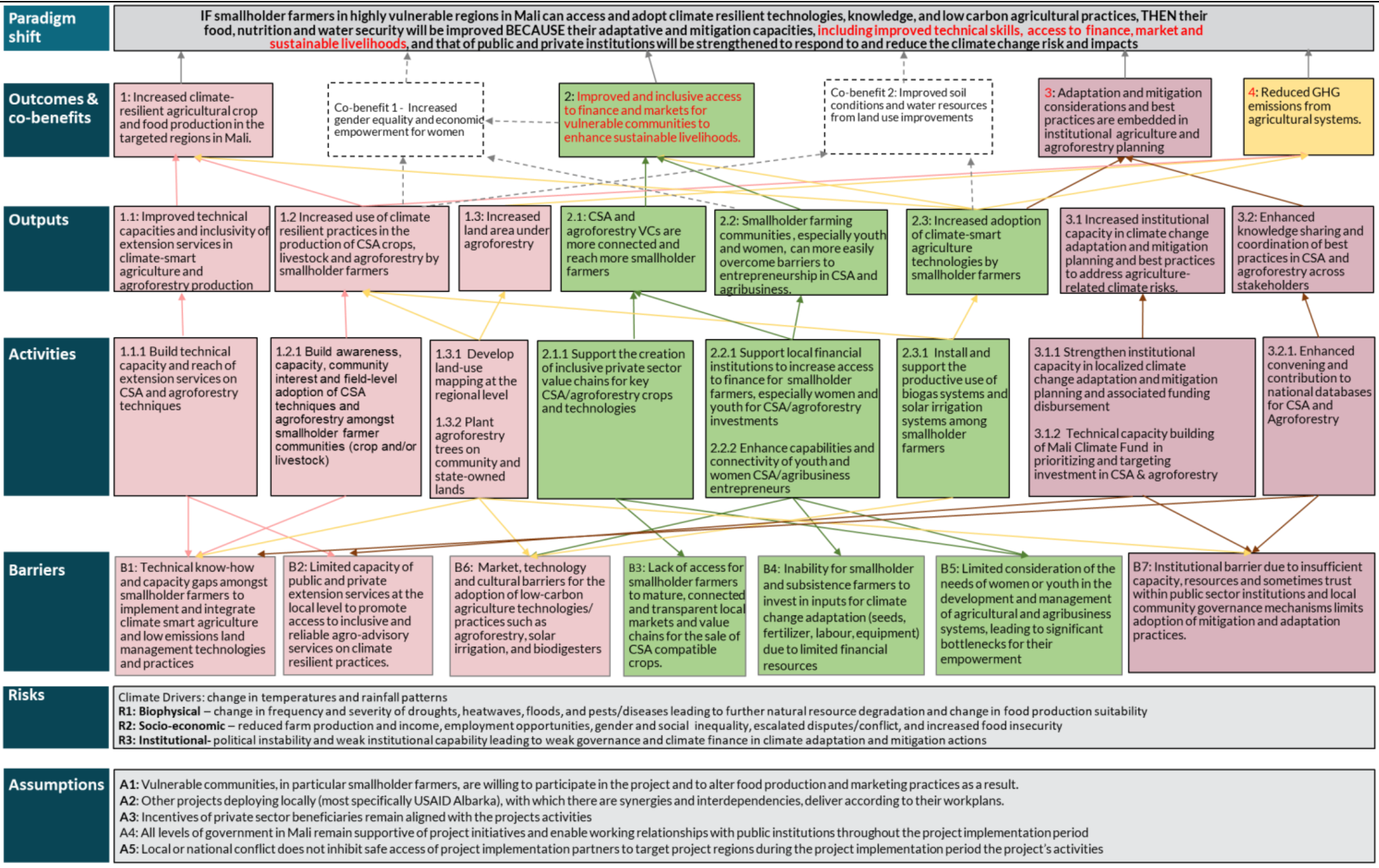
("*technical know-how and capacity gaps amongst smallholder farmers to implement and integrate climate-smart agriculture and low emissions land management technologies and practices*") respectively.

75. **Outcome 4: Reduced GHG emissions from the agricultural system.** To achieve outcome **4**, **if** the project i) increases the adoption of low-carbon agriculture technologies by smallholder farmers (Output 2.3 and Activity 2.3.1) and ii) increases the land area under agroforestry within the project's target regions (Output 1.3 and Activities 1.3.1 and 1.3.2) **then** the net emissions associated with smallholder farms will decrease **because** both low emission technologies and practices will be deployed in replacement of higher emissions technologies and practices (solar pumps replace diesel pumps for irrigation, biodigesters replace biomass as a source of energy) and carbon "sinks" will be created in the form of 91,285 Ha of agroforestry. The project will both support the direct adoption of 1,000 solar irrigation and 5,000 biodigester systems as well as support the cultivation of their supporting value chains and services. IAAT will collaborate with the private sector solar and biogas technology suppliers in Mali (e.g., EcoTech, EMICOM, EnDev, EKOenergy, and PEG Africa) to scale out these technologies through the PAYG approach. Some of these companies have already started expanding this financial model in Mali. This approach will bring a significant amount of private sector investment in solar pump and biodigester systems in the IAAT locations. Additionally, the installation of solar irrigation systems will contribute to enhancing productivity for smallholder farmers. The lower cost of operation compared to diesel pumps, which require traveling long distances from remote rural areas to purchase fuel, coupled with the support to effectively harness the potential of solar irrigation systems, will allow farmers to improve water management for enhanced productivity. Similarly, agroforestry, a technique shared with smallholder farmers as part of the CSA techniques in component 1, will be explicitly supported to increase its current deployment through support to the surrounding infrastructure such as nurseries and transportation of saplings as well as increased capacities amongst regional governments to conduct land use mapping to identify and support appropriate selection of trees. This will directly address barrier *6 "Market, technology and cultural barriers for the adoption of low-carbon agriculture technologies/ practices such as agroforestry, solar irrigation, and biodigesters"* as well as support component 3 in addressing barrier 7, *"Institutional barrier due to insufficient capacity, resources and sometimes trust within public sector institutions and local community governance mechanisms limits the adoption of mitigation and adaptation practices",* and component 1 in addressing barrier 1 **"***technical know-how and capacity gaps amongst smallholder farmers to implement and integrate climate-smart agriculture and low emissions land management technologies and practices".*

76. When combined, the 4 project outcomes can create a systems-level change through re-enforcing outcomes at the on-farm, market, and institutional level:
- **On-farm** adoption of new climate-resilient, low-carbon practices by smallholder farmers (components 1 and 2)
- Increased **market** inclusivity to key inputs for the adoption of CSA and low-carbon farming, and access to destination markets for new goods (component 2)
- Increased **institutional** capacity and interest in climate resilient and low emissions practices and therefore prioritization in institutional discourse and policies, enabling increased interest at the market and on-farm levels (components 3 and 1)

77. **In order to achieve the paradigm, shift through the above project components, the following assumptions are included** below:
Assumptions included in the Theory of Change:
- Vulnerable communities, in particular smallholder farmers, are willing to participate in the project and to alter food production and marketing practices as a result.
- Other projects deploying locally (most specifically USAID Albarka), with which there are synergies and interdependencies, deliver according to their work plans.
- Incentives of private sector beneficiaries remain aligned with the project's activities.
- All levels of government in Mali remain supportive of project initiatives and enable working relationships with public institutions throughout the project implementation period.
- Local or national conflict does not inhibit safe access of project implementation partners to target project regions during the project implementation period.

Figure B. 2. 1: IAAT Theory of Change Diagram



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	**GCF Mitigation Results Area (MRA 1-4)**				**GCF Adaptation Results Area (ARA 1-4)**			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, cities, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 3	☒	☐	☐	☒	☐	☐	☐	☐
Outcome 4	☐	☐	☐	☒	☒	☐	☐	☐

In addition to the direct outcomes achieved, this project will also delivery co-benefits across the environmental social, economic, and gender areas:

Co-benefit number	**Co-benefit**					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1	☐	☐	☒	☒	☐	☐
Co-benefit 2	☒	☒	☐	☒	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Project Component and Activities

78. IAAT implemented intensive consultations with regional and communal councils, community beneficiaries (i.e. farming cooperatives and unions, community leaders, youth and women, community committees, and local farmers in the targeted circles of the IAAT project. The selected IAAT interventions (climate-smart agriculture, agroforestry, solar pump irrigation, biodigester included in the activities are based on the consultation with the regional and communal councils, community beneficiaries i.e. farming cooperatives and unitions, community leaders, youth and women, community committees, and local farmers (also include smallholder, marginal, and vulnerable populations). IAAT will conduct further consultations with the community members in all 12 circles during the initial planning phase of the project to integrate the interests and needs of the Indigenous People (if identified during the implementation), marginal and smallholder farmers, agropastoralists[201], and climatically highly vulnerable people during the selection of 48 communes that will ensure application of free, prior and informed consent through their community members and/or their representative institutions.

Component 1: Improving Extension Services and Increasing On-farm CSA Adoption

79 Component 1 will focus on i) increasing the awareness of climate risks and ii) the understanding and technical capacity of smallholder farmers to adopt CSA techniques by firstly building the capacity of agro-advisory services, the key educational service for smallholder farmers, and secondly by supporting the on-farm adoption of CSA techniques. It will therefore achieve outputs 1.1 *"Improved technical capacities and inclusivity of extension services in climate-smart agriculture and agroforestry production* "and output 1.2 "*Increased use of climate resilient practices in the production of CSA crops, livestock, and agroforestry by smallholder farmers".*

Output 1.1 Improved technical capacities and inclusivity of extension services in climate-smart agriculture and agroforestry production

80 **Output 1.1 will be delivered through activity 1.1.1 "Build technical capacity and reach of extension services on CSA and agroforestry techniques"** which will directly address barrier 1, "Limited development of public and private extension services at the local level to promote access to inclusive and reliable agro-advisory services on climate resilient practices". This activity seeks to expand the reach of extension services and to improve the transfer of climate-smart agriculture (CSA) and agroforestry knowledge and skills to smallholder farmers. The activity will develop a tailored curriculum of practices for extension services that address climate risks and deliver productivity, mitigation, and adaptation benefits and implement it through a "train the trainers" approach to refine public and private extension services. It will also involve local organizations and leverage digital platforms to increase the reach and inclusivity of training practices, ultimately enhancing the effectiveness and accessibility of extension services in promoting CSA and agroforestry among smallholder farmers. Beneficiaries of activity 1.1.1 will be extension services in the 5 regions of the project, reaching approximately 500 extension agents across Gao, Koulikoro, Tombouctou, Segou and Mopti.

81 **Activity 1.1.1 will be delivered through 3 sub-activities:**

1.1.1.1 Develop an updated curriculum each for public and private extension services including Rural Resource Centers (CRRs), and Farmer Field Schools in project regions (Gao, Mopti, Koulikoro, Segou and Tombouctou) and tailor to the circle level via workshops.

IAAT will use a baseline study conducted including a stakeholder consultation workshop at project inception to create a tailored curriculum for localized CSA practices, selection of direct beneficiaries, prioritize CSA interventions based on local climate risks, location and farmers' characteristics, and selection of communes and villages for CSA scaling. IAAT will closely work with existing CRRs and agriculture extension's farmers' field school program to update the curriculum. The new curriculum will be focused on i) overall causes and impacts of climate change on Malian agriculture, ii) benefits and implementation methods of the 22 priority CSA and agroforestry techniques identified as having the most relevance for the project's target regions across the three CSA criteria of climate change adaptation, climate change mitigation, and increased productivity in agriculture (see table B.3.2 below) and iii) analysis and interpretation of land use and production data to provide tailored advisory services to farmers and improve on-farm decisions. For more information on how these practices were prioritized amongst the larger number of potential CSA practices, see **Annex 2: Feasibility Study, Chapter VII, Section 7.3.** Workshops will also be held in each of the 12 target circles to support the tailoring of the new CSA curriculum to the contexts of

201 The pastoralists in IAAT circles are not nomadic and use community grazing pasture lands for livestock production and do not move to different locations. They are agropastoralists (Source: consultation with the stakeholders in the IAAT locations).

the circles, and their communes. This will focus on i) including perspectives from local NGOs and stakeholders on traditional knowledge, practices and how best to embed CSA alongside these, ii) how to analyze and interpret land use and production data to further select and prioritize CSA techniques at the circle and, where relevant, commune level.

Table B.3.2 Priority CSA Techniques Selected for Inclusion in the updated extension services curriculum

Priority CSA Technique	Focus Area	Description of the technique
Contour bunding	Crop Farming	A form of micro-catchment technique, a simple and cheap form of water control. The bunds are created along the contour lines. There are also small earth ties, perpendicular to the bunds, that subdivide the system into micro-catchments.
Cultural sowing technics	Crop Farming	Enhance rational use of soil water by deploying specific sowing techniques
Improved varieties adapted to different agro-climatic conditions	Crop Farming	Use of seeds that are specifically adapted to offer higher and more stable yields and deliver greater tolerance to stressors (e.g., flood, pests)
Production and use of on-farm compost	Crop Farming	Farm effluent and manure along with leftover materials such as wasted feed, wood chips, rice hulls, are used to produce compost and be recycled back on the farm.
"Rational" management of land (management of flooded and dewatered areas)	Crop Farming	Management of flooded and dewatered areas
Development of inland valley for rice cultivation	Crop Farming	Creation of simple, low-cost water management structures in the rice fields (e.g. weirs, dykes, canals, drains and basins/ plots by the farmers) for growing rice in inland valleys
Half-moon technique	Crop Farming	A water harvesting intervention that consists of half-moon shaped basins dug in earth.
System of rice intensification (SRI)	Crop Farming	Rice cultivation practice includes the early establishment of healthy plants; low plant density; soil enrichment; and the sparing application of water. Provides mitigation and adaptation benefits.
Zaï pit technique	Crop Farming	Small basins in which seeds of annual or perennial crops are planted. They are beneficial for soil conditions because they increase termite activity which leads to a higher water infiltration when it rains.
Crop diversification	Crop Farming	Reduce risk to crops from threats (e.g. pests, climate events), increase productive and improve water use efficient
Integrated pest and disease management	Crop Farming	Focus on long-term prevention of pests and diseases through a combination of techniques such as biological control, habitat manipulation, modification of cultural practices, and use of resistant varieties.
Fertilization of fields by animal corralling	Crop Farming / Livestock farming / Agroforestry	Penning livestock into a field so that their waste fertilizes the field
Fodder crops	Livestock farming	Planting of crops that serve as food for livestock
Feed supplements	Livestock farming	Feed supplements are phosphate, calcium and trace mineral mixtures that can be given to grazing animals to supplement grazing when it is deficient in minerals and trace minerals.
Dual purpose crops (Food and fodder)	Livestock farming / Agroforestry	Planting of crops that serve as food for livestock and people. These crops can also include trees.
Assisted natural regeneration of trees	Agroforestry	Increase the number of trees by protecting and pruning living tree stumps, and by creating the environmental conditions for tree seeds to sprout
Hedgerows	Agroforestry	Planting of wild shrubs and occasional trees, typically bordering a road or field.
Tree nursery and transplanting	Agroforestry	An area where trees are grown for use as stocks for budding and grafting. At the appropriate maturity, trees are then transplanted to the destination location.
Climate Information & Warning Services	Resilience Service	Climate information and early warning systems share information on climate hazards and observed trends in climate, supporting smallholder farmers as well as public officials to anticipate and prepare for climate risks.
Crop Insurance	Resilience service	Crop insurance provides a risk mitigation mechanism for farmers by insuring crops against risks such as flooding or pests.
Solar irrigation systems	Mostly crop farming (Resilience & low emission technology)	Solar panels provide a clean energy source to a power pump to extract water (ground or surface); water connects to an irrigation system (primarily for use on crops)
Biodigester systems	Mostly livestock farming (Resilience & low emission technology)	A system that uses anaerobic digestion to convert organic matter into a gas (for fuel) and create additional valuable outputs (e.g. slurry, manure, and fertilizer). The organic matter is inserted into a sealed digester, and, in the absence of oxygen, anaerobic bacteria consume the organic matter to multiply and produce biogas and other byproducts.

1.1.1.2 Sub-Activity: Train extension agents on the new curriculum through a "train the trainer" approach.

"Train the trainers" will allow the new curriculum to be rapidly and cost-effectively rolled out across the target regions, by operating a cascading approach to introducing the new curriculum across the services. The approach will first introduce universities and technical schools to the new curriculum through a national workshop and training day. One university and technical school per region that are recognized by the AEDD and Department of

Agriculture will be selected to provide training to the 400 public extension staff and 100 private agriculture extension agents. Secondly, 1 workshop will be organized for the selected universities and technical schools to provide a detailed presentation of the new curriculum and how to implement it to the trainers. These 500 public and private extension staff (agro-input suppliers and livestock health services providers) will be expected to be trained through universities and tech schools (100 per region, approximately 10 per commune). The selected universities and technical schools will provide training to the public agriculture extension staff and private agriculture extension agents (maintain a 50:50 men and women ratio wherever possible) in the training workshops organized by IAAT at the circle level. In total, IAAT will organize 12 training workshops in collaboration with circle-level agriculture extension offices (1 per circle and 40-45 participants/training). These participants will provide training to the farmers and other value chain actors through public extension programs (agriculture ministry's regular farmers training programs and farmers-field schools) and training and services provided by private sector actors (training centers, universities, agro-advisory service providers, input suppliers, etc.). In addition, the curriculum will be reviewed and updated every year in the annual project review and progress assessment meetings. AEDD in collaboration with the Department of Agriculture will facilitate the integration of the CSA curriculum into the government agriculture extension system to continue the national scaling-up of the curriculum.

1.1.1.3 Sub-Activity: Train extension services on inclusivity practices and expand utilization of existing resources, in particular digital tools, and educational platforms.

In addition to revising the curriculum, extension services will be supported to increase their reach by improving their methods and techniques for training smallholder farmers. This will cover 2 key areas: i) inclusivity of training approaches (including how to ensure a broader demographic reach is achieved) and ii) how to incorporate and increase access to digital tools in their training. This training will be delivered through workshops (estimated at 12, 1 per circle) as well as the creation and publication of a guide which will include: i) recommended inclusive practices and how to integrate them in service offering; ii) the benefits of using digital tools and educational platforms both for extension agents and farmers; iii) list of key digital tools and educational platforms, their respective objectives, functioning, and where and how to access them. Currently, 98% of the population have mobile connectivity and despite the current low internet connectivity in rural areas (1/3 population have internet connectivity), the use of digital tools in agriculture extension has long-term applications with rapidly increasing internet connectivity in the rural Mali, thereby resulting in increased access and inclusivity of vital CSA and agroforestry climate-related information.

Output 1.2: Increased use of climate-resilient practices in the production of CSA crops, livestock, and agroforestry by smallholder farmers

82 **Output 1.2 will be delivered through activity 1.2.1 "Build awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry amongst smallholder farmer communities (crop and/or livestock)"** which will directly address barrier 1, "Awareness, technical know-how and capacity gaps amongst smallholder farmers to implement and integrate CSA and low emissions land management technologies and practices". This activity focuses on the promotion and adoption of CSA and agroforestry practices among smallholder farmers by increasing their awareness, interest, and capacity through locally tailored training packages developed as part of 1.1.1. Several channels will be leveraged to increase the potential to reach more farmers through extension services (including farmer field schools, and Rural Resource Centers (CRRs)). This activity will also provide technical support to local community organizations, notably by (i) enhancing the capacity of Early Warning Groups to support the implementation and dissemination of national EWS and (ii) strengthening Community Action Cycles (CACs) established by Albarka and replicating the model in other regions. The CAC, based on an appreciative community mobilization approach, is the process through which communities organize for action, explore climate adaptation and mitigation issues and set priorities, and plan, act, and evaluate successful interventions. Utilization of this approach in 297 villages under Albarka in Mali contributed to significant improvements in participation of communities and farming households in agriculture and natural resource management. The CAC includes five major steps: 1) Deepen community understanding, build mutual trust and gauge willingness to contribute, 2) households and community entry in the process, 3) community visioning and selection of community priorities, 4) develop community action plan for community-owned interventions, 5) Use community level data to refine community action plan and inform social accountability in climate adaptation and mitigation actions. Adopting the CAC model will promote collective resource management, (enhancing and reducing conflict between livestock and crop farmers on community resources). Beneficiaries of activity 1.2.1 will be smallholder farmers in the 48 targeted communes of the project. They are estimated to be 431,700 beneficiaries (men: 215,850, women: 215,850).

83 **Activity 1.2.1 will be delivered through 4 sub-activities, 3 of which will at least in part be delivered by the Albarka team:**

1.2.1.1 Sub-activity: Deliver tailored local CSA and agroforestry training packages through dissemination networks, including field schools, extension agents, and Rural Resource Centers (CRRs), and cooperative leaders trained by Albarka.

Through the extension agents trained as part of 1.1.1, the finalized and tailored training program will be disseminated across the 12 circles, focusing on the 48 project-targeted communes, by using a variety of dissemination approaches listed above. As part of this dissemination, Albarka will also train 250 cooperative leaders on the new curriculum who will then train their members.

1.2.1.2 Sub-activity: Strengthen operational capacities of key local organizations (such as GAP-RUs (Early Warning Group/Emergency Response) to support the implementation and dissemination of the national Early Warning Systems

Existing and new Early Warning Groups (GAP-RUs) will be supported to improve their understanding of and capacity to efficiently disseminate existing EWS[202] and provide timely information to smallholder farmers and communities in general. Clear guidelines regarding the procedures and measures to be taken in response to potential climate risks and threats will be compiled in contingency plans distributed to GAP-RUs. The contingency plan will aim to enhance the readiness and resilience of GAP-RUs in the face of challenges, such as extreme weather events, that could prevent timely and accurate information. The procedures will include protocols for communication, coordination, and response mechanisms, as well as the allocation of resources and responsibilities among the group members. GAP-RU members will be trained in their respective roles and responsibilities and will receive kits to enhance the timeliness and effectiveness of their responses.

1.2.1.3 Sub-activity: Strengthen Community Action Cycles (CACs) established by Albarka to ensure their sustainability beyond project implementation and enhance their capacity to manage community assets for CSA technologies.

Albarka has established 250 CACs in Gao, Mopti, and Koulikoro, with 1 CAC in each village targeted through the project. IAAT will select 250 villages based on the commune's priorities within 48 communes to establish 250 additional CACs. These CACs are mechanisms based on a systematic and iterative approach to engaging communities and supporting them to autonomously identify, plan, implement, and evaluate actions to address specific needs or challenges relating to their village/neighborhood. This approach facilitates community empowerment by fostering their active involvement in decision-making, problem-solving, and the implementation of initiatives aimed at improving their socio-economic and environmental well-being. Albarka has effectively used the CACs approach to minimize and eliminate conflicts in common resource use by agropastoralists and crop farmers which provides a successful model to transfer and apply to the IAAT, see **Annex 23**.

CACs will be supported to improve their management capacities and ensure sustainability beyond the implementation of Albarka. This will include providing management kits to each CAC and supporting them to develop and implement self-assessment plans. CACs will be the key point of reference for implementing farmers' training and capacity building by the extension staff and agents, farmers field schools, GAP-RU, producers associations, and other interest groups.

1.2.1.4 Sub-activity: Extend the Albarka CAC model in other communes in Gao, Mopti, and Tombouctou, and replicate it in Segou and Koulikoro

GCF IAAT will replicate the same model of CACs launched by Albarka (Sub-activity 1.2.1.3) in other communes in Albarka regions, as well as Segou and Koulikoro which are not targeted by Albarka. This will include establishing a CAC in each village and strengthening their management capacities by providing management kits and developing self-assessment plans.

Output 1.3: Increased land area under agroforestry trees

89 **Output 1.3 will be delivered through activity 1.3.1 "Develop land-use mapping at the regional level" and activity 1.3.2 "Plant agroforestry trees on community and state-owned lands".** These activities will collectively address barrier 6 "Immature markets (for relevant inputs and outputs) for low-carbon agriculture technologies/ practices such as agroforestry, solar irrigation, and biodigesters, alongside cultural barriers". Activity 3.2.1 will also partially address barrier 7: "Public sector institutions and community governance mechanisms cannot sufficiently include adaptation and mitigation considerations and best practices in their work, particularly with regards to planning, and remain uncoordinated across levels and with other stakeholder groups."

90 Activity 1.3.1 aims to facilitate the adoption and effective utilization of regional-level land use mapping to support the development of agroforestry and inform the implementation of locally relevant CSA and agroforestry practices. By assisting regional authorities in mapping and classifying land types and uses in the target regions, it seeks to enhance their understanding of land resource distribution, enabling improved land-use planning and decision-making at the commune level. Additionally, a land-use database will be developed to enhance the knowledge of

[202] The Government of Mali in collaboration with World Bank, GCF, and World Meteorological Organization (WMO) have invested to modernize Mali's EWS. by strengthening the data use and forecasting of droughts, heavy rains and floods, locust infestation. They have established a common alert portal, using mainly cell phones, but also using TV and Radio. Realtime data are collected from regional and local hydrological and metrological stations to develop locally relevant early warning information. However dissemination and coordinated response at a local level is lacking and will be supported by the IAAT.

extension services regarding land characteristics, allowing them to provide tailored advisory services to farmers on suitable crops and techniques for enhanced productivity. This land-use database will also be shared with national authorities to promote wider access to agriculture and agroforestry data in Mali. Beneficiaries of activity 1.3.1 will be regional and local authorities and public and private extension agents.

91 **Activity 1.3.1 will be delivered through 2 sub-activities:**

1.3.1.1 Sub-activity: Create land-use maps at the regional level to identify best suited areas for agroforestry production. Regional authorities will be supported to develop comprehensive land-use maps providing detailed information on types of land cover and land use categories, including agricultural fields, forests, degraded lands, water bodies, infrastructure, and other land designations, as well as the spatial distribution of all land categories. Clear identification of the types and uses of land will allow further classification and determination of areas reserved for agroforestry. IAAT's activity will focus on developing land use plan for sustainable management of current land use categorise such as crop, pasture and degraded lands. The land use plan will not focus on changing crop land to pasture lands and vice versa. In the crop lands, land use plan will identify sustainable crop lands (i.e. private lands) management practices including suitable locations for CSA and planting agroforestry trees. For promoting agroforestry in public lands, land use plan will identify the location where land (pasture and other category) is rapidly degrading and possibility of rehabilitation by planning locally suitable grasses and trees and useful for both agropastoralists and agriculturalist.

1.3.1.2 Sub-activity: Establish a project land use database. A project land use database will be established to collect specific data on the agricultural land in the targeted regions, including land use categories, status of land ownership (public or private), land parcel information, land use activities and changes over time, environmental features, soil characteristics, etc. This will provide valuable information to all stakeholders, including extension services, national authorities, research institutes, and farmer groups. Accessing this information will allow extension agents to tailor their services to the needs of and existing opportunities for farmers in each area, improving the quality of information provided on the most adapted crops and techniques to increase productivity. This land use database will also provide an opportunity for the project to contribute to improve access to data in Mali, in line with 3.2.1.2. This land use database will be owned and managed by AEDD under the Ministry of Environment and Sanitation. IAAT will build data management and maintenance capacity within the AEDD.

92 Activity 1.3.2. seeks to promote enhanced adoption of agroforestry by supporting the on-farm planting of 91,285 Ha of agroforestry, targeting specific value chains including acacia Senegal, and shea. The capacities of several stakeholders will be leveraged to ensure successful implementation of this activity (e.g., nurseries for production and distribution of inputs, regional and commune local authorities for the identification of suitable land, farmers for planting, communities for the management of trees and shared resources). The efficient management of these planted areas by communities will be critical to ensure sustainability after project implementation, and avoid conflict, notably between crop and livestock farmers (see **Annex 23**).

93 **Activity 1.3.2 will be delivered through 3 sub-activities:**

1.3.2.1 Sub-activity: Support the production and distribution of agroforestry tree saplings (e.g., acacia Senegal and shea) in partnership with nurseries. Nurseries and Rural Resource Centers will be leveraged to identify and produce the seeds and tree saplings that are best suited to each area targeted by the project. The nurseries will be supported to produce and distribute tree saplings to cover 91,285 Ha of land dedicated to agroforestry in the five regions. Agroforestry activity in IAAT will only include a) in private crop land – integration of fodder and fruit trees in the existing crop and livestock farming system for better soil and land management, b) in public land – plantation of fodder, fruit and other useful trees in degraded lands for sustainable rehabilitation without land transferring to crop cultivation and other structural changes. The GCF funds will be utilized for training, nursery development, and the distribution and planting of tree saplings to enhance existing agroforestry systems on private lands that will cover 21,585 ha land. Meanwhile, the AEDD co-finance will focus on raising awareness, providing training, and building the capacity of local communities and extension systems. This will help scale up the government's efforts in reforesting and rehabilitating degraded lands (targeted 69,700 ha) in the project areas.

1.3.2.2 Sub-activity: Plant agroforestry trees (e.g., acacia Senegal and shea) in collaboration with smallholder farmers. The land use maps developed in 1.3.1.1 will support the identification of the areas allocated to agroforestry tree plantation by regional and local authorities. These areas will be used to plant 91,285 Ha of agroforestry trees in farmers' lands and state-owned public lands. Out of 91,285 ha., 21,585 Ha will be allocated by farming households (86,340 farming households with an average allocation of 0.25

ha) to plant locally suitable agroforestry trees. The government of Mali is promoting agroforestry tree plantation in the degraded forests and grasslands in the regions through different projects. IAAT in collaboration with AEDD will mobilize communities in the targeted 48 communes to expand agroforestry tree plantation in the public grasslands that will cover 69,700 Ha degraded lands without soil disturbance and structural changes in the forests and grasslands. AEDD will support seedling nursery development and distribution of tree saplings to the communities and mobilize local communities for plantation. In partnership with local government, extension services and CACs, Smallholder farmers will be encouraged and participate in agroforestry tree plantation in additional communal and/or state-owned lands. Note that as part of 1.2, farmers will receive training on how best to manage and maintain agroforestry lands. Tree plantation in public grass/pastureland will be done around communities where they are historically used for livestock production. 48 communes will select areas in their periphery where the territorial collective and local government granted permits to use these lands for sustainable management and use.

1.3.2.3 Sub-activity: Develop plans to manage saplings and growing trees in crop, grass/pasture, and community lands in partnership with CACs. The project will foster community contribution to ensure sustainable management of cultivated areas, especially in community lands. Community Action Cycles will be supported to autonomously monitor, maintain, and use the plants in their respective villages. They will also participate in the allocation and management of pastures, to avoid conflict between crop farmers and agropastoralists. Workshops will be organized to build the capacity of CACs on how to ensure the development of plants, efficiently manage shared resources, and handle conflict.

Component 2: Supporting the development of CSA and agroforestry value chains

84 Component 2 aims to create sustainable markets for CSA and agroforestry products and ultimately more diverse and sustainable livelihoods for the target communities by fostering the development of connected value chains (from input stage, including financing and technologies, to product sales and distribution) and lifting the barriers disproportionately faced by women and youth entrepreneurs for starting and running businesses, including access to finance. Component 2 will achieve outputs 2.1 *"CSA and agroforestry VCs are more connected and reach more smallholder farmers"* and 2.2 *"Smallholder farmers, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness"*

Output 2.1: CSA and agroforestry VCs are more connected and reach more smallholder farmers.

85 **Output 2.1 will be delivered through activity 2.1.1 "Support the creation of inclusive private sector value chains for key CSA/agroforestry crops and technologies";** which will directly address barrier 4 "Lack of access for smallholder farmers to mature, connected and transparent local markets and value chains for CSA compatible crops", and it will lay the foundations to address barrier 5 "Limited consideration of the specific and distinct needs of women and youth in the development and management of agricultural and agribusiness systems, leading to significant bottlenecks for their empowerment (e.g., limited access to finance and land, exclusion from decision-making processes)". In addition, in combination with component 3, it will address barrier 6: "Immature markets (for relevant inputs and outputs) for low-carbon agriculture technologies/practices such as agroforestry, solar irrigation, and biodigesters, alongside cultural barriers".

86 Activity 2.1.1 will support the development of smallholder accessible value chains across CSA and agroforestry outputs by improving connectivity in the value chains (VCs) and fostering access to markets for smallholder farmers. This will be achieved by leveraging digital marketplaces and existing innovation platforms, improving access to market information (e.g., prices) and fostering the development of a network of community-built service providers. Additionally, the activity seeks to enhance the capacity of businesses and producer organizations to reach a larger market through the development and adoption of inclusive business models, capturing the needs of smallholder farmers, women, and youth. The activity will mainly target businesses that deliver added productivity and climate mitigation/adaptation benefits, and are willing to invest in reaching smallholders, offering job opportunities for women and youth, and/or are connected to the mitigation and adaptation technologies prioritized in component 3 (solar irrigation, biodigester systems).

87 Beneficiaries of activity 2.1.1 will be farmers with access to extension services (overlapping with the beneficiaries of 1.1.2), private sector companies, CSA technology suppliers, and producer associations. The activity will specifically aim to reach 22,000 farmers (5% of farmers receiving the CSA and agroforestry training), 100 private sector companies, 48 Economic Interest Groups regrouping technology suppliers (one in each commune) and 240 producer associations (5 in each commune).

88 **Activity 2.1.1 will be delivered through 4 sub-activities:**

2.1.1.1 Sub-activity: Increase linkages in CSA value chains by connecting service providers (including community-built service providers embedded in cooperatives) and processors with producer groups and cooperatives using digital marketplaces and existing innovation platforms – Albarka

Linkages will be developed across crop and livestock value chains to increase farmers' access to markets and information for CSA and agroforestry technologies and related support from suppliers. Establishing linkages will also allow processors to have access to additional high-quality produce. The linkages will be created through existing digital platforms in Mali, focusing on value chain connectivity (e.g., Sugu Sud Kunkan developed by Feed the Future (FtF) and USAID). The networks of producers and private sector companies created by Albarka will be extended to include more stakeholders and will be leveraged to increase subscriptions in identified platforms. Current internet access is a limiting factor to the reach of the initiative resulting in a target of 22,000 farmers and 100 service providers through the platform. However, the expansion of internet access is rapidly growing in the rural areas of Mali which can connect large numbers of the rural population during the 5-year project period. IAAT will showcase the use of digital platforms to improve and expand the agriculture value chains. The 100 service providers will correspond to the same cohort of businesses identified and supported in 2.1.1.2, ensuring seamless continuity and alignment in the provision of assistance and resources.

2.1.1.2 ***Sub-activity: Deliver technical assistance to private sector businesses that are involved in CSA and agroforestry value chains, to enhance the ability to develop youth and gender-inclusive service/product offering and business plans.*** 100 private sector companies providing CSA and agroforestry solutions to farmers will be selected and supported to increase their reach by developing inclusive services and marketing strategies to specifically target women and youth. The technical assistance will be offered through three steps: 1) discussions will be held with the companies to identify the main gaps in their service offering and therefore identify the specific support and/or adjustments to the business model needed to enhance the adoption of an inclusive approach. 2) The results of these conversations will be documented and reviewed to identify patterns that would lead to the development of specific guidelines tailored to CSA and agroforestry companies in Mali, in accordance with the context in each region. 3) Workshops will be conducted to share the new guidelines and provide them with clear instructions to efficiently integrate the recommendations in their operation and planning, and continuously assess their progress and additional support required. The 100 companies will be selected for inclusion in this activity on the basis that they (i) offer services related to increased CSA practices and crop yields (examples include input providers for modern seed varieties, pest reduction services, and services related to the processing of agroforestry outputs) (ii) commit to making investments that integrate a large number of smallholders, including job opportunities for women and youth, and (iii) offer a clear path to climate resilient economic, environmental, and social sustainability. IAAT will report the number of jobs (sex disaggregated) created by the 100 private companies and their level of investment after building their capacity during the project period.

2.1.1.3 Sub-activity: Develop technical guidance for private sector technology companies working in solar irrigation and biodigester systems on inclusive business growth.

Private sector technology providers working in solar irrigation and biodigesters, especially the informal operators, will be assisted to improve their business capacities and develop their distribution network. This will include grouping technology providers into Economic Interest Groups in each commune to support the market development for these technologies by streamlining and consolidating the current supply available and as per the tried and tested approach taken by IFAD’s complementary biodigester project - MERIT.[203] This IFAD project does not cover IAAT project areas and only focuses on Kayes, Sikasso, and part of Koulikoro and Segou. The IAAT’s targeted 12 circles avoid overlapping with MERIT’s locations. Workshops will be held in each region to train businesses on (i) expansion of distribution channels, (ii) how to refine product model and price to be affordable by smallholder farmers and (iii) how best to implement and scale relevant financing models such as leasing.

2.1.1.4 Sub-activity: Implement training to producer associations to support them to deliver aggregation and value addition services to their members.

Producer associations are well-suited to share knowledge and build the capacity of their members due to the proximity and continuous collaboration within the organisations. The activity will leverage cooperatives and other producer associations to support access to broader markets for crops. This will be achieved by supporting producer associations to provide services such as cleaning, grading, and quality control, to ensure that the produce is compliant with the international quality standards. Workshops will be organized to train producer associations on these services to allow them to transfer the knowledge to their members and support them in implementing best practices. These workshops will aim to reach 240 producer associations, corresponding to 5 in each commune.

[203] IFAD’s MERIT project was consulted during the design of IAAT, for further information on it see here

Output 2.2: Smallholder farming communities, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness.

94 **Output 2.2 will be delivered through activity 2.2.1 "Support local financial institutions to increase access to finance for smallholder farmers, especially women and youth, for CSA/agroforestry investments"** and **2.2.2 "Enhance capabilities and connectivity of youth and women CSA/agribusiness entrepreneurs".** These activities will collectively address barrier 4 "Inability for smallholder and subsistence farmers to invest in inputs for climate change adaptation (seeds, fertilizer, labour, equipment) due to limited financial resources" and barrier 5 "Limited consideration of the specific and distinct needs of women and youth in the development and management of agricultural and agribusiness systems, leading to significant bottlenecks for their empowerment (e.g., limited access to finance and land, exclusion from decision-making processes)"

95 Activity 2.2.1 will increase access to finance for smallholder farmers and/or agribusiness entrepreneurs by working directly with existing financial institutions to increase their reach as well as creating new VSLAs for women and youth. Creating new and supporting existing VSLAs will provide opportunities for women and youth to receive micro-loans to invest in revenue generative activities, with a particular focus on new CSA techniques. VSLA facilitators will be established and trained to extend support and ensure sustainability beyond implementation. In addition, existing microfinance organizations active in CSA and agroforestry will be targeted to increase their reach by delivering investment guidelines that are inclusive of women and youth entrepreneurs and supporting the expansion of risk adjusted lending (e.g., through the inclusion of crop and weather insurance in loan decisions). Beneficiaries of activity 2.2.1 will be existing VSLAs created by Albarka, and new VSLAs established by GCF IAAT, including within youth associations, and microfinance institutions.

96 **Activity 2.2.1 will be delivered through 3 sub-activities.**

2.2.1.1 Sub-activity: Support the sustainability and capability of Albarka-established VSLAs by training VSLA facilitators and members on best management practices and income opportunities.

During its initial implementation years, Albarka established new women-focused VSLAs to increase access to finance for women. During IAAT, Albarka will build the capacity of existing VSLAs by establishing and training VSLA facilitators to ensure sustainability and improve the management capacities of these VSLAs. Albarka will create 76 VSLA facilitators in Gao, Mopti, and Tombouctou who will gradually be established as the main points of contact for VSLAs. Training will be provided to facilitators to improve their understanding of their roles and responsibilities and build their capacity for Income Generating Activity (IGA) development. The facilitators will provide VSLAs with relevant training, offering advisory services, and supporting them in their investment procedures. The capacities of VSLAs will be reviewed to monitor the implementation of management guidelines and identify additional areas of support. The findings will be documented and summarized in follow-up reports, providing information on the VSLA cycle and governance, especially credit and savings management.

2.2.1.2 Sub-activity: Extend and replicate the Albarka VSLA model (particularly in Segou and Koulikoro which are not addressed by Albarka), with a particular emphasis on youth such as youth associations to increase participation of youth in savings initiatives.

GCF IAAT will apply the model developed by Albarka to establish new VSLAs in the 33 communes that are not covered by Albarka. IAAT will target to the creation of 900 new VSLAs, corresponding to ~27 VSLAs per commune. In addition to the women focus of Albarka's existing VSLAs, a youth focus will be implemented with 30% of the new VSLAs established dedicated to youth and youth associations. All VSLA members will benefit from improved financial capacity and access to new opportunities to receive loans and engage in revenue-generating activities. Capacity building will also be provided to the 900 VSLAs to improve their management capacities for enhanced efficiency of procedures and sustainability of activities.

The funding sources in the initial stages of VSLAs are purely internal. The new VSLA groups will not be an immediate source of credit, however over time the shared accumulated savings by group members can be drawn on as a source of credit for productive uses. The well-functioning VSLA's can start issuing credit within the first 12 months of start-up. The amount of credit depends on the level of savings typically not more than three times the savings amount. The interest on loans is channeled back into the total accumulated resources increasing the available funds for group members. As the VSLA corpus grows, the transaction history serves as a basis for financial institutions to assess credit worthiness of VSLA members and the accumulated savings as collateral for issuing additional financing to the VSLA members as a group or to individual VSLA members.

2.2.1.3 Sub-activity: Support extension of microfinance reach by developing investment guidelines that are inclusive of women and youth entrepreneurs and smallholder farmers.

In addition to the establishment and strengthening of VSLAs to improve access to finance for communities at the village level, microfinance institutions across the project regions will be supported to improve their reach and bridge the gap with communities. For example, the Credit Union Federation established in 1997 provides services in IAAT's regions. A detailed assessment of financial institutions available in Mali and IAAT locations is presented in the Feasibility Report (Chapter III, Section 3.1.4, Sub-heading Mali's Financial Institutions). The microfinance institutions will be supported to develop investment guidelines that will allow them to reach women and youth entrepreneurs, even in remote rural areas. Specific guidelines will be developed in partnership with the communities and relevant stakeholders to overcome the barriers preventing microfinance institutions to reach and serve more women and youth entrepreneurs. In addition, microfinance organisations will be encouraged to consider risk management techniques to reduce the risk profile of smallholder farmers, such as crop insurance. Workshops will be organized to discuss barriers identified and potential solutions with microfinance institutions and provide them with clear directions regarding how to develop and implement inclusive guidelines.

The project does not expect to make significant investments in providing specialized technical services to financial institutions to establish credit scoring systems or develop new financial products. However, there are opportunities for leveraging on other projects in Mali that are designed to increase access to inclusive/green finance. These include Inclusive Finance in Agricultural Value Chain Project (INCLUSIF) and Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries - Phase I. The project will hold in-depth conversations with project representatives and specific financial institutions involved in these projects to determine the most effective opportunities for leverage and develop the detailed activities to support this collaboration.

97 Activity 2.2.2 aims to enhance the skills and connectivity of women and youth smallholder farmers and/or agribusiness entrepreneurs, empowering them to effectively manage their enterprises and facilitate their access to financial resources, thus strengthening their role within a sustainable supply chain. This initiative will provide comprehensive training on essential business competencies, while also establishing linkages between beneficiaries and local financial institutions mentioned in section 2.2.1, as well as leveraging funding opportunities from complementary projects like PRAPS II and engaging with other actors in the value chain supported in section 2.1.1. PRAPS II (the World Bank Funded Project) targets women and youth to provide support in agriculture and related entrepreneurship development. IAAT will support women and youth farmers in IAAT's project locations to connect with PRAPS II and similar other projects to access financial resources. Beneficiaries of activity 2.2.2 will include 5,250 youth and women entrepreneurs. IAAT will target minimum level of direct cash distribution to the beneficiaries. Women and youth entrepreneurs will buy equipment and/or services to start their business and project team will verify the activities/process before providing cash support. The cash support will be reimbursement based (as much as possible) after verification of activities/process started for the entrepreneurship.

98 **Activity 2.2.2 will be delivered through 3 sub-activities.**

2.2.2.1 Sub-activity: Strengthen the business capacity of youth entrepreneurs. 1,500 youth entrepreneurs will be selected by Albarka in Gao, Mopti, and Tombouctou to enhance their business capacity and improve their potential to launch and sustain viable products on the market aligned with climate climate-resilient approach. These entrepreneurs will be trained in core business skills, including business plan development and market research. Emphasis will be made on the capacity to develop well-structured loan applications to improve access to finance, which is one of the main barriers hindering youth and women entrepreneurship in Mali. IAAT will purchase tools/equipment or processing units on behalf of the entrepreneurs (20 progressive youths) to start and expand their chosen business activity after receiving the training. The tools/equipment or processing units will be provided based on the co-evaluation of the business proposals including scope, rate of return and possibility of drawing additional and alternative funds. The co-evaluation will include the entrepreneurs as part of the training, as well as IAAT business trainers delivering the capacity building. These business proposals will provide information about the type of entrepreneurship and financing and products needed to support them. There is a potential of youth involvement in business such as CSA technology supply (e.g., seed, feed, irrigation, farm tool/equipment, and livestock health services), farm product aggregation and supply to market, maintenance services for solar pumping and biodigester systems.

2.2.2.2 Sub-activity: Strengthen the business capacity of women entrepreneurs, including woman-led processing units. IAAT will replicate the same model as 2.2.2.1 in the communes that are not covered by Albarka. 3,750 women will be targeted and provided the same support as youth entrepreneurs, including training on business skills, and provision of processing units. This sub-activity will include a strong focus on women-led processing units, which grant the opportunity to women in rural areas to lead revenue-generating activities in the agribusiness sector. IAAT will directly support 40 progressive and innovative women to startup new business with some financial support. Like sub-activity 2.2.2.1, the provision of processing units will be provided based on the co-evaluation of the business proposal including scope, rate of return, and possibility of drawing additional and alternative funds. Women's entrepreneurship potential exists in the processing and value addition of agricultural products (e.g. fruit jams and juices, peanuts products, milk products). Products such as leaves for sauces, shea nuts for oil/butter, seeds for condiments and nuts/seeds for soap are the most frequently mentioned products collected by women from the agroforestry systems in Mali. IAAT will increase the scale of production through the promotion of improved agroforestry systems as well as women's business skills to enhance their entrepreneurship.

2.2.2.3 Sub-activity: Connect youth and women entrepreneurs with the local financial institutions and private sector businesses supported in 2.2.1, and other programs in the region offering financing opportunities, in order to increase access to finance, production services and distribution networks. IAAT will foster the creation of opportunities for youth and women entrepreneurs to improve access to finance and integrate sustainable value chains. This support will be provided through connections established with financial institutions (large micro-finance and investment banks operating in the regions), and private sector service providers (e.g. agriculture input suppliers and CSA technology suppliers). All the youth and women entrepreneurs trained and supported by Albarka and GCF IAAT will be targeted for this sub-activity.

IAAT project will collaborate with local financial institutions to address gaps in knowledge, information and/or access that limit the ability of youth, women, and young entrepreneurs to access finance. These activities would include (based on assessed need and demand) orientation on specific financial products provided by financial institutions relevant to the target HH needs; trainings on the application process and specific requirements for availing credit; linking to financial institution agent network and advisory services. The activity may also support financial institutions to identify and train agents to expand their network based on anticipated business opportunities.

The activity will also engage with private sector companies such as agro-input suppliers or CSA technology suppliers to explore the potential for extending value chain finance directly by the companies to the farmers or in partnership with financial institutions for vetted farmers. If meaningful financing opportunities are established, the activity would follow a similar process as described in the previous paragraph to orient farmers on these options and build their capacity to successfully avail themselves of the financing.

Output 2.3: Increased adoption of climate-smart agriculture technologies by smallholder farmers

99 **Output 2.3 will be delivered through activity 2.3.1 "Install and support the productive use of biodigester systems and solar irrigation systems amongst smallholder farmers";** which in combination with output 1.3 will directly address barrier 6 "Market, technology and cultural barriers for the adoption of low-carbon agriculture technologies/ practices such as agroforestry, solar irrigation, and biodigesters".

100 This activity will increase the productivity and revenue of smallholder farmers whilst reducing GHG emissions of farms by supporting the adoption of mitigation and adaptation technologies and their associated markets in Mali. The activity will install 1,000 solar irrigation systems to install new and replace existing diesel generator irrigation pumps and install 5,000 biodigester systems and improved stoves for livestock farmers to provide an additional income source (slurry, manure, or fertilizer) and an alternative, lower emissions, source of energy. The installation of 1,000 solar irrigation systems and 5,000 biodigesters will result in 131,566 tCO2e of GHG mitigation in 15 years lifetime. The activity will ensure that the full potential of these technologies is harnessed by (i) developing the understanding and capacity of farmers to implement them, and (ii) fostering the creation of sustainable value chains to support their long-term productive use. In order to enhance the development of supply, the project will organize existing service providers into Economic Interest Groups. This will allow them to collaborate and pool their resources to provide tailored and high-quality services to smallholder farmers. Beneficiaries of activity 3.1.1 will include 100 farmer groups, 5,000 livestock farmers, ~5,000 crop farmers, and 48 Economic Interest Groups

101 **Activity 2.3.1 will be delivered through 4 sub-activities:**

2.3.1.1 Sub-activity: Install 1,000 solar irrigation systems for smallholder farmers from the providers supported in output 2.1 and 2.2

The project prioritizes replacing diesel pumps with solar irrigation systems wherever possible to reduce emissions while maintaining or improving the efficiency of irrigation systems. IAAT prioritizes replacement of existing diesel fuel-based water pumps (small sized 3-5 hp) by solar pumps which is economically beneficial. IAAT will conduct detailed profiling of soil, water availability, and irrigation needs to finalize the location and size of the solar pumping systems during the initial preparation phase of the project. Current cost estimation includes 0.4 and 5-hp solar pump irrigation systems. This size may vary depending on the location and farming type. A detailed profiling of solar pumps during the initial stage of the project will provide this information.

All 100 solar pump irrigation will not replace diesel fueled based pumps. IAAT will target at least 25% solar pumps installed by the project to replace existing diesel fuel-based pumps that provide higher economic return than diesel pumps. The replacement of diesel pump is based on the business model (based on cost-benefit analysis) that varies with pump size and locations. In addition, IAAT will support the installation of new solar pump systems in smallholder farming households where currently irrigation is not available, and potential of solar pump irrigation exists within the five project regions using the PAYG model.

IAAT will scale 100 solar pumping systems with its investment (75% project and 25% beneficiary contributions). The additional 900 solar pumping systems will be promoted by using Pay-As-You-Go (PAYG) model of the solar pumping system in collaboration with private solar technology suppliers (e.g., ECOTECH, EnDev, EKOenergy, and PEG Africa) to scale out solar irrigation systems in the project locations. In this model, solar technology providers cover initial investment, and customers pay for electricity supply (in kilowatt-hours)[204]. IAAT will only cover the initial start-up cost of all target beneficiary system users (5-10% depending on farmers' income status). Such a business model gives farmers the ability to pay for what they can afford to use as they need. This allows farmers to pay for solar pumping system instalments, with each payment contributing to the total purchase price of the system. This model will support private sector investment in scaling up solar pumping systems in Mali (see scaling approach in Feasibility Study, Chapter VII, Section 7.1.4). IAAT will develop partnerships with private solar technology suppliers and financial institutions based on their outreach regions in Mali in the first year of the project. IAAT will provide a platform to make investments by solar and biodigester technology suppliers and financial institutions in the project locations. The investment from private solar technology suppliers and financial institutions will be tracked and play a key role in delivering 1,000 multipurpose solar pumping systems through IAAT. Key selection criteria will be households currently using diesel-powered generators to irrigate crops and households' potential to use pumping irrigation but not currently using it and willing to install the solar pump. Within these two groups, an assessment will be conducted to determine the suitability of farmer lands and the potential efficiency of solar irrigation systems based on solar irradiation and water access (depth) in the targeted villages. Households for the installation of new solar pumps or replacing diesel pumps will be selected based on their willingness to install or replace the pumps. IAAT will determine the level of co-financing based on the financial condition of the selected households, however, at least a 25% beneficiary cost share will be expected for the initial 100 solar pump irrigation systems. The PAYG model will be used to reach the target number (1,000 solar pumps[207]). The additional criteria may be included after discussion with the key stakeholders during the project implementation IAAT will bring private sector solar pump suppliers and investors to scale out 900 solar pump irrigation systems by using the PAYG model. IAAT will cover the initial start-up cost of all 1,000 solar pump users (5-10% depending on farmers' income status), including training and capacity building of solar pump users and private sector technology suppliers. Farmers will pay the total cost of the solar pump irrigation system to the private solar technology suppliers on an installment basis. IAAT will facilitate financial institutions including VSLAs and other investors to link with farmers/group of farmers and private solar technology suppliers for access to loan, concessional interest rates, and financial guarantees for PAYG model. IAAT will also work with financial institutions to implement the possibility of other business models such as out-grower (commercial agribusiness companies particularly those involved in out-grower schemes, might be interested in investing in solar irrigation pumps for farmers), and cost-sharing model (a group of farmers pool their funds to invest in a solar irrigation pump), The private solar suppliers will receive start up business of 100 solar pump irrigations from the project cost.

2.3.1.2 Sub-activity: Install 5,000 biodigester systems and improved stoves for smallholder livestock farmers from the providers supported in output 2.1 and 2.2.

5,000 livestock farmers will be selected and provided with biodigesters improving access to clean energy for households and providing an additional income stream. The biodigester system will provide an additional source of income for farmers either through the sale of the biodigester output (fertilizers), or the alleviation of the cost of fertilizers in case the livestock farmer can use the fertilizer on their own crops. Furthermore, the biodigester will provide an opportunity for clean cooking for smallholder households. Improved stoves will be provided to beneficiaries to ensure that the system is harnessed at its full potential. In order to be selected for this component, the household will require access to 20-30kg of dung/day (estimated at 2-4 cows) and equal amounts of water to ensure the efficient functioning of the system. Additional criteria to select the households for biodigester include their willingness to install biodigester and co-financing (cash or in-kind). IAAT will determine the level of co-financing based on the financial condition

[204] IRENA. 2020. Pay-As-You-Go Models. Available here

of the selected households, however, at least a 25% cost share will be expected for the 5,000 biodigester users[205]. Similar to the-project scaling of solar pump systems, the PAYG model will be used to reach the target number of biodigesters, from an initial pool of 1,000. IAAT will bring private sector biodigester suppliers and investors to scale up the remaining 4,000 by using the PAYG model. IAAT will cover the initial start-up cost of all target beneficiary system users (5-10% depending on farmers' income status), including training and capacity building of biodigester users and private sector technology suppliers. Like solar irrigation systems, farmers will pay the total cost of the biodigester system to the private biodigester technology suppliers on an instalment basis.

SCI Mali, as the Executing Entity, will implement activities 2.3.1.1 and 2.3.1.2 through private sector suppliers for solar systems and biodigesters through an open tender procurement. Once the suppliers have been selected, they will implement under SCI Mali's supervision.

2.3.1.3 Sub-activity: Deliver targeted training for those receiving new technologies in 2.3.1.1 and 2.3.1.2 on the use of biodigester and solar irrigation systems via local farm schools and extension agents supported in component 1. IAAT will invest in training activities to ensure that the beneficiaries of solar irrigation systems and biodigesters have the required knowledge to operate the technologies and avoid breakdowns by providing tailored training through farmer field schools and extension agents. Training will be phased in semi-annual instalments applicable to the stage of the equipment including (i) requirements during installation, (ii) requirements for productive use (e.g. identification of appropriate inputs for solar irrigated VCs (e.g., appropriate crops), and relevant livestock management practices for biodigesters), and (iii) establishment of businesses related to the system e.g., compost/fertilizer sale, irrigated crop cultivation. All materials will be made locally available for ease and affordability of repair via activity 2.3.1.4 (development of service support system). The training will create the opportunity for beneficiaries to harness the full potential of these technologies and reduce reliance on external operators for repairs that they can directly perform to maintain continuity of activities. All farmers who are beneficiaries of 2.3.1.1 and 2.3.1.2 will be supported through this sub-activity.

2.3.1.4 Sub-activity: Generate technical guidance and deliver commune-level workshops on the installation of an O&M for the long-term productive use of equipment for local tradespeople and construction companies.

The project seeks to build a sustainable market for solar irrigation systems and biodigesters by emphasizing the development of a strong supply chain for the technologies. Informal operators, including local tradespeople and construction companies, will be organized into Economic Interest Groups (see 2.1.1.3) and trained on technical requirements for the installation of and O&M for the long-term productive use of equipment. The training will support the development of services to support technology beneficiaries of 2.3.1.1 as well as develop the required networks to disseminate the technologies in the future. Training will be provided through workshops at the commune level.

2.3.1.5 Sub-activity: Generate technical guidance and deliver commune-level workshops on the installation of an O&M for the long-term productive use of equipment for local tradespeople and construction companies.

The project seeks to build a sustainable market for solar irrigation systems and biodigesters by emphasizing the development of a strong supply chain for the technologies. Informal operators, including local tradespeople and construction companies, will be organized into Economic Interest Groups (see 2.1.1.3) and trained on technical requirements for the installation of and O&M for the long-term productive use of equipment. The training will support the development of services to support technology beneficiaries of 2.3.1.1 as well as develop the required networks to disseminate the technologies in the future. Training will be provided through workshops at the commune level.

Component 3: Increasing Institutional Capacity and Knowledge

102 Component 3 will work to strengthen the institutional capacities of government entities at the local, regional, and national levels, as well as communities through Community Action Cycles (CACs), improving their adaptation planning capacities and management systems and procedures. Component 3 will also enhance stakeholder collaboration for knowledge-sharing for scaling out CSA, enhancing the impacts and reach of Components 1 and 2. Component 3 will achieve outputs 3.1 "Increased institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks" and 3.2 "Enhanced knowledge sharing and coordination of best practices in CSA and agroforestry across stakeholders".

[205] This is a conservative estimate as biodigester systems may benefit multiple households

Output 3.1: Increased institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks.

103 **Output 3.1 will be delivered through activity 3.1.1 "Strengthen institutional capacity in localized climate change adaptation and mitigation planning"** and activity **3.1.2 "Technical capacity building of Mali Climate Fund in prioritizing and targeting investment in CSA and agroforestry";** which will collectively address barrier 7 " Public sector institutions and community governance mechanisms lack the capacity to sufficiently include adaptation and mitigation considerations and best practices in their work, particularly with regards to planning, and remain uncoordinated across levels and with other stakeholder groups." Activity 3.1.1 will also address the agroforestry component of barrier 6: "Market, technology and cultural barriers for the adoption of low-carbon agriculture technologies/ practices such as agroforestry, solar irrigation, and biodigesters".

104 Activity 3.1.1 will strengthen institutional capacity for scaling-out CSA and agroforestry by enhancing the priority given to and capability for climate change adaptation and mitigation planning at the local level, leveraging the village level community action plans and the commune-level Economic, Social, and Cultural Development Plans (PDESC) as the main entry points for improved adaptation planning. At the commune level, the activity will provide tools to local community groups for an effective and timely review of community action plans to foster efficient management of local resources and inclusion of adaptation planning. The activity will also seek to bridge the knowledge gap preventing communal councils and other local stakeholders to efficiently coordinating initiatives at the local level, by embedding adaptation and mitigation planning into the Economic, Social, and Cultural Development Plans (PDESC) with a particular focus on agroforestry. Beneficiaries of activity 3.1.1 will be 48 communal councils and 250 CACs.

105 **Activity 3.1.1 will be delivered through 2 sub-activities:**

3.1.1.1 Sub-activity: Organize regional workshops with stakeholders who participate in the development of the PDESC at the commune-level to improve their understanding of CSA, agroforestry, and adaptation planning.

Communal councils oversee the development and implementation of the key local government planning tool in each commune: Economic, Social, and Cultural Development Plans (PDESC). The PDESC at the commune level integrates community (village level) action plans developed through CACs. Community Action Groups play a broader role as they are involved in establishing CACs, developing community action plans, and implementing them. The community action plans are informed and available to all members of the communities. The project will build their capacity to design and integrate adaptation, and mitigation plans in the PDESC to support the adoption of a clear and common strategy at the local level to reduce the vulnerability of communities to climate change and shocks. The training will also aim to bridge the knowledge gap between CSA and agroforestry, hindering the capacity of local authorities to effectively design, implement, and coordinate adaptation and mitigation activities. Representatives of the 48 communal councils will be trained on (i) adaptation plan development and best practices for how to incorporate into local government decision-making, (ii) overview of CSA and agroforestry to improve local authorities understanding of challenges and importance of the techniques, and iii) introduction to land-use planning to enhance the development of agroforestry. Municipalities will be subsequently supported to review and update their respective PDESC, effectively integrating adaptation and mitigation planning.

3.1.1.2 Sub-activity: Support new and existing Community Action Groups in the five regions to periodically review and update community action plans for the management of local resources and adaption planning.

Community Action Groups are essential for the empowerment of communities and their involvement in decision-making processes at the local level. Through sub-activity 3.1.1.2, the project will aim to strengthen the capacity of communities to autonomously identify, plan, implement, and evaluate actions to address specific needs or challenges relating to their villages. Community Action Groups will be supported to develop and implement inclusive community action plans for the management of shared resources and the implementation of activities for the empowerment of communities, especially women and youth, and the use of traditional knowledge and skills. Workshops will be organized to train them on the specific requirements for and benefits of the development of inclusive plans. The Community Action Groups will also be trained on the design and implementation of adaptation and mitigation activities at the community level, as well as their integration into community action plans. Community Action Groups will be supported through a process to self-evaluate their action plans with specific guidelines developed to lead them through the process of reviewing and updating their plans. Learning sessions will be organized every quarter at the commune level to enhance collaboration and experience sharing between Community Action Groups. This will allow us to identify best practices and coordinate how neighbouring villages can collaborate to collectively empower their communities. All-inclusive community action plans will be developed in the selected villages before starting the implementation of CSA and agroforestry-related intervention.

106 Activity 3.1.2 supports streamlining the funding of the Mali Climate Fund in CSA and agroforestry systems. The Mali Climate Fund and National Climate Change Committee (CNCCM) will be supported to improve its capacity to prioritize and allocate funds to CSA and agroforestry projects in Mali and efficiently manage and monitor these funds post-allocation. The activity will also aim to improve the contribution of the public sector in co-financing schemes with development partners through the refinement of funding processes. Beneficiaries of activity 3.1.1 will be the Mali Climate Fund and National Climate Change Committee (CNCCM) estimated at 10 representatives of these institutions.

Activity 3.1.2 will be delivered through 1 sub-activity:

3.1.2.1 Sub-activity: Enhance the capacity of the Mali Climate Fund to target adaptation and mitigation actions in agriculture. The project will support members of the Mali Climate Fund (MCF) and the National Climate Change Committee (CNCCM) to prioritize and target their investment in CSA and agroforestry technologies, practices, and services. Workshops will be organized to train staff of MCF and CNCCM on specific guidelines for prioritizing and designing adaptation and mitigation actions in agriculture and forestry. The workshops will provide a platform to discuss the required improvements to enhance the participation of government institutions in co-financing schemes.

Output 3.2: Enhanced knowledge sharing and coordination of best practices in CSA and agroforestry across stakeholders.

107 Output 3.2 will be delivered through activity 3.2.1 “Enhanced convening and contribution to national databases for CSA and Agroforestry”; which will directly address barrier 7 “Public sector institutions and community governance mechanisms cannot sufficiently include adaptation and mitigation considerations and best practices in their work, particularly with regards to planning, and remain uncoordinated across levels and with other stakeholder groups.”

108 Activity 3.2.1 will enhance knowledge-sharing and coordination of best practices among relevant stakeholders. The activity will compile the insights gained from the IAAT project and disseminate them to national and regional stakeholders, extending the reach of valuable knowledge and lessons learned. This will involve active participation in regional and local gatherings, conferences, and knowledge networks. A special emphasis will be placed on involving traditional knowledge groups to ensure a diverse range of perspectives and expertise are included in the knowledge-sharing process. Activity 3.2.1 will contribute to building an enabling environment supporting the enhanced reach of outputs and outcomes targeted through the other components of the project. This knowledge and lessons learned support the Government of Mali and other stakeholders (in Mali and beyond) to scale up after completion of IAAT implementation.

109 **Activity 3.2.1 will be delivered through 3 sub-activities:**

3.2.1.1 Sub-activity: Attend national and regional conferences and organize regional stakeholder convenings to share lessons on project implementation and insights as they emerge for overall collaborative learning. GCF IAAT will actively contribute to knowledge-sharing and policy-making platforms at the national and regional levels, especially on the improvement of adaptation and mitigation capacities in agriculture and agroforestry. Participating to these platforms will allow us to remain aware of the most recent developments and future trends in the areas of interest, and influence thought leadership and policy-making towards better solutions for enhanced adaptation and mitigation of smallholder farmers. The insights collected during these events will be documented and discussed with relevant stakeholders to inform project design updates. Workshops will be organized in a year to regroup key stakeholders in the climate, agriculture, and agroforestry space, to share lessons learned during project implementation, align on key priorities in the regions of interest, and identify further areas of collaboration to enhance the impact of existing initiatives.

3.2.1.2 Sub-activity: Join existing knowledge sharing and academic networks and contribute to existing databases (e.g., with land use data) to allow project data and findings to be leveraged in academic research. IAAT project staff will participate in existing knowledge sharing and academic networks to contribute and learn from research and innovations related to climate change, agriculture and agroforestry in Mali and the region. These networks will support the discussion and testing of new ideas, for example in support of streamlining of the CSA technology market or identify and replicate solutions that have been successfully implemented in countries with a similar context. The data generated through project implementation, including land use maps and land use database, will be shared with relevant stakeholders to contribute to the development of national data platforms, national/sub-national planning to scale up CSA and agroforestry and enhance the development of research in climate change, agriculture, and agroforestry in Mali.

3.2.1.3 Sub-activity: Organize annual knowledge-sharing event to report best-practices on CSA learned from the project. Annual meetings will be organized to share the knowledge generated through the GCF IAAT project. These meetings will regroup all relevant stakeholders, including government, communities (including women, youth, smallholder farmers and agropastoralists and children), NGOs, CSOs, private sector, and research institutes and will provide an opportunity to reflect on project activities and impact, risks and potential mitigation, and identify solutions to improve project implementation. Importantly these stakeholders will include representatives from other regions outside of those targeted by the project and where relevant, neighbouring countries, to allow for a broader reach of the lessons learned.

110 By acting within the same communities and regions but across different areas of the agricultural system, the components and activities also directly interlink to create a magnified impact in support of the targeted paradigm shift in Mali's agricultural system. Examples of how the project components are coherent and interlinked are outlined below:

111 Component 1 and Component 2: within the same target communes, component 1 will work to first increase agricultural output and Component 2 will then provide mechanisms for the sale of the increased products (access to value-added services and to buyers). In addition, component 2 will increase access to finance to fund the inputs needed to cultivate the crops in component 1. These components will build on the introduction to agroforestry techniques for smallholder farmers delivered in Component 1, by actively supporting the planting of 91,285 ha of agroforestry through increased use of land-use maps to identify suitable locations, cultivation of tree saplings and their distribution across project regions.

112 Component 2: component 2 will support the businesses that sell priority mitigation technologies (solar irrigation and biodigester systems) to expand their reach to new smallholder farmer market segments.

113 Component 3 in support of component 1 and 2: component 3's institutional support will support an overall shift in the prioritization of CSA and agroforestry at the community, commune, and national level, serving to support the local interest and investment in the techniques advocated as part of the training for smallholder farmers in component 1 and implemented in agroforestry in component 2.

114 As outlined in section B.1, the project will focus on the 12 circles, and within them 48 communes, in the Tombouctou, Gao, Mopti, Koulikoro and Segou regions in Mali on the basis of their increased climate vulnerability. Within these regions, circles and communes' beneficiaries will include smallholder farmers and their households, women and youth entrepreneurs, private sector businesses, extension services members and members of public institutions.

Table B.3.3: Project Target Beneficiaries

Direct Beneficiaries	Project interventions	Primary Selection Criteria	Target Number	Adaptation benefits (GCF Result Area)
Farming households or farmers/individuals	Receiving CSA training via extension services in project target communes (Output 1.2)	Farming households within the 48 target communes of the project and able to access extension services	86,340 (Farming households)	Minimizing yield and income loss due to changing climate through better planning and implementation of CSA GCF Result Area: ARA1- Vulnerable people, ARA2 – Well-being and food security
	Connected to broader services and markets through access to online platforms (Output 2.1)	Farmers within the 48 target communes of the project; interested in joining platforms; able to access by extension services and the internet	22,778 (Farmers/individuals)	Improving farm productivity and minimizing income losses by informed decision-making using information provided by extension and online platforms GCF Result Area: ARA1- Vulnerable people, ARA2 – Well-being and food security
	Receive & use Biodigester systems (output 2.3)	Farmers within the 48 target communes of the project and able to access by extension services; access to 20-30kg of cow dung (breeding 2-4 cows) and have access to the same quantity of water, willingness to engage in the project	35,000 (Farmers/individuals)	Minimize the increasing risks of firewood supply under climate change by using biodigester. GCF Result Area: MRA1- Energy generation and access

	Receive & use solar irrigation systems (output 2.3)	Farmers within the project's target communes of the Mopti, Segou, and Koulikoro regions and able to access by extension services. Household already utilizes a diesel pump for irrigation purposes and households' potential to use pumping irrigation but not currently using it, and willing to install the solar pump, Suitability will also depend on solar irradiation and water access (depth)	7,632 (Farmers/individuals)	Minimize the increasing risk of drought and water scarcity for agriculture and livestock production. Increase the potential of agriculture diversification to build a resilient agriculture production system GCF Result Area: MRA1- Energy generation and access ARA2 – Well-being and food security
	Plant Additional Agroforestry (output 1.3)	Farming households within the 48 target communes of the project and able to access extension services. Able to access at least 0.25 Ha of land for the plantation of agroforestry (either their own or state-owned lands)	64,755 (Farming households)	Minimize the risk of decreasing fodder supply under climate change for livestock, minimize the risk of firewood supply, and diversify household income portfolios. GCF Result Area: MRA1- Energy generation and access ARA2 – Well-being and food security
Entrepreneurs (women and youth)	Training for overcoming barriers faced specifically by women and youth entrepreneurs (output 2.2)	Women and youth farmers within the 48 target communes of the project Fully or partially "unemployed" and interested in entrepreneurship OR already active in entrepreneurship and interested in additional training	5,250 (Farmers/individuals)	Diversification of livelihood opportunities and enhance economic resilience to climate change. GCF Result Area: ARA1- Vulnerable people, ARA2 – Well-being and food security
Private Sector Businesses	Trained and received guidance on business model extension to new segments (output 2.1)	Active in the 5 target regions of the project Currently offer services related to increased CSA practices and crop yields (examples include input providers for modern seed varieties, pest reduction services, and services related to the processing of agroforestry outputs) and commit to making investments that integrate a large number of smallholders, including job opportunities for women and youth, and offer a clear path to climate-resilient economic, environmental, and social sustainability	100 (Farmers/individuals)	Enhancing and diversification of livelihood opportunities for economic resilience to climate change. GCF Result Area: ARA1- Vulnerable people, ARA2 – Well-being and food security
Extension Service Members	Trained on new CSA curriculum (output 1.1)	Public and private sector extension agents active in the 48 target communes of the project (estimated 10 per commune, or 100 per region)	500 (Farmers/individuals)	Build institutional adaptive capacity to climate change that will support farmers and their communities GCF Result Area: ARA1- Vulnerable people,
Individual Institutional Beneficiaries	Receive training on integrating mitigation and adaptive capacity considerations into planning and development (output 3.1)	Involved in PDESC planning in the 48 target communes (48) Work as part of disbursement or impact measurement at NCF (10)	58 (Farmers/individuals)	B Build institutional adaptive capacity to climate change that will support farmers and their communities GCF Result Area: ARA1- Vulnerable people,

Indirect Beneficiaries	Project interventions and Criteria	Target Number	
Farmers and their household members	Farmers who are resident within and have access to extension services within the 5 project regions (Output 1.2)	700,980 (Farmers/individuals)	Enhance climate change adaptive capacity through knowledge transfer GCF Result Area: MRA1- Energy generation and access ARA2 – Well-being and food security

117. For further details on the project beneficiaries and their selection criteria, please refer to section E of the funding proposal and **Annex 2, Section VIII-Vulnerability and Targeting Assessment**.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

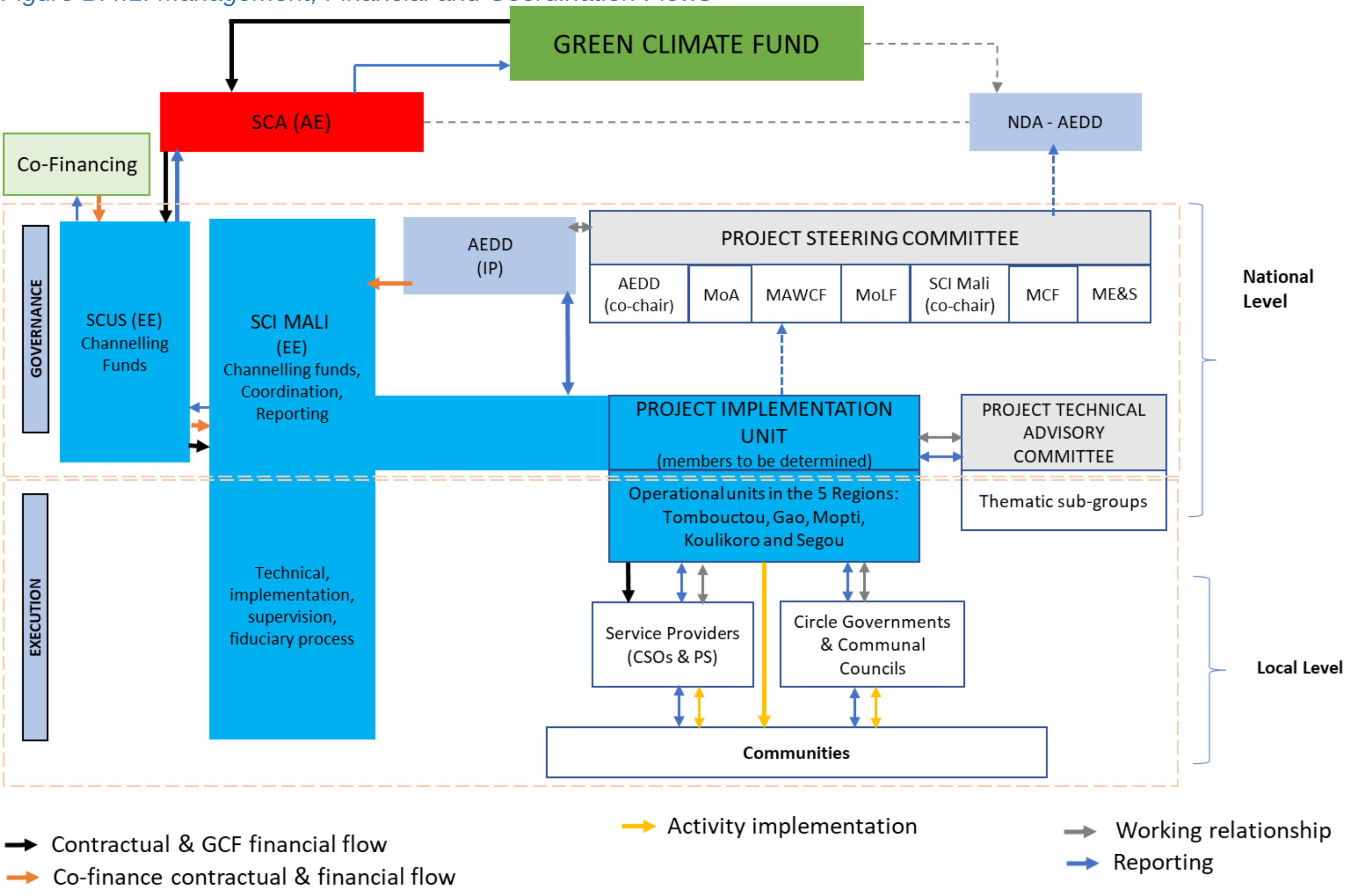
118. This project will have two Executing Entities (EEs): Save the Children Federation, Inc. (SCUS), and Save the Children International (SCI) au Mali. SCUS will a channelling EE with responsibilities for supervision and oversight of the utilisation of the funds they are channelling in line with the Save the Children Association ways of working. SCUS will provide also technical support in designing, implementing, monitoring, evaluating, and reporting. SCI Mali will serve as EE channelling funds in Mali and will be responsible for the project implementation in collaboration with the GoM. As implementing EEs, SCI Mali will be responsible for the execution and supervision of technical and minor infrastructure activities (e.g. solar irrigation system, biodigesters, etc). The project will be implemented in close consultations with the Government of Mali's ME&S through AEDD, which acts as Mali's NDA to the GCF. Save the Children Australia (SCA) as the Accredited Entity (AE), will maintain reporting channels to the GCF to ensure compliance with the fiduciary, environmental, social, and other relevant standards. SCI Mali will maintain reporting channels to ME&S to maintain and increase the country's project ownership. SCUS will provide account management support to SCI Mali in accordance with the account management system employed by the wider Save the Children movement to ensure compliance and high-quality delivery of the project and financial management and reporting.
119. All field-level activities which need direct engagement with farmers and communities, local government and civil society organizations, and the private sector will be implemented by the local NGOs. Implementing partners (local NGOs) will be procured through the open bid process. SCI Mali will be the procuring entity, and the procurement will be done in accordance with the requirements under the AMA. For, technology supply (i.e solar irrigation, biodigesters, and other CSA technologies) SCI Mali will follow an open bid process to purchase from the private suppliers/companies.
120. A high-level Project Steering Committee (PSC) will be responsible for the project's strategic direction and oversee activity implementation, including steering activities implemented by the Project Implementation Unit (PIU), which will manage day-to-day operations. Representatives of SCI Mali and the AEDD will co-chair the PSC, which will also include representation of senior officials from ME&S. Ministries represented on the PSC will include the Ministry of Environment and Sanitation, Ministry of Agriculture, Ministry of Livestock and Fisheries, Ministry of the Advancement of Women, Children, and Families, Mali Climate Fund. No financial flows will take place between SCI Mali and the PSC; the working relationship will be solely based on reporting and consultation. The PSC will also consult with and report to ME&S as the Mali NDA. While the wider group co-chairs will make decisions, consensus will be required for any key decisions with contractual ratifications. The AE will ensure that decisions taken by the PSC are consistent with the AMA, FAA and FP by regularly holding meetings with the core project team, including the PIU team lead and the Country Director of SCI Mali. The AE will also run a learning and knowledge session during the project inception period on the AMA and the FAA so that all partners understand their responsibilities ahead of project implementation and understand what amendments may need AE and / or donor prior approval. This is no different to the delivery of any other donor funded project
121. The PIU will be led by the Team Leader, who will hold decision-making authority, and co-managed in partnership with the AEDD and SCI Mali. The PIU will manage project implementation, support implementing entities, and be supported by a technical supervisory team, including a Team Leader, Deputy Team Leader, Finance Director, Monitoring and Evaluation Advisor, Technical Leads, and support staff. It will work directly with the technical implementing teams of the AEDD, SCI Mali, SCUS and local NGOs, and directly manage the implementation of the project at circle and commune levels. The PIU will draw on technical expertise from the broader SC movement (specifically climate change adaptation and mitigation in agriculture and NRM) as well as from executing and implementing entities. The PIU will report to SCI Mali and will also have accountability to the PSC. The technical

advisory committee will include subject matter specialist in agriculture, climate change, agroforestry, gender and social inclusion, conflict management, market system development and other experts from SCI Mali, SCUS, and AEDD. This committee will provide technical backstopping to the PIU to implement the IAAT project activities.

122. The PIU will monitor and oversee technical and financial implementation, fiduciary processes, auditing, measurement reporting and verification system, and internal evaluation of the project. The PSC will oversee the PIU and provide it with guidance and direction while receiving regular reports on project implementation. The PIU will receive funds (both GCF and co-financing) distributed according to the Subsidiary Agreements with the EEs and procurement contracts for the provision of goods and services. The PIU will operate according to the Annual Work Plan approved by the PSC of the project. All the administrative matters of the project: project implementation plans, procurement plans, financial plans, periodic reports, will be approved by the PSC. The funds will be distributed to EE and implementing partners, based on work plans and budgets approved by the PSC. The PIUs will collaborate closely with national ministries and implementing partners to ensure efficient use of resources, in line with project objectives

123. The management oversight of the PIU will be held by SCI Mali, which has the legal standing necessary to implement GCF activities in Mali. The implementing partners, including AEDD, local NGOs, Solar and Biodigester Suppliers, have no discretion in implementing funded activities or in the usage of GCF funds. All such discretions shall lie with SCI Mali, the Executing Entity.

Figure B.4.1: Management, Financial and Coordination Flows



AEDD = Agency for Environment and Sustainable Development, **ME&S** = Ministry of Environment and Sanitation, **MoA** = Ministry of Agriculture, **MoLF** = Ministry of Livestock and Fisheries, **MCF** = Mali Climate Fund, **MAWCF** = Ministry for the Advancement of Women, Children and Families, **NDA** = National Designated Authority, **SCA** = Save the Children Australia, **SCI Mali** = Save the Children Mali, **CSOs** =Civil Society Organizations, **PS** = Private Sector

124. To meet the fiduciary requirements and standards of the GCF, legal and management oversight of the PIU will be held by SCI Mali, which has the legal standing necessary to implement GCF activities in Mali. It is also legally and organizationally bound to SCA, which acts as AE and holds the AMA with the GCF on behalf of the SC movement.

125. Based on the assessment of the Financial Management and Capacity Assessments (FMCAs), as AE, SCA will channel GCF resources to a project-specific bank account managed by the PIU in Mali. The PIU will use these

resources and co-financing to manage project governance and implementation and will manage all downstream flows implementing partners (local NGOs), and any other procured partners. (Note: Implementing partners will be selected and subcontracted as procured parties in accordance with the AE's procurement guidelines).

126. The PIU will deliver results through a combination of direct and indirect interactions with the 48 communities in 12 circles across 5 regions. The operational units in the five regions will be responsible for overseeing the daily operations of IAAT activities in the field. They will organize training, meetings, and conferences; liaise with service providers and communities; conduct monitoring and evaluation (M&E) activities; and engage with local stakeholders to ensure the smooth implementation of project activities. While the PIU will deliver technical and project assistance directly to the beneficiary communities, it will rely on the AEDD, SCI Mali, and local NGOs for the delivery of specific outputs (as depicted in Figure B 4.1).

All activities will be implemented in line with the project's Gender Assessment and Action Plan (see **Annex 8**), which is aligned with the GCF's Gender Policy and SC's gender, inclusive development, and child safeguarding policies and Prevention of Sexual Exploitation and Abuse policy and guidelines. Per the Gender Action Plan, all project activities prioritize building the resilience of the most vulnerable community members (e.g., women, children, smallholder farmers, and youth). Activities will contribute to gender transformation and ensure community participation.

127. All activities will be implemented in line with the project's Environmental and Social Action Plan (ESAP), which is aligned with the GCF's safeguards policies and includes a Grievance Redress Mechanism.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

128. There is a significant gap in climate finance in Mali that the GCF is well-positioned to fill and address critical areas of climate change vulnerability that the Government of Mali has prioritized and committed to, however is unable to fund these due to limited domestic and international financing sources. As an African LDC – a strategic priority for GCF investment – Mali has very limited public resources, and as a country recovering from a series of increasing droughts, floods, and heat events during the past 30 years due to rising temperatures and erratic rainfall, Mali's ability to invest in its agriculture and natural resource management sector is restricted. At the same time, the estimated adaptation cost is very high due to the population's dependency in climate sensitive sectors such as agriculture and natural resources (NDC 2021).

129. This project will benefit 431,700 people directly (86,340 farming households) and 700.980 people indirectly; all of whom are amongst the most vulnerable to climate change in Mali. There is a significant gap in climate change adaptation capacity and action, particularly amongst smallholder farmers who constitute many direct beneficiaries, resulting in extreme poverty and food and water insecurity amongst households. The current and future needs of smallholder farmers are underserved by the financial and technical funding and capacity-building gaps by the Government. Also, the perceived poor economic returns and high-risk nature of investment in smallholder farmers are unattractive to the private sector. As a result, concessional funding from the GCF, in the form of grants, is critical to enable the IAAT to catalyse change in five regions with scalable results for a full paradigm shift in other regions of Mali and internationally to countries with similar contexts, such as Niger.

130. Ongoing conflict limited domestic resources, the impact of COVID and sanctions from the ECOWAS have hindered the government's ability to allocate funds for agricultural development aligned to climate adaptation and mitigation commitments. As a result, only USD 0.5 billion and USD 67 million are respectively dedicated to agriculture and environmental protection in the 2023 national budget. The National Strategy for Climate Change and the Mali Climate Fund, which is the main vehicle for climate finance in Mali, have struggled to mobilize sufficient funding with the latter obtaining funding of USD 29 million. The estimated funding needs for climate-resilient agriculture and agroforestry are in the order of magnitude of billions resulting in a significant gap to drive transformative change. In 2022, Mali's Public Treasury raised only 71.9% of its resource objectives and the AFDB identified new sanctions by the Economic Community of West African States resulting from changes to the consensus timetable for elections, the lack of security, and the impact of climate change as headwinds for the country's economic outlook.[206] Where public finance has been able to be directed to climate change adaptation, agriculture or agroforestry the total amount has often been limited. The government's National Strategy on Climate Change has not been able to mobilize sufficient funding.[207] Although the MCF was in part established to resolve challenges on access to finance, the investment it has received is woefully insufficient: with a budget of USD 29 million largely from Swedish and Norwegian bilateral funding.[208] The IAAT seeks to address this national and international gap in climate financing through Activity 4.1.2. Although the exact level of funding needed for climate-resilient, smallholder agriculture and agroforestry in Mali is difficult to determine, the Growth and Poverty Reduction Framework estimates that an additional investment of US$307 million will be needed to improve land management strategies

[206] AFDB (2023), "Mali Economic Outlook", accessible here
[207] Mali Climate Fund, (2012) Presentation, Link
[208] Government of Mali, 2022 Annual report on the activities of the Mali Climate Fund, Link

and infrastructure, increase access to technologies, structured partnerships, and advisory services, especially for women. The National Agricultural Sector Investment Plan (PNISA) estimates implementation for agricultural transformation at US$6.7 billion, of which more than 80% is anticipated to come from donors.[209]

131. Similarly, the total estimated cost of Mali's NDC implementation is US$1.257 billion, that includes water management and system of rice intensification (US$590 million), soil erosion control through tree plantation (US$ 500 million), sustainable crop nutrient management (US$2.5 million, and GHG emissions reduction from the livestock sector (US$165 million). Major investments are also needed in research and institutional capacity building to address systemic issues related to the design and delivery of climate-resilient interventions that fit local needs and conditions, something the IAAT is delivering through locally-led approaches to mitigation and adaptation solutions. The implementation of Mali's NDC is conditional on technical and financial support from international donors and financial institutions. Although development and climate funders are present in Mali there is insufficient support to close the funding gap, and the complex political and conflict situation in Mali also pose challenges. For example, after France withdrew its military operation in 2022, it announced plans to withdraw public aid funding, which is estimated at EUR470 million between January 2013 and September 2017.[210] In addition, there are prominent barriers to private sector financing in Mali including a lack of training in procedures for accessing climate finance, low participation in the development of climate change strategies, and limited access to international finance.[211] A detail of Mali's financial needs and absence of alternative sources of finance is presented in the **Feasibility Study, Chapter IX, Section 8.1.18**. The IAAT is seeking to address private-sector partnerships through Output 2.1.

132. **Climate finance to smallholder agriculture in Mali is also disproportionately low when compared with the importance of agriculture to the economy**. Reflecting global challenges in financing smallholder agriculture, barriers limiting climate financing in this area include the low investment readiness of agri-businesses; the need, at the farmer level, for financing to be associated with technical assistance to reduce the risks involved in transitioning to new climate-smart practices; and the difficulty of aggregating assets to make them attractive to large scale investors.[212] In addition, the low-profit and high-risk nature of smallholder agriculture has made lending conditions onerous, created limited exposure to external investment, and attracted higher interest rates and high collateral requirements, which severely impact agricultural production and weaken small-scale agribusinesses. As a result, the alternative funding options for supporting investment at the scale proposed by the IAAT project for smallholder farmers are limited outside of donors such as the GCF. The IAAT is also seeking to address locally led financing options (VSLA and microfinancing) through Output 2.2.

133. **The $5 million grant co-financing from USAID is provided through complementarity with its Albarka project, a US$60 million total grant project on Resilience and Food Security in Gao, Tombouctou and Mopti.** GCF IAAT and USAID Albarka have high levels of complementarity across timing, location and interventions. USAID Albarka commenced implementation on October 01, 2020, and is expected extend to 2026, resulting in an anticipated overlap of 2.5 years with IAAT (2024Q2, 2025 and 2026). The project aims to improve food security and resilience of communities in conflict-affected areas in Mali through strengthening local systems and community participation. Examples of overlapping interventions include the establishment and strengthening of strategic organizations for community development and enhanced adaptation and resilience capacities as well as enhancing business capacities of women and youth and creating sustainable markets for producers. The IAAT project has therefore been designed to build on and complement the existing work of the Albarka project, maximizing the impact of the two projects. For further clarity on the activities implemented by the Albarka project please refer to section B.3, and for the corresponding breakdown of the split of the co-financing by component please refer to section C.1. The GOM has also prioritised just over US$5 million in-kind co-financing towards the IAAT in recognition of the value it brings to the climate resilience of vulnerable Malians. There is a strong justification for the GCF grant request which in the absence of alternative sources, will support the GoM to advance progress on its national climate change adaptation and mitigation commitments through IAAT focused on local and scalable CSA and agroforestry.

134. Progress on climate action in Mali, including gender equity, clean water and sanitation, and sustainable communities is moderately increasing[213]. Without the GCF funding to de-risk climate investments, especially with private sector engagement, not only the climate change agenda is compromised, but any progress made on wider SDGs is likely to fail to achieve ambition as the climate-vulnerable population is likely to increase without warranted financial support.

B.6. Exit strategy (max. 500 words, approximately 1 page)

[209] Government of Mali, (2014), National Investment Plan for the Agriculture Sector (PNISA), Available here
[210] RFI (2022), "France halts development aid to Mali", accessible here
[211] AFDB (2023), "Mali Economic Outlook", accessible here
[212] CPI (2020), "Examining the Climate Finance Gap for Small-Scale Agriculture", accessible here
[213] SDSN. 2023. Sustainable Development Report 2023. Available here.

135. Project strategies and activities include a strong focus on sustaining and continuing results and benefits beyond the implementation period; enabling long-term ownership, operations, and maintenance of investments; and supporting MoA to replicate and scale climate-smart agriculture and agroforestry technologies and practices across the project regions and beyond. The key elements of the exit strategy are:

136. **Long-term sustainability and systems-level change is one of the guiding principles of the project design and implementation** (see **Annex 2, Section II**). IAAT will provide the building blocks of a shift in the agricultural and agroforestry system in Mali as well as an opportunity to create transferable learnings and models across communities and regions in Mali, and neighboring countries with similar contexts. The long-term sustainability of IAAT includes capacity building and institutional strengthening; beneficiary ownership; investment in suitable and sustainable technologies; and compatibility with existing government policies.

Capacity building forms part of all components, supporting the long-term sustainability and impact if IAAT's outcomes. The project directly works with the national government, national extension services, local financial institutions (e.g. VSLAs, Community Action Groups), private sector businesses, and smallholder farmers to help them build capacity in techniques, technologies and processes directly or indirectly relevant to the delivery of climate-smart agriculture and agroforestry. Within this capacity building, there are a number of activities and sub-activities from the GCF IAAT project that are easily transferable beyond both the direct beneficiaries of the project during and after project implementation. The AEDD (also the NDA) is at the heart of the project and offers a critical opportunity for institutional beneficiary ownership that will help deliver government buy-in, the smooth exit of the GCF project, and the long-term outcomes of the project. In addition, as outlined in section B.4, the Project Steering Committee also includes 5 government ministries and directorates, representing a broad range of government areas to foster inter-departmental collaboration and government ownership.[214]

137. **New knowledge generation and sharing** through the organization of annual forums for all key actors (national, regional, and local bodies, other agricultural and climate change projects, researchers, and traditional knowledge groups) to share and discuss the key technical learnings of the project and how these could be applied to additional geographic areas. In these forums, organizations will be invited from both the regions of the project and the regions not covered in order to extend the reach of the findings and build synergistic relationships. IAAT will also serve to create long-term sustainability of the project through its focus on land use mapping which will provide an enhanced data environment and understanding of land use in Mali. This can support the creation of future climate and agricultural policies, research, and interventions.

138. **IAAT invests in and promotes technologies and their supporting value chains that have been selected as sustainable and suitable for the local context**. IAAT supports the installation of solar irrigation and biodigester systems that ensure long-term sustainability by increasing households'/communities' access to reliable energy and water sources through private sector technology providers. In addition, the technology providers will be supported to expand their businesses further and develop women and youth-inclusive business models, supporting the development of the technology market and the long-term expansion of the technologies in Mali. IAAT will support the creation of improved value chains for CSA and agroforestry products to create long-term market system development.

139. **Local and beneficiary ownership at both the community and institutional levels is built into all project components supporting long-term buy-in for the project.** At the community level, all activities implemented within communities have been designed in collaboration with beneficiaries and will be informed by community stakeholder engagement during implementation. For example, within the target community's stakeholder engagement will be conducted to collectively determine the distribution and fair use of all new technologies, equipment and practices distributed (e.g. solar irrigation systems, biodigester systems, on-farm agroforestry planting) increasing beneficiary ownership and reducing risk of local conflict. Community Action Groups will be supported to develop and monitor plans for local adaptation practices and the management of local community assets. Similarly, in the support delivered to extension services, linkages between extension services and local community organisations are developed and supported in order to ensure extension services are collaborative and tailored to local contexts as well as able to leverage local community knowledge.

140. **The project supports and is in turn strengthened by existing climate change, agricultural and environmental policies in Mali.** IAAT will support Mali's policies that promote climate smart agriculture, sustainable land management practices and reforestation. Many IAAT's activities are prioritized by Mali's Nationally Determined Contributions (NDC), National Adaption Programme of Action (NAPA), National Climate Policy (PNCC) and National Climate Change Strategy (SNCC). The key components of climate policies include mainstreaming climate change into sectoral policies, capacity building, climate finance, and private sector engagement in climate change mitigation efforts. Various actions are outlined within these policies, including reforestation, enhancement

[214] The PSC provides project oversight and strategic guidance, and members may comprise: Save the Children Mali as a executing entity, the Ministry of Environment and Sanitation (ME&S is mandated to coordinate climate change action in Mali) as co-executing entity, Environment and Sustainable Development Agency (ESDA, as an executing entitiy), the Ministry of Agriculture, the Ministry of Livestock and Fisheries, Ministry for the Advancement of Women, Children and Families, regional government and Mali Climate Fund.

of weather and climate information systems for agriculture, climate resilience in the agriculture and forestry sectors, agricultural diversification, and sustainable land management, among others.[215]

141. **IAAT is closely aligned with the above policies across its components which collectively seek to increase the climate resilience of Mali's agricultural system and increase the adoption of low-carbon livelihoods**. IAAT will achieve this through activities that deliver increased awareness and usage of Climate Smart Agriculture techniques and technologies *(highlighted in NAPA, NDC),* the corresponding on-farm production of CSA crops/livestock, agroforestry *(SNCC)* and the creation of integrated and inclusive CSA value chains – ultimately generating an adaptive and mitigation shift in Mali's agricultural system. For example, IAAT focuses on increasing the awareness of climate risks and adoption of CSA techniques amongst smallholder farmers by firstly building the capacity of agro-advisory services, the key educational service for smallholder farmers, and secondly by supporting the on-farm adoption of CSA techniques, with a particular emphasis on agroforestry. This component also includes integrating existing work from the Malian government, NGOs and the GCF on climate information and early warning systems to broaden awareness and usage *(highlighted in SNCC and NAPA).*

[215] Ibid.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	33,717,918	USD ($)

GCF financial instrument	Amount	Currency	Tender	Pricing
(i) Senior loans	Enter amount	Options	Enter years	Enter %
(ii) Subordinated loans	Enter amount	Options	Enter years	Enter %
(iii) Equity	Enter amount	Options		Enter % equity return
(iv) Guarantees	Enter amount	Options		
(v) Reimbursable grants	Enter amount	Options		
(vi) Grants	**33,717,918**	USD ($)		
(vii) Results-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	10,000,000	USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Government of Mali (AEDD)	In kind	**5,000,000**	USD ($)	Enter years Enter years	Enter%	Options
Albarka Resilience Food Security Activity (USAID)	Grant	**5,000,000**	USD ($)	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	43,717,918	USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	
	Co-finance from the Government of Mali. AEDD is a legal entity under the Ministry of Environment and Sanitation (ME&S). AEDD oversees the Government of Mali's climate change-related projects. The government of Mali's co-finance will be channeled through AEDD. Currently, AEDD is implementing climate adaptation projects in the IAAT target regions.

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Component	Output	Indicative cost (USD)	GCF financing		Co-financing		
			Amount (USD)	Financial Instrument	Amount (USD)	Financial Instrument	Name of Institutions
Component 1	Output 1.1	763,087	263,087	Grant	500,000	In-kind	GoM
	Output 1.2	17,892,454	14,460,454	Grant	3,432,000	In-kind + Grant	GoM + USAID
	Output 1.3	5,608,100	3,818,100	Grant	1,790,000	In-kind + Grant	GoM + USAID
Component 2	Output 2.1	1,939,342	1,939,342	Grant	-	-	-
	Output 2.2	4,066,226	2,356,226	Grant	1,710,000	Grant	USAID
	Output 2.3	4,712,614	4,712,614	Grant	-	-	-
Component 3	Output 3.1	2,755,017	1,005,017	Grant	1,750,000	In-kind	GoM
	Output 3.2	945,677	945,677	Grant	-	-	-
M&E	M&E	2,046,204	1,746,204	Grant	300,000	In-kind + Grant	GoM + USAID
Project Management		2,007,122	1,489,122	Grant	518,000	In-kind + Grant	GoM + USAID
Contingency		982,075	982,075	Grant			
Indicative total cost (USD)		**43,717,918**	**33,717,918**		**10,000,000**		

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

142. **IAAT invests in technologies and their supporting value chains that have been selected as sustainable and suitable for the local context, specifically solar irrigation systems, and biodigester systems** Outcome 3, and part of Outcome 1 and 2, support the installation of solar irrigation and biodigester systems. Through stakeholder consultation and a review of the existing academic literature these technologies have been identified as particularly suitable for the regions of Mali where the project is being implemented (for more information see **Annex 2, Section VII, Sub-section 7.3**).

143. IAAT supports their deployment in a long-term, sustainable way by ensuring that: i) beneficiaries are selected carefully for technical compatibility (for example, applicable livestock ownership for biodigester systems, applicable groundwater for solar irrigation), ii) beneficiaries are knowledgeable of best practices for the equipment's usage through delivery of targeted training to beneficiaries on usage at each stage in the technology deployment (installation requirements, relevant inputs, opportunities to create businesses with the technologies, and required maintenance practices), and iii) the supporting infrastructure has capacity to ensure the equipment's long-term productive use through capacity building for local tradespeople on construction and operations and maintenance.

144. In addition, the technology providers will be supported through Outcome 2 to expand their businesses further and develop women and youth inclusive business models, supporting the development of the technology market and the long-term expansion of the technologies in Mali.

145. Farmland use mapping and monitoring techniques will also be supported by Outcome 3 by training on usage for farmer associations, and the creation of project land use database, integrated into the existing databases and connected to researchers through Outcome 4.

Table C.3.1: Technology Funding for technology transfer within IAAT

Outcome	Output	Activity or Sub-Activity
2	*CSA and agroforestry VCs are more connected and reach more smallholder farmers*	Sub-activity 2.1.1.3 Develop technical guidance for private sector technology companies working in solar irrigation and biodigester systems on inclusive business growth
3	*Increased adoption of climate-smart agriculture technologies by smallholder farmers*	Activity 2.3.1 Install and support the productive use of biodigester systems and solar irrigation systems amongst smallholder farmers
3	*Increased land area under agroforestry*	Activity 1.3.1 Develop land-use mapping at the regional level

146. **IAAT also invests in capacity building within component 1, 2 and 3.** The project directly works with the national government, national extension services, local financial institutions, private sector businesses and smallholder farmers to help them build capacity in techniques, technologies and processes directly or indirectly relevant to the delivery of climate smart agriculture. Capacity will be built up at three levels: on-farm (e.g. smallholder farmers), agriculture market (e.g. solar irrigation and biodigester companies) and institutional (e.g. extensions services, National Climate Fund).

Table C.3.2: Capacity Building Funding for technology transfer within IAAT

Outcome	Output	Activity or Sub-Activity
1	*Improved technical capacities and inclusivity of extension services in climate-smart agriculture and agroforestry production*	Activity 1.1.1 Build technical capacity and reach of extension services on CSA and agroforestry techniques
1	*Increased use of climate resilient practices in the production of CSA crops, livestock and agroforestry by smallholder farmers.*	Activity 1.2.1 Build awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry amongst smallholder farmer communities (crop and/or livestock)
2	*Smallholder farmers, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness.*	Activity 2.2.1 Support local financial institutions to increase access to finance for smallholder farmers, especially women and youth, for CSA/agroforestry investments
		Activity 2.2.2 Enhance capabilities and connectivity of youth and women CSA/agribusiness entrepreneurs
3	*Increased institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks*	Activity 3.1.1 Strengthen institutional capacity in localized climate change adaptation and mitigation planning
		Activity 3.1.2 Technical capacity building of Mali Climate Fund and National Climate Change Committee (CNCCM) prioritization and fund allocation on CSA and agroforestry technologies, practices and services.

147. Further description of the activities and sub-activities is provided in section B.3

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

148. The project will contribute towards the GCF's mitigation and adaptation goals of supporting climate-resilient sustainable development and reduced GHG emissions by supporting its direct and indirect beneficiaries, 431,700 and 700,980 people respectively, whilst achieving an estimated mitigation reduction 2,446,543 tCO2eq in 15 years. The project will support the development of low carbon, climate-resilient and sustainable livelihoods amongst communities in Mali who are increasingly climate vulnerable and food, nutrition and water insecure. As a result, the project will support smallholder farmers to be increasingly resilient to climate-related disasters, such as drought, heat stress, floods and pests and diseases which are increasingly common in Mali due to climate change, whilst also lowering the GHG emissions of the agricultural systems. The project will aspire to reduce loss of life related to climate impacts primarily through food security improvements, help to increase and diversify climate-resilient livelihoods, and restore environmental assets through i) the planting of trees in agroforestry which will create a carbon "sink" and ii) improve land management practices through CSA techniques such as zai

pits, increasing soil fertility and reducing desertification. The project will also reduce GHG emissions through the installation of biodigester systems and solar irrigation, which provide both adaptation and mitigation benefits for smallholder farmers.

Direct Beneficiaries

149. The estimated direct beneficiary population will be 431,700 people (86,340 farming households) across 48 communes, corresponding to 2.6% of the national population.[216] . Communities in 48 communes across 12 circles in the 5 targeted regions will receive locally tailored inputs based on their evolving needs and local context. Direct beneficiaries have been selected primarily on the basis that they are the most vulnerable to climate change, combining an assessment of climate risk, exposure, and adaptive capacity. As outlined in section B.3, smallholder farmer direct beneficiaries will benefit from select activities in components 1, 2 and 3 that will increase the resilience of their farms to climate change. Beneficiary numbers were calculated via a systematic approach to beneficiary identification at the Circle level based on published statistics and data, with further detail on beneficiary selection and targeting provided in **Annex 2, Feasibility Study, Section VIII- Vulnerability and Targeting Assessment**217. The project will provide direct support (improved seed, manure, insect/pest management, and capacity building training) of implementing CSA for all direct beneficiaries. The direct beneficiary population split into 215,850 men and 215, 850 women; a 50:50 gender split is targeted across the project's overall beneficiaries except for additional activities for women and youth beneficiaries included in component 2.

Indirect Beneficiaries

150. The 700,980 indirect beneficiaries account for the 3.8% of the total national population who will benefit indirectly from the updated curriculum for public extension services Component 3 will also further increase the capacity of national and local institutions to adapt to and mitigate climate change, resulting in long-term indirect benefits.

Mitigation Impact

151. An estimated 91,285 Ha of Agroforestry will be developed through components 1 and 2, through increased capacity amongst smallholder farmers for cultivation of agroforestry, the production and distribution of agroforestry tree saplings (acacia Senegal and shea), planting and management samplings and growing trees in crop and grass/pasture lands (either farmer-owned or state/community lands). IAAT will directly support improving the agroforestry system in 21,585 ha of private land corresponds to 1,668,656 tCO2e of emissions reduction in 15 years.
152. 42,632 farmers/individuals (> 6,000 households) will obtain increased access to clean energy sources through biodigester and solar pump systems, which will also provide a further mitigation outcome of 131,566 tCO2eq of emissions reduction in 15 years (solar pump = 54,512 tCO2eq, and biodigester = 77,054 tCO2eq).
153. An estimated 64,755 farming households (75% of 86,340 farming households) will also benefit from reduced carbon emissions associated with the adoption of climate-smart agriculture practices. This will result in the further mitigation of 646,320 tCO2e over 15 years.
154. For further information on the assumptions behind the mitigation impact of the project, please refer to **Annex 22**.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

155. As described in the ToC section, this project aims to be a catalyst for a broader shift in agriculture across Mali, from its baseline state of extreme vulnerability—due to high exposure to climate hazards and low adaptive capacity of smallholder farmers—to an alternative paradigm in which smallholder farmers alongside other agriculture sector stakeholders have increased capacity and resources to adopt low-carbon climate resilient livelihoods providing increased food and water security and lower emissions of the agricultural systems, with ongoing support from public institutions, market actors and the private sector.

Potential scale up and replication

156. The project will effectively contribute to low emission and climate-resilient development pathways in Mali by increasing the long-term capacity of smallholder communities and their surrounding environment to recognize and respond to climate risks, increasing sustainable and low-carbon crop and livestock

[216] It is widely recognized that population estimates are not accurate in Mali as the last census was performed in 2009. The data here has been taken from the OCHA official estimate from 2022 which is available here.
217 Beneficiaries are estimated based on the population's access to agriculture extension services and current adoption range of improved agricultural technologies and practices in the project locations. Adaptation benefits are mapped based on the recent CSA literature.

production and land management by smallholder farmers, and hence, both increase the resilience and reduce the GHG emissions of the agricultural system.

157. IAAT will facilitate building a strong partnership between private sector solar and biodigester technology suppliers through a PAYG financial and technology transfer model in the IAAT locations. This approach will bring a large amount of private sector finance from technology suppliers and financial institutions. The private sector will continue scaling up and replicating this business model in and beyond IAAT locations during and after the project completion.

158. IAAT's approach is consistent with Mali's existing policy priorities and devolved governance approach increasing its long-term paradigm shift potential as well as its support within the country. For example, the project's emphasis on Climate Smart Agriculture is directly aligned with NDC, NAPA, and PDESC.

159. IAAT also has strong institutional foundations for scale-up within Mali. The USAID Albarka project, IAAT's co-funder and co-implementing partner, has existing field infrastructure and already established relationships with the AEDD, the Ministry of Agriculture, other governmental and non-governmental entities and vulnerable communities. These existing relationships will provide a platform for the project to scale-up actions effectively and efficiently both during and after the project implementation. Replication examples include: establishment of CACs and development of Community Action Plans, development process for contingency plans for EWS information dissemination and response, extension officer curriculum, national land use database, youth and women business and financial training, guidelines for financial institutions (including microfinance), establishment and capacity building of private sector to engage and provide services to small holder farmers and conflict management response.

Potential for knowledge exchange and sharing lessons learned

160. IAAT seeks to enhance the knowledge and data environment around climate smart agriculture and agroforestry at the community, regional and national level in Mali. The project will also enhance the local knowledge network including communities, public and private extension agents, and market agents. This knowledge exchange directly supports the scale-up up climate-resilient agricultural practices and reinforces the resilience of farmers' livelihoods. For example, CSA curriculum, social mobilization, and integration of traditional knowledge and experience in climate change adaptation are key tools for scaling up CSA and agroforestry.

161. At the regional and national level, IAAT explicitly targets knowledge exchange and the sharing of lessons learned through yearly forums for all key actors (national, regional and local bodies, other agricultural and climate change projects, researchers and traditional knowledge groups) to share and discuss the key technical learnings of the project and how these could be broadened to additional areas.

162. The IAAT project will also create an enhanced land use database and deliver training on interpretation of land use data to extension services. By increasing access to reliable information at extension service level, the project will facilitate improved climate adaptation by improving the tailoring of adaptive techniques to the circle level. In addition, the land use database will enhance the national data environment for future mitigation and adaptation planning and research at the local and national level, including for policies, research and interventions.

163. In support of this knowledge sharing, the Monitoring and Evaluation plan (see **Annex 11**) will also develop a detailed view on the realized outputs of the project during implementation to support the on-going development of implementation insights and "lessons learnt". The project's knowledge management and learning activities will therefore develop and enhance knowledge availability and knowledge resource management competence at the national level, as well as farmers' ability to propose solutions to changing climate circumstances.

Contribution to the creation of an enabling environment

164. IAAT will contribute to the creation of an enabling environment for a low carbon and climate resilient agricultural system in Mali by i) addressing existing systemic barriers to adaptation and mitigation at the farm and community levels, ii) closing market-level knowledge gaps on how private sector businesses can serve smallholder farmers with low-carbon and climate resilient products and services and iii) building institutional capacity for long-term adaptation and mitigation planning and implementation. The project will address a broader range of systemic barriers as outlined and shown in the ToC diagram, including financial, informational, social, market, and technical as discussed in section B. 2 (a). Addressing these barriers will also help to create the enabling environment for scaling-up CSA and agroforestry.

165. IAAT will significantly contribute to institutional capacity building. The institutional capacity building at the community, regional and national level for mitigation and adaptation planning and implementation will contribute to shift in the institutional priorities to address climate change challenges across the country.

D.3. Sustainable development (max. 500 words, approximately 1 page)

166. The IAAT will make direct and indirect contributions to Mali's efforts to meet the Sustainable Development Goals, particularly in addressing SDG13 "strengthen resilience and adaptive capacity to climate-related disasters, integrate climate change measures into policies and planning, build knowledge and capacity to meet climate change"; and SDG 2 "End hunger, achieve food security, improve nutrition, and promote sustainable agriculture". It will also support the following SDGs:
    - SDG1 (end poverty in all its forms everywhere)
    - SDG2 *(end hunger and achieve food and nutrition security)
    - SDG8 (promote inclusive and sustainable economic growth, employment, and decent work for all)
    - SDG12 (ensure sustainable consumption and production patterns)
    - SDG15 (sustainably manage forests, combat desertification, halt and reverse land degradation, and halt biodiversity loss).

167. The project activities will result in the following 2 co-benefits:
    - **Co-benefit 1 Gender and social inclusion:** Increased gender equality and economic empowerment for women. Outcomes 1, 2 and 3 will together deliver gender co-benefits. Output 2.2 will specifically seek to reduce barriers to entrepreneurship with a focus on women by increasing access to finance, delivering training on entrepreneurship, and making connections to other value chain actors. In component 1, the improved extension services targeted in output 1.1 will include a provision to increase the gender inclusivity of the existing services and output 1.2 will target to reach 50% of women to ensure women are equally able to benefit from the new training (in contrast to their existing reduced access). In addition, the biodigester systems installed and promoted in component 3 will increase gender equality by reducing the time taken for households to collect wood for existing energy sources, a task often conducted by women. Taken together, these outcomes will deliver increased gender equality and empowerment by supporting women to have access to services and inputs currently more easily accessible by men, increasing the capacity of women beneficiaries and the time available to generate a sustainable income. In addition, increased gender equality and economic empowerment will also generate economic co-benefit by improving women participation in food value chain and business activities. While the promotion of solar pumping system from Outcome 3 will improve women's access to water for household and agriculture use that reduces women's time to fetch water which can result in more time available for other productive or non-productive activities.
    - **Co-benefit 2 Environmental:** Improved soil conditions and water resources from landuse improvements. Outcomes 1 and outcome 3 will alter landuse by smallholder farmers and together result in improved soil and water quality as a result of i) the adoption of improved water and soil management practices by smallholder farmers during crop and livestock production (output 1.2), and ii) the soil and water benefits provided by agroforestry such as improve soil biota and water flow (output 1.3). These activities will enhance healthy ecosystem services and create public goods for the communities. Several recent studies show that sustainable intensification of agroforestry and agriculture systems have positive impacts on diversification of pollination, pest control, nutrient cycling, GHG emissions reductions, water regulation, and soil health.[218,]

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Vulnerability of the country and specific vulnerable groups and economic and social development level of the country and the affected population

168. Mali is highly exposed to the impacts of climate change with GAIN ranking it as the 24th most vulnerable to climate change and the 47th least ready for climate change out of 197 countries[219]. The country's climate change vulnerability stems from its erratic rainfall and rising temperatures resulting in a continuous cycle of droughts, floods, increased heat stress and pests and diseases, Droughts and floods have affected nearly 7 million people in the 1980-2014 period and generated an economic cost of USD 140 million. In many locations, women and youth are more sensitive to climatic risks due to loss of jobs, production and income and discrimination which means are less likely to participate in and express their needs for integration in decision-making forums. As the frequency and severity of floods, droughts, heat stress and pests and disease increase, Mali is forecast to generate losses of up-to USD 300 million per year from 2030 onwards

[218] Snapp et al. 2021. Agroecology and climate change- rapid evidence review. Agroecology and climate change rapid evidence review: Performance of agroecological approaches in low- and middle- income countries. Wageningen, the Netherlands: CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS)
[219] GAIN (2021), Vulnerability Index, available here

(around 15% of the value created by agriculture and breeding).[220] This needs urgent action for climate change adaptation in Mali.

169. Mali is an LCD with limited finances to currently address all of its climate change commitments, nevertheless, climate change impacts are exacerbating other socio-economic aspects, especially for the most vulnerable people. For example, extreme poverty is widespread throughout Mail and increased to 19.1% in 2022 (up from 15.9% in 2021).[221] Given agriculture is the main source of livelihood for 80% of the population, with the majority of farmers practicing small-scale, rainfed, subsistence agriculture and agropastoralism, diminishing incomes result in harsh trade-offs for households which weigh-up the prospects of surviving as best they can by choosing whether to eat, migrate for work, buy medicines, abandon education, enter into forced marriages or possibly other harmful or maladaptive practices.[222,223][224]

170. In addition, Mali's growing population compounds food security challenges placing additional strain on limited natural resources. A growing and developing population also has consequences for energy use and GHG emissions – for small holder farmers the opportunity is to replace expensive diesel and unsustainable charcoal with renewable sources of energy, thereby meeting needs while curtailing possible increases in domestic GHG. Mali's vulnerability is also increased by the presence of historical and ongoing armed conflict at both the national and local level. IAAT has been designed to be conflict sensitive by considering the effects of the conflict on livelihoods, food security and the rural economy. For further information on Mali's overall vulnerability please see **Annex 2: Feasibility Study, Chapter VIII**, see **Annex 23** for conflict management response.

Absence of alternative sources of financing

171. As outlined in the NDC and other policy documents, Mali requires climate finance support from international development and financial institutions. Despite the urgent need for climate funding for Mali's agriculture, there is a clear lack of alternative financing sources for this IAAT cross-cutting project. Mali currently has no viable source of domestic financing to implement the IAAT, in part due to ongoing conflict and humanitarian crises. There are barriers to accessing sufficient financing from the government in Mali, the private sector, as well as from international donors. There are also prominent barriers to overall private sector financing in Mali including: a lack of de-risking of financial investment – especially in small holder farmer operations, training in procedures for accessing climate finance, or participation in the development of climate change strategies, and limited access to international finance.[225] In addition, the Government of Mali had to redirect a large amount of financial resources to address the COVID-19 pandemic, which further reduced the capacity to invest in climate action and set back the climate agenda.

172. Although development and climate funders are present in Mali there is insufficient support to close the funding gap, and the complex political and conflict situation in Mali also poses challenges regarding the availability of international donor funding. For example, after France withdrew its military operation in 2022, it also announced plans to withdraw public aid funding, which is estimated at 470 million euros (equivalent to US$495 million) between January 2013 and September 2017.[226]

Strengthening institutions and implementation capacity

173. Although CSA and adaptation in agriculture have been prioritized by the national government, there is limited capacity to implement much of the government's policy and initiatives targeting smallholder farmers, due to institutional capacity gaps in climate adaptation and mitigation planning and implementation at both the national and sub-national levels. There is a need to improve coordination between national and local level governance institutions for effective implementation of climate change adaptation and mitigation planning and programming In addition, there is a strong need to transfer global best practices on adaptation and mitigation planning to Mali's local governments, whilst also tailoring practices by integrating inclusive and traditional knowledge to support locally-led adaptation and mitigation approaches. The presence of many other competing local priorities such as conflict, and lack of resources also prevents local authorities from effectively leveraging the PDESC to implement adaptation practices.[227]

[220] Makougoum C. T. P., (2015) Changement climatique au Mali : Impact de la sécheresse sur l'agriculture et stratégies d'adaptation
[221] World Bank, Mali Presentation, Link
[222] N'Diaye, I., Aune, J.B., Synnevåg, G., Yossi, H., & Hamadoun, A. (Eds.). (2020) Adaptation de l'Agriculture et de l'Élevage au Changement Climatique au Mali: Résultats et leçons apprises au Sahel. Available here
[223] Douyon A, Worou ON, Diama A, Badolo F, Denou RK, Touré S, Sidibé A, Nebie B and Tabo R (2022) Impact of Crop Diversification on Household Food and Nutrition Security in Southern and Central Mali, Available here
[224] CIAT, ICRISAT, BFS/USAID, (2021), "Climate-Smart Agriculture in Mali", Available here
[225] AFDB (2023), "Mali Economic Outlook", accessible here
[226] RFI (2022), "France halts development aid to Mali", accessible here
[227] Ministère du Développement Rurale, Plan Triennale de Campagne Agricole Consolidé et Harmonisé (2023)

In addition, implementation of local initiatives is limited by decision-making structures that rely on the male heads of households resulting in reduced accessibility for women.[228,229,230]

174. Lower trust in formal institutions than local informal institutions can also be found in some regions of Mali, in part derived from backdrop of conflict.[231] Consequently, there is a need to both strengthen trust in formal institutions and leverage informal institutions further when implementing adaptation interventions. Informal institutions rely on social capital, kinship, and religious networks for authority with their support required to be able to implement many community-level changes.[232]

D.5. Country ownership (max. 500 words, approximately 1 page)

175. IAAT was developed and designed in collaboration with stakeholders from across all 5 regions of the project (Gao, Tombouctou, Segou, Koulikoro and Mopti), including farmers and farmer organisations, private sector businesses, local and international NGOs, and national and local government representatives. Project activities, outputs, outcomes, and impacts were evaluated and validated by the stakeholders at the national workshop organized in Bamako in May 2023 (see **Annex 7 Summary of consultation and stakeholder engagement**)

176. The project builds upon a letter submitted by the NDA to the GCF in 2021 endorsing the submission of this project idea. It has been designed in collaboration with the Malian government, in particular the Environmental and Sustainable Development Agency (AEDD), Mali's NDA. Throughout project design, over 350 stakeholders were engaged to collaboratively design the project, including government ministries, private sector businesses, local community beneficiaries and local NGOs. These stakeholder consultations have significantly shaped the project (with further detail provided in **Annex 7: Stakeholder Engagement Plan**) and stakeholders in Mali will continue to be engaged to refine the project during project implementation.

177. The project is a high priority for the Government of Mali because of its high potential to contribute towards meeting national priorities for climate-resilient agricultural development as espoused in the NDC, Growth and Poverty Reduction Strategic Framework 2012 to 2017, the Agricultural Orientation Law, the National Food Security Strategy, the Strategic Framework for Economic Recovery and Sustainable Development in Mali 2016 to 2018, National Adaptation Programme of Action (NAPA), National Agricultural Sector Investment Plan (PNISA), National Climate Change Policy (PNCC) and the National Climate Change Strategy (SNCC), the National policy for Land Use Planning, the National Strategy for the Prevention and Management of Disaster Risks, and the 2019 Climate-Smart Agriculture Investment Plan. Together, these policies and plans call for improving institutional development and governance, enhancing food security and nutrition, fostering national and regional integration processes, and promoting intensive, diversified, and sustainable agriculture.

178. Furthermore, in its updated Nationally Determined Contribution (NDC), Mali committed to reducing emissions by 25% for agriculture, 31% for energy, 39% for LUCF, and 31% for waste by 2030 compared to projected business-as-usual emissions.[233] Priority actions in Mali's NDC include agroforestry, sustainable rice cultivation, and crop nutrient management, all of which are actively included in this project.

179. In addition to the alignment with national agricultural and climate policies, the GCF IAAT project is intricately linked to the priorities delineated by the Malian government in its GCF country program, which embodies the collective vision established by all pertinent stakeholders in Mali concerning climate adaptation and mitigation and represents the main inception point for GCF-funded initiatives. Aligned with the GCF country program in Mali, the project aims to enhance the accessibility of alternative energy sources, bolster the adaptation and mitigation capacity of farmers, with special attention to women and youth, enhance capacities across all levels, and facilitate collaboration, knowledge exchange, and experience sharing among stakeholders. The harmonization with the country program bestows legitimacy and credibility upon the project in the eyes of the government and major stakeholders involved in agriculture, agroforestry, and climate sectors in Mali and has fostered extensive and smooth collaboration with and endorsement of the project design from key stakeholders in Mali, who recognize the pertinence of the identified barriers and acknowledge the accuracy of the proposed activities to address them. These stakeholders will play a pivotal role as indispensable partners during the project's implementation phase.

180. The project will together be implemented by Save the Children Mali and the AEDD. AEDD involvement will continue to enable strong country ownership, further supported by the project's steering committee which includes the Ministry of Environment and Sanitation, the Ministry of Rural Development (including

[228] USAID (2017) *Decentralized governance and climate change adaptation: A case study on Mali*. rep. Available here
[229] AGRA, "Value 4 Her" Synthesis Report on Agri-enterprise event held in Bamako Mali, May 2021, Available here
[230] https://knowledge4food.net/wp-content/uploads/2016/12/161130_youth-inclusiveness-agri_mali.pdf
[231] Ahmadou Aly Mbaye, Nancy Benjamin, "Institutions, informality and conflict in the Sahel: the case for Mali" 2022, Available here
[232] Ahmadou Aly Mbaye, Nancy Benjamin, "Institutions, informality and conflict in the Sahel: the case for Mali" 2022, Available here
[233] Government of Mali, NDC. Available here.

agriculture, livestock, and fisheries), Mali Climate Fund, the Ministry for the Advancement of Women, Children and Families, civil society organisations as well as AEDD in its capacity as the NDA and the implementing entity. .

Accredited Entity

181. Save the Children is the world's leading independent organization for children, with 30 national organizations working together to deliver programs in approximately 120 countries via a network of 24,000 professionals. In 2019, Save the Children delivered programs worth over USD 2.2 billion across 117 countries and directly reached over 38.7 million children. The organization currently has a global health portfolio valued at USD 700 million that spans 50 countries. SC International, the implementation structure for the global movement, oversees a portfolio of approximately 700 contracts including a portfolio of 100+ resilience-related projects and programs valued at more than USD 200 million, with explicit objectives to reduce climate and disaster risks as well as increase adaptive capacity and those who seek the social and economic empowerment of women and youth and the amplification of the voices of the most marginalized.

182. SCA was accredited to the GCF in November 2019 on behalf of the global SC movement. SCA was chosen to lead on the GCF for SC due to its longstanding leadership role in climate change and disaster risk reduction and its robust fiduciary and compliance systems and processes. The AMA was made effective in May 2020. SCA currently has six GCF projects under implementation and numerous others at various stages of development.

183. Save the Children, including SCI, SCUS, and SCA, has extensive experience designing, delivering, evaluating, and documenting approaches to community-based adaptation, including in Mali, where SCI has been working for more than three decades. Our approach is to support governments to deliver against their adaptation policy objectives and needs, bringing a consultative approach to engaging a range of stakeholders, including communities, in the design of climate change interventions. As we work as a confederated movement, our experience in delivering relevant projects in Mali and in the agriculture and nature resource management sector more broadly sits with SCI Mali and SC US.

Executing Entities

SCUS

184. SCUS is one of 30 national organizations that implement non-domestic operations through a single program delivery unit, Save the Children International (SCI). With a 2022 annual budget of $955 million, SCUS oversees a portfolio of approximately 968 awards in 116 countries, financed by corporations, foundations, the private sector and major multi-lateral and bilateral institutions, including the U.S. Government. Our network of climate change, agriculture and natural resource management, gender and social inclusion, and food security professionals offers expertise in climate risks assessment, prioritization and designing interventions in agriculture and natural resource management, food and nutrient security management under climatic and economic shocks, community capacity strengthening, social and behavior change (SBC), inclusive market system development and monitoring and evaluation (M&E). We are a trusted partner to governments, NGOs, the private sector, other local actors, and institutions and regularly collaborate with a wide range of local communities to strengthen capacity and accelerate progress towards development outcomes in the countries where we work. SCUS serves as the lead Resilient Food Security Activities implemented in many African countries such as Mali, Niger, Malawi, Ethiopia, and Tanzania funded by USAID.

SCI Mali

185. SCI Mali has been working in Mali in collaboration with the GoM since 1987 and supporting vulnerable children, youth, women, and their communities through implementing programs in food and health security, water, sanitation and hygiene, agriculture, and natural resource management sectors. Most of its current programs focus on food and nutrient security, health, natural resource management, and climate change adaptation in agriculture and health. FY 2023, Save the Children has 32 humanitarian development projects/programs and active developments for a total portfolio of USD 33 million.

186. 2012-2013, SC implemented a food security and livelihoods (SAMS) and nutrition project in the Diéma district, Kayes region funded by the OFDA (USD 999,665) and KOICA (USD 700,000); a SAMS and nutrition intervention in the Gao district funded by the M.A Cargill Foundation (USD 999,992). A "NEMA" food security project, a consortium with CRS subsidized by USAID (1998-2003, USD 35 million) in the circles of Gao and Bourem, Gao region targeted 124,859 beneficiaries in 130 communities across two districts. Thanks to integrated interventions. The MYAP 2008-2013 improved the living conditions of vulnerable rural households, strengthened their human capacities and increased the resilience of

communities to shocks. From 2010 to 2013, Save the Children conducted a search for savings, loan and money transfer activities targeting women in Sikasso in the circles of Bougouni and Yanfolila, Sikasso region. The Food Facility subsidy supported by the EC (2010-2011, USD 884,601) focused on social protection (cash transfers and agricultural support to very poor and poor farmers) in the Sikasso region.

187. In Diéma's circle, a Food Security and livelihood restoration project for the most vulnerable countries in Switzerland (USD 511,289), in 2013. Support for the livelihoods of communities affected by the conflicts in Mali, USD 1,479,055, funded by USAID / FFP & OFDA in the Gao circle in 2014. From 2017 to 2020, Save the Children benefited from the Program for Strengthening Resilience in Food Security (PRORESA) baptized "Lafia" in 5 municipalities in the circles of Gao and Ansongo, Gao region. In 2018, the emergency food security assistance project in 28 communes in the Kayes circle and the project to strengthen resilience in food and nutritional security made it possible to make four cash transfers (USD 76 / Transfer) to 5,000 beneficiaries. In 2020, a key priority is to provide livelihood opportunities to IDP families and their host, children affected by the conflict and young people affected by the armed conflict in central and northern Mali through "Resilience and social cohesion Liptako-Gourma cross-border communities" RECOLG, funding from USAID / FFP and the EU. RESILIENCE ACTIVITY & FOOD SECURITY – RFSA IN MALI (ALBARKA) is a five (05) year project (2020 to 2025) in 12 circles in the regions of Timbuktu, Gao and Mopti, with targeted beneficiaries: 500,000 (210,000 direct); It is financed at USD 60 million by BHA/USAID.

188. Save the Children in Mali actively participates in national and regional thematic groups for FSL, in addition to being an active member of the food security cluster in Bamako, SC is a member of the Cash and Vouchers working group and sits on the steering committee of the Cash Learning Partnership (CaLP).

Implementing Partner
AEDD

189. Established in July 2010, AEDD is a public administrative institution under the Ministry of Environment and Sanitation. AEDD targets to achieve sustainable development through effective environmental management that focuses on the preservation of biodiversity, fight against desertification, and climate change. AEDD focus areas include: i) strengthening the capacity of actors involved in environmental management, sustainable development and climate action through the development of modules, information, education and communication materials, sessions, and training, ii) mobilization and monitoring financial mechanization in climate change and environmental protection, including Mali Climate Fund, Green Funds, Clean Development Mechanisms, Global Environmental Fund, etc., and iii) ensure coordination and monitoring of the implementation of international Conventions, Agreements and Treaties ratified by Mali on climate change and sustainable development.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

190. The project's total proposed budget is estimated at USD 46.05 million, of which 78% (USD 36 million) is proposed to be a GCF grant, and 11% (USD 5 million) is the co-financing component from USAID (in grant form), and 11% (USD 5 million) co-financing from AEDD (in-kind) over the project period.

Financial adequacy and appropriateness of concessionality:

191. The project's proposed concessionality level is rationalized through i) the lack of alternative funding sources (as explained in Section B.5), ii) the IAAT's provision of largely "public" goods, such as improved extension, social and ecosystem services and iii) the IAAT's focus acclimate vulnerable beneficiaries who have contributed the least to the climate change problem, yet suffer the most from impacts and future risk, while being unable to increase their adaptive capacity. The project will contribute to improving the management capacity of local communities for natural resources, directly benefiting 431,700 people and indirectly benefiting 700,980. Furthermore, the project will result in the plantation of 91,285 Ha of agroforestry in the five targeted regions. The adoption of improved agroforestry through the direct IAAT support in private land will result in the reduction of 1,668,656 tCO_2eq net through improved agroforestry, a further 646,320 tCO_2eq of mitigation due to the adoption of CSA techniques, and 131,566 tCO_2eq through the adoption of low-carbon technologies (solar pumps and biodigester) in 15-year lifetime. The public good of carbon sequestration associated with agroforestry is not valued sufficiently by the existing market, meaning that it is only through grant financing that the agroforestry component can be implemented.

192. The IAAT will provide a successful model for enhancing climate resilience and reducing emissions in targeted areas, offering opportunities for replication in other regions of Mali and countries with similar contexts. Based on the high financing needs and vulnerability to climate change in Mali, and the public

nature of goods and services provided through the project, the IAAT will act as a catalyst for transforming the agriculture and agroforestry sectors in Mali, generating economic benefits for smallholder farmers and creating an environment conducive to future private sector funding. It is not viable for the Government of Mali or its' local communities to fund this transformation currently, therefore concessionality is requested through the GCF for this cross-cutting project that has far-reaching results for an African LCD facing current and future climate change. For a detailed financial assessment of the key interventions in the project locations see **Annex 3 Economic and Financial Analysis.**

Cost effectiveness of mitigation component

193. The total emissions mitigated through IAAT's activities are 2,446,543 tCO_2eq over the 15 years (potential long-term contribution). The total project funds associated with emissions reduction is USD 17,678,010 and is funded by the GCF. The economic public value of the negative carbon balance (reduction of GHG emissions) generated by the project is estimated at USD 4.9 million per year when using USD 30 as the social price for one ton of CO_2. Accounting for both additional production and GHG co-benefits, the economic NPV reaches USD 31.5 million while the Economic IRR is up to 26%. The numerical values of the social value of carbon recommended for use by the World Bank Group in USD per 1 metric tonne of CO_2 equivalent, are ranging between USD 20 (low) and USD 60 (high) in 2020 and increase to a range of USD 40-120 in real terms by 2040. On this basis, it is relevant to propose a simulation of economic analysis with a higher social price per TCO_2e of USD 60. Using this price, the GHG benefits are priced at USD 9.8 million. The analysis below now provides an EIRR of up to 39% and an ENPV of over USD 58.6 million on the next 15 years (2025-2040).

Expected financial internal rate of return and long-term financial viability.

194. The assessment of these three components demonstrates promising financial viability for the project over the 15 years of the project lifetime, i) agroforestry – value chain of Gum Arabic (130% IRR and NPV USD 146), and Shea Butter 66% IRR and NPV USD 51), ii) use of solar irrigation – Mango (IRR 134% and NPV USD 5,736), Vegetables (IRR 246% and NPV USD 2.540), Moringa (IRR 233%, NPV USD 4,962), iii) use of CSA - cereal production (IRR 220% and USD 65). The average expected financial internal rate of return in combination of all products considered in the study is 171% IRR and NPV USD 2250. These analyses highlight the strong financial returns of the various value chains within the project, before considering the economic value of the project. The financial analysis results show a strong financial return profile for the farmers after project support, showing the compelling financial case for delivering the project's activities. Included in the accompanying financial analysis excel, is a modelled scenario of financial return without project investment which also in some cases presents a strong return over 15 years. Additionally, as shown by the sensitivity analysis, the returns profile remains uncertain for farmers. The modelled variation in two key inputs (price obtained for crops and yield of crops) shows a significant range in the potential return for farmers, with uncertainties in climate as well as political and security context resulting in a wide range of potential return scenarios.

Expected Economic internal rate of return

196. This economic analysis is based on cash flow analysis with a project investment of USD 43.7 million and Economic return of agriculture productions as benefit. The project budget of USD 43.7 million is distributed between year 1 and year 5 (16% in year 1, 24% in year 2, 24% in year 3, 20% in year 4, and 16% in year 5). It results in an Economic Net Present Value (ENPV) of USD 4.4 million and an Economic Internal rate of return (EIRR) of 12% without valuing the economic value of GHG mitigated by the project.

Application of best practices and degree of innovation

197. IAAT will strongly rely on innovation and technology to drive change across the agricultural landscape. Innovation platforms will be leveraged to enhance adoption of CSA and agroforestry technologies and practices and extension services will be trained and encouraged to use digital platforms to extend their reach. The project will also leverage existing digital marketplaces to create linkages between farmers and other players and enablers of the value chain, including service providers, processors, financial institutions, etc. Furthermore, land use maps will be developed to enhance the capacity of regional authorities to effectively map and classify public assets. A land use database will also be created, with further training provided to extension agents for its utilization to foster the development of digital services, implying more accurate and specifically tailored advisory services for smallholder farmers in Mali, and setting pre-eligibility criteria for women and youth to employ in agriculture extension system.

Efficiency and Cost Metrics

Table D.6.1: Efficiency and cost metrics

	Project Total Budget Metric	GCF Grant-specific metrics
Ratio of co-financing (Expressed as GCF funding: co-financing funding)	*3.2:1*	*n/a*
Total cost per direct beneficiary (USD)	*101.26*	*78.1*
Total cost per indirect beneficiary (USD)	*62.36*	*48.1*
Total cost per ton of Carbon Dioxide equivalent (USD)	*17.86*	*7.22*
Expected financial internal rate of return	*171%*	*n/a*
Expected economic internal rate of return	*12%*	*n/a*
Expected economic internal rate of return (with Eco value of GHG impact at USD 30 / tCO_2	*26%*	*n/a*

LOGICAL FRAMEWORK

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☒ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario	How the project/programme will contribute)
	Description	Rating		
Scale	Smallholder farmers in Mali are highly vulnerable to climate change due to a combination of high exposure and sensitivity to climate risks (such as drought and flooding) combined with low adaptive capacity with limited access to finance or information to support the adoption of climate-resilient and low-carbon farming techniques. Reliant on their own agricultural outputs and with limited cash or asset reserves, the prevalence of climate hazards and other risks (such as conflict) results in high food insecurity for smallholder farmers. Extension services which are designed to share information and training on agricultural practices that can help reduce this vulnerability have limited reach and experience disseminating.	<u>Low</u>	IAAT presents an evidenced, successful model for increasing climate resilience and reducing the emissions of the agricultural systems in the targeted communes. (By increasing smallholder farmers ability to implement CSA techniques by supporting their direct capacity as well as building a supporting market environment including access to finance, agriculture increases resilience and reduces emissions). This successful model leads to the scale up of the project by expanding to neighboring communes as institutional actors realize the benefits of the approach. As national extension services see the greater impacts of the new training programs, they roll it out utilizing the tailoring to the circle approach, across regions. Greater emphasis is placed on the recruitment of extension agents, resulting in greater reach of the service. As CSA businesses and entrepreneurs deliver sustained growth by targeting new market segments, additional businesses utilize the guidelines created by the project to	IAAT will help to achieve the target (431,700 direct beneficiaries/86,340 farming households and contribution to reducing 2,446,543 tCO2eq in 15 years lifetime) through its multi-level approach, capacity building, and strong relationships with key scale-up stakeholders. Component 1 will improve the public and private extension system by strengthening the capacity of 500 public and private sectors extension staff in the IAAT locations. These trained extension experts will provide training on CSA and agroforestry including providing information on CSA technologies availability and use in 48 communes of 12 circles in 5 regions (Component 2). They will deliver tailored local CSA and agroforestry training packages through dissemination networks, including farmers' field schools, and local organizations (e.g.GAP-RUs, CRRs, and CAG) to cover 431,700 direct beneficiaries. IAAT will demonstrate CSA and agroforestry technologies and practices (22 prioritized by the stakeholders) in 48 communities in collaboration with AEDD and the agriculture department.

	information on the most up to date CSA techniques, as well as the best local/traditional practices. Institutional capacity and knowledge of CSA (both adaptation and mitigation technologies and practices) for extension services need to strengthen to promote wide adoption of CSA. Markets for CSA crops and technologies are hard to access for smallholder farmers, and value-added services are insufficiently targeted at this market segment. Existing funding and interest in increasing the adaptive capacity of smallholder farmers whilst targeting low-carbon development pathways is currently insufficient to meet the scale of the problem.		access these segments and smallholder famers in project regions are better served by the private sector, particularly by service providers. IAAT targets directly reaching over 431,700 beneficiaries and indirectly over 700,980 beneficiaries across the project regions with a contribution to reducing 2,446,543 tCO2eq in 15 years.	Component 2 creates sustainable markets for CSA and agroforestry products by fostering the development of connected value chains (from the input stage, including financing and technologies, to product sales and distribution) and lifting the barriers disproportionately faced by women and youth entrepreneurs for starting and running businesses. IAAT will connect 22,778 rural farmers to an online platform resulting in increased connection to other value chain actors and training 18,000 women and youth in financial management and business skills. IAAT will contribute to GHG emissions reduction (2.446.543 tCO2eq in 15 lifetime) by promoting and scaling solar pump irrigation systems (target 1000), biodigesters (5,000), and improved agroforestry systems in 21,585 ha of private lands Institutional capacity building for climate action is a key component of the IAAT (Component 3) that will contribute to increasing institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks in 48 communes in 12 circles at local level and the capacity of AEDD, Department of Agriculture and Mali Climate Fund at national and regional level.
Replicability	Climate smart agriculture has limited adoption amongst smallholder farmers and the market for low carbon technologies is nascent in the country, and region more generally. Implementation of CSA technologies and practices is challenged by stakeholders	<u>Low</u>	IAAT will actively engage in CSA knowledge building and sharing across all regions in Mali, through organising and attending existing regional and national conferences. Capacity will be created amongst institutional stakeholders at the national, regional (extension department) and commune (e.g. CACs, GAP-RUs, CRRs, and CAG) levels which will also allow for the	IAAT will strengthen locally led adaptation and mitigation model in collaboration with farmers, community-based organizations (e.g., CACs, GAP-RUs, CRRs, and CAG), local government, and private sector value chain actors. In Mali, this model will support to design and implementation of PDESC by integrating climate action at the local level. IAAT will contribute to the increased replicability of PDESC with the integration

	limited technical and financial capacity. Government interest in CSA, agroforestry, and low carbon development is demonstrated through policies and strategy documents, but funding for implementation remains limited.		transfer of new skills to additional institutional stakeholders beyond the immediate geographic remit of the project. IAAT will also provide an example of a successful model for achieving a shift in the agricultural system towards climate resilience and low-carbon growth through the value chain and market system development. For instance, use of the PAYG financial and technology transfer model for solar and biodigester systems with active participation of private sector technology suppliers and financial institutions. Improved value chains for CSA and agroforestry products model can spill over across the regions and Mali through the private sector leadership with support from the Mali government. IAAT will also demonstrate options for growing nascent CSA technology markets and encouraging the growth of new often under-served market segments (women and youth) which can also be used as a model for multiple market sectors and technology types both within Mali and other neighbouring countries.	of locally led-led adaptation and mitigation actions across the project regions and Mali. A majority of agriculture development programs/project focuses on adaptation with less or no focus on mitigation component. IAAT will demonstrate mitigation potential under climate change adaptation-related interventions that reduce GHG emissions without compromising farm production and income. IAAT will showcase this project integrated model for climate projects in agriculture and natural resource management and can be scaled out/up in Mali and beyond. IAAT will also provide an example of a successful model for achieving a shift in the agricultural system towards climate resilience and low-carbon growth through the value chain and market system development. Improved value chains for CSA and agroforestry products model can spill over across the regions and Mali through the private sector leadership with support from the Mali government. IAAT will also demonstrate options for growing nascent CSA technology markets and encouraging the growth of new often under-served market segments (women and youth) which can also be used as a model for multiple market sectors and technology types both within Mali and other neighbouring countries.
Sustainability	Climate resilience and adaptation considerations are factored into national level policies, plans and commitments (e.g. NAPA, CSAIP, NDC, PNISA and	<u>Medium</u>	The IAAT has a high likelihood of achieving sustained impact well beyond the implementation period. IAAT will demonstrate a long-term benefit generated by increasing climate resilience in tandem with lower emissions	The capacity building undertaken by IAAT in combination with the underlying shift in incentives provided by access to new approaches and markets will support the long-term sustainability of the project.

	PNCC). For example, CSA and agroforestry are key priorities of NDC and it also targets to reduce 25% of agriculture GHG emissions for the current level by 2030. However, resources for the implementation of these commitments remain limited, compounded by conflict risks and budgetary challenges. At the commune level, adaptation and mitigation capabilities are limited and are rarely integrated into PDESC planning. Farmers lack the capacity to implement CSA/agroforestry practices, access to supporting services, finance and a market to sell their crops. Private sector businesses, especially those focusing on new technologies, have limited knowledge of market segments, or understanding of the ways to reach marginalized customer segments (such as smallholder farmers).		for smallholder farmers, the continued shift towards low-carbon, climate resilient development pathways maintained in Mali's agricultural system after the project closes. Farmers will continue to deploy new farming practices and technologies as they both benefit from increased food security and resilience to increasing droughts, floods, heatwaves and pest/diseases as well as have obtained long-term resources to implement them (increased individual capacity as well as access to key inputs and output markets). IAAT will also achieve a reduced risk profile for the CSA /agroforestry crops markets, as well as the supporting services that serve them by increasing resilience to climate shocks, and the volume of goods that flow through them. This will result in a shift in market incentives that will continue to foster growth in both i) products produced by smallholder farmers and ii) businesses that serve this segment. Government / institutional stakeholders' increased capacity (for example public extension, local adaptation planning) to include mitigation and adaptation in planning will mean that the impacts of the project's training programs are integrated into institutional behavior after the project closes (e.g PDESC).	National, circle and commune government stakeholders will first develop further skills integrating climate adaptation and mitigation into government planning/processes (For example PDESC at commune and CACs at community). Institutionalization of PDESC and CACs supports long-term climate finance from the government. Strengthening the public extension system by capacity building of extension officers provides knowledge and skill to continue beyond the project period. They can use the CSA curriculum and training modules to implement similar projects. Farmers' will obtain increased capacity across a number of locally applicable CSA techniques and technologies through extension agent training, leading to the practices being able to be deployed after the project. The incentives to continue deploying these techniques will be provided by the financial and food security benefits delivered by transitioning to this mode of production. For private sector businesses supported by the project to expand into new market segments (youth and women), the growth this generates will lead to continued emphasises on these under-served segments and a long-term source of new customers. .

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline*	Target		Assumptions / Note
				Mid-term	Final	

		Sources of information and methods used to collect and report data /information to measure progress against targets.	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact the outcomes* *Data sources and methodologies applied for estimating baseline and targets*
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	• Project baseline, intermediate, and final surveys • Household surveys • Uptake of livelihood options • Field visits (including interviews and field surveys • National/sub-national statistics	Total: 0	Direct Total: *172,680 (40% of total)* Female: 69,072 (40% of total) Male: 69,072 (40% of total) Indirect Total: *210,294(30% of total)* Female: 105,147 (30% of total) Male: Female: 105,147 (30% of total)	Direct Total: 431,700 Female: 215,850 (50% of total) Male: 215,850 (50% of total) Indirect Total: 700,980 Female: 350,490 (50% of total) Male: 350,490 (50% of total)	• A low level of conflict risk and no conflict between farmers and agropastoralists exist in the selected 12 circles that will continue during the project period. • Direct beneficiaries cover 40% of the population in the 48 rural communes targeted by the project, based on estimated extension service coverage. 5,350 additional direct beneficiaries from training delivered to entrepreneurs and businesses through additional channels, and 58 from government planning training. Note other beneficiaries of activities are assumed to be sub-sets of these groups. • Indirect beneficiaries: 40% of the population in 12 circles has access to agriculture extension services. Among this population, 60% will be the CSA adoption rate. (Detail methodology is presented in Annex 02 Section 8.2)

ARA1 Most vulnerable people and communities	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	• Project baseline, intermediate, and final surveys • Household surveys • Inventory of livelihood options • Field visits (including interviews and field surveys	Total: 0 Female: 0 Male: 0	Total: *34,536(40% of total)* Female: 17,268 *(40% of total)* Male: 17,268 *(40% of total)*	Total: 86,340 Female: 43,170 Male: 43,170	• Low level of conflict risk and no conflict between farmers and agropastoralists exist in the selected 12 circles that will continue during the project period. • Project will provide direct support (improved seed, manure, insect/pest management, and capacity building training) of implementing CSA for all direct beneficiaries and 75% direct beneficiaries will adopt at least one CSA technology or practice for long-term
ARA1 Most vulnerable people and communities	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience	• Project baseline, intermediate, and final surveys • Inventory of households adopting CSA and agroforestry technologies and practices managed by the project team • Field visits (including interviews and field surveys • Reduction in loss of farm production and income or increase in farm production and income	Total: 0 Female: 0 Male: 0	*Total:* *40,000 (40% of total)* Female: 20,000 *(40% of total)* Male: 20,000 *(40% of total)*	Total: 100,000 Female: 50,000 Male: 50,000	• Low level of conflict risk and no conflict between farmers and agropastoralists exist in the selected 12 circles that will continue during the project period. • Corresponds to the adoption of low emission and climate-resilient technologies (solar irrigation systems and biodigester systems). Note that these beneficiaries are a sub-set of those in core 2: direct beneficiaries. • Covers direct recipients of technology and their households (average 7 people per household).

ARA2 Health, well-being, food and water security	Supplementary 2.2: Beneficiaries (female/male) with improved food security	• Project baseline, intermediate, and final surveys • Household food security surveys • Field visits (including interviews and field surveys) • HH food security survey (FIES)	Total: 0	Total: 172,680 *(40% of total)* Female: 86,140 *(40% of total)* Male: 86,140 *(40% of total)*	Total: 431,700 Female: 215,850 (50% of total) Male: 215, 850 (50% of total)	• A low level of conflict risk and no conflict between farmers and agropastoralists exist in the selected 12 circles that will continue during the project period. • Assumes gender split is 50% women and 50% men, in line with target
ARA2 Health, well-being, food and water security	Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security	• Project baseline, intermediate, and final surveys • Inventory of households adopting water management technologies and practices managed by the project team • Field visits (including interviews and field surveys)	Total: 0	Total: 2,308 *(40% of total)* Male: 1,154 *(40% of total)* Female: 1,154 *(40% of total)*	Total: 5770 Male: 2885 Female: 2885	• A low level of conflict risk and no conflict between farmers and agropastoralists exist in the selected 12 circles that will continue during the project period. • Corresponds to the adoption of solar irrigation systems (1000 installed) • Covers direct recipients of technology assuming 50:50 gender split • Note that these beneficiaries are a sub-set of those in core 2: direct beneficiaries.
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	• Project baseline, intermediate, and final surveys Household surveys • Uptake of livelihood options • Field visits (including interviews and field surveys	Total: 0	Direct Total: *172,680 (40% of total)* Female: 69,072 (40% of total) Male: 69,072 (40% of total)	Direct Total: 431,700 Female: 215, 850 (50% of total) Male: 215, 850 (50% of total)	Direct beneficiaries cover 40% of the population in the 48 rural communes targeted by the project, based on estimated extension service coverage. By adoption of CSA technologies/practices, they will improve well-being and food security through

		• National/sub-national statistics		Indirect Total: *210,294(30% of total)* Female: 105,147 (30% of total) Male: Female: 105,147 (30% of total)	Indirect Total: 700,980 Female: 350,490 (50% of total) Male: 350,490 (50% of total)	increased farm production and/or income Indirect beneficiaries: 40% of the population in 12 circles has access to agriculture extension services. Among this population, 60% will be the CSA adoption rate. (Detail methodology is presented in Annex 02 Section 8.2) By adoption of CSA technologies/practices, they will improve well-being and food security through increased farm production or/and income
MRA1 Energy generation and access	Core 1: GHG emissions reduced, avoided or removed/sequestered	• Energy inventory generated by installed sources and locations maintained by field staff • Field visits (including interviews and field surveys)	0 MtCO2eq	13,156 tCO2eq (10% of total)	131,566 tCO2eq (in 5 years)	• GHG emissions are avoided through the adoption of solar irrigation and biodigester systems. • See Annex 22 for the assumptions behind this calculation and baseline scenario
MRA4 Forestry and land use	Core 1: GHG emissions reduced, avoided or removed/sequestered.	• Inventory of land use change and tree plantation under the agroforestry systems maintained by field staff	0 MtCO2eq	166,865 tCO2eq (10% of total)	1,668,656 tCO2eq (direct)	• GHG emissions sequestered through plantation of 21,585 Ha of improved agroforestry in private lands. • See Annex 22 for the assumptions behind this calculation and baseline scenario.
	Core 1: GHG emissions reduced, avoided or removed/sequestered.	• Inventory of CSA adoption in the croplands maintained by field staff	0 MtCO2eq	64,632 tCO2eq (10% of total)	646,320 tCO2eq	• GHG emissions avoided through the adoption of CSA practices by 64,755 farming households (75% of 86,340 farming households)

MRA4 Forestry and land use	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice.	• Project baseline, intermediate, and final surveys • Inventory of households adopting improved agroforestry technologies and practices managed by project team • Field visits (including interviews and field surveys) • Land use database maintain by AEDD	0 Ha	6,476Ha improved agroforestry (30% of total)	21,585 Ha (improved agroforestry in private land)	For agroforestry: • Farmers are interested in utilizing agroforestry techniques and services provided by the project • Assumes that all beneficiaries that, adopt improved agroforestry on 0.25 ha of their land (or add extra land to do so).
	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice.	• Project baseline, intermediate, and final surveys • Inventory of households adopting CSA technologies and practices managed by project team • Field visits (including interviews and field surveys) • Land use database maintain by AEDD	0 Ha	9,713 Ha (30% of total)	32,378 Ha (adoption of CSA technologies practices)	For CSA adoption: 75% of 86,340 direct beneficiaries adopt new techniques and practices in their farming land

*IAAT's baseline survey and assessment will be completed in the first 6 months of the project (preparation phase). GCF indicators will be updated based on baseline survey assessment.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	**Baseline context (description)**	**Rating for current state (baseline)**	**Target scenario (description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Limited knowledge of the regulatory and legislative framework regarding climate change, as well as the existing methods and technologies for climate change adaptation and mitigation by local authorities and community action groups Existing planning documents/strategies at the local and community levels (e.g., Economic, Social, and Cultural Development Plan at the commune-level). However, none of these planning documents/strategies include climate adaptation and mitigation. Limited technical knowledge and capacity of local government authorities and communities for adaptation and mitigation planning Existing mechanisms and strategies for climate finance, but limited capacity to mobilize and deploy	*Low*	A regulatory framework for climate-smart agriculture and agroforestry is in place that incentivizes the shifting of public and private investment in adaptation and mitigation actions in agriculture and natural resource management. The government of Mali focuses on the localization of climate adaptation by integrating community priorities into the PDESC. Communal councils are actively engaging to develop PDESC through local stakeholders' participation. Within IAAT's 48 target communes, community action groups can effectively integrate it into their planning. These community action groups can develop action plans and review periodically and collaborate between them to share learnings and best practices.	Strengthen institutional capacity in localized climate change adaptation and mitigation planning and associated funding disbursement (Component 3)	National level (one country)

	resources for the design and implementation of initiatives for climate *change adaptation and mitigation*				
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Limited existing use of new climate smart technologies within the agricultural system, and by smallholder farmers that make up the majority of the nation's farmers (e.g. only 1,600 biodigesters in Mali in 2019 and the market is still nascent, high post-harvest losses, and few farmers use improved or certified seeds). There are some examples of businesses with expertise in climate-smart technology, but these focus on resource rich farmers. The majority of deployment is through international NGO-run projects (e.g. MERIT). There are limited financial incentives to adopt new technology, with a key barrier to adoption being high upfront cost and limited access to loans to finance initial investment.	low	Within IAAT's 48 target communes, smallholder farmers are aware of the use of biodigester and solar irrigation systems, and there are ~6,000 households with successful income generating usage of the technologies. There is also adequate service provision for the maintenance and productive use of the technologies delivered by local tradespeople. Private sector solar and biodigester technology suppliers (e.g., EcoTech, EMICOM, EnDev, EKOenergy, and PEG Africa) are available to scale up these technologies through the PAYG approach. Agrovet and animal health service providers are available to uptake services for CSA technologies in the IAAT-selected 48 communes. The commercial markets for improved seeds, farm machinery, and other inputs are available to support CSA technology supply chains. Agriculture extension offices are available at the circle level that provide services to communes and village through extension officers and field staff. They farm	Installation of solar irrigation and biodigester systems for long-term sustainable use (Component 2, Output 2.3, and Activity 2.3.1) Incorporation of technologies in CSA extension service curriculum to build awareness. (Component 1, Output 1.1, and Activity 1.1.1) Delivery of training to technology recipients on productive use at each stage in the technology deployment (installation requirements, relevant inputs, opportunities to create businesses with the technologies, and required maintenance practices) (Component 2, Output 2.3, and Activity 2.3.1) Technology providers supported to expand their businesses further through development of technical guidelines and development of women and youth inclusive	Multiple sub-national areas within a country

	There are limited pathways for smallholder farmers to access these businesses or the knowledge required to leverage them. Extension services lack training in new technologies and therefore do not disseminate it to farmers.		field schools and other thematic training on a regular basis.	business models. Technology providers linked up to other members of the agricultural value chain. (Component 2, Output 2.3, and Activity 2.3.1) Development of supporting services through capacity building for local tradespeople on construction and operations and maintenance. (Component 2, Output 2.3, and Activity 2.3.1)	
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Government policies and agricultural plans identify priority value chains for development, including by considering climate impacts. Government programs provide access to reduced cost inputs for agricultural markets (e.g. fertilizer, seeds). However, smallholder farmers still have limited access to required inputs for CSA value chains due to i) high risk perception results in low access to finance – e.g. denial of loans due to high risk profile of farmers; ii) limited knowledge of input providers actors. Some	low	A vibrant commercial market indicates multiple private sector operations shifting to more climate-smart production, processing, and distribution. In IAAT'S 48 target communities, women, are able to utilize greater volumes of inputs to produce higher volume and value CSA crops (e.g. fertilizer, modern seeds) as a result of enhanced access to finance and other value chain actors. In the IAAT locations, markets for CSA crops (including agroforestry products) are more transparent, connected, and able to be accessed by smallholder farmers – supporting the national development of these value chains and markets.	Local financial institutions receive technical assistance to increase women and youth access to finance. New financial institutional are also established (VSLAs specifically for women and youth). (2.2) Connections between smallholder farmers and other local market actors will be increased through local platforms. (Component 2, Output 2.1 and Activity 2.1.1) Producer associations receive training on how to deliver and/or increase access to aggregation and value	Multiple sub-national areas within a country

	existing programs target reduction in risk e.g. through support for crop insurance. Limited (market or subsidized) incentives for growing highly resilient / low emissions value chains, with limited demand /access to markets where smallholders can sell highly resilient / low emissions agricultural crops.			addition services for their members. (Component 2, Output 2.1 and Activity 2.1.1)	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Communication of agriculture extension including climate information and agro advisory services is not well advanced, largely use of farmers field school and community sharing, but limited use of digital tools. Smallholder farmers lack the technical knowledge and capacity to implement climate resilient or low-carbon techniques. There is limited knowledge of "fair prices" for agricultural outputs amongst smallholder farmers. Extension services have limited reach and approaches for sharing latest innovations or	low	There is enhanced knowledge transfer on CSA practices (including those based on local and traditional knowledge) from extension services (public and private) to smallholder farmers, resulting in enhanced adoption of CSA techniques. Farmers have a greater understanding of a "fair price" for their products, and which processes may enhance their value. Government, private sector organisations and NGOs have greater clarity on the existing deployment of biodigester and solar irrigation systems, supporting the development of new policies. Local institutions (at circles and communes) incorporate climate change mitigation and adaptation	Updated curriculum for extension services and improved training approaches. Curriculum is enhanced by co-design with local organisations and leverages local and traditional knowledge (Output 1.1). Actors across the value chain are connected to smallholder farmers, to access services as well as market knowledge. (2.1) Increased understanding of biodigester and solar irrigation system usage in target communes enhances government and NGO ability to assess success and	Multiple sub-national areas within a country

	knowledge. Local/traditional knowledge is not adequately incorporated into the extension service curriculum. Researchers and NGOs lack comprehensive knowledge of how CSA techniques are deployed. There is not an up-to-date inventory of the existing biodigester or solar irrigation deployment. Climate change adaptation or mitigation is not consistently included in regional or commune level planning.		in annual planning cycles (e.g. CACs and PDESC). **National and regional conferences** for exchange on CSA best practices increase adoption of CSA techniques, as well as sharing of “lessons learned” from IAAT.	plan additional technology deployment. (Output 2.3) Local and national institutions receive capacity building training to incorporate climate change mitigation and adaptation in annual planning cycles. (3.1) New and existing national and/or regional knowledge on best practices for CSA will be shared across public, private and educational institutions during twice yearly forums (Output 3.2).	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	**Project/programme specific Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Note**
				Mid-term	**Final**	
		Sources of information and methods used to collect and report data/information to measure progress against targets	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact on the Component* *Data sources and methodologies applied for estimating baseline and targets*
Output 1.1 *Improved technical capacities and inclusivity of extension services in climate-smart agriculture and agroforestry production*	# of extension agents that increase capacity in CSA training	• Pre- and post-training assessment • Training reports • Field surveys on the application of training outcomes by extension officers • Field inspections and reports	0	Total: 250 (50% of total) Men 175 Women: 75	Total: 500 Men: 350 (70%) Women: 150 (30%)	• Farmers actively engage in climate-resilient agricultural training initiatives because they perceive benefits in mitigating climate risk and assume positive results in agricultural product pricing, yields and access to inputs. • Assumes 100 trained per project target region, corresponding to approximately 10 per project target commune
Output 1.2 *Increased use of climate resilient practices in the production of CSA crops, livestock and agroforestry by smallholder farmers .*	# of farmers who have attended enhanced and updated training on CSA topics though extension services,	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports	0 (enhanced and updated training to be formulated during the project)	Total: 69,072(40% of total) Men: 34,536 34,536	Total: 172,680 Men: 86,340 (50%) Women: 86,340 (50%)	• Farmers actively participate in project interventions 40% of total estimated population of 48 implementation communes (% based on AGRA statistic of smallholder access to extension services in Mali) • 50% women and 50% male target

	% trained targeted farming households implementing agroforestry and at least one additional CSA technique on-farm	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Platform tracking reports • Field inspections and reports	0		75%	Farmers actively participate in project interventions • % of farming households who adopt new CSA techniques is based on USAID Albarka logframe target and cross-referenced with order of magnitude of change achieved in a CGIAR climate smart village program over a 5 year period
	# of farming households adopting the CSA technologies and practices	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Platform tracking reports • Field inspections and reports	0	25,902 farming households (40% of total adaptors) Men: 19,726 (75% of male headed households) Women: 6,476 (25% female headed households) Note: this number will be disaggregated by type of CSA technologies.	64,755 farming households (with 75% adoption rate) Men: 48,566(75% of male headed households) Women: 16,188 (25% female headed households) Note: this number will be disaggregated by type of CSA technologies.	Farmers actively participate in project interventions • % of farmers who adopt new CSA techniques is based on USAID Albarka logframe target and cross-referenced with order of magnitude of change achieved in a CGIAR climate smart village program over a 5-year period
	Area under CSA adoption (Ha)	• Baseline, intermediate, and final surveys • Project activity implementation		12,951ha (40% of total)	32,378 ha crop land (Note: data collection will	Farmers actively participate in project interventions • % of farmers who adopt new CSA techniques is

		inventory maintained by the project team • Field inspections and reports • Regional reports/GIS data			disaggregate land area by CSA technologies and practices)	based on USAID Albarka logframe target and cross-referenced with order of magnitude of change achieved in a CGIAR climate smart village program over a 5 year period
Output 1.3 *Increased land area under agroforestry*	Ha of improved agroforestry in private lands	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports • Regional reports/GIS data	8,634	9,360 Ha (40% of total)	21,585 Ha	• Farmers use agroforestry techniques and services provided by the project • Assumes that all 1.2.1 beneficiaries that adopt CSA, adopt agroforestry on 0.25 ha of their land (or an add extra land to do so). • Corresponds to 2.5% of grass and cropland area in the targeted communes or 1% of the total area in the targeted circles. • For further detail please reference the Annex 2 Feasibility Study
	Ha of tree plantation in public lands	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports • Regional reports/GIS data	0	27,880 Ha (40% of total)	69,700 Ha	• Community use tree plantation techniques and services provided by the project
	# of land use data bases developed to capture increase in CSA and agroforestry interventions	• Regional reports/GIS data		1	1	• Land use data base support CSA and agroforestry planning and monitoring by agriculture, forestry and other organizations working at the regional and local level

Output 2.1 *CSA and agroforestry VCs are more connected and reach more smallholder farmers*	# of farmers connected to an online farming platform resulting in increased connection to other value chain actors	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Platform tracking reports • Field inspections and reports		9,111 (40% of total)	22,778	• Farmers actively participate in project interventions. • Phone and internet connectivity improved at the current annual growth rate of > 3% and cover large number of rural farmers
	# of local producer groups representing small holder farmers established	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Platform tracking reports. • Field inspections and reports		120 (50% of total)	240	• Business opportunities exist for local producer groups
Output 2.2 *Smallholder farming communities, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness.*	# of women and youth engaged in VSLAs in targeted regions	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports	0	Total: 8,000 (40% of total) Women: 8,800 (40% of total) Men: 1,200 (40% of total)	Total: 20,000 Women: 17,000 Men: 3,000	• IAAT will promote 1000 women and youth targeted VSLAs (split 70:30, women:youth focus, with youth having 50% male:female gender split) • Each VSLA includes on average 20 people. • Note that these farmers are a sub-set of those benefitting from 1.1 and 1.2
	# of women and youth entrepreneurs who receive business training	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports	0	Women: 1,800 (40% of total) Men: 300 (40% of total)	Women: 4,500 Men: 750	• Women entrepreneurs receiving training through IAAT with target based on Albarka precedent of training 750 entrepreneurs / year, (for 5 years) • Youth entrepreneurs will also continue to be trained by Albarka for 2 years (~750/year) with an

						assumed gender split of 50:50 • These entrepreneurs are additional to the beneficiaries of 1.2, and either unemployed or employed in non-farm livelihoods.
	# of investment guidelines for financial institutions produced to include women and youth entrepreneurs and smallholder farmers	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports	0	1	1	• Guideline will attract financial institutions for women and youth sensitive investments
Output 2.3 *Increased adoption of low-carbon agriculture technologies by smallholder farmers*	# of new and functioning solar irrigation systems	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team. • Field inspections and reports	0	250 (30% of total)	1,000	• Farmers utilizing new technology, and conflict does not prevent projects from reaching beneficiaries. • Note that these farms are a sub-set of those benefitting from 1.2 • For further detail on these targets please see the Technology Options Assessment in the feasibility study
	# of new and functioning biodigester systems		0	1,500 (30% of total)	5,000	
	“% of targeted farmers successfully trained to adopt low carbon technologies	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports		2,400 (40% of total)	6,000	• Farmers actively engage in carbon reduction technologies training because they perceive benefits in adopting these technologies provide benefit in yield, price and access to inputs
Output 3.1 *Increased institutional capacity in climate change adaptation and mitigation planning and best practices to*	# institutional actors that increase knowledge on adaptation and mitigation through capacity building	• Baseline, intermediate, and final surveys • Project activity implementation	0	24 (40% of total)	58	• 1 local government representative per commune is trained (48 commune level beneficiaries)

address agriculture-related climate risks		inventory maintained by the project team • Field inspections and reports				• 10 national level representatives are trained • These beneficiaries are additional to those in 1.1
	# of PDESCs updated with climate change adaptation and mitigation related to CSA and agroforestry	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports		20	48	
	# of Community Action Groups strengthen to develop CACs	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports		100	250	• At least 5 Community Action Groups in 1 commune are trained (48 commune level beneficiaries)
Output 3.2: *Enhanced knowledge sharing and coordination of best practices in CSA and agroforestry across stakeholders*	# of knowledge-sharing events and/or workshops organized dedicated to sharing best-practices on CSA arising from IAAT learnings	• Baseline, intermediate, and final surveys • Project activity implementation inventory maintained by the project team • Field inspections and reports	0	1	2	• Conflict or security risk does not prevent the event from taking place • One national knowledge sharing event to take place per year • 2 workshops / year with key project stakeholders to share lessons learnt
Project/programme co-benefit indicators						
Co-benefit 1 Gender: Increased gender equality and economic empowerment	% of women who feel their contributions are integrated	Beneficiary surveys	40%[234]	No mid-line measurement	50%	- Conservative assumption of 2% increase per year - Baseline figures are indicative only, will need to be updated to cover exact project locations
	% of women empowered due to increased gender equality	Beneficiary surveys	40%	No mid-line measurement	50%	- Conservative assumption of 2% increase per year - Baseline figures are indicative only, will need

[234] https://www.sipri.org/sites/default/files/2019-12/sipriinsight1912_6.pdf (p. 22)

						to be updated to cover exact project locations
Co-benefit 2 Environmental: Improved soil conditions and water resources from landuse improvements	% of farmers reporting reduction in fertilizer use	Baseline, intermediate and final surveys Beneficiary surveys	0	No mid-line measurement	50%	- At least 50% of direct beneficiaries will adopt improved soil nutrient management practices in their agricultural fields
	% of farmers reporting use of less water after application of CSA	Baseline, intermediate and final surveys Beneficiary surveys	0	No mid-line measurement	50%	- At least 50% of direct beneficiaries will adopt improved water management practices in their agricultural fields

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Build technical capacity and reach of extension services on CSA and agroforestry techniques	This activity seeks to improve and expand extension services to support the transfer of climate-smart agriculture (CSA) and agroforestry knowledge and skills to smallholder farmers., The activity will develop a tailored curriculum of practices that address climate risks and deliver productivity, mitigation, and adaptation benefits and implement it through a "train the trainers" approach to refine public and private extension services. It will also involve local organizations and leverage digital platforms to increase the reach and inclusivity of training practices, ultimately enhancing the effectiveness and accessibility of extension services in promoting CSA and agroforestry among smallholder farmers.	1.1.1.1 Develop an updated curriculum for public and private extension services including Rural Resource Centers (CRRs), and Farmer Field Schools in project regions (Gao, Mopti, Koulikoro, Segou and Tombouctou) and tailor to the circle level via workshops	• Training program developed on: ○ overall causes and impacts of climate change on Malian agriculture ○ Benefits and implementation methods of 22 priority CSA and agroforestry techniques to enhance climate change mitigation, adaptation, and productivity (see section Selection of CSA techniques in the Feasibility Study for more detail on the regional applicable options and why these will be implemented) ○ Analysis and interpretation of land use and production data to provide tailored advisory services to farmers and improve on-farm decisions • At least 12 workshops organized at the circle level to tailor CSA and agroforestry curriculum to local conditions, by i) emphasizing the importance of analysis and interpretation of land use and production data to provide tailored advisory services to farmers and improve on-farm decisions, and ii) including local NGOs and stakeholders in the workshops to include inputs from traditional knowledge holders

			Contribution to project specific Indicator: # of extension agents that increase capacity in CSA training
		1.1.1.2 Train extension agents on the new curriculum through a “train the trainer” approach	• At least 1 workshop organized at the national level to introduce universities and technical schools to the new curriculum • At least 1 regional workshop organized in each region to provide a detailed presentation of the new curriculum and how to implement it to the trainers in universities and technical schools • At least 1 national workshop organized each year to review and update the curriculum and share learnings • 500 public and private sector extension agents targeted and trained through universities and tech schools (100 per region, approximate 10 per commune) Contribution to project specific Indicator: # of extension agents that increase capacity in CSA training
		1.1.1.3 Train extension services on inclusivity practices and expand utilization of existing resources, in particular digital tools and educational platforms	• At least 12 workshops organized for 500 public and private sector extension agents, with 1 in each target circle to increase adoption of inclusivity practices • At least 12 workshops organized to train extension agents on digital tools and educational platforms • Guides to extend reach of extension services through the adoption inclusive practices and digital learning platforms developed and shared with 500 public and private sector extension agents. This guide will comprise: o Recommended inclusive practices and how to integrate them in service offering o List of key digital tools and educational platforms, their respective objectives, functioning, and where and how to access them o The benefits of using digital tools and educational platforms both for extension agents and farmers Contribution to project specific Indicator: # of extension agents that increase capacity in CSA training

Activity 1.2.1 Build awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry amongst smallholder farmer communities (crop and/or livestock)	This activity focuses on the promotion and adoption of CSA and agroforestry practices among smallholder farmers by increasing their awareness, interest, and capacity through locally tailored training packages developed as part of 1.1.1. Several channels will be leveraged to increase the potential to reach more farmers through extension services (including farmer field schools, and Rural Resource Centers (CRRs)) as well as innovation platforms such as "Djoloko Nafama". This activity will also provide technical support to local community organizations, notably by (i) enhancing the capacity of Early Warning Groups to support the implementation and dissemination of national EWS and (ii) strengthening Community Action Cycles (CACs) established by Albarka and replicating the model in other regions. Adopting the CAC model will provide an opportunity to promote collective resource management, (enhancing and reducing conflict between livestock and crop farmers on community resources).	1.2.1.1 Deliver tailored local CSA and agroforestry training packages through extension agents and associated dissemination networks, including field schools, and Rural Resource Centers (CRRs), and cooperative leaders trained by Albarka	• Finalized training programs distributed to farmer field schools, extension agents, and CRRs • 455,557 farmers trained, of which 45,566 through farm field schools • 1,310 cooperative members trained by cooperative leaders (a sub-set of totals) • 50% of farmers trained are women (all farmers, not just landowners are targeted by the training) Contribution to project specific Indicator: # of farmers who have attended enhanced and updated training on CSA topics though extension services,
		1.2.1.2 Strengthen operational capacities of key local organizations (such as GAP-RUs (Early Warning Group/Emergency Response) to support the implementation and dissemination of the national Early Warning Systems - Albarka	• 19 contingency plans developed and shared with GAP-RUs at the commune level • 88 kits provided to GAP-RUs • 704 GAP-RU members trained on roles and responsibilities Contribution to project specific Indicator: # of farmers who have attended enhanced and updated training on CSA topics though extension services,
		1.2.1.3 Strengthen the 250 Community Action Cycles (CACs) established by Albarka to ensure their sustainability beyond project implementation and enhance their capacity to manage community assets for CSA technologies - Albarka	• Management kits distributed to each of the 250 CACs • Self-Assessment Plans developed to ensure autonomy and sustainability of CACs supported during project implementation Contribution to project specific Indicator: # of farmers who have attended enhanced and updated training on CSA topics though extension services,
		1.2.1.4 Extend the Albarka CAC model in other communes in Gao, Mopti, and Tombouctou, and replicate in Segou and Koulikoro	• 627 CACs established in communes of the project not covered by Albarka (with 1 CAC per village established) • Management kits distributed to each of the 627 CACs • Self-Assessment Plans developed to ensure autonomy and sustainability of CACs supported during project implementation Contribution to project specific Indicator: # of farmers who have attended enhanced and updated training on CSA topics though extension services,

Activity 1.3.1 Develop land-use mapping at the regional level	This activity aims to facilitate the adoption and effective utilization of regional-level land use mapping to support the development of agroforestry and inform the implementation of locally relevant (CSA) and agroforestry practices. By assisting regional authorities in mapping and classifying land types and uses in the target regions, it seeks to enhance their understanding of land resource distribution, enabling improved land-use planning and decision-making at the commune level. Additionally, a land-use database will be developed to enhance the knowledge of extension services regarding land characteristics, allowing them to provide tailored advisory services to farmers on suitable crops and techniques for enhanced productivity. This land-use database will also be shared with national authorities to promote wider access to agriculture and agroforestry data in Mali	1.3.1.1 Create land-use maps at the regional level to identify best suited areas for agroforestry production to improve land-use planning of communes in 3.1.1.1, and support the selection of areas for activity 2.3.2 (biodigesters and solar irrigation systems)	• Detailed land-use map developed for each region.Types and uses of land mapped and classified by regional authorities, including areas dedicated to agroforestry and best-suited areas for the installation of biodigesters and solar irrigation systems Contribution to project specific Indicator: # of land use data bases developed to capture increase in CSA and agroforestry interventions.
		1.3.1.2 Establish a project land use database to i) improve the quality of information provided to farmers by public and private extension services regarding the most adapted crops and techniques to increase productivity and ii) enhance access to data in line with 3.2.1.2	• Land-use database developed and provided to extension services, national authorities, research institutes, and other development partners Contribution to project specific Indicator: # of land use data bases developed to capture increase in CSA and agroforestry interventions.
Activity 1.3.2 Plant agroforestry trees on community and state-owned lands	This activity seeks to promote enhanced adoption of agroforestry by supporting the on-farm planting of 91,285 Ha of agroforestry, targeting specific value chains including acacia Senegal and shea. The capacities of several stakeholders will be leveraged to ensure successful implementation of this activity (e.g., nurseries for production and distribution of inputs, farmers for planting, communities for the management of trees and shared resources). The efficient management of these planted areas by communities will be critical to ensure sustainability after project implementation, and avoid conflict, notably between crop and livestock farmers.	1.3.2.1 Support the production and distribution of agroforestry tree saplings (e.g., acacia Senegal and shea) in partnership with nurseries - Albarka	• Agroforestry tree saplings produced and distributed to farmers Contribution to project specific Indicator: Ha of agroforestry trees planted
		1.3.2.2 Plant agroforestry (e.g., acacia Senegal and shea) in collaboration with smallholder farmers - Albarka	• 91,285 Ha of agroforestry (e.g., acacia Senegal and shea) planted by smallholder farmers across the five regions • This area under the agroforestry will be reported disaggregated by types of tree plantation (fodder, grass, timber and species), restoration of degraded land areas, area of farming size, etc Contribution to project specific Indicator: Ha of agroforestry trees planted
		1.3.1.3 Develop plans to manage samplings and growing trees in crop, grass/pasture, and community lands in partnership with CACs - Albarka	• 250 plans developed and shared with CACs for the sustainable management of samplings and growing trees, especially in community lands • At least 1 workshop organized in each commune to train CACs on the implementation of management plans Contribution to project specific Indicator: Ha of agroforestry trees planted

Activity 2.1.1 Support the creation of inclusive private sector value chains for key CSA/agroforestry crops and technologies	This activity will support the development of smallholder accessible value chains across CSA and agroforestry outputs by improving connectivity in the value chains and fostering access to markets for smallholder farmers. This will be achieved by leveraging digital marketplaces and existing innovation platforms, improving access to market information (e.g., prices) and fostering the development of a network of community-built service providers. Additionally, the activity seeks to enhance the capacity of businesses and producer organizations to reach a larger market through the development and adoption of inclusive business models, capturing the needs of smallholder farmers, women, and youth. The activity will mainly target businesses that deliver added productivity and climate mitigation/adaptation benefits, and are willing to invest in reaching smallholders, offering job opportunities for women and youth, and/or are connected to the mitigation and adaptation technologies prioritized in component 3 (solar irrigation, bio digester systems)	2.1.1.1 Increase linkages in CSA value chains by connecting service providers (including community-built service providers embedded in cooperatives) and processors with producer groups and cooperatives using digital marketplaces and existing innovation platforms - Albarka	• Existing digital tools identified (e.g., Sugu Sud Kunkan developed by FtF and USAID) and leveraged to connect service providers and processors with smallholder farmers, including producer groups and cooperatives • ~22,000 new subscribers to the digital marketplace (5% of farmers trained) • 100 service providers linked with cooperatives in target regions Contribution to project specific Indicator: - # of farmers connected to an online farming platform resulting in increased connection to other value chain actors - # of local producer groups representing small holder farmers established
		2.1.1.2 Deliver technical assistance to private sector businesses that are involved in CSA and agroforestry value chains, to enhance ability to develop youth and gender inclusive service/product offering and business plans	• 100 private sector companies selected and supported. These companies will be selected on the basis that they: ○ offer services related to increased CSA practices and crop yields (examples include input providers for modern seed varieties, pest reduction services, and services related to the processing of agroforestry outputs) commit to making investments that integrate a large number of smallholders, including job opportunities for women and youth, and offer a clear path to climate resilient economic, environmental, and social sustainability Contribution to project specific Indicator: - # of farmers connected to an online farming platform resulting in increased connection to other value chain actors - # of local producer groups representing small holder farmers established
		2.1.1.3 Develop technical guidance for private sector technology companies working in solar irrigation and biodigester systems on inclusive business growth	• 1 Economic Interest Group facilitated in each commune for each technology (solar irrigation systems and biodigesters) to regroup informal suppliers • Workshops organized at the regional level to train operators on: ○ expansion of distribution channels ○ product model/price refinement to be affordable by smallholder farmers ○ scaling and implementing relevant financing models such as leasing

			Contribution to project specific Indicator: - # of farmers connected to an online farming platform resulting in increased connection to other value chain actors - # of local producer groups representing small holder farmers established
		2.1.1.4 Implement trainings to producer associations to support them to deliver aggregation and value addition services to their members, for example cleaning, grading, and quality control, to support access to broader markets for crops	• Regional workshops to train producer associations • Targeting 5 producer associations trained per commune (240 producer groups) Contribution to project specific Indicator: - # of farmers connected to an online farming platform resulting in increased connection to other value chain actors - # of local producer groups representing small holder farmers established

Activity 2.2.1 Support local financial institutions to increase access to finance for smallholder farmers, especially women and youth, for CSA/agroforestry investments	This activity will increase access to finance for women and youth smallholder farmers and/or agribusiness entrepreneurs by working directly with financial institutions. This will be achieved by enhancing the contribution of women and youth in savings groups, especially VSLAs, which will provide opportunities to receive micro-loans to invest in revenue generative activities. The activity will develop and strengthen VSLAs in the five target regions and enhance their capacity to make informed investments, especially in CSA techniques. VSLA facilitators will be established and trained to extend support and ensure sustainability beyond implementation. Furthermore, existing microfinance organizations active in CSA and agroforestry will be targeted to increase their reach by delivering investment guidelines that are inclusive of women and youth entrepreneurs and supporting the expansion of risk adjusted lending (e.g., through crop and weather insurance and land-use mapping).	2.2.1.1 Support the sustainability and capability of Albarka established VSLAs by training VSLA facilitators and members on best management practices and income opportunities - Albarka	• Establishment and training of 76 VSLA facilitators on:
			  ○ The roles and responsibilities of VSLA facilitators
			  ○ Income Generating Activity (IGA) development
			• VSLA management tools provided to all committee members of VSLAs created by Albarka
			• Management committee members trained on VSLA management tools
			• All members of the VSLAs created by Albarka trained on IGA development
			• Follow-up report on VSLA cycle and governance (credit and savings management)
			
			Contribution to project specific Indicator: # of women and youth engaged in VSLAs in targeted regions

		2.2.1.2 Extend and replicate the Albarka VSLA model (particularly in Segou and Koulikoro which are not addressed by Albarka), with a particular emphasis on youth such as youth associations to increase participation of youth in savings initiatives	• 900 VSLAs established in communes not covered by Albarka, of which 30% are dedicated to youth and youth associations. • VSLA management tools provided to the VSLAs • Management committee members trained on VSLA management tools Contribution to project specific Indicator: # of women and youth engaged in VSLAs in targeted regions

		2.2.1.3 Support extension of microfinance reach by developing investment guidelines that are inclusive of women and youth entrepreneurs and smallholder farmers	• Develop technical guidelines for business expansion to women & youth market segments. • At least 1 workshop in each region to train microfinance institutions on the development of inclusive investment guidelines to improve reach Contribution to project specific Indicator: - # of investment guidelines for financial institutions produced to include women and youth entrepreneurs and smallholder farmers
Activity 2.2.2 Enhance capabilities and connectivity of youth and women CSA/agribusiness entrepreneurs	This activity will build the capacity and connectivity of women and youth smallholder farmers and/or agribusiness entrepreneurs to improve their capacity to efficiently manage their business and enhance their capacity to access finance and contribute to a sustainable supply chain. The activity will deliver core business skills training and connect beneficiaries to local financial institutions in 2.2.1, funding programs from complementary projects (PRAPS II) and other value chain actors supported in 2.1.1.	2.2.2.1 Strengthen the business capacity of youth entrepreneurs - Albarka	• 1,500 youth entrepreneurs selected and trained on core business skills such as business plan development and market research, including on loan application • Seed fund provided to youth and women entrepreneurs Contribution to project specific Indicator: - # of women and youth entrepreneurs who receive business training
		2.2.2.2 Strengthen the business capacity of women entrepreneurs, including woman-led processing units	• 3,750 women entrepreneurs selected and trained on core business skills such as business plan development and market research, including on loan application • Business kits and cash distributed to women entrepreneurs Contribution to project specific Indicator: - # of women and youth entrepreneurs who receive business training
		2.2.2.3 Connect youth and women entrepreneurs with the local financial institutions and private sector businesses supported in 2.2.1, and other programs in the region offering financing opportunities, in order to increase access to finance, productions services and distribution networks.	• 5,250 youth and women entrepreneurs connected to local financial institutions and private sector service providers Contribution to project specific Indicator: - # of women and youth in entrepreneurship
Activity 2.3.1 Install and support the productive use of biodigester systems and solar irrigation systems amongst smallholder farmers	This activity will increase the productivity and revenue of smallholder farmers whilst reducing GHG emissions of farms by supporting the adoption of mitigation and adaptation technology and their associated markets in Mali. The activity	2.3.1.1 Install 1,000 solar irrigation systems from the providers supported in Component 2 to smallholder farmers that are currently using diesel pumps for irrigation	• Initial 100 solar irrigation systems scaled up within the project implementation period to 1,000 by applying PAYG model. Contribution to project specific Indicator:

	will install 1,000 solar irrigation systems to replace existing diesel generator irrigation pumps and 5,000 biodigester systems and improved stoves for livestock farmers to provide an additional income source (slurry, manure, or fertilizer) and an alternative source of energy. The activity will ensure that the full potential of these technologies is harnessed by (i) developing the understanding and knowledge of farmers, and (ii) fostering the creation of sustainable value chains. In order to enhance the development of supply, the project will organize existing operators into Economic Interest Groups. This will allow them to collaborate and pool their resources to provide a tailored and high-quality services to smallholder farmers		- # of new and functioning solar pump irrigation system
		2.3.1.2 Install 5,000 biodigester and improved stoves systems from the providers supported in component 2 to smallholder livestock farmers	• Initial 1,000 biodigester systems scaled up within the project implementation period to 5,000 biodigester systems by applying PAYG model. Contribution to project specific Indicator: - # of new and functioning biodigester systems
		2.3.1.3 Deliver targeted training for those receiving new technologies in 2.3.1.1 and 2.3.1.2 on the use of biodigester and solar irrigation systems via local farm schools and extension agents supported in component 1	• 1,000 smallholder farmer groups and 5,000 livestock farmers trained on the use of biodigesters and solar irrigation systems. Training to be phased in semi-annual installments applicable to the stage of the equipment including: ○ requirements during installation ○ requirements for productive use (e.g. identification of appropriate inputs for solar irrigated VCs (e.g appropriate crops) and relevant livestock management practices for biodigesters) ○ establishment of businesses related to the system e.g., compost/fertilizer sale, irrigated crop cultivation. ○ ongoing maintenance and refresher of the best practices to deliver system benefits Contribution to project specific Indicator: - # of targeted farmers successfully trained to adopt low carbon technologies
		3.1.1.4 Generate technical guidance and deliver commune-level workshops on the installation of and O&M for the long-term productive use of equipment for local tradespeople and construction companies, building on the experience of the MERIT Biodigester project	• Technical guides for the installation, operation, and maintenance developed and shared with Economic Interest Groups facilitated in 2.1.1.3 • At least 1 workshop in each commune to train Economic Interest Groups on technical requirements for the installation of and O&M for the long-term productive use of equipment Contribution to project specific Indicator: - # of targeted farmers successfully trained to adopt low carbon technologies
Activity 3.1.1 Strengthen institutional capacity in localized climate change adaptation and mitigation planning	This activity will strengthen institutional capacity for scaling-out CSA by enhancing climate change adaptation and mitigation planning at the local level, leveraging community action plans and the Economic, Social, and Cultural Development Plans (PDESC) as the main entry points for improved adaptation planning. At the commune level, the	3.1.1.1 Organize regional workshops with stakeholders who participate in the development of the PDESC at the commune-level to improve their understanding of CSA and adaptation planning	• At least 1 representative from each communal council of the 48 communes trained on: ○ adaptation plan development and best practice for how to incorporate into local government decision making. ○ high-level overview of CSA and agroforestry to improve local authorities'

	activity will provide tools to local community groups for an effective and timely review of community action plans to foster efficient management of local resources and inclusion of adaptation planning. The activity will also seek to bridge the knowledge gab preventing communal councils and other local stakeholders to efficiently coordinate initiatives at the local level, by embedding adaptation and mitigation planning into the Economic, Social, and Cultural Development Plans (PDESC)		understanding of challenges and importance of the technique. ○ introduction of land-use planning to enhance the development of agroforestry. • Integration of adaptation planning into the PDESC of 48 communes Contribution to project specific Indicator: - # of institutional actors that increase knowledge on adaptation and mitigation through capacity building - # of PDESCs updated with climate change adaptation and mitigation related to CSA and agroforestry
		3.1.1.2 Support new and existing Community Action Groups in the five regions to periodically review and update community action plans for the management of local resources and adaption planning - Albarka	• At least 1 workshop organized in each commune to train community groups on the development and implementation of inclusive action plans • 250 Community Action Groups supported to conduct the self-evaluation of their action plan • Quarterly learning sessions between community action groups organized at the commune level Contribution to project specific Indicator: - # of community action groups strengthen to develop CACs
Activity 3.1.2 Technical capacity building of Mali Climate Fund and National Climate Change Committee (CNCCM) in prioritizing and targeting investment in CSA and agroforestry	The project will support Mali Climate Fund and National Climate Change Committee (CNCCM) to prioritize and target their investment in CSA and agroforestry technologies, practices and service. Workshops will be organized to train Mali Climate Fund on specific guidelines for prioritizing and designing adaptation and mitigation actions in agriculture and forestry. The workshops will provide a platform to discuss the required improvements to enhance the participation of government institutions in co-financing schemes.	3.1.2.1 Technical capacity building of Mali Climate Fund and National Climate Change Committee (CNCCM) in prioritizing and targeting investment in CSA and agroforestry	• 1 Technical guide developed to build the capacity of Mali Climate Fund and National Climate Change Committee (CNCCM) in prioritizing and targeting investment in CSA and agroforestry. • At least 1 national workshop with at least 10 attendees organized to train Mali Climate Fund on technical guidelines. • At least 1 national workshop with at least 10 attendees organized to monitor implementation of recommendations, share experience, and identify areas of and solutions for improvement Contribution to project specific Indicator: - # of institutional actors that increase knowledge on adaptation and mitigation through capacity building
Activity 3.2.1 Enhanced convening and contribution to	This activity aims to enhance knowledge-sharing and coordination of best practices among relevant stakeholders. The activity will compile the insights	3.2.1.1 Attend national and regional conferences and organize regional stakeholder convenings to share	• Active contribution to national and regional conferences on adaptation and mitigation for smallholder farmers

national databases for CSA and Agroforestry	gained from the IAAT project and disseminate them to national and regional stakeholders, extending the reach of valuable knowledge and lessons learned. This will involve active participation in regional and local gatherings, conferences, and knowledge networks, while also organizing new events as needed. A special emphasis will be placed on involving traditional knowledge groups to ensure a diverse range of perspectives and expertise are included in the knowledge-sharing process.	lessons on project implementation, and insights as they emerge for overall collaborative learning	• Summary of learnings from each national and regional conference and how they will benefit project implementation. • At least 1 workshop organized each year with key stakeholders to share lessons on project implementation and insights for collaborative learning Contribution to project specific Indicator: - # of knowledge-sharing events and/or workshops organized dedicated to sharing learnings -
		3.2.1.2 Join existing knowledge sharing and academic networks and contribute to existing databases (e.g., with land use data) to allow project data and findings to be leveraged in academic research	• Join existing knowledge sharing and academic networks • Project database, including land-use maps and database shared with national authorities, research institutes, and other relevant stakeholders Contribution to project specific Indicator: - # of knowledge-sharing events and/or workshops organized dedicated to sharing learnings
		3.2.1.3 – Organize annual knowledge sharing event to report best-practices on CSA learnt from project	• 1 annual report developed each year • 2 knowledge sharing event organized one in mid-year and one final year of the project Contribution to project specific Indicator: - # of knowledge-sharing events and/or workshops organized dedicated to sharing learnings

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

198. The Monitoring and Evaluation Plan of the GCF IAAT is designed to closely align with GCF's integrated results management framework (IRMF) for existing projects/programs, monitoring and evaluating core indicators such as reduced emissions, increased resilience, and systemic change, which the IAAT ultimately aims to achieve. The monitoring and evaluation process will be overseen by the Accredited Entity and Executing Entities and implemented by the Project Management Unit. However, the M&E arrangements will be further developed at the onset of project implementation by the project M&E staff, to determine in more detail the roles and responsibilities for data collection and management, project components' impact chains, information flows and reporting systems.

Monitoring

199. The project monitoring plan includes two levels of evaluation: GCF-level performance and project-level performance. The IAAT M&E team will conduct surveys to establish baseline levels for GCF core indicators, project-level results, and natural resource-related factors impacting food security and livelihoods. The monitoring process will consider diverse population segments and aim to capture effects based on gender, age, household head status,

and vulnerability. The monitoring structure allows flexibility for unforeseen incidents, with regular monitoring of project indicators, objective progress reporting, and alignment of project components with higher-level results and GCF goals.

200. The monitoring plan will employ diverse and comprehensive data sources to facilitate an informed assessment. Surveys and questionnaires will enable broad engagement with stakeholders, encompassing farmers, consumers, and community representatives, to gather valuable insights and accurately measure baseline conditions and project achievements, including the project's transformative impact. The field staff will maintain inventories, informed through field visits and inspections, to ensure meticulous tracking of progress, such as land use changes and tree plantations, while pre- and post-training evaluations will gauge the effectiveness of training programs. Additionally, specialized models will be utilized to estimate the extent of greenhouse gas emissions reduction, avoidance, or sequestration. The data collected through these varied sources will contribute to the development of a comprehensive project database, fostering knowledge sharing and facilitating collaboration with stakeholders.

201. The project team is committed to regular monitoring of indicators outlined in the project results framework, providing quarterly and annual assessments to objectively report project progress. Implementation of monitoring and evaluation activities will be discussed and agreed upon between the Accredited Entity (AE) and Executing Entities (Save the Children Mali), ensuring a structured and coordinated approach. Annual reviews, led by the project management unit (PMU), will involve local partners and government ministries, potentially organized at the regional level due to the project covering five regions. These reviews will synthesize progress against outcomes, including gender equality, social inclusion, and youth aspects, using a Collaborative Outcomes Reporting approach to identify adaptive management requirements.

Evaluation

202. The evaluation plan for the IAAT project will ensure consistent methodology and quality across baseline, midterm, and final evaluations, involving the AE, Executing Entities, and local implementing partners. The plan will guide the selection of external evaluation firms, inform the project's learning agenda, and facilitate coordination of learning among stakeholders for the promotion of CSA and agroforestry in Mali. The impact evaluation will utilize a quasi-experimental design called difference-in-differences, comparing outcomes for direct and indirect project-supported farmers with a comparison group, while regression analysis will control for time-varying differences.

203. The evaluation will encompass performance and impact evaluations, utilizing beneficiary and value chain questionnaires. Quantitative data collection will employ electronic tools with embedded consent provisions, while documentation of cleaning and analysis code will ensure result replication and future support. Qualitative data will be gathered through focus group discussions and key informant interviews, complemented by written notes and recordings. The evaluation aims to foster learning, inform decision-making, and improve performance by engaging stakeholders in a participatory process, presenting findings with contextualization and strong evidence.

204. Performance evaluations will be conducted at baseline, midterm, and endline, while impact evaluations will occur at baseline and endline, involving quantitative data collection from both direct and indirect project beneficiaries. The sample design will be carefully designed to encompass key subpopulations such as women, youth, smallholders, and vulnerable groups, ensuring representative inclusion in the evaluations.

205. Save the Children will engage an independent consultant firm for the baseline, midterm, and final performance evaluations, as well as the impact evaluation. IAAT staff, led by Save the Children's MEAL Manager, will oversee project monitoring and evaluation, ensuring access to respondents during fieldwork. Save the Children's technical advisory staff in Food Security and Livelihoods and MEAL, based in the USA., will contribute expertise to evaluation design, including questionnaires and sampling, and offer feedback on draft evaluation reports.

206. **Annex 11** presents a detailed IAAT’s monitoring and evaluation plan.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The IAAT will be implemented in 12 circles across five regions. These circles were selected based on their vulnerability to climate change, minimal or no ongoing conflicts, and the absence of transhumance corridors. Additionally, there is little to no conflict or competition between agriculturists and agropastoralists in these areas, making them ideal for implementing the initiative.

Selected Risk Factor 1

Category	Probability/Impact (before mitigation measures)	Probability/Impact (after mitigation measures)
Technical and operational	Low	Low

Description

Active conflict reduces the ability to successfully implement the project.
All regions of Mali are impacted by the effects of conflict, through live conflict itself or through an increase in internally displaced people. The Conflict has previously impacted the ability to access specific areas of Mali, resulted in the closure of financial institutions, closed public services, and resulted in injury and death for residents, including humanitarian workers. For IAAT, this could result in reduced access to project locations, a reduced pool of financial institutions to work with, and ultimately a personal security risk for those involved in the implementation of the project.

Mitigation Measure(s)

The project will mitigate the risk of conflict by i) minimizing exposure to conflict, ii) minimizing the risk of inflaming existing conflict tensions, and iii) minimizing the impacts of conflict if exposure does occur.

Minimizing exposure to conflict:

- Circles that have a high-security risk have not been selected for this project (based on the assessment by SCI Mali security expert).
- Project implementation will also be designed to require minimal travel for the implementation team, reducing the risk of conflict exposure due to travel.

Minimising risk of inflaming existing conflict tensions, resulting in reduced implementation ability:

- A local (commune-level) security and political-economy analysis will be undertaken to identify and then limit the risk of inflaming local tensions. Activities and sub-activities will then be reviewed under the lens of “do no harm” and “conflict sensitivity” to ensure all activities are adapted to local contexts.
- To avoid beneficiary selection inflaming existing local tensions, beneficiaries will be selected using the same approach as the Albarka project which is: “a community led participatory process, which involved community leaders and government technical services.”
- Utilisation of redress mechanisms and community feedback to allow any tensions arising to be swiftly resolved.

Minimizing the impacts on the project if conflict does occur:

- During the implementation phase, contingency plans will be developed to identify the activities that can occur or will be halted during different levels of local conflict.
  In addition, where possible and appropriate, GCF IAAT will utilize online methods of project activity delivery.

Selected Risk Factor 2

Category	Probability/Impact (before mitigation measures)	Probability/Impact (after mitigation measures)
Technical and operational	Low	Low

Description

Limited engagement or interest in project activities from local communities leads to non-realization of project outputs. In order for the project outputs to be achieved, smallholder farmers need to voluntarily implement new CSA practices. Limited interest or engagement from smallholder farmers could lead to the poor implementation of the project for example, reducing the adoption rate of CSA techniques in Component 1 or limited use of the new technologies supported in Component 3 underdeveloped projects due to limited interest in ongoing maintenance after installation of equipment.

Mitigation Measure(s)

○ Smallholder farmer engagement has been key to design the project and CSA practices selected for promotion by the project have been prioritised based on their suitability for smallholder farmers in the project locations and approved through stakeholder engagement. ○ Smallholder farmers cite that one of the biggest barriers to existing adoption of CSA practices is the lack of technical capacity rather than lack of interest. As a result, the project will support improvements in technical capacity through the creation of a new CSA curriculum for and delivered by extension agents. To ensure it is appropriate at the local level, it will be tailored at the circle level including through collaboration with local interest groups. ○ In addition, to support interest in the adoption of new techniques the project will support the economic return of CSA practices by supporting the relevant sale of the new goods in component 2, and by training biodigester and solar irrigation system users about the additional income opportunities from the technologies.		
Selected Risk Factor 3		
Category	**Probability/Impact (before mitigation measures)**	**Probability/Impact (after mitigation measures)**
Technical and operational	Low	low
Description		
Some beneficiary farmers revert to pre-project (climate -vulnerable) farming practices and/or stop using new technologies after project closure. Smallholder farmers show reduced interest in CSA practices after project implementation closes. Biodigester and solar irrigation systems are not used by farmers due to challenges with operations and maintenance after implementation support closes.		
Mitigation Measure(s)		
○ This risk, thoroughly considered during project design, is mitigated by ensuring that project-promoted CSA practices are financially viable and generate sufficient income, in partnership with the access to more reliable project-support value chains, so that farmers will be incentivized to continue using these practices beyond project closure. ○ In addition, the new extension service curriculum created during the project will continue to be utilized after the project closure, supporting the long-term and continued prioritization of these practices at the local level, and the adoption of CSA practices after the implementation period is finished. ○ For new technologies specifically, operations and maintenance skills are built out in project communities to ensure that there are accessible service providers after the implementation period.		
Selected Risk Factor 4		
Category	**Probability/Impact (before mitigation measures)**	**Probability/Impact (after mitigation measures)**
Governance	Low	Low
Description		
Limited interest from and high administrative burden on the government Limited implementation of commitments by political decision-makers in the implementation of the project. Complex and restrictive administrative procedures faced by governmental organizations result in a high and potentially obstructive administrative burden for the government organisations involved in the project		
Mitigation Measure(s)		
○ AEDD and the Department of Agriculture actively participate in the project with co-financing and this project largely works with local organizations. ○ A majority of interventions including training and capacity building, procurement, and delivery of technologies will be carried out by contracting third party on a competitive basis.		
Selected Risk Factor 5		
Category	**Probability/Impact (before mitigation measures)**	**Probability/Impact (after mitigation measures)**

Governance	Low	Low
Description		
Political instability or elections result in a change of central authorities during the implementation of the project. Although delayed vs 2022, presidential elections are scheduled to be held during project implementation (currently February 2024). This could result in a reduction in support for the project from the national government, including the implementing entity. Alternatively, a major internal government reshuffle could lead to a change of key stakeholders.		
Mitigation Measure(s)		
○ The risk that new officials and/or new members of the AEDD will not support the implementation of the project will be mitigated through information sessions to update any new stakeholders on the project interventions and outcomes, and how these are aligned with national priorities and local community needs. ○ Additionally, the locally led, commune and circle-level implementation plans and agreements will be prioritized to ensure that the local communities that benefit from the Project will establish a permanent and continuous work agenda, regardless of the national government.		
Selected Risk Factor 6		
Category	**Probability/Impact (before mitigation measures)**	**Probability/Impact (after mitigation measures)**
ML/FT	Low	Low
Description		
Risk of project funds being used for money laundering, terrorist financing, prohibited practices		
Mitigation Measure(s)		
○ Completed assessment of the financial management capabilities of the governmental implementing entity, AEDD (**Annex 20: AML/KYC**). ○ Set up a financial control mechanism specific to the IAAT project, including procedures for verifying financial transactions and annual audits of implementing partners. ○ Include compliance with SCI's anti-money laundering policy in agreements to be signed between SCI and the government. ○ Annually monitor and evaluate the effectiveness of anti-money laundering measures in place in order to make adjustments where necessary and continually improve anti-money laundering practices. ○ Provide training on fraud management policy and procedure in procurement processes. ○ SCI Mali uses compliance monitoring and verification measures for procurement process, service providers and field monitoring and verification of any materials or technology use to process the payment process.		
Selected Risk Factor 7		
Category	**Probability/Impact (before mitigation measures)**	**Probability/impact (after mitigation measures)**
Other	Low	Low
Description		
Climate hazard events (such as prolonged extreme drought or flood), and/or macro-economic events (such as higher or volatile market prices for agricultural inputs in international markets or tightened financing conditions from the banks due to global financial turmoil), result in reduced impact of project interventions		
Mitigation Measure(s)		
○ The project is designed to operate within the context of recurrent drought and floods. The occurrence of such climate extremes will inevitably have consequences for project delivery but will also strengthen support for the role of the project in mitigating such risks in the long term. ○ Project interventions are specifically designed to reduce the impact of climate hazards on project beneficiaries, such as improved water management (zai pits, solar irrigation), plantation of agroforestry to improve soil quality and enhanced information and early warning systems will reduce the impact of drought.		
Selected Risk Factor 8		
Category	**Probability**	**Impact**

	(before mitigation measures)	(after mitigation measures)
Other	Low	Low
Description		
Gender inequality in access to resources and decision making for adopting climate-smart agriculture and agroforestry related interventions, result in low participation of women and adoption of gender sensitive interventions		
Mitigation Measure(s)		
○ Actively inclusion of women in training and capacity building activities based on suitability of women farmers by location and timing of the training events ○ Provide project grant for CSA, agroforestry, solar pump and biodigester when farming household ensure women's participation in the project activities ○ Increase women access to financial resources and other group activities		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

207. Eight performance standards of the GCF Environmental and Social Policy were evaluated during the project preparation, and none is assessed to have been activated in accordance with the Category C risk ranking. : The project is assessed as low risk due to small-scale infrastructure which may create short-term disturbances to households or health and safety risks during installation works. These are expected to have extremely limited to no significant impacts and will be readily mitigated through appropriate measures as reflected in the ESAP. The project will implement strategies to ensure that risks remain low. Through detailed site assessments and consultations, it has been confirmed that there are no transhumance corridors intersecting the IAAT project locations (12 circles). Additionally, the project locations do not overlap with areas traditionally inhabited or utilized by Indigenous People. Therefore, the IAAT project activities will not impact transhumance routes and Indigenous Peoples in the project areas. In addition, there will be no land acquisitions and involuntary resettlement to implement the IAAT activities. The conflict risk analysis in the IAAT selected 12 circles that indicate low conflict risk. IAAT will select 48 communes within these 12 circles based on a further comprehensive conflict-risk assessment conducted in the project's initial phase. This approach will help exclude communes affected by conflict from implementing IAAT project activities.
208. As part of their AE status, SCA implements all projects in accordance with their own Environmental and Social Management System (ESMS), which formed the basis for their Category C GCF accreditation. The risk categorisation table within the SCA ESMS has been tailored to this project and includes expanded aspects of the applicable GCF PSs to ensure that all ongoing environmental and social screening of activities as they are further developed continue to capture all potential impacts that are outside the Category C rating. The GCF Simplified Approval Process (SAP) was utilized as the foundation for customizing the ESMS.
209. SCA has zero tolerance for any abuse and exploitation committed by representatives against adults or children in the communities where the projects are implemented in. A key priority is safeguarding all children and adults who come into contact with our organisation from all forms of abuse and harm including sexual exploitation, abuse, and harassment (SEAH). The SRA ensures that safeguarding risks including SEAH are identified, and adequate controls are developed and monitored.
210. **Annex 6** (Environmental and Social Assessment and Residual Risk Management Plan) in section 3.2 describes the GCF Environmental and Social Safeguard (ESS) screening including the results. The proposed project has minimal or no adverse environmental or social risks or impacts. The overall project design, as well as the list of eligible activities at the farm, community, and institutional levels, have been screened using the tailored Environmental and Social Management System (ESMS) screening tool of Save the Children. The screening tool indicates a low degree of concern, i.e., category C.
211. The project is confirmed to fit within the thresholds of Category C (low risk) because of the Environmental and Social screening which entailed a project development discussion, stakeholder meetings, a desk study of similar projects in the region, and a review of potential options. Save the Children implements all projects according to the ESMS through a screening and management process based on their Category C GCF accreditation.
212. In undertaking this screening, the potential risks and impacts have been considered that include direct and indirect, induced, long-term and cumulative impacts, and considered the proposed activities' area of influence, residual risks (if any) and mitigation measures toward these. All IAAT activities have been assessed, and initial risk levels are classified as low. The project will be implementing mitigation strategies to either eliminate these low risks or maintain them at a non-significant, low level (Annex 6, Table 4)
213. Activity 1.2.1 (Build awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry amongst smallholder farmer communities) and Activity 3.1.1 (Install and support the productive

use of biogas systems and solar irrigation systems among smallholder farmers) does not activate water use and waste related risk such as new construction of water sources (e.g. boreholes, ponds) and discharge of slurry from the biodigesters. These activities fall under low-risk category and IAAT will implement risk mitigation strategies that include: i) promoting solar pumping in good water table and recharge areas, ii) promoting water-efficient technologies and practices (e.g. micro irrigation and irrigation scheduling methods), iii promoting manure production using slurry.

214. The SCA SRA tool is used to screen for safeguarding and SEAH risks caused by project activities, staff, and representatives, including consultants, volunteers, partners, and suppliers. Also safeguarding and SEAH risks in the project context and external environment. The full tool is attached as Appendix 1 in Annex 6).

215. Another identified risk is associated with training activities with the risk of SEAH incidences due to a lack of participants' sensitization. Risk level for SEAH is low in the IAAT selected circles and the project will implement mitigation measures that include: all staff and participants (including foreign nationals) will be sensitised on existing national, and local if applicable, legislation, ESAP and GCF policy related to SEAH (Annex 6, Table 4). Monitoring of SEAH incidences will are included in the Grievance Redress Mechanism, as outlined in this document

216. There is a potential risk of invasion of foreign planting materials and soil erosion in the agroforestry system is low. The project will promote only locally available species, avoid the import of foreign genetic materials and GMO, and use minimum soil disturbance and loss technologies and practices.

217. Save the Children Australia (SCA), as the Accredited Entity (AE), Save the Children International (SCI) Mali as the Executing Entity and the Agency of Environmental and Sustainable Development (AEDD), as the Implementing Entity , are responsible for ensuring that the requirements of this Environmental and Social Assessment and Residual Risk Management Plan (ESARRM) are fully integrated into the Project. It is the responsibility of the AE to ensure that suitable ESS processes and reporting are in place to guarantee that the Project is completed with little or no negative environmental or social effect.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

218. Women in Mali play a pivotal role in the agricultural sector, environmental management, and biodiversity conservation. With over 80% of the population engaged in agriculture, women are primarily responsible for crucial food crops, undertaking cultivation and processing throughout the agricultural value chain. However, they face discrimination and therefore numerous challenges such as: limited access to productive inputs and technologies, underdeveloped markets, and the disproportionate impacts of climate change. Despite the presence of equity and equality principles in laws and policies, a lack of awareness and implementation focused on removing barriers for women and youth still prevails. Traditional social structures influenced by religion and customary practices further contribute to gender inequalities, restricting women's and girls' rights and opportunities for empowerment.

219. A detailed analysis was conducted to ensure comprehensive integration of gender in the GCF IAAT project, in accordance with Save the Children's principles and GCF guidelines, see **Annex 8**. The study examined the division of labor, women's access to resources and benefits, their participation in decision-making, education, and the impact of gender-based violence and the impacts of the COVID-19 pandemic. Multiple data collection methods, including literature review, interviews, regional consultations, and key informant surveys, were employed. The analysis covered national and international documents, policies, reports, and project-related materials. Survey questions cover assessment of gender gaps including vulnerability of different groups of people in the community such as women and girls, men, youth and elder. The study was carried out in Bamako, Koulikoro, Ségou, Mopti, Gao, and Tombouctou, involving various institutions, women leaders, and women in the communities and the findings were used to draw comprehensive conclusions on gender inequalities in roles, responsibilities, resource access, and control.

220. The gender assessment conducted in Mali reveals a range of challenges and disparities faced by women, highlighting the need for gender equality and empowerment. Women's illiteracy, limited access to climate change information, and low sensitivity of women's associations to adaptation further exacerbate their vulnerability. For instance, while men dominate decision-making positions, women are relegated to reproductive and community roles, limiting their agency and economic opportunities. Moreover, women's limited access to land, resources, and agricultural equipment curtails their productivity and economic empowerment. Discriminatory practices and socio-cultural norms prevent women from inheriting land, relegating them to marginal plots and hindering their livelihood potential. They have less control over livestock and forest resources, perpetuating their dependence on men and limiting their economic autonomy. Men have access to benefits provided by governments and other organizations and income generated by the household members, while women must seek approval from male household heads, further entrenching gender-based power imbalances. The gender disparities extend to information access and control, as women are often

excluded from village meetings and public announcements that led to limited use of women and girls' potential to enhance households' prosperity. Limited literacy and dependence on male family members for access to resources impede women's access to information and technology. The unequal control over communication devices, such as phones owned and controlled by husbands, further marginalizes women's voices and limits their opportunities. These gender disparities are particularly evident in rural areas, where women have lower exposure to media and limited access to climate, environmental, and agricultural information, leaving them more vulnerable to the impacts of climate change. The repercussions of conflict further compound the issue, with displaced women often left as heads of households, facing precarious conditions and heightened risks.

221. Women are also involved in agricultural production downstream. They are present to varying degrees throughout the agricultural value chain and participate in the processing and marketing of the products produced. However, they are not paid, nor do they control the income from this production. In addition to reproductive tasks, women, with the help of their daughters, also take part in income-generating activities, enabling them to generate their own income, which they manage and control, even if they have to inform the men[235] . Men are involved in agricultural production upstream. They are only involved in productive tasks and control all income linked to agricultural production.

222. Water scarcity due to climate change has a big social impact, particularly on women and girls in Mali. Women and girls are the most negatively affected due to their traditional responsibility for water collection. Recurrent droughts have required women and girls to travel longer distances to find water, forcing them to quit or curtail economic activities, school, and participation in local organizations. The energy supply burden is also directly transferred to women and girls, who are the prime household energy users. Decreasing access to energy mainly affects women because they are usually responsible for providing energy for the households and must spend a large amount of time and physical energy obtaining traditional fuels (such as wood, charcoal, dung, and agricultural waste). In addition, women and girls were impacted much by the severe droughts and floods through production and income pathways. Women in Mali are largely dependent on agriculture, and the reduction in labor demand due to floods and droughts directly impact women's job. Second, household food supply in the agricultural-dependent households is largely contributed by women, and any reduction of farm production due to floods, droughts, heatwaves and pests and diseases, increases women's burden to provide food to their household members.

223. Promoting women's empowerment and addressing gender disparities in Mali is essential for their active participation and benefits in agricultural interventions, as well as their resilience to climate change impacts. This involves ensuring women's access to resources, increasing their representation in decision-making processes, and involving them in conflict prevention and resolution. Gender assessment reveals opportunities to bridge these gaps, including recognizing specific roles within households and communities to drive positive change. Sensitisation of the communities is needed to secure women's control over resources and benefits, such as land tenure and natural resource management. Additionally, initiatives like improved access to agricultural equipment and training in climate-smart techniques are crucial for climate change adaptation. Benefitting from innovation and private sector engagement will empower women through activities such as processing and support their fuller engagement in diverse agricultural and livestock livelihoods. Mobilizing women's action requires providing them with necessary resources, land access, training, and resilience-building skills, ultimately fostering national reconciliation and social cohesion. These opportunities, combined with capacity development, collective actions, awareness-raising, and financial support, have the potential to drive gender equality and women's empowerment in Mali.

224. The IAAT aims to bring a meaningful contribution to the effective creation of opportunities for women through several activities, embedding gender across the 4 components. The project will seek to reach at least 50% of women, emphasizing specific avenues for women's empowerment, especially through support to enhance access to finance, improve contribution to decision-making processes, and foster involvement in revenue generating activities. Equal representation of women and men is targeted in all capacity building and awareness-raising activities, as well as decision-making processes and committees such as Community Action Cycles (CACs). IAAT will focus on increasing the participation of women and equal contribution in developing CACs. Access to finance will be streamlined by creating Village Savings and Loans Associations (VSLAs) specifically dedicated to women and 3,750 women entrepreneurs will be supported to enhance involvement in revenue generating activities. The project will ensure contribution of women in the development of community action plans through CACs and adaptation and mitigation planning at the commune-level. Furthermore, private sector companies and extension officers will be supported to integrate women (at least 50:50 men and women) in their service offerings.

[235] Source: Gender analysis report Albarka Programme zones

225. IAAT will closely work with the Regional Directorates for the Proportion of Women, Children and Family to leverage its technical knowledge and experience of mobilizing women for information transfer at the local level. This institution implements program and projects that support the advancement of women and gender equity and develop women entrepreneurship in the agriculture and other sectors.

226. The Gender Action Plan highlights in more detail how gender will be integrated into project components, as well as key objectives and indicators to monitor and evaluate the successful empowerment of women. The plan also maps the potential risks to create and/or exacerbate gender gaps and the corresponding mitigation strategies.

227. The detailed Gender Assessment and Action Plan is provided in **Annex 8**.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

228. **Financial structure**. The project's financial structure draws lessons learned from SCI Mali's decades-long experience as an implementing partner in Mali and is designed to achieve the objectives in the most cost-effective and efficient way. As described above, funds will flow from SCA to SCI Mali, then will be managed through the PIU with disbursements to implementing entities. Where SCI Mali directly implements, such as at the circle and commune levels, all activities will be carried out in conjunction with government partners (AEDD, local governments) to facilitate buy-in, ownership, and experience in achieving all project outputs and outcomes. This structure will allow the project to balance support to national and subnational levels of government and engage appropriate GoM counterparts at the right levels in implementing each activity while minimizing potential financial and procurement risks. While implementation in close partnership with the government can create bottlenecks, the PIU will manage this by coordinating daily with government partners and the project steering committee.

229. **Procurement arrangement**: All goods and services to implement IAAT activities will be procured according to SCI Global Procurement Policy, GCF policy, applicable Mali laws, sustainability, and value for Money Principles. Where there is a discrepancy between these, the most stringent requirement must be applied. SCI Mali strives to meet and remain compliant with the highest ethical, legal, and professional standards. SCI Mali expects and is legally obliged to ensure suppliers adopt similar Codes of Conduct. The IAAT's financial management and procurement approach follows SCA's approved processes to which they were accredited and hold the AMA with GCF. A detailed procurement plan for goods, works, and non-consulting services and the selection process is presented in Annex 10 (Procurement Plan). All procurement activities will be incurred by SCI Mali. AEDD will not receive GCF funding and is not responsible for GCF funding related procurements. AEDD will manage procurement for co-funding activities based on its own procurement policy and not liable for SCI and GCF's procurement policies.

G.4. Disclosure of funding proposal

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if it contains confidential information.

C. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s)
☒	Annex 2	Feasibility study
☒	Annex 3	Economic and/or financial analyses
☒	Annex 4	Detailed budget plan
☒	Annex 5	Implementation timetable including key project/programme milestones
☒	Annex 6	Environmental and social assessment and residual risk management plan
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and action plan
☒	Annex 9	Legal due diligence
☒	Annex 10	Procurement plan
☒	Annex 11	Monitoring and evaluation plan
☒	Annex 12	AE fee request
☒	Annex 13	Co-financing commitment letter, if applicable
☒	Annex 14	Term sheet

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval
☒	Annex 16	Maps for proposed location for interventions
☐	Annex 17	Multi-country project/programme information
☐	Annex 18	Appraisal, due diligence, or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☒	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (operations and maintenance)
☒	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting
☒	Annex 23	Conflict analysis and management strategy

No-objection letter issued by the national designated authority(ies) or focal point(s)

MINISTERE DE L'ENVIRONNEMENT, DE L'ASSAINISSEMENT ET DU DEVELOPPEMENT DURABLE

Agence de l'Environnement et du Développement Durable (AEDD)

N° 00696 /MEADD/AEDD.

REPUBLIQUE DU MALI
Un Peuple - Un But - Une Foi

Bamako, le 09 NOV 2023

The Director General of the Environment and Sustainable Development Agency

To

Ms. Mafalda Duarte
Director Executive of Green Climate Funds
G-Tower, 24-4 Songodo dong Yeonsu gu
Incheon City Republic of Korea

Re: Funding proposal for the GCF by The Save the Children Australia regarding the Project " **Intensification of agriculture and agroforestry techniques (IAAT) to increase climate resilence of food and nutrition security in Koulikoro, Ségou, Mopti, Tombouctou, Gao regions in Mali"**

Dear, Sir,

We refer to the Project Intensification of agriculture and agroforestry techniques (IAAT) to increase climate resilence of food and nutrition security in Koulikoro, Ségou, Mopti, Tombouctou, Gao regions in Mali as included in the funding proposal submitted by the Save the Children Australia to us on the 10th of October 2023.

The undersigned is the duly authorized representative of Environment and Sustainable Development Agency, the National Designated Authority of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that :

(a) The government of Mali has no-objection to the Project as included in the funding proposal ;

(b) The Project as included in the funding proposal is in conformity with Mali's national priorities, strategies and plans ;

(c) In accordance with the GCF's environmental and social safeguards, the Project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the Project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the project.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

The Director General,

Le Directeur Général

Zantigui Boua KONE
Chevalier de l'Ordre national

Secretariat's assessment of FP256

Proposal name:	Intensification of Agriculture and Agroforestry Techniques (IAAT) for Climate Resilient Food and Nutrition Security: Tombouctou, Gao, Mopti, Koulikoro and Segou regions of Mali
Accredited entity:	Save the Children Australia
Country:	Mali
Project size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The Intensification of Agriculture and Agroforestry Techniques (IAAT) project targets Mali, a highly vulnerable country impacted by conflict and climate change. The accredited entity, Save the Children Australia, is well positioned to lead this project and address urgent climate challenges, given its strong track record of implementing climate projects in conflict-affected countries with solid conflict-sensitive analysis.	There is a risk of active conflict spillover affecting public service delivery, business activities and project continuity, leading to reduced access to project locations and limited availability of extension services, financial institutions and private companies, as well as delays in implementation and diminished impact. To mitigate this risk, the accredited entity will conduct continuous monitoring and implement adaptive measures. Furthermore, the project includes a contingency budget and plans for reallocating resources between regions as needed.
The IAAT project targets the agriculture sector, critical for Mali's food security and livelihood, given that it employs 80% of the population. Beneficiaries have been selected primarily on the basis that they are the most vulnerable to climate change, through an assessment of climate risk, exposure and adaptive capacity.	There is a risk that some beneficiaries may revert to climate-vulnerable farming practices, leading to limited adoption, reduced impact and diminished sustainability and paradigm shift. To mitigate this, climate-smart agriculture practices are designed to be financially viable, generate sufficient income and include multiple intervention entry points to support income diversification.
The IAAT project leverages the Albarka project funded by the United States Agency for International Development, especially its existing field platform, namely its relationship with the Government of Mali and knowledge and lessons learned in other communities. It	

also mobilizes co-financing by the Albarka project for a further scale-up.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity (AE) and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.41/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Mali is highly vulnerable to climate change, ranking 11th in vulnerability and 23rd in readiness for climate change adaptation and mitigation out of 192 countries according to the Notre Dame Global Adaptation Initiative country index, with significant impacts from rising temperatures, erratic rainfall and frequent climate hazards like droughts and floods affecting up to 500,000 people annually. The socioeconomic context in Mali, as a least developed country with high levels of poverty and food insecurity, dependence on subsistence agriculture and a fast-growing young population, further increases its vulnerability to climate hazards. The largely subsistence-based agriculture sector, employing 80 per cent of the workforce and contributing 36 per cent of gross domestic product, is particularly threatened, with projected decreases in crop suitability and significant yield losses by 2050. Climate change induced stressors, including heat and reduced fodder availability for livestock, exacerbate food insecurity, with severe food shortages impacting up to 24 per cent of the population and malnutrition affecting 1.2 million children.

4. The goal of the Intensification of Agriculture and Agroforestry Techniques (IAAT) project is to increase the climate resilience in agricultural systems and reduce greenhouse gas (GHG) emissions, in close collaboration with the Government of Mali, civil society, the private sector and communities by increasing capacity and providing support for climate risk informed decisions, including scaling locally led adaptation actions to address vulnerability linked to food and water insecurity.

5. The focus will be on improving extension services for climate-smart agriculture (CSA) adoption; supporting the development of CSA and agroforestry value chains; reducing GHG emissions from agricultural systems; and increasing institutional capacity and knowledge for climate action. Five regions (Gao, Koulikoro, Mopti, Segou and Tombouctou) and 12 highly vulnerable cercles covering 48 communes were selected as target areas on the basis of national and regional workshops, stakeholder consultations and climate change vulnerability assessment.

6. In targeting Mali, a country highly vulnerable to the impacts of conflict and climate change, the IAAT project holds significant strategic value. The AE, Save the Children Australia, is well positioned to lead this initiative, leveraging its strong track record of implementing projects in conflict-affected regions with effective conflict-sensitive analysis. Additionally, the project focuses on the agriculture sector, which is critical for food security and livelihoods in Mali. Beneficiaries have been selected on the basis of their high vulnerability to climate change, using an assessment based on climate risk, exposure and adaptive capacity.

7. The proposed funding for the project is around USD 43.7 million, utilizing grant instruments. The GCF contribution to the project is around USD 33.7 million in grant. Co-financing of USD 10 million will be made, consisting of USD 5 million by the Environment and Sustainable Development Agency (AEDD) of the Government of Mali and USD 5 million by the

United States Agency for International Development through its Albarka project (a USD 60 million grant project on resilience and food security in Gao, Mopti and Tombouctou).

8. The project targets 431,700 direct beneficiaries and 700,980 indirect beneficiaries. Together, these beneficiaries make up 6.2 per cent of the whole population of Mali. The mitigation impact over the project's lifetime of 15 years is 2.4 million tonnes of carbon dioxide equivalent (t CO_2 eq).

9. GCF categorizes the project as C (minimal) for its environmental and social safeguards (ESS).

2.2 Component-by-component analysis

10. The project is structured into three components:

(a) **Component 1**: Improving extension services and increasing on-farm CSA adoption;

(b) **Component 2:** Supporting the development of CSA and agroforestry value chains; and

(c) **Component 3:** Increasing institutional capacity and knowledge.

Component 1: Improving extension services and increasing on-farm CSA adoption (total cost: USD 24,263,640; GCF cost: USD 18,541,641 in grants)

11. This component aims to increase (i) the awareness of climate risks and (ii) the understanding and technical capacity of smallholder farmers to adopt CSA techniques by building the capacity of agro-advisory services, the key educational service for smallholder farmers, and by supporting the on-farm adoption of CSA techniques.

12. Activities under this component include (i) building the technical capacity and reach of extension services on CSA and agroforestry techniques; (ii) building awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry among smallholder farmer communities (crop and/or livestock); (iii) developing land-use mapping at the regional level; and (iv) planting agroforestry trees on community and State-owned lands.

Component 2: Supporting the development of CSA and agroforestry value chains (total cost: USD 10,718,182; GCF cost: USD 9,008,182 in grants)

13. This component aims to create sustainable markets for CSA and agroforestry products and ultimately more diverse and sustainable livelihoods for the target communities by fostering the development of connected value chains and lifting the barriers disproportionately faced by women and youth entrepreneurs for starting and running businesses, including access to finance.

14. Activities under this component include (i) supporting the creation of inclusive private sector value chains for key CSA/agroforestry crops and technologies; (ii) enhancing capabilities and connectivity of youth and women CSA/agribusiness entrepreneurs; and (iii) installing and supporting the productive use of biodigester systems and solar irrigation systems among smallholder farmers.

Component 3: Increasing institutional capacity and knowledge (total cost: USD 3,700,693; GCF cost: USD 1,950,694 in grants)

15. This component aims to strengthen the institutional capacities of government entities at the local, regional and national level, as well as communities through the Community Action Cycles, improving their adaptation planning capacities and management systems and procedures. It also aims to enhance stakeholder collaboration for knowledge-sharing for scaling up CSA, enhancing the impacts and reach of components 1 and 2.

16. Activities include (i) strengthening institutional capacity in localized climate change adaptation and mitigation planning; (ii) building the technical capacity of the Mali Climate Fund in prioritizing and targeting investment in CSA and agroforestry; and (iii) Strengthening national databases for CSA and agroforestry.

Monitoring and evaluation (total cost: USD 2,046,204; GCF cost: 1,746,204)

Project management (total cost: USD 2,007,122; GCF cost: 1,489,122)

17. Project Management budget will finance project management expenses in various fields, such as finance, procurement, internal control, supply chain and communication.

Contingency (total cost: USD 982,075; GCF cost: 982,075)

18. The contingency budget is intended to address unforeseen challenges, such as unexpected cost increases due to higher inflation or security costs. It also covers unforeseen emergency situations that could potentially disrupt project activities or require changes to scope and location, which could include shifting activities to another region due to heightened conflict, or extreme weather events that undermine project gains such as an event that leads to crop failure.

III. Assessment of performance against investment criteria

19. The project consistently scores medium to high against GCF investment criteria. A detailed assessment of the project's fit with GCF investment criteria is provided below.

3.1 Impact potential *Scale: N/A*

20. The project aims to reach 431,700 direct beneficiaries (2.4 per cent of the population), which includes 172,680 small-scale farmers, including youth and women agripreneurs, and 700,980 indirect beneficiaries. Direct beneficiaries have been selected primarily on the basis that they are the most vulnerable to climate change, through an assessment of climate risk exposure and adaptive capacity. The targeted agriculture sector employs 80 per cent of the population and is highly climate sensitive.

21. It also mitigates GHG emissions amounting to 2.4 million t CO_2 eq over its 15-year lifetime through capacity development of smallholder farmers for cultivation of agroforestry, the production and distribution of agroforestry tree saplings (Acacia Senegal and shea), planting and management of saplings and growing trees in cropland, grassland or pasture.

3.2 Paradigm shift *Scale: N/A*

22. The IAAT project represents a paradigm shift in the agriculture sector by scaling up CSA and agroforestry practices. It also facilitates building strong partnerships among private sector supply chains and incubation for small-scale farmers. Additionally, it will strengthen institutional arrangements with the Government of Mali and communities for upscaling.

23. The project leverages the Albarka project of the United States Agency for International Development, especially its existing field infrastructure and established relationship with AEDD. This will provide a platform for the project to scale up actions effectively and efficiently.

3.3 Sustainable development potential *Scale: N/A*

24. The project will contribute to several Sustainable Development Goals (SDGs): SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 8 (Decent work and economic growth), SDG 12 (Responsible consumption and production) and SDG 15 (Life on land).

25. The project expects to realize gender and social inclusion co-benefits and environmental co-benefits.

3.4 Needs of the recipient *Scale: N/A*

26. Mali faces heightened vulnerability to climate change owing to frequent climate hazards, conflict-affected situation and a limited economic base. In particular, the agriculture sector, which is a critical industry for the country for food security and livelihood, faces severe challenges such as drought and floods, impacting the overall economy. Mali requires urgent mitigation and adaptation support to address environmental, social and economic vulnerabilities.

3.5 Country ownership *Scale: N/A*

27. No-objection letters have been issued by the Government of Mali.

28. The national designated authority actively participated in major consultations with GCF and the AE, and the IAAT project has been designed in collaboration with the national designated authority, in addition to over 350 other stakeholders from across all five regions of the project, including farmers and farmer organizations, private sector businesses, local and international non-governmental organizations, and national and local government representatives.

29. This proposal is fully aligned with a high priority of the Government of Mali as outlined in the nationally determined contribution, the Growth and Poverty Reduction Strategic Framework 2012 to 2017, and other laws and strategic documents.

3.6 Efficiency and effectiveness *Scale: N/A*

30. The project incorporates an innovative Pay-As-You-Go financing mechanism for solar irrigation systems, reducing upfront costs for farmers. The alignment of repayment schedules with seasonal income cycles makes this approach particularly suitable for smallholder farmers. Complementary measures, such as financial literacy training and partnerships with microfinance institutions, enhance beneficiaries' capacity to access and manage finance effectively.

31. The adaptation cost of USD 58 per direct beneficiary (and USD 22 per direct and indirect beneficiary) and the mitigation cost of USD 3.44/t CO_2 eq, represents cost efficiency competitive within the GCF portfolio for both mitigation projects and benchmarks for adaptation in agriculture. The co-financing ratio is 1:0.3, which is below the GCF target but reflects the financial constraints of the implementation context in Mali. An analysis of the project shows that it is economically viable as a whole, resulting in an economic net present value of USD 4.4 million and an economic internal rate of return of 12 per cent without valuing the economic value of GHG emissions mitigated by the project. If the economic public value of the negative carbon balance (reduction in GHG emissions) generated from the project is taken into consideration, when using USD 30 as the social price for 1 t CO_2 eq, the economic net present value reaches USD 31.5 million and the economic internal rate of return is 26 per cent, reflecting robust economic viability and societal value.

32. The farm-level financial analysis results show high profitability for beneficiaries across key value chains, such as millet, mango and moringa, indicating that vegetables have the highest financial internal rate of return (246 per cent) and Shea butter production has the lowest at 66 per cent. This highlights that this project will enable optimal business decisions and the selection of CSA technologies to minimize costs and maximize production. The findings also indicate that there is potential for long-term sustainability and attractiveness to commercial financing post GCF support. The AE also conducted a robust sensitivity analysis, addressing critical risks such as price fluctuations, yield variability and climate impacts. This enhances confidence in the project's financial and economic resilience under various scenarios.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

33. The project is categorized as C for environmental and social impacts owing to minimal environmental and social risks. The categorization is in accordance with the GCF Environmental and Social Policy and the accreditation level of the AE. The anticipated risks and impacts mostly emanate from the small-scale installations under component 2, involving rainwater harvesting systems, solar micro-irrigation systems and household-level biodigester systems. Other activities to be supported by the project include awareness-raising and capacity-building on CSA and agroforestry techniques, institutional strengthening, and planting agroforestry trees on community and State-owned lands. The project will utilize locally available species, avoid the use of genetically modified organisms, and use minimum soil disturbance and loss technologies and practices. The project is not expected to involve physical or economic displacement of people and the activities within private lands will be voluntary in nature. The identified environmental and social risks and impacts can be readily avoided by using basic management actions as outlined in the environmental and social assessment and residual risk management plan. This plan also provides for further screening of proposed activities and management of the potential environmental and social risks and impacts. The plan has an exclusion list that would screen out activities which may result in a medium or high environmental or social risk categorization. The conflict sensitivity assessment prepared by the AE reflects that low-risk areas for conflict have been selected. Nevertheless, the conflict situation in the target areas will continuously be monitored by Save the Children International (SCI) Mali to ensure the safety of stakeholders. The project interventions, along with continuous community engagement, are seen to contribute to minimizing the impact of conflict and climate change in the communities. The project's environmental and social co-benefits include improved soil conditions from reduced reliance on fertilizers and less water usage, GHG emission reductions, and increased gender equality and women empowerment.

34. A Project Implementation Unit (PIU), including an Environmental and Social Safeguards Specialist, will be established by SCI Mali to implement safeguards measures on a day-to-day basis. The Project Steering Committee and the AE will provide oversight functions. Stakeholder consultations were undertaken with community beneficiaries (i.e. farming cooperatives and unions, community leaders, youth and women, community committees and local farmers in the targeted cercles of the IAAT project), regional and communal councils, local non-governmental organizations and civil society organizations, international civil society organizations, academia, private sector representatives, and regional and national government agencies. Working with community stakeholders and decentralized technical services plays an important role in ensuring acceptance, ownership and sustainability, and in strengthening social cohesion through the project interventions. The Stakeholder Engagement Plan will continue to be implemented during implementation and an activity-level grievance redress mechanism will be

established in consultation with the communities, in addition to the AE and the GCF Independent Redress Mechanism.

35. **Sexual exploitation, abuse and harassment safeguarding:** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The AE provided SEAH safeguarding in its submission for this funding proposal. In alignment with GCFs approach to SEAH, the AE, SCA, has zero tolerance for any abuse and exploitation committed by representatives against adults or children in the communities where the projects are implemented in. Annex 6 of the project, the Environmental and Social Assessment and Residual Risk Management Plan assessed the potential SEAH risks of this specific project, identified measures of preventing SEAH from happening and provided measures on how the project will respond and report on SEAH cases. An environmental and social action plan has been developed to address the identified potential SEAH risks. A range of capacity building activities are included in the plan ranging from, building technical capacity and reach of extension services on CSA and agroforestry techniques - training activities associated with the risk of SEAH incidences due to a lack of participants sensitization; and development of land-use mapping at the regional level which will include training activities associated with the risk of SEAH incidences. Periodic screening of environmental and social screening, including SEAH, will be done and any mid-course correction of the SEAH measures will be employed to ensure that SEAH management is robust. While the AE has indicated that the SEAH related GRM as an outcome to the capacity building initiatives they have put in place, the secretariat recommends that a robust survivor-centred and survivor centred SEAH GRM is well thought and planned in the initial stages of the project implementation. The AE may leverage on other existing and likeminded institutions in the GBV/SEAH sector and collaborate where possible.

36. **GCF Indigenous Peoples Policy and ESS 7 on Indigenous Peoples**: the AE does not foresee any involvement of Indigenous Peoples, and none of its activities involve any adverse impact on Indigenous Peoples. The AE has excluded any activity that will lead to involuntary resettlement or land acquisition (including non-physical displacement and involuntary restrictions to economic activities and land use), as well as any activity in the transhumance corridors and locations customarily occupied and used by Indigenous Peoples. If any Indigenous Peoples are identified during implementation as a result of the systematic screening that the AE will undertake throughout implementation, the AE will consider, through its grievance redress mechanism and the customary institutions available in the communities, traditional methods of resolving disputes. The AE is recommended to engage with the GCF Indigenous Peoples Advisory Group on relevant Indigenous Peoples' issues that may be encountered.

4.2 Gender policy

37. The AE provided a gender assessment and action plan with the funding proposal and therefore, complies with the requirements of the GCF Updated Gender Policy. The gender assessment highlights significant barriers to gender equality in Mali and the agriculture and agroforestry sector. There are various constraints faced by women in pursuing their practical and strategic gender needs and interests. Among others, the AE found that some of these challenges have an impact in realizing gender equality; social roles that are attributed to each sex; illiteracy levels particularly in women; women's lack of access to sources of information on climate change; low levels of awareness among women's associations on climate change adaptation ; inadequate knowledge on sustainable management and use of natural resources ; low representation of women in conflict prevention and management frameworks and mechanisms.

38. The Gender Action Plan (GAP) outlines measures to address these challenges by integrating gender equality and women's empowerment into the project with a GAP goal of mobilization and utilization of project resources effectively and efficiently to tackle the current normative gender barriers, and biases, and contribute towards improved gender equality. In order to achieve this goal and the impact area of enhancing climate resilience of both women and men in farming households, the following output areas will be key in this project (i) Improved technical capacity and inclusiveness of extension services in climate-smart agricultural and agroforestry production; (ii) Increased use of climate-resilient practices in the production of CSA crops, livestock, and agroforestry by smallholder farmers; (iii) Increased land area under agroforestry; (iv) Smallholder farming communities, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness; and (v) Increased institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks. The secretariat recommends to the AE to have a SMART GAP with detailed indicators and targets. It further recommends that grievance mechanisms should be access to women and any other marginalized groups identified in the gender assessment.

4.3 Risks

39. **Overall project assessment is medium risk.**

40. **Project-specific execution risks (medium risk). Political risk:** Mali is rated Caa2 with a stable outlook by Moody's. Based on Moody's latest report, the country is highly susceptible to event risk which may have spillovers on public service delivery, business activity and project continuity, resulting in reduced access to project locations, reduced pools of extension services, and financial institutions and private companies to work with, with delays in implementation and reduced impact. The risk of conflict will be mitigated through the continuous monitoring by the AE and implementation of adaptative measures agreed with GCF in the term sheet (contingency budget, reallocation of funds between regions in the case of exposure).

41. **Operational risks**: some beneficiaries revert to pre-project climate-vulnerable farming practices and/or have insufficient access to finance/disposable income for the adoption of solar irrigation, thus resulting in reduced sustainability and paradigm shift. The AE identified the risks and designed out mitigation measures in the funding proposal to an acceptable level of residual risk.

42. **GCF portfolio concentration risk (low risk)**: in the case of approval, the impact of this proposal on the GCF concentration risk remains within the monitoring thresholds of the risk appetite statement in terms of results areas, single proposal or AE concentration.

43. **Compliance risk (low risk)**: many of the project activities that focus on the implementation of IAAT across the target five regions in Mali through interventions, including technical assistance/capacity-building and enabling the development of CSA and agroforestry value chains, pose relatively low risks of money-laundering and financing terrorism. However, infrastructure activities, such as the installation of solar irrigation and biodigester systems, present more elevated money-laundering and financing terrorism risks. Preventive measures under the project include the following:

(a) The AE has assessed the money-laundering and financing terrorism risk as low on the basis of factors including the establishment of a financial control mechanism specific to the IAAT project, which includes financial transaction verification procedures and annual audits of implementing partners, as well as other internal controls such as monitoring and verification measures, including for procurement and payment processing. Moreover, the AE will also monitor and evaluate anti-money-laundering and countering the financing of terrorism control effectiveness on an annual basis;

(b) SCI Mali and Save the Children Federation, Inc. will co-execute as executing entities in collaboration with an implementing entity, Mali's AEDD. The capacity assessment of AEDD by the AE determined that AEDD anti-fraud/bribery/mismanagement and anti-money-laundering and countering the financing of terrorism management meets the requisite criteria; and

(c) On the basis of the relatively low inherent money-laundering and financing terrorism and other prohibited practices risk posed by most of the proposed project activities and the various controls and mitigation measures to be employed by the AE to mitigate the elevated risks associated with certain infrastructure activities, the overall compliance risk is deemed to be low.

4.4 Fiduciary

44. Save the Children Australia, as the AE, will be responsible for project supervision, oversight and ensuring compliance with GCF standards. SCI Mali and Save the Children Federation, Inc. will act as executing entities, jointly managing implementation. AEDD will serve as an implementing partner, coordinating with local non-governmental organizations and solar and biodigester suppliers. AEDD will not manage or receive GCF funds directly.

45. The project will establish a PIU within SCI Mali, dedicated to project coordination, fund disbursement and reporting. The PIU is led by a Project Manager and includes specialized financial and procurement staff to ensure adherence to fiduciary standards. The PIU will report to a Project Steering Committee chaired by AEDD and SCI Mali. The Project Steering Committee will provide strategic guidance, review annual work plans and oversee project progress to enhance accountability and coordination with government priorities.

46. Save the Children Australia will ensure that SCI Mali manages GCF funds in compliance with SCI global financial procedures and GCF standards. All project funds will be held in segregated accounts to facilitate precise tracking and reporting of GCF resources. SCI Mali will utilize its financial management systems to provide robust controls, including regular budget reviews, financial reconciliations and internal audits to ensure transparent fund utilization.

47. The PIU will oversee financial reporting, with periodic financial updates provided to the AE for review and submission to GCF according to the funded activity agreement and accreditation master agreement. Annual independent external audits will be conducted in line with international standards to assess compliance and identify any discrepancies, reinforcing the project's financial integrity.

48. Procurement activities will adhere to the SCI Global Procurement Policy, which mandates competitive bidding, multi-level approvals and transparent documentation to prevent conflicts of interest and ensure value for money. For local procurement, SCI Mali will follow Malian regulations where compatible with GCF guidelines, and all procurement processes will be monitored by the PIU to ensure alignment with both SCI and GCF procurement standards.

49. To maintain transparency and compliance, SCI Mali will conduct procurement training for staff and partners, covering anti-fraud, anti-corruption and procurement best practices. Independent reviews of procurement processes and an annual audit of procurement records will further support accountability and mitigate risks associated with procurement activities.

4.5 Results monitoring and reporting

50. The theory of change of the proposed project showcases the logical pathways through which the project's outputs will contribute to the identified climate impact outcomes and co-benefits. The project has outlined two adaptation outcomes that contribute to climate-resilient agriculture and livelihood resilience, while the mitigation outcome will be reduced GHG emissions from the agricultural system. A cross-cutting outcome contributing to enhanced institutional capacity and application of best practices of mitigation and adaptation practices is also identified. The paradigm-shifting pathways of the project are described in the goal statement, which is to improve the food and water security of vulnerable farmers.

51. The project's logical framework follows the Integrated Results Management Framework, which proposes core and supplementary indicators to measure the project's outcomes on mitigation and adaptation. The enabling environment and paradigm shift potential indicators selected by the AE are consistent with the requirements and allow for comprehensive results monitoring. The project-specific indicators deal with context-specific results from the output level, ensuring that project performance is adequately measured.

52. The monitoring and evaluation plan submitted by the AE provides information on how the project is planning and budgeting its monitoring and evaluation activities. Monitoring project outputs and outcomes will be conducted through periodic beneficiary surveys, allowing the AE to report on results and implementation performance to GCF annually. A baseline survey will be conducted to validate the proposed baseline and apply any necessary target calibrations of the logical framework indicators. The AE has proposed to evaluate the project through performance and impact evaluation. The evaluation plan follows the GCF evaluation policy, in which an interim and a final evaluation will be conducted by independent evaluators and managed by the AE.

4.6 Legal assessment

53. The Accreditation Master Agreement was signed with the Accredited Entity on 20 December 2019 and it became effective on 20 May 2020.

54. The Accredited Entity has provided a legal opinion confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

55. The proposed project will be implemented in the Republic of Mali, country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

56. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities for GCF and its personnel.

57. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP256

Proposal name:	Intensification of Agriculture and Agroforestry Techniques (IAAT) for Climate Resilient Food and Nutrition Security: Tombouctou, Gao, Mopti, Koulikoro and Segou regions of Mali
Accredited entity:	Save the Children Australia
Country:	Mali
Project size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. The Intensification of Agriculture and Agroforestry Techniques (IAAT) project is a five-year initiative targeting five regions in Mali: Tombouctou, Gao, Mopti, Koulikoro and Segou. Its overarching aim is to strengthen climate resilience in agriculture and food systems while reducing greenhouse gas (GHG) emissions. This cross-cutting, small-scale project pursues a dual objective: (a) to enhance the resilience of Mali's agriculture against climate risks, including droughts, floods and extreme heat; and (b) to mitigate climate change by reducing GHG emissions from agriculture and agroforestry systems.

2. With a total lifespan of 15 years, the proposed project focuses on four core outcomes: (a) increased climate-resilient agricultural crop and food production in the target regions in Mali; (b) improved and inclusive access to finance and markets for vulnerable communities to enhance sustainable livelihoods; (c) adaptation and mitigation considerations are embedded in institutional agriculture and agroforestry planning; and (d) reduced GHG emissions from agricultural systems, including agroforestry practices.

3. To achieve these outcomes, the proposed project will deliver the following eight outputs: 1.1) improved technical capacities and inclusivity of extension services in climate-smart agriculture (CSA) and agroforestry production; 1.2) increased use of climate-resilient practices in the production of CSA crops, livestock and agroforestry by smallholder farmers; 1.3) increased land area under agroforestry; 2.1) CSA and agroforestry value chains are more connected and reach more smallholder farmers; 2.2) smallholder farming communities, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness; 2.3) increased adoption of CSA technologies by smallholder farmers; 3.1) increased institutional capacity in climate change adaptation and mitigation planning and best practices to address climate-related risks in agriculture; and 3.2) enhanced knowledge-sharing and coordination of best practices in CSA and agroforestry across stakeholders.

4. The environmental and social category of the project is C, reflecting minimal potential environmental and social risks.

5. The total cost of the project is USD 43.7 million, of which USD 33.7 million is requested from GCF as a grant. The remaining USD 10 million will be co-financed by national and international partners (USD 5 million in-kind contribution from the Government of Mali and

USD 5 million from the Albarka project funded by the United States Agency for International Development).

6. Implementation arrangements involve Save the Children Australia as accredited entity; Save the Children International and Save the Children Federation, Inc. as executing entities; and the Environment and Sustainable Development Agency, a legal entity under the Ministry of Environment and Sanitation and the NDA, as implementing entity, coordinating with national and regional bodies, local councils and private sector stakeholders. Local coordination committees will oversee planning, activity approvals and monitoring.

7. This initiative is critical, as agriculture forms the backbone of Mali's economy, employing 80 per cent of the population and contributing 36 per cent of gross domestic product. By focusing on CSA and agroforestry, the IAAT project aligns with Mali's nationally determined contributions for climate action, strengthens food security and reduces poverty under changing climate conditions.

1.2 Impact potential *Scale: Medium*

1.2.1. Adaptation impact potential

8. The IAAT project targets 431,700 direct beneficiaries, including 172,680 smallholder farmers, with the emphasis on vulnerable groups such as women and youth. Additionally, 700,980 indirect beneficiaries representing approximately 3.8 per cent of Mali's total population will benefit from strengthened public and private extension services. This comprehensive reach addresses the most climate-vulnerable populations and significantly enhances equity in resilience-building efforts. However, maintaining inclusive participation of marginalized groups, particularly women and youth, remains a key challenge in fully realizing the expected adaptation benefits.

9. To enhance the inclusive participation and empowerment of such marginalized groups, the project should build on its existing mechanisms by establishing a dedicated gender and youth inclusion task force within the local coordination committees. This task force would leverage the project's targeted measures, such as gender-sensitive training programmes, Village Savings and Loan Associations (VSLAs) and locally led adaptation (LLA) plans, to identify and address barriers to participation while ensuring equitable representation in decision-making processes. Enhancing these mechanisms with tailored outreach strategies or youth- and gender-sensitive incentives would further reduce barriers for women and youth.

10. The proposed project aims to avoid the lock-in of vulnerable infrastructure by promoting CSA and agroforestry systems that prioritize sustainable soil and water management practices. These interventions reduce reliance on conventional, unsustainable infrastructure such as traditional irrigation methods. A potential concern is that these systems could remain underutilized if institutional and technical support diminishes post-implementation. The project's strategy of integrating LLA plans and fostering private sector partnerships creates a foundation for sustainable resource allocation post-project. To strengthen this, formalizing long-term commitments with financial and institutional partners would ensure continued support and engagement.

11. While the project provided a list of prioritized CSA practices selected for inclusion in the updated extension services curriculum, it offered few details on how their resilience to future climate conditions will be addressed. Moreover, the CSA practices for livestock are limited to feed production. Given the drought conditions in Mali, it would be important to consider CSA practices that also support better livestock management during prolonged periods of drought.

12. The project reduces vulnerability by enhancing adaptive capacity through CSA, agroforestry and renewable energy solutions such as solar pumps and biodigesters. These interventions stabilize livelihoods and are likely to provide resilience against recurrent climate risks such as droughts and floods. However, long-term adaptive benefits will require ongoing training, capacity-building and monitoring to sustain progress.

13. Along with the proposed CSA and agroforestry practices aiming to enhance resilience, the project must consider the integration of transhumance corridors in the target areas, particularly in regions like Mopti and Gao. These regions often experience tensions between farmers and pastoralists over land and water resources. Agroforestry systems and CSA practices that expand agricultural land use could inadvertently encroach on established transhumance routes, exacerbating such conflicts. To ensure adaptability and inclusiveness, the project should incorporate multi-use landscape planning approaches, balancing the needs of farmers and pastoralists. Specific measures, such as delineating and preserving transhumance corridors within agroforestry zones, should be included in the project's planning framework.

14. Institutional and regulatory systems for climate-responsive planning are strengthened through the integration of LLA plans into local governance frameworks. These plans, developed with stakeholder input, aim to embed adaptation considerations in planning processes.

15. The project increases the generation and use of climate information through training and agrometeorological advisories, enabling farmers to make informed decisions. However, gaps in access to reliable climate data and stakeholder-tailored information could limit the effectiveness of these advisories in some regions, requiring ongoing investment in reliable climate data and adequate information systems. The project lacks a clear strategy on this issue. To address this, it should foster long-term commitments from national and international stakeholders, which can be integrated into a long-term commitment strategy.

1.2.2. Mitigation impact potential

16. The project is expected to achieve a total reduction in GHG emissions of 2,446,543 tonnes of carbon dioxide equivalent over 15 years through the adoption of CSA practices, agroforestry systems and renewable energy technologies (solar pumps and biodigesters).

17. The project's projected GHG emission reductions are based on robust methodologies and tools, such as EX-ACT (Environmental eXternalities ACcounting Tool) of the Food and Agriculture Organization of the United Nations. However, these estimates rely on linear adoption models, assuming consistent and widespread uptake of CSA, agroforestry practices and climate technologies. Such assumption may overestimate actual reductions, as adoption rates could be slower owing to economic or climatic barriers. To address this, the project should implement a dynamic monitoring system that tracks real-time adoption rates and validates emission reductions. This system will enhance accuracy and provide insights that could inform any necessary adjustments to interventions, ensuring realistic and verifiable mitigation outcomes.

18. By promoting solar irrigation systems and biodigesters, the proposed project avoids the lock-in of high-emission fossil fuel based infrastructure. These renewable energy solutions offer sustainable alternatives for agricultural activities.

19. Agroforestry initiatives under the project will cover over 91,000 hectares, significantly improving land management and enhancing carbon sequestration. These interventions are expected to contribute substantially to reducing land-use emissions. A key consideration is ensuring land tenure clarity to sustain the mitigation benefits of these activities. To ensure long-term sustainability, the project should formalize agreements with local communities and authorities through land-use management plans, supported by ongoing capacity-building initiatives. Additionally, establishing community-led monitoring committees to oversee

rehabilitated lands post-project would help to maintain engagement and prevent disputes or underutilization.

20. The project also fosters low-carbon value chains by strengthening market linkages for CSA-compatible crops and promoting low-emission agricultural practices. Robust market infrastructure and connectivity are critical for achieving this goal, but challenges in establishing and maintaining these structures must be addressed to ensure long-term success.

21. The independent Technical Advisory Panel (iTAP) considers the proposed project's impact potential to be medium.

1.3 Paradigm shift potential *Scale: Medium to high*

22. The IAAT project introduces innovative CSA and agroforestry practices, including the deployment of solar pumps and biodigesters, to enhance agricultural productivity and reduce GHG emissions in Mali. These solutions target smallholder farmers, emphasizing women and youth, creating opportunities for new market segments and business models. By leveraging private sector involvement to scale these technologies, the project reduces adoption costs and risks, thereby shifting market incentives towards sustainable practices. However, sustaining private sector engagement beyond the project period may require additional efforts to institutionalize these market-driven incentives.

23. To ensure sustained financial support and market-driven incentives post-project, the IAAT initiative should operationalize the long-term commitment strategy mentioned in paragraph 15 above. This includes formalizing partnerships with private sector actors, financial institutions and government agencies through memorandums of understanding and embedding financial sustainability measures, such as a co-financing scheme linked to VSLAs, in national policies and frameworks.

24. The project's theory of change provides a clear pathway for scaling up its impact without proportionally increasing implementation costs. Through capacity-building, community-led approaches and LLA plans, the project creates replicable models that can be adopted in other sectors and regions with similar vulnerabilities. While the framework is robust, ensuring the adaptability of these models to diverse contexts will require a focus on knowledge-sharing and capacity-building frameworks that prioritize inclusivity and local context.

25. Therefore, to further tailor the project's knowledge-sharing and capacity-building mechanisms to diverse local contexts, the project should incorporate localized knowledge hubs within target regions. These hubs would focus on collecting and sharing region-specific data, traditional practices, and challenges, while adapting training modules to reflect the unique socioeconomic and environmental conditions of each area. Engaging local leaders and community organizations in co-designing these modules would ensure cultural relevance and increase stakeholder buy-in.

26. The security situation in Mali, particularly in the target regions of Tombouctou, Gao, Mopti, Koulikoro and Segou, remains relatively precarious. These conditions pose significant risks to project implementation, including disruptions to planned activities, delays in execution and potential safety threats to project staff and beneficiaries. The current risk assessment framework seems to underestimate the probability and impact of such security challenges and may need to be updated. Furthermore, to mitigate these risks, the project, which is critically important for the country, should consider context-specific strategies, such as establishing secure project sites, strengthening coordination with local authorities, and integrating adaptive measures to sustain operations during periods of heightened insecurity. Additionally,

contingency plans should be developed to ensure that project outcomes are safeguarded even under adverse security conditions.

27. A comprehensive monitoring and evaluation (M&E) framework ensures that lessons learned from the IAAT project are documented and shared. The use of baseline, midline and endline surveys, coupled with real-time data collection, provides a robust system for assessing progress and identifying areas for improvement. These insights will not only inform future iterations of the project but also contribute to the global body of knowledge on CSA practices. However, ensuring sustained funding for the M&E system post-project remains a critical challenge.

28. The project outlines potential strategies to address the challenge of sustaining the M&E system post-project. However, these strategies lack sufficient detail and firm commitments, leaving uncertainty about their ability to secure long-term funding and institutional support. To address this gap, the sustainability of the M&E system should be explicitly integrated into the project's long-term commitment strategy. This integration should include detailed financial mechanisms, institutional road maps and capacity-building measures to ensure the system's continued functionality and impact post-project.

29. The project emphasizes creating an enabling environment for long-term sustainability through partnerships with local governments, private sector actors and community groups. By institutionalizing CSA and agroforestry practices within national frameworks, the project promotes mainstreaming climate-resilient development in Mali. However, as stated above, the absence of explicit provisions for formalizing long-term financial agreements could limit the continuity of key project outcomes, particularly in the maintenance of climate information systems and agroforestry initiatives.

30. By facilitating the development of markets for CSA products and technologies, the project generates new business activities at the local and national level. The integration of VSLAs and private sector partnerships enhances access to finance and reduces barriers for market participants. These activities not only stimulate economic growth but also shift incentives towards low-carbon and climate-resilient solutions. The challenge lies in ensuring that these markets remain competitive and inclusive, particularly for marginalized groups, after the project concludes.

31. The IAAT project addresses systemic barriers to low-carbon and climate-resilient development by aligning its activities with Mali's national climate policies and strategies. It advances regulatory frameworks by embedding CSA practices in local governance through LLA plans, which integrate climate considerations into decision-making processes. These efforts catalyse broader investments in low-emission and climate-resilient development. However, ensuring sustained government and institutional commitment will be crucial to fully realizing these systemic changes.

32. The iTAP assesses the paradigm shift potential of the proposed project as medium to high.

1.4 Sustainable development potential *Scale: Medium to high*

33. The project demonstrates significant potential for positive environmental externalities through its adoption of CSA and agroforestry practices. By incorporating technologies such as solar pumps and biodigesters, the project reduces reliance on fossil fuels, which also improves air quality. Additionally, agroforestry practices enhance soil health and biodiversity by integrating tree cover into agricultural landscapes, which provides habitats for various species. These activities contribute to global climate goals and align with Sustainable Development Goals (SDGs) 13 (Climate action), 15 (Life on land) and 12 (Responsible

consumption and production). However, sustaining these environmental benefits post-project will require consistent adoption by farmers and adequate technical and financial support from local and national institutions.

34. The project partially addresses the concern regarding sustaining environmental benefits post-project. While it builds a strong foundation through training, policy alignment and initial financial mechanisms, the lack of detailed and formalized long-term strategies for institutional and financial support weakens its sustainability.

35. The project is poised to deliver significant social co-benefits by improving the health and well-being of both women and men. It achieves this by promoting access to clean energy solutions, such as solar irrigation systems, and reducing exposure to harmful agrochemicals. Furthermore, capacity-building programmes empower communities to adopt safer and more resilient practices, contributing to healthier living conditions. Gender-responsive strategies embedded within the project aim to address social inequities, fostering inclusion and participation. These outcomes align with SDGs 3 (Good health and well-being) and 5 (Gender equality). However, a challenge remains in ensuring equitable access to these benefits for marginalized groups, particularly after the project ends.

36. From an economic perspective, the project has the potential to transform local economies by creating jobs along the CSA and agroforestry value chains. Employment opportunities in the installation, maintenance and supply chains for technologies like solar pumps and biodigesters are expected to enhance economic capacity. Strengthened local industries and markets for CSA products will improve income generation and financial stability for smallholder farmers, especially for women and youth. Additionally, the VSLAs expand financial inclusion in underserved rural areas. These efforts contribute to SDGs 8 (Decent work and economic growth) and 1 (No poverty). However, maintaining these economic gains post-project will depend on private sector engagement and the successful integration of financial mechanisms into national frameworks.

37. Furthermore, to maximize the economic and social benefits of renewable energy solutions like biodigesters while minimizing potential conflicts, the project should promote clear crop residue allocation guidelines. These guidelines should ensure that crop residues are appropriately distributed between biogas production and livestock feed to meet the needs of both sectors. Capacity-building programmes should include training for farmers on optimizing residue use and implementing sustainable practices.

38. The project's private sector engagement strategy has promising elements but lacks the robustness required to ensure sustained involvement post-project. To strengthen the strategy, the project should formalize private sector partnerships through binding agreements, provide clearer financial incentives for long-term engagement and outline a more comprehensive plan for scaling these efforts within national frameworks.

39. The proposed project takes a proactive approach to gender equality and social inclusion, addressing gender disparities in climate vulnerability and access to resources. By improving women's access to training, financial services and leadership opportunities in CSA practices, the project ensures their active participation and reduces gender gaps. Community-level participation mechanisms include women as key stakeholders, enhancing inclusivity and cultural relevance. These actions align with SDGs 5 (Gender equality) and 10 (Reduced inequalities). However, ensuring sustained institutional and cultural support for gender-focused interventions remains a challenge after the project life cycle has ended.

40. The iTAP rates the sustainable development potential of the proposed project as medium to high.

1.5 Needs of the recipient *Scale: High*

41. As a least developed country, Mali faces severe exposure to climate risks, including prolonged droughts, erratic rainfall and rising temperatures, which are exacerbating vulnerabilities in key sectors like agriculture. The northern regions, such as Gao and Tombouctou, are particularly impacted by these risks, with frequent droughts and limited access to water resources undermining livelihoods. Agriculture, which employs 80 per cent of the workforce and contributes 36 per cent to gross domestic product, is heavily reliant on rain-fed systems, making it highly sensitive to these climate impacts. The project addresses these vulnerabilities by introducing CSA and agroforestry practices to improve resilience.

42. The project targets a significant proportion of Mali's population, including 431,700 direct beneficiaries, with a focus on smallholder farmers, women and youth. These groups are disproportionately affected by climate risks owing to their dependence on climate-sensitive livelihoods and limited access to resources. The rural population, where poverty is most concentrated, faces additional barriers in accessing adaptive technologies and financial services. By addressing agricultural and water resources as well as critical social and economic assets, the proposed project will contribute to preserving livelihoods and national food security. However, challenges remain in ensuring that benefits are equitably distributed, especially in remote areas. Strengthening the M&E framework to track equitable benefit distribution across all target groups is recommended.

43. Mali's low-income status, high unemployment and underdeveloped financial infrastructure present significant barriers to financing climate-resilient development. Smallholder farmers, who make up most of the target population, often lack access to credit and adaptive technologies. The project seeks to overcome these barriers by leveraging co-financing mechanisms and strengthening local institutions, such as extension services and community organizations. However, reliance on project-specific funding and limited integration into national financial systems may hinder the sustainability of these interventions. It is recommended that the project develop a clear road map to integrate its outcomes into Mali's institutional frameworks and national budgetary processes to secure long-term financial support. This can be part of the long-term commitment strategy mentioned in paragraph 15 above.

44. The iTAP assesses the needs of the recipients of the proposed project as high.

1.6 Country ownership *Scale: High*

45. The project aligns closely with Mali's national climate priorities, as reflected in its nationally determined contributions and its national adaptation programme of action. The national adaptation programme of action identifies agriculture and water management as critical sectors for enhancing climate resilience, and the project's activities, such as the promotion of CSA and agroforestry practices, directly support these priorities. Additionally, the integration of technology-based solutions like solar pumps and biodigesters aligns with Mali's technology needs assessment, which highlights sustainable energy and agriculture technologies.

46. The project leverages Mali's existing enabling policy and institutional frameworks, involving key institutions like the Ministry of Environment and other sectoral ministries. By integrating CSA practices into LLA plans, the project ensures alignment with decentralized governance structures, which strengthens its institutional foundation. However, the absence of explicit policy commitments to institutionalize CSA and agroforestry practices within national frameworks presents a potential gap in sustainability. Formalizing agreements to integrate these practices into national legislation would enhance the enabling environment created by the project and ensure its continuity beyond the project duration.

47. The accredited entity and the implementing entities have a track record of implementing climate-resilient initiatives in Mali and similar contexts. While the proponents'

operational expertise is evident, the project could better ensure long-term ownership by focusing on capacity-building initiatives for local governments and institutions. Strengthening local implementation capacity would enhance the sustainability of project outcomes and reduce reliance on external actors post-project.

48. The project has been developed through a participatory process that included intensive consultations with government agencies, local communities, private sector actors and civil society organizations. Special attention has been paid to gender considerations, with the project emphasizing women's empowerment through targeted capacity-building and financial inclusion mechanisms. However, maintaining active stakeholder engagement mechanisms post-project remains a challenge. To address this, the project should establish institutionalized platforms for ongoing stakeholder collaboration and input, ensuring continued support and inclusivity beyond its life cycle.

49. The iTAP assesses the country ownership of the proposed project as high.

1.7 Efficiency and effectiveness *Scale: Medium*

50. The proposed project employs a grant-based financial structure, which is appropriate given the significant upfront costs and long timelines needed to achieve results in CSA and agroforestry practices. With a total investment of USD 43.7 million phased over five years, the project's structure supports both immediate implementation and, at some instances, long-term scalability. Mechanisms such as VSLAs and subsidies for solar irrigation systems ensure accessibility for smallholder farmers, with farmers expected to contribute 15 per cent of costs for larger solar irrigation systems during deployment. However, the economic and financial analysis (EFA) does not explicitly account for the potential impacts of extreme climate events, such as droughts or floods, which could affect the project's financial soundness and the farmers' ability to co-finance. This was acknowledged during a call with the accredited entity, which also confirmed that the proposal primarily considered gradual changes in climate variables. While it is easier to adopt an EFA approach based on gradual changes, there is nonetheless a risk of underestimating the significant and devastating impacts of extreme climate events or slow onset events, which are also key features of climate change and variability assessments in the project area. Efforts in this regard for climate financing should be encouraged.

51. The project demonstrates strong cost-effectiveness through its competitive emissions mitigation costs and co-benefits. While these results highlight the project's viability, the exclusion of extreme climate events in the EFA as stated in paragraph 50 above poses a critical limitation. One single slow onset or extreme event such as prolonged droughts or flooding could disrupt production systems and significantly affect projected returns. Regular updates to the EFA to include the potential financial impacts of extreme climate events would enhance its accuracy and robustness.

52. The project aims to leverage additional finance through partnerships with microfinance organizations, agribusiness investments and farmer contributions. Co-financing mechanisms, such as VSLAs, are designed to reduce financial barriers for women and youth while encouraging private sector engagement. However, the EFA does not assess how extreme climate events might influence co-financing contributions or the willingness of private sector actors to participate. Clear risk-sharing agreements and contingency measures could mitigate these concerns and maintain stakeholder confidence in the project's financial sustainability.

53. Best practices and innovative approaches are central to the project's design. The adoption of globally recognized CSA practices, such as compost application, contour farming and solar irrigation systems, reflects alignment with industry standards. Additionally, the use of payment schemes for scaling solar irrigation and digital platforms for market access highlights the project's innovative strategies for overcoming systemic barriers. However, the absence of a

strategy to adapt these practices to extreme climate conditions may limit their effectiveness during periods of severe weather stress. Incorporating adaptive measures and regularly updating practices based on local climate trends would ensure that the project remains resilient under extreme conditions.

54. The project shows strong financial viability for farmers, with internal rates of return ranging from 66 per cent for shea butter to 246 per cent for vegetables across value chains. Incremental cash flow improvements further incentivize adoption of CSA and agroforestry practices. However, sensitivity analyses do not sufficiently consider how extreme climate events (even a single event) could affect yields, market prices or farmer income. This oversight presents a risk to the economic soundness of the project, especially for vulnerable farmers with limited capacity to absorb shocks. Integrating these risks into financial models would better protect farmers and enhance the project's resilience to climate variability.

55. The iTAP considers the efficiency and effectiveness of the proposed project to be medium.

II. Overall remarks from the independent Technical Advisory Panel

56. The IAAT project is aligned with Mali's climate and development priorities, addressing key challenges in agriculture and food security. Its dual focus on enhancing climate resilience through CSA and agroforestry, and mitigating GHG emissions through renewable energy adoption, positions it as a transformative intervention. The theory of change and institutional arrangements are comprehensive, with clear pathways for achieving systemic change. However, gaps in the explicit and inclusive assessment of extreme climate risks, security risks in the country and challenges related to long-term financial and institutional sustainability, as well as land-use conflicts, could hinder its full potential. Strengthening these areas through a thorough risk evaluation will enhance the project's effectiveness, sustainability and replicability, ensuring greater impact for Mali's vulnerable populations, particularly women and youth.

57. Based on the analyses presented above, the iTAP recommends that the Board approve this funding proposal with the following conditions precedent to the second disbursement under the funded activity agreement:

58. Delivery to GCF by the accredited entity, in a form and substance satisfactory to the GCF Secretariat:

(a) A comprehensive and detailed strategy to achieve institutional and financial sustainability of the project, including mechanisms for long-term viability and resource management. This involves formalizing agreements and partnerships with public and private stakeholders to secure their commitment, support and collaboration, ensuring continued investment, resource allocation and active engagement even after the implementation period has concluded; and

(b) A strategy to preserve transhumance corridors within agroforestry zones to ensure sustainability and inclusiveness of the project.

59. Furthermore, the iTAP recommends that, during the first two years of the project, the accredited entity undertake the following actions:

(a) Establish clear mechanisms for tailoring CSA and agroforestry practices to current and future local conditions, leveraging localized knowledge hubs and stakeholder feedback;

(b) Institutionalize platforms for continuous stakeholder consultation, with a particular focus on ensuring inclusivity for marginalized groups and gender-responsive measures;

(c) Establish a dedicated gender and youth inclusion task force within the local coordination committees;

(d) Update the risk assessment and management strategy for existing and any potential emerging risks;

(e) Assess the potential additional financial and economic impacts of at least one specific climate extreme event in the target regions and share the findings with project partners for any additional measures; and

(f) Consider flexible and appropriate monitoring and feedback approaches into the project's M&E framework to track adoption trends and make timely adjustments to interventions.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP256)

Proposal name:	Intensification of Agriculture and Agroforestry Techniques (IAAT) for Climate Resilient Food and Nutrition Security: Tombouctou, Gao, Mopti, Koulikoro and Segou regions of Mali
Accredited entity:	Save the Children Australia
Country:	Mali
Project size:	Small

Impact potential
Thank you for the feedback and the medium rating.
Paradigm shift potential
Thank you for the feedback and the medium to high rating.
Sustainable development potential
Thank you for the feedback and the medium to high rating.
Needs of the recipient
Thank you for the feedback and the high rating.
Country ownership
Thank you for the feedback and the high rating.
Efficiency and effectiveness
Thank you for the feedback and the medium rating.

Overall remarks from the independent Technical Advisory Panel:

Thank you iTAP for the endorsement, overall feedback and the specific conditions and recommendations put forward to improve our response during implementation, to strengthen the GCF investment criteria. We have discussed as a team and agree to deeper anchoring of engagement and commitment for longer term sustainability and paradigm shifting opportunities. Although there are some resource implications on the approved budget, we feel confident that by the second disbursement, we should be well-positioned to initiate and take up the:

A) **two conditions** on strategic institutional and financial sustainability supported by formalized agreements; and strategic preservation of transhumance corridors; and

B) **six recommendations** around: CSA/agroforestry mechanisms for local conditions and knowledge hubs; platform for continued stakeholder engagement; gender and youth peer groups within local committees; emerging risks assessment; EFA on extreme event in the target geography; adaptive M&E.

Thank you again, very much for the endorsement. We are very excited to get to work on IAAT in Mali in 2025.

Intensification of Agriculture and Agroforestry Technologies (IAAT) for Climate Resilient Food and Nutrition Security: Tombouctou, Gao, Mopti, Koulikoro and Segou Regions of Mali

Annex 8: Gender Assessment and Action Plan

Accredited Entity: Save the Children Australia

Version: B.41– 2025/01/21

Contents

Acronyms

AEDD	Agency for Environment and Sustainable Development
CAEB	Advice and Support for Basic Education (CAEB)
CAFO	Coordination of Women's Associations and NGOs
CCD	Convention to Combat Desertification
CCA	Climate Chagne Adapation
CCM	Climate Change Mititation
CNT	National Transition Council (Conseil National de la Transition)
UNFCCC	United Nations Framework Convention on Climate Change
CDDR CCS	Disarmament, Demobilisation and Reintegration Commission Local safety committees
ECOWAS	Economic Community of West African States
CMAE	Conference of African Environment Ministers
CoFo	Property Commission
CREEDD	Strategic Framework for Economic Recovery and Sustainable Development
CRSS	Security Sector Reform Commission
CSA	Agreement Monitoring Committee (Comité de suivi de l'accord)
CVJR	Truth, Justice and Reconciliation Commission
DRPSIAP	Direction Régionale de la Planification, de la Statistique, de l'Informatique et Regional Planning
EDSM V ERAR	Mali Demographic and Health Survey V Regional Reconciliation Support Team
FAFE	Fond d'Appui à l'Autonomisation des Femmes et l'Épanouissement des Filles (Support fund for the empowerment of women and the development of girls)
FAO	Food and Agriculture Organization of the United Nations
FENAFER	National Federation of Rural Women
GBV	Gender Based Violence
GRB	Gender Responsive Budgeting
GBVIMS	Gender Based Violence Information Management System
GII	Gender Inequality Index
GDI	Gender Development Index
MPI	Multidimensional Poverty Index
IAAT	Intensification of Agricultural and Agroforestry Techniques
IGAs	Income Generating Activities
LOA, MARN	Agricultural Orientation Law, National Reconciliation Support Mission
MGF	Female Genital Mutilation
MPFEF	Ministry for the Promotion of Women, Children and the Family
MARN	National Reconciliation and Social Cohesion Process

NEPAD	New Partnership for Africa's Development
NGO	Non-Governmental Organization (reconciliation and social cohesion process at national)
ODD	Sustainable Development Objectives
OP	Political orientation
OS	Strategic Option
STP/CIGQE	Permanent Secretariat of the Institutional Committee for the Management of Environmental Issues
PBSG	Gender Responsive Budgeting and Planning
PFA	Agricultural Land Policy
PNG	National Gender Policy
PRAAPS	Regional Support Project for Pastoralism in the Sahel
PRODAFF	Ten-Year Plan for the Empowerment of Women, Children and the Family
SCI	Save the Children International
RTS	Flexible hose ramp
SDGs	Sustainable Development Goals
TDR	Terms of reference
UEMOA	West African Economic and Monetary Union

1. Introduction

Mali is currently divided into 8 Regions, plus one capital (Bamako). The Regions are in turn subdivided into 49 Cercles and 703 Communes. Mali has enormous potential and abundant resources, but it faces major security, institutional, governance, and economic challenges, which are hampering progress toward peace and sustainable development. Development and security in the country are also threatened by the effects of climate change, including drivers such as rising temperatures and erratic rainfall, resulting in hazards such as prolonged droughts, floods, heat waves, and pests and diseases. This has led to other impacts such as soil erosion, groundwater salinity, crop damage and livestock illnesses etc. These impacts greatly affect vulnerable people which is exacerbâted by the limited adaptability of human response to climate change.

Regional and local researchers, as well as the international community, have agreed for more than a decade that the impacts of climate change are affecting the natural resources essential to livelihoods and food security in the Sahel Region, and particularly in Mali, where these problems are exacerbating the risks associated with forced migration and insecurity in the country. While the links between inter-community conflicts over access to and control of natural resources, the rise of violent extremism and the country's governance challenges are complex, the resurgence of conflicts between farmers and pastoralists (and their exploitation by various actors) illustrates the security risks exacerbated by climate change and constitutes one of the country's most pressing challenges[1] .

A large proportion of Mali's population (over 80%) works in agriculture and related sectors, and this sector holds great potential for stimulating economic growth and food security. Low productivity with limited farmer access to productive inputs and technologies, underdeveloped agricultural markets and high vulnerability to climate change are some of the main challenges that need to be addressed to build climate-resilient food systems in the country. These challenges are disproportionately distributed between climate-vulnerable farming communities and inland locations, such as the regions of Gao, Timbuktu, Mopti, Segou and Koulikoro. Climate variability, extremes and change are already affecting the target regions, reducing agricultural productivity, increasing land degradation, changing traditional transhumance patterns, increasing pressure on fodder resources and escalating biodiversity loss. The impacts of climate change are felt disproportionately by the poor, women, and subsistence farmers.

An innovative project is therefore needed to articulate interventions and catalyze a transformation of smallholder agricultural production in Mali, focusing on the sustainable use of agricultural and agroforestry systems to ensure food and nutritional security and support equitable income increases and adaptive learning for farmers and other value chain actors; while adapting and building resilience to climate change.

This IAAT project aims to achieve these objectives by sustainably increasing the agricultural and agroforestry production of vulnerable households to contribute to food diversification using innovative and climate-resilient techniques and practices; supporting the development of climate-resilient livelihood options and developing adaptive local governance mechanisms; and strengthening the enabling environment for smallholder farmers, as well as public and private agroforestry institutions. To ensure that the IAAT project takes gender more fully into account, a detailed analysis was carried out of gender inequalities in professional fields and in the rural economies of the target regions. The analysis also looked at the division of labour, women's access to and control over resources and benefits, particularly land, women's participation in decision-making and conflict management bodies, their access to formal education, extension and

[1] **Source**: Integrated Environmental Assessment of Mali, Assessment Report, UN Environment Programme, UNDP, December 2022

financial services, as well as the impact of GBV and the COVID pandemic and the presence or absence of male supporters of women. The gender assessment is provided in more detail following the introduction, in summary, the assessment results show that:

- In terms of the status of women, despite the existence of a legal and regulatory arsenal to strengthen the position of women in Malian society, women's representation remains marginal in political and administrative bodies, as well as in rural institutions and decision-making bodies. In 2023, women accounted for 28.5% of members of parliament2.. Similarly, women farm managers have less access to farmers' organisations: 17% of them belong to a farmers' organisation, compared with 32% of men. The proportion of women among local councilors was 8.3% in the 2009 elections, up 1.9 points in 2004 and 4.3 points on 1999. In 2020 this rate rose sharply to 25.5%. This result was boosted by the provisions of Law No. 2015-052, which requires political parties to include one in three (i.e. one-third) women when registering voters. This provision was complied with, and out of 79,238 candidates, 26,436 were women, i.e. 33.4%[3] . The Conseil National de la Transition (CNT) has twenty-eight (28) women out of one hundred and forty-seven (147) members, i.e. 28.57%. The current transitional government has only 6 women out of 29 members, i.e. 20.68%. Of the ten - nine - governors, only one is a woman, i.e. 5.26%.
- Women are much more present in child care and household activities and much less present than men in food production and income generation activities, which are mainly reserved for men. Gender analysis also shows that access to and control over resources, benefits and services, participation in decision-making, and how women and men are treated by customary legal mechanisms all depend on gender.
- Women play a vital role in agriculture, environmental management and biodiversity in Mali (particularly agro-biodiversity), as they are primarily responsible for crops that are essential to nutritional security. Socio-cultural factors have generally conditioned the development, use and transfer of the knowledge and skills needed to manage small farms according to gender-related social codes. On these farms, household members have different knowledge and skills. The work and activities carried out by each household member vary according to gender and age. The same applies to roles and responsibilities within each household. These differences between women's and men's spheres of activity and expertise foster the development of distinct but intertwined knowledge, preferences and priorities in the field of agricultural, wildlife and forest biodiversity.
- In the agricultural sector, women are the main operators and processors of staple foods (maize, rice, sorghum, sorrel, monkey bread, shea butter, etc.) and are present throughout the agricultural chain. In addition, women are generally involved in small-scale livestock farming and all fish processing and marketing activities; they are the main collectors, processors and sellers of forest food resources. They also have specialised knowledge of forest plants, which are also used for fodder and traditional medicines. However, women generally lack the means to increase their production and productivity. In particular, they are discriminated against in terms of land management (especially access and control), due to socio-cultural factors which mean that in most communities, a woman cannot inherit land. Only 3.1% of farms in Mali are managed by women (compared with 17% for men), and these tend to be small: 54% have less than one hectare. The average size of women's plots are also much smaller than that of men (0.5 ha compared with 1.5 ha).

2 World Bank. 2023. Proportion of seats held by women in national parliament (%). Available here.

[3] **Source**: Election report/synthesis, 2016

- Furthermore, men and women do not have the same opportunities when it comes to supporting production, such as access to finance. Nationally, men have more access to credit than women, and generally borrow larger amounts (e.g. men receive almost 100% of loans over 250,000 FCFA (equivalent to US$400 in 2023) ,), as agricultural credit services do not deal directly with women who have no guaranteed source of income. These various discriminatory factors make women and girls particularly vulnerable to the impacts of climate change, especially as economic migration by men is a frequent adaptation strategy that leaves women who remain behind with the responsibility of maintaining agricultural production.
- SCI Mali, SCUS, and AEDD are aware that :
  - Climate adaptation and mitigation pathways are not gender-neutral
  - Women and girls are more vulnerable tothe effects of climate change,
  - Genndered needs and vulnerabilities of marginal groups needs to be mainstreamed into adaptation design, resilience capacity-building and mitigation services, and
  - Gender-transformative impact can be driven by allocating robust financial means and dedicated resources to create a more comprehensive and effective approach to addressing climate change in a way that empowers women, promotes gender equality.

1.1 Access to technologies and information: Despite the equity efforts of support partners (national and international NGOs), women have little access to technologies and services. The constraints to this access are: (i) land issues; (ii) methods of accessing land that do not encourage women farmers to invest; (iii) illiteracy among women which limits their access to technologies and services. The literacy rate of women is just 22% compared to 44% for men; (iv) customs and traditions which make them dependent on the men in the family, v) low access to financial resources, and vi) low investment decision-making power in the household. Women have access to the telephone. However, they do not control it. It is the husband who buys the telephone. In the household, it is always the man who controls the device (telephone, TV or radio)[4] . In terms of time availability (daily, monthly, annually), the analysis shows that women invest more time than men, whatever the season. So men are more available than women.

1.2 Involvement in value chains: Gender equality is an essential dimension of inclusive agricultural and economic growth, food and nutrition security and resilient food systems. Women play an important role in agriculture and most value chains. However, their work often remains invisible or insufficiently supported.

Gender inequalities limit women's development and hurt the economic performance of value chains. In Mali, women are present to varying degrees throughout the agricultural value chain and are involved in the processing and marketing of produce. In the agricultural sector, women are present throughout the agricultural chain, but they do not have the necessary means to increase their production and productivity (lack of access to and control and ownership of land and means of production, etc.). They are generally involved in small-scale livestock farming and all the activities associated with processing and marketing agricultural products. Their role as producers in their own right is not always valued or accounted for. Women are mainly involved in growing food crops and market gardening for the family's subsistence. The challenge of food security cannot be met without counting on their contribution to food production or seeking to improve their access to land. So, despite the non-discriminatory legislation, in practice, the influence of men dominates in terms of leadership. Decisions are generally taken by men in public, and women are only consulted when necessary.

[4] **Source**: Regional consultations and Albarka Gender Analysis

1.3 Participation in decision-making bodies and conflict prevention/management: In Mali, social norms and the sexual division of labour largely exclude women from areas of public decision-making or conflict resolution, despite Law 052 of December 2015. The NGP-Mali includes gender indicators in its action plan.

COVID - 19 has further weakened the most vulnerable groups, namely women and girls. In particular, those who sold food in markets, near offices and in schools saw their incomes fall and found themselves trapped in their day-to-day activities, in most cases taking no appropriate measures to protect themselves. The configuration of Malian society meant that they were the ones in constant contact with other members of the community, so they ran a greater risk of being contaminated and transmitting the disease to the rest of their family. In times of conflict, the pandemic has only increased the vulnerability of women, who have been deprived of their homes and whose means of subsistence have been undermined, the arrival of a pandemic on the scale of COVID-19 exacerbated their vulnerabilities. Also, with the pandemic, women suffered more gender-based violence due to the general social stress combined with increasing tensions surrounding the fact that the family lived in unusual proximity, in addition to limited access to food and basic supplies.

As far as the level of information was concerned, many people, especially women, knew that there were health arrangements in place, but did not have precise and clear information about containment and health regulations. They were not aware of the care facilities, let alone the care protocol. The telephone and local radio stations were effective means of communication for reaching the most vulnerable women and girls, but e-mail was not within their reach. Several actions were taken, including the involvement of women in community dialogue and the creation of income-generating activities to empower them. A number of measures have been taken, including involving women in community dialogue via social networks and WhatsApp groups (particularly in the Mopti region, with the organisation of E-tontines for women and E-grins for young people to break the chain of contamination) and creating income-generating activities (IGAs) to empower them. As a result, women have been able to assess the scale of COVID-19 and become more resilient to the pandemic.

The scale of gender-based violence (GBV) has been exacerbated by the political, institutional and security crises that began in 2012, the persistence of inter-community conflicts, and the atrocities committed by armed groups and violent extremism. According to the MICS 2015, female genital mutilation is the most widespread form of GBV in Mali (83% of women aged between 15 and 49) and 48.8% of girls were married before the age of 18. GBVIMS recorded 4,617 incidents in 2019, 4,411 incidents were recorded from January to September 2020. These figures only partially reflect the real situation of GBV in the country, given the weakness of the judicial system and impunity. Women have very little access to legal services (4%), unlike men (9%), and family pressure often prevents them from speaking out about the violence they have suffered. A draft special law drawn up and validated in 2017 by a national multi-sectoral and multi-party committee under the leadership of the MPFEF is still in the process of being adopted internally at the executive level. According to the Regional Directorates for the Promotion of Women, Children and the Family, 126 cases of GBV were recorded in Ségou in 2022, 3,419 cases in Koulikoro, including Dioila and Nara, and 88 cases in Timbuktu during the same period.

Concerning the involvement of women in conflict resolution, conflict prevention and resolution issues in Mali remain under the almost exclusive management of men in the communities and are therefore highly discriminatory for women, even beyond any religious or cultural considerations, even though they have proven potential and capabilities in conflict resolution and prevention and more generally in peace-building. Since 2012, women have accounted for less than 10% of peace negotiators, and this figure has hardly changed since the adoption of resolution 1325 in 2000. The government has shown the political will to take account of the gender dimension in conflict prevention and management, for example through the provisions of the national strategy for reconciliation and social cohesion and its 2022-2026 action plan (priority 1: Promote peace, national reconciliation and social cohesion, sub-priority 1.2: Strengthen the participation of women and young people in the peace process).

Although there is a clear political will to take the gender dimension into account in the reconciliation and social cohesion process at the national (MARN), regional (ERAR) and local (CCR, COFO, CCS) level, it is clear that women are poorly or not at all represented in conflict prevention and management bodies. Through participatory approaches that value women's contribution, the IAAT project will encourage a change in behaviour towards a culture of inclusiveness in formal and informal conflict resolution and prevention mechanisms at local level, as well as improving women's access to sustainable and resilient economic empowerment opportunities. In addition to the contestation of local leaders, opinion leaders and religious leaders, the exclusion and discrimination of women and ignorance of their role in revitalising conflict management mechanisms exacerbate women's vulnerability. This programme presents an opportunity to reverse this trend by relying on women who have proven skills in social mediation and by strengthening their role in family and community units on issues of social cohesion and conflict management. Past experience, particularly with peace huts in Mali, has shown that if women are mobilised and involved, this can have a positive impact on conflict prevention and resolution at community level, especially as they themselves can play an active role in triggering these conflicts. The translation of Law 052 of December 2015 and its dissemination in the IAAT project areas should also be considered.

1.4 **Displaced people**: With the security situation, women and girls are becoming increasingly vulnerable. Many women have become heads of household (the men have either been killed, abducted or fled) and are left to fend for themselves, mainly in the Mopti and Tomboctou regions. The displaced are scattered in camps, in families and houses under construction. At the national level, for example, there are 412,387 displaced returnees and 120,298 internally displaced persons, of whom 224,124 are women, i.e. 54.34% are women[5].

1.5 **Associations**: The Regional Directorates for the Promotion of Women, Children and the Family in the gender analysis regions listed 102 associations/groupings in Ségou, 255 in Koulikoro, 251 associations in Mopti, 1,700 in Gao and 455 in Timbuktu. Many women's associations are also involved in village saving and landing groups (VSLAs) and producer groups. Despite the lack of material and financial resources, these associations can mobilize women and raise their awareness of climate change. The leaders of some associations are resource persons, leaders who can be put to good use in promoting women in value chains, climate security and the sustainable management of natural resources. They are also involved in self-help and support for women, socio-economic development and the fight against poverty. To this end, an organisational audit of these groups and associations is more than necessary.

1.6 **Women's allies and supporters**: The coordination of neighborhood and village chiefs, the High Islamic Council, the Catholic and Protestant churches, youth leaders and traditional communicators all support gender inclusion/gender equity and equality. The IAAT project will have to consider each category of actor as a strategic ally and supporter of women and young people in the development of CACs and community action plans for climate chagne adaptation and mitgation. To do this, each of these actors will have to benefit from a reinforcement and support programme based on the needs identified above, so that they are in a position to fully

[5] **Source**: DNDS, Population Monitoring Matrix Report (DTM)_ April 2022

play their roles as allies and supporters. Women face a number of constraints identified by the analysis that merit particular attention in the implementation of the IAAT project.

1.7 Project goal and objectives

This proposed IAAT project will support climate-vulnerable smallholder agricultural production systems and rural livelihoods in Mali to increase climate resilience and low-emission development pathways. The project has four components: 1) increase adoption of improved and climate-smart agroforestry and agricultural technologies, practices, and services, 2) increase connectivity of smallholders, women and youth to the agriculture input and output markets, 3) reduce GHG emissions through the promotion of agroforestry, use of solar pump irrigation, and manure management through biogas plants, 4) strengthen institutional and stakeholders' capacities for scaling-up improved and climate-smart agroforestry and agriculture.

The project's paradigm shift goal is: **IF** smallholder farmers in highly vulnerable regions in Mali can access and adopt climate resilient technologies, knowledge, and low carbon agricultural practices, **THEN** their food, nutrition and water security will be improved **BECAUSE** their adaptative and mitigation capacities including improved technical skills, access to finance, markets and sustainable livelihoods, and that of public and private institutions, will be strengthened to respond to and reduce the climate change risk and impacts. The project will drive immediate and medium-term impacts through farm and community-level actions that improve localized climate-resilient agricultural yields; reduce the time, energy, and labor effort by food producers, especially women, on farming activities; increase women's access to and control of agricultural inputs; and increase incomes. Long-term impacts will be achieved by increasing the climate resilience of farming systems, applying land use management, and reducing greenhouse gas emissions. Other long-term impacts will include enhanced access to climate-smart technologies to reduce climate-compromised (drought, heat seasonality, and flood-impacted) yield gaps, improve access to resources and support for women farmers to raise yields and improve food security, and reduce emissions through improved livestock and fodder production and reduced fertilizer use.

The proposed project will be implemented in five climate-vulnerable regions of Mali selected based on the climate risk assessment and stakeholder consultations. These regions include from north to south: Tombouctou, Gao, Mopti, Koulikoro, and Segou. In Tombouctou, Gao and Mopti, the project will build on the work of Albarka's Resilience Food Security Activity (Albarka funded by USAID). The Albarka will co-finance to implement some climate action and capacity-building activities of this GCF project in the regions. This project will focus on the 12 circles and within them 48 communes, in the Tombouctou, Gao, Mopti, Koulikoro and Segou regions in Mali based on their increased climate vulnerability. Within these regions, circles, and communes, beneficiaries will include smallholder farmers and their households, women and youth entrepreneurs, private sector businesses, extension services members, and members of public institutions. The Agency of Environmental and Sustainable Development (AEDD in French acronym) under the Ministry of Environment and Sanitation will serve as the Implementing Entity and Save the Children Mali (SCI Mali) will serve as the Executing Entity.

This annex outlines the gender assessment and action plan of the IAAT. The assessment presents how the project resources will be mobilized and utilized to tackle the persisting gender barriers, and biases, and contribute towards improved gender equality.

Table 1: IAAT components, outputs and activities

Component	Output	Activity
Component 1: Extension Services and On-Farm CSA Adoption	1.1: Improved technical capacities and inclusivity of extension services in climate-smart agriculture and agroforestry production	Activity 1.1.1 Build technical capacity and reach of extension services on CSA and agroforestry techniques
	1.2: Increased use of climate resilient practices in the production of CSA crops, livestock and agroforestry by smallholder farmers	Activity 1.2.1 Build awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry amongst smallholder farmer communities (crop and/or livestock)
	1.3: Increased land area under agroforestry	Activity 1.3.1 Develop land-use mapping at the regional level
		Activity 1.3.2 Plant agroforestry trees on community and state-owned lands
Component 2: CSA and Agroforestry Value Chains	2.1: CSA and agroforestry VCs are more connected and reach more smallholder farmers	Activity 2.1.1 Support the creation of inclusive private sector value chains for key CSA/agroforestry crops and technologies
	2.2: Smallholder farmers, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness	Activity 2.2.1 Support local financial institutions to increase access to finance for smallholder farmers, especially women and youth, for CSA/agroforestry investments
		Activity 2.2.2 Enhance capabilities and connectivity of youth and women CSA/agribusiness entrepreneurs
	2.3 Increased adoption of climate-smart agriculture technologies by smallholder farmers	Activity 2.3.1 Install and support the productive use of biodigester systems and solar irrigation systems amongst smallholder farmers
Component 3: Increasing Institutional Capacity and Knowledge	3.1: Increased institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks	Activity 3.1.1 Strengthen institutional capacity in localized climate change adaptation and mitigation planning
		3.1.2 Technical capacity building of Mali Climate Fund in prioritizing

		and targeting investment in CSA and agroforestry
	3.2: Enhanced Knowledge sharing and coordination of best practices in CSA and agroforestry across stakeholders	Activity 3.2.1 Enhanced convening and contribution to national databases for CSA and Agroforestry

2. Methodological approach

Our methodological approach was participatory at all stages, involving all stakeholders and in particular the SCI Mali team leading the assignment. There were six (6) main phases in the implementation of the evaluation mission. These were briefly as follows (i) scoping meeting; (ii) document review; (iii) production of data collection tools; (iv) data collection and analysis; (v) production and submission of the gender report and action plan.

2.1 Assessment steps:

Stage 1: Scoping the study: a scoping meeting was held with the SCI Mali team responsible for technical supervision and validation of the methodology and data collection tools. This meeting was an important step, in that it enabled us firstly to gain a better understanding of SCI Mali's expectations and also to gather additional specific objectives not covered in the terms of reference. It also provided an ideal framework for: (i) introducing ourselves to the rest of the team; (ii) harmonising our understanding of the objectives and expectations of the study in order to adopt a consensual work programme for conducting the survey; (iii) gathering information and specific recommendations on how to conduct the assignment; (iv) providing us with all the documents and tools we needed to carry out the documentary review and conduct the assignment.

Stage 2: Literature review : Another key stage in our approach was the literature review, which will continue until gender is effectively integrated into the project document. Without being exhaustive, this stage covered all the national and international documents relevant to the issue in question. These documents include national policies and documents on national standards, reports on similar studies, the project document and all other documentation. Documents were also collected from other structures (technical services, AEDD) and other relevant organisations. The reports available and the factual information provided by the project's stakeholders were analysed, bearing in mind that the primary source of documentation is SCI Mali. Internet documentation, which has already begun, is underway. Finally, this stage was completed by interviews with the SCI Mali staff in charge of the gender evaluation to ensure that SCI Mali's expectations were taken into account.

Stage 3: Determination of data collection targets and development of the methodological guide for the work (including the guides for collecting data broken down by sex): For the actual data collection, it was decided to supplement the secondary data with data collected at regional consultations and from identified key informants. Secondly, sex-disaggregated data collection questionnaires were drawn up.

Stage 4: Meetings held in Bamako, Koulikoro, Ségou, Mopti, Gao and Timbuktu: Meetings were held in these regions in the form of national and regional consultations, and the information gathered forms an integral part of this report. Additional information was gathered through interviews with key informants.These guides were used to gather information and identify gender inequalities in the Mopti region in terms of the roles and responsibilities of men and women, and access to and control over resources and benefits.

These guides helped us to gain information on the division of labour, access to and control of natural resources and benefits in terms of: Who does, what, how, where, why, for whom, with whom? who has access? who controls. Who decides how natural resources are used. They (the guides) will also enable us to know: (i) Who has access to information related to NR, agriculture, climate change, agroforestry; (iii) What are the constraints to the participation/emergence of women and young people in the value chain; (iv) What **is the legal status of** women; What are the conditions that must be met for women and young people to **perform optimally** in value chains; (v) What are the **challenges that** men and women may face in order to derive **optimum benefit** from the value chain? (vi) **Where** are the men, women and young people among producers, retailers, wholesalers and processors? (vii) **Who controls** the marketing channels? (viii) Who has access to equipment and infrastructure for post-harvest management, processing, etc.? (ix) **Who controls** the means of production, equipment and infrastructure? (x) How are men, women and young people positioned among aggregators?

Stage 5. Methodology and data collection tools

Target audience and geographical coverage as proposed in the terms of reference, the analysis was carried out in Bamako and the regions of Koulikoro, Ségou, Mopti, Gao and Timbuktu. The study involved the national and regional directorates for: Development Planning; Promotion of Women, Children and the Family; Water and Forests; Livestock; Agriculture; Fisheries; Meteorology; the Rural Economy Institute; the National Agency for the Great Green Wall; l'Assemblée Permanente des Chambres d'Agriculture du Mali ; Point Focal Opérationnel du Fonds pour l'Environnement Mondial (GEF); le Point focal du Fonds Vert pour le Climat au Mali; les femmes leaders transformatrices des régions de Koulikoro, Ségou, Mopti, Gao et Tombouctou. In Bamako, the AEDD and its **gender unit.** Both women and men representatives from these organizations were included. The information collection also included members of village saving and loan associations and women representatives from NGOs, and the private sector.

Step 6. Study methods and data collection

We opted for one-to-one interviews with key informants and the search for additional information at regional consultations.

3. Description of the reference situation

3.1. Gender issues and inequalities in Mali

Mali has a long and rich history of successive kingdoms and empires, which have exerted a strong influence in West Africa, and the country is known for its cultural diversity and its more than twenty main ethnic groups, which are traditionally characterised by strong social, gerontocratic and patriarchal hierarchies, with a few exceptions in the north. The main religions practiced in Mali are Islam (92.5%), Christianity (3.2%) and Animism (4.3%). Customary and traditional rights and the Muslim religion have a major influence on the exercise of human rights, in particular the misinterpretation of messages that prevent women from fully enjoying their rights. As a result, the legal status of women in Mali is characterised by the coexistence of modern and customary laws or rules, which are much more conservative in terms of the roles, responsibilities and power relationships between men and women. Table 2 provides some country-level indicators about gender.

Table 2. Some of the country's indicators

Designations/indicators	Rate for men	Rate for women	Global
Maternal mortality rate		13,7%[6]	
The infant and child mortality rate[7]	10,5%	9,7%	
The educational status of girls and boys			
Gross enrolment rate in the first cycle of basic education[8]	78,3%	66,0%	
Gross enrolment rate in upper basic education[9] (%)	55,6%	46,8%	
The adult literacy rate in urban areas[10]	71 %	52 %	
The adult literacy rate in rural areas	38 %	19 %	
The poverty line rate[11]			42,1%
Population living in extreme multidimensional poverty			44,7%
Population vulnerable to multidimensional poverty			15,3%
Labour force participation rate (participation rate)[12]	80,6%	61,2%	
The employment rate[13]	90 %	55 %	
The unemployment rate	4,3%	4,0%	
Political participation rates			
Parliamentary representation rate [14]	72,7%	27,3%	
Rates of representation on communal councils[15]	62, 2%	33.8%)	
Life expectancy in the country of intervention and/or the area where the project/programme is based [16]	57.6 years old	60.3 years old	

Under Malian customary law, women are excluded from access to land ownership, even though they play a major role in land use. In the increasingly widespread land disputes, women are the main victims, with consequences for family food security. The Agricultural Policy Law of August 2006 provides for the protection of women following extensive consultations in which they took

[6] INSTAT REPORT 2020 MALI IN FIGURES 2013 - 2017
[7] REPORT Demographic and Health Survey 2018
[8] INSTAT REPORT 2020 MALI IN FIGURES 2013 - 2017
[9] INSTAT REPORT 2020 MALI IN FIGURES 2013 - 2017
[10] REPORT Demographic and Health Survey EDSM-VI 20119
[11] UNDP IDH 2021 REPORT
[12] IDH-MALI 2020 REPORT
[13] REPORT Enquête Démographique et de Santé EDSM-VI 20119 : ceux qui ont déclaré qu'ils travaillaient au cours des 7 jours avant l'enquête
[14] UNDP IDH 2021 REPORT
[15] UN-WOMEN https://africa.unwomen.org/fr/gouvernance-et-participation-des-femmes
[16] UNDP IDH 2021 REPORT

part, but it has yet to be applied, even though power is still largely vested in men[17] . According to the World Bank[18], only 5% of Malian women own land. Mali is one of the countries with low human development, with an HDI of 0.434 in 2019. This value places Mali 184ème out of 189 countries and territories ranked[19] .

- **Gender Development Index (GDI)[20]** : The GDI measures gender inequalities in three fundamental dimensions of human development: health (measured by the life expectancy of women and men at birth), education (measured by the expected length of schooling for boys and girls and the average number of years of study for adults aged 25 and over) and control over economic resources. The HDI is calculated for 167 countries. In 2019, Mali's HDI is 0.388 for women, compared with 0.473 for men, giving a GDI of 0.821 and placing the country in group 5. By comparison, Burkina Faso and Niger have an HDI of 0.867 and 0.724 respectively.[21]

- **Multidimensional poverty index (MPI)**: The multidimensional poverty index determines the deprivations faced by people in the areas of education, health and standard of living. It provides the incidence of non-monetary multidimensional poverty, i.e. the number of people living in multidimensional poverty and the intensity of this poverty (relative number of deprivations suffered simultaneously by these people). The MPI, i.e. the proportion of the population living in multidimensional poverty, adjusted for the intensity of deprivation, is 0.376. Burkina Faso and Niger have MPIs of 0.519 and 0.590 respectively[22] .

- **Gender Inequality Index (GII)[23] :** This measures inequalities in the achievements of men and women in three dimensions : reproductive health, empowerment and the labour market. The GII should enable analysis of policies and efforts. A high value indicates strong inequality between men and women. Mali's GII of 0.671 places it 158th out of 162 countries in the 2019 index. In Mali, 9.5% of parliamentary seats are held by women and 7.3% of adult women have completed secondary education, compared with 16.4% of men. Out of every 100,000 live births, 562 women die from pregnancy-related causes, and the adolescent fertility rate is 169.1 births per 1,000 women aged 15 to 19. The activity rate for women is 61.2%, compared with 80.6% for men.

3.2. National, regional and international policies on gender equality and the environment, particularly in areas relating to women's rights, land rights, climate change, forest resources and agriculture.

- At the national level, the preamble to the Malian Constitution of 25 February 1992 affirms the sovereign people's subscription to the Universal Declaration of Human Rights of 10 December 1948 and the African Charter on Human and Peoples' Rights. The principle of the primacy of universal rights over national laws is affirmed in the Constitution. Thus, it guarantees the same rights to citizens of both sexes without discrimination and proclaims in its preamble the defence of the rights of women and children as well as the cultural and linguistic diversity of the national community.

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[17] **Source**: Joint women's empowerment programme, United Nations agencies and MINUSMA, December 2021.
[18] **Source**: Gender Report, Kayes and Sikasso, ACF, 2022
[19] Source: HDI Report, Mali 2021
[20] **Source**: HDI Report, Mali 2021
[21] **Source**: HDI Report, 2021
[22] **Source**: HDI Report, 2021
[23] **Source**: HDI Report, 2021

- **The National Gender Policy:** The National Gender Policy, with its six (6) Strategic Orientations and nineteen (19) Strategic Axes, is based on a set of legal, political, socio-economic and socio-cultural foundations. Mali's National Gender Policy is based on respect for universal rights while combining the values associated with a tolerant society that is open to the world and keen to bring about a favourable change in traditions and mentalities towards greater justice, equity and equality.
- In addition, Mali has ratified, often without reservation, a number of regional and international conventions and treaties relating to the protection and promotion of women's rights, such as the Convention on the Elimination of All Forms of Discrimination against Women since September 1985, and ratified the Additional Protocol to the said Convention in September 2000 and its Additional Protocol, as well as the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol).
- On 25 September 2015, governments adopted a new international development agenda, comprising a set of Sustainable Development Goals (SDGs) with the aim of eradicating poverty, protecting the planet and ensuring prosperity for all by 2030. Goal 5 of this agenda is to achieve gender equality and empower all women and girls, and Goal 2 is to eradicate hunger, ensure food security, improve nutrition and promote sustainable agriculture. Concrete actions have been identified as critical conditions for achieving the MDGs, including: (i) undertake reforms to give women equal rights to economic resources, as well as access to ownership and control of land and other forms of property, financial services, inheritance and natural resources in accordance with domestic law ; (ii) strengthen the use of key technologies, in particular information and communications technology, to promote the empowerment of women; (iii) adopt and strengthen appropriate policies and enforceable legislation for the promotion of gender equality and the empowerment of all women and girls at all levels.
- **Mali also has a new reference framework for formulating, implementing and monitoring the various programmes, policies and strategies that the country has developed in recent times, at both national and sectoral levels**. This 2ème generation covers the period 2019 - 2023. Its objective is to make it possible to achieve the Sustainable Development Goals (SDGs) by 2030, building on the country's potential and resilience to promote inclusive development to reduce poverty and inequality in a peaceful and united Mali. The new Government Action Programme for the transition period was adopted on Monday 02 July 2021. All public policies will have to integrate gender, employment and climate change into the development and monitoring-evaluation of public policies.
- At sector level, a national strategy to combat gender-based violence (GBV) was drawn up in 2018. In addition, Law No. 2015-052 was adopted in 2015 to promote gender equality in elective and nominative public posts. This law increased the percentage of women in appointed positions from 10.4% in 2014 to 15.73% in 2017[24] . However, the minimum representation of 30% provided for by this law has not yet been achieved.

3.3 The legal framework

The legislative framework for environmental protection is marked by important international and national provisions. Mali has ratified the conventions of the Rio de Janeiro generation (UNFCCC, CBD, CCD), other earlier conventions (Ramsar Convention, CITES) and is a party to numerous international and sub-regional agreements relating to the management of forest ecosystems.

At the continental level, Mali has taken part in major initiatives such as the Great Green Wall project, the development of the NEPAD Environment Initiative, the Poverty-Environment Initiative, the Global Climate Change Alliance and the development of a common vision on the fight against climate change under the aegis of the Conference of African Ministers of the Environment (CAME).

Law no. 95-50 of 16/10/96 on the principle of establishing and managing the domain of local authorities identifies the conditions under which these authorities may appropriate and manage parts of the national land domain, and sets out their responsibilities and rights in this area. This law governs the creation and management of local authority land, namely forest land, agricultural land, pastoral land, wildlife land, fish farming land, mining land and housing land.

Civil society is also represented in consultation frameworks and decision-making bodies. SECO/NGO is the environmental leader for national NGOs and CAFO is the environmental leader for women's organisations and associations. These two organisations have participated in all the activities of the CIGQE Advisory Committee. This active participation by Malian civil society is also in line with the conclusions of the Rio Conference and Agenda 21.
The main legislative and regulatory texts relating to forests and wildlife are 6 Laws, 18 Decrees and 6 Orders. The six (6) laws are :

- **Forestry**
  - Law no. 95-003 of 18 January 1995 on the organisation of timber exploitation, transport and trade
  - Law No. 95-004 of 18 January 1995 laying down the conditions for the management of forest resources
  - Law N°10-028 of 12 July 2010 determining the principles of management of the resources of the national forest estate.

- **With regard to wildlife**
  - Law No. 18-036 of 27 June 2018, setting the conditions for the management of wild fauna and its habitat

- **Institutional framework**

At institutional level, the Ministry of the Environment has set up six central departments:
- Direction Nationale des Eaux et Forêts ;
- Direction Nationale de l'Assainissement, des Pollutions et des Nuisances ;
- Environment and Sustainable Development Agency ;
- Niger River Basin Agency ;
- Agence Nationale de Gestion des Stations d'Épuration du Mali.

In addition to these legal and institutional frameworks, there are :
The National Policy on Climate Change (PNCC**)** has three objectives:
- Promotion of actions to adapt to the impacts of climate change (OPN2).
- Promotion, extension and transfer of appropriate technologies as developed by research (OPN5).
- Capacity building on climate change (OPN6).
- The National Environmental Protection Policy (PNPE) was adopted in 1998 and revised in 2018. This policy ensures the protection of natural resources including water, forest and land.

In most of the policies developed over the last ten years, the principles of equity and/or equality are announced. In strategies and/or programs, gender is specifically or indirectly integrated through the targeting of women, young people or mixed, marginalized, and vulnerable groups.

The Fonds d'Appui à l'Autonomisation de la Femme et à l'Épanouissement de l'Enfant (**FAFE) is** a fund initiated by UN Women as part of the implementation of the NGP in support of the Ministry for the Promotion of Women, Children and the Family. It is aimed at financing initiatives to develop women's entrepreneurship in Mali. **Gender-responsive budgeting** (**GRB)** is now an integral part of Mali's budget planning system, making it possible to take gender-specific concerns into account in various development sectors with a view to creating opportunities in the fight against women's poverty. In 2011, Mali adopted its National Gender-Sensitive Planning and Budgeting Strategy (PBSG), which is part of the implementation of strategic guideline No. 6 of the National Gender Policy on good governance in public policies and reforms in the context of decentralisation and deconcentration.

4. Results of the gender assessment

The consultations conducted at the national and sub-national level were essential for the design of the project in a way that is responsive to the needs of women as the target populations, and to identify the key climatic and non-climatic challenges and vulnerabilities faced by women in the target areas. The present assessment includes information collected from all the consulted stakeholders on key gender aspects, including:

- Gender and social norms and division of labour
- Access to assets, material and immaterial resources
- Decision making power at household and community level
- Impact of climate change, particularly drought, on the pre-existing gender inequalities
- Coping strategies are currently in place affecting women and girls to overcome climate change impacts
- Existing awareness, understanding and adoption of gender mainstreaming for social protection and climate change adaptation activities.

4.1. Division of labour between men and women and decision-making models

The roles and responsibilities of men and women are specific to each culture. They can also vary from one community to another. In the targeted regions, the results show that both women and men, as well as girls and boys, may engage in activities, but most of the time their responsibilities are different. The domestic production process obeys the rules of land sharing and the division of tasks and working time for each member according to age and gender. In terms of the social division of labour, men and women both carry out agricultural work, but household chores remain the role of women.

The social or sexual division of labour also obliges women to take care of housework, the education and health of their children, the daily chore of supplying drinking water and firewood, the upkeep of the farmyard and small livestock, the sale of foodstuffs and other basic necessities, and so on. All the labour and resources available to the extended family are mobilised to cultivate the common or family field. From the point of view of the organisation of work in the Koulikoro, Ségou, Mopti, Timbuktu and Gao regions, a large part of the week (generally five or six days; Christians do not go to the fields on Sundays and Muslims on Fridays, but family members also do not go to work in the fields on fair days) is reserved for work on the collective field. The rest of the time is reserved for rest and work in the individual fields, especially for the women, on the plots of land granted to them by the male heads of household. The produce from the family fields is used to feed the family, pay each man's first marriage expenses, buy collective equipment and solve other collective problems. This agricultural economy, which is not very mechanised, requires a lot of humanpower.

Women are also involved in the agricultural value chain and participate in the processing and marketing of the products produced. However, they are not paid, nor do they control the income from this production. In addition to primary care and household tasks, women, with the help of their daughters, also take part in income-generating activities, enabling them to generate their own income, which they manage and control, even if they have to inform the men[25] . Men are involved in agricultural production upstream. Men are only involved in food production and income generating tasks and control all income linked to agricultural production.The movement of men in connection with transhumance creates additional constraints for women, such as economic restrictions, fewer meals and a double workload.

The assessment also revealed that in the target regions, the social division of labour results in an arbitrary division of agricultural activities between men and women. This division is often based on traditions and customs. It should be noted, however, that it is increasingly common to see men collecting water and firewood, where the increase in insecurity has led to these changes aimed at protecting women and girls from sexual violence. The gendered division of labour does not exclude women and girls from taking part in agricultural, livestock and income-generating activities. Women's entrepreneurship is encouraged by their husbands.

- **Land:** Women and men, girls and boys plant the seed; after the fields have been prepared, the men and boys are responsible for weeding. At harvest time, everyone is involved, but at different levels. The men and young men cut the millet, rice and sorghum with knives, while the women and girls, young men and boys pack and sack the produce. Women and girls, sometimes helped by men and boys, transport the produce home in carts. They do this for their families. An analysis of agricultural production shows that:
  - **In Mopti,** women are much more involved in the production of groundnuts, cowpeas, millet and rice, while men produce rice (especially in flooded areas), millet and sorghum. Pottery is an activity reserved exclusively for women, who use the earth as a resource.
  - **In Koulikoro:** women produce groundnuts, sesame and red "da" (hibiscus)
  - **In Ségou:** women produce groundnuts, da, okra, red da, sesame, especially in Bla, and calabash.
  - **In Gao:** women are heavily involved in rice production
  - **In Timbuktu:** women are involved in rice production and increasingly in leguminous crops and some fruit trees.

- **Pasture / livestock:** in the traditional environment of all the regions analysed, livestock farming is mainly extensive. Men rear cattle for the needs of their families. They also fatten cattle for the Ramadan festivities, either individually or as part of a family cooperative, in stalls or on pasture. The men market the live animals locally, to private individuals and to wholesalers, some of whom come from outside the area. As for the women, they rear small ruminants (goat and sheep), but are increasingly improving their family or bush rearing (fattening), as well as rearing their own poultry. The by-products of this farming belong to women, but they inform their husbands before using them, and the same applies to organic manure. They are also much more involved in harvesting at this level. With the support of certain partners, such as PRAAPS, women are increasingly involved in cattle fattening. Traditionally, cattle pens are the property of the whole family, and roles are shared between men and women. Generally, the man is responsible for grazing and transhumance. The woman is responsible for the whole process of milking and its management. The young

[25] **Source**: Gender analysis report Albarka Programme zones

men raise pigeons, quail and some poultry for themselves. The young boys act as shepherds for their families or for other families in return for wages in-kind or in cash. The men are responsible for veterinary care of all livestock.

- **Water points / reservoirs / Fishing**: women use water points and reservoirs for household consumption including drinking and sanitation,. In many locations, men are involved in fisheries where enough water is available. Some women also farm fish in their farm ponds, often with the support of projects. The women and girls are productively engaged in the processing and marketing of fish.
- **Market gardening**: Women and girls, men and boys take part in market gardening. It's worth noting that there are more women than men, and that young girls in particular and young boys help their mothers on the market-gardening sites. Marketing is carried out by the women with the help of their daughters. Given the level of illiteracy among the women, the men in these women's groups often act as administrative secretaries (keeping administrative documents). In general, the women are supported by certain development partners such as the WFP, FAO, UN Women and CAEB in various ways, such as obtaining land-security deeds from customary/land-owning chiefs, elected representatives and the administration; developing sites; installing drainage systems; and, in some cases, providing technical assistance; installation of drainage and solar systems, small farm equipment such as wheelbarrows, picks, watering cans, shovels, inputs (improved seeds); capacity building/training in cultivation techniques and intermediation with input suppliers and potential buyers. Market gardening is one of the interventions in the Albarka regions, where the project promotes vegetable gardening as a part of income-generating activity.
- **Forests**: women and girls are involved in collecting firewood for household consumption all year round. They sell surplus firewood in the local markets to generate some income for the houshold. With increasing insecurity due to conflict, men are gettinginvolving in collecting firewood in Mopti, Gao, Timbuktu and parts of Ségou and Koulikoro. Women also source shea butter and néré in Mopti; shea butter, néré and Zaban in Ségou and Koulikoro; leaves such as Fakouhoye, jujube, doumia fruit and water lilies in Timbuktu; henna in Banamba and Koulikoro; and wild dates, jujube and doum in Gao, for sale in raw or processed form.
  Indigenous Populations and Peoples (IPP)" level[26] : land management methods, for example, are heavily based on customary norms and women are absent. Also, the political reforms undertaken at national level since the advent of democracy in 1992 have had major repercussions on certain traditional chieftaincies, which totally exclude women, even though older women are consulted. The inheritance of traditional chieftaincy, in the event of the death of the legitimised chief, passed, in the absence of consensus, by electoral means under the supervision of the State delegate, does not make room for women. Indigenous peoples and commodity chains. The division of labour is as follows:
  - **Men**: the honey industry, agricultural/range production, cattle and sheep fattening (food, animal care), bourgou production in Mopti, Timbuktu and Gao: fattening industry, beekeeping (sale of honey), market gardening (production and sale of produce). In Timbuktu, bourgou is increasingly exported to Algeria and Mauritania.
  - **Women**: market gardening (production and sale of produce), sheep/goat and poultry fattening (feeding and watering);
  - **Young boys**: honey (production), market gardening (labour), fattening (grazing animals, watering); poultry (pigeons);

[26] **Source**: Rapport de mission sur les peuples autochtones et communautés locales dans le cadre de la formulation du "Projet PNUD-GEF ID 10687 - Sécurité climatique et gestion durable des ressources naturelles dans les régions centrales du Mali pour la consolidation de la paix" (Project Preparation Grant (PPG)), Dr ONGOIBA Salomon, PhD June 2021

- **Young girls**: helping the family in the market gardening and fattening sectors.

IAAT will engage with local men, women and youth based on their current involvement in the value chain and use their knowledge and experiece to expand the value chains.

4.2. Access to and control of resources

Men and women, girls and young people can access and use all resources, but only with the agreement and/or permission of the heads of household. Women in particular, through their groups, can easily obtain plots of land for their market gardening activities, with the contribution of elected representatives, village chiefs/customary chiefs/counsellors and land chiefs, and the support of certain development partners.

Control of the land : is often subject to certain social constraints. It is the responsibility of heads of families, chiefs or landowners. Women have only indirect access to land. They generally benefit from plots of land acquired through a third party (usually a man: husband, head of lineage, other relative with the spouse's agreement, etc.) for their production, but have no control over the resource. Discrimination against women in terms of land management is linked more to socio-cultural constraints, which mean that in most communities, women cannot inherit land. They generally only have precarious access to land that is usually marginal.

Control of grazing: In Mopti, for example, in the flood zone, the management of the bourgoutières[27] is the responsibility of the djoros[28], who decide/determine the order of passage of the animals for grazing. It should be noted that their animals are the first to go out to graze. The sale of household animals is generally the responsibility of the men. When it comes to selling their own livestock, women consult their husbands first. When it comes to marketing or selling, women entrust their animals to the men in the family. This is because women do not go to livestock markets to sell their animals. It should be noted that the emergence of armed groups has led to the emergence of new actors who have taken the place of the Djoros, thereby reducing their level of control over bourgoutier resources. These new actors are the armed groups, who are increasingly managing the bourgoutières, claiming that they belong to God, allocating them to whomever they please and levying tithes[29] and zakat[30] on their use[31].

Forests : are controlled by the management committees, village chiefs/councillors and landowners. As for the women, they have control over harvested products such as gum arabic, néré, shea, zaban, certain leaves, and acacia albida pods, which are very profitable to sell and earn income from, with the advice of the men or after consulting them. But the production of these pods is still insufficient to build a value chain. In the agro-pastoral regions covered by the gender assessment, the exploitation of natural resources is generally considered to be abusive by some and even anarchic by others. These resources are exploited in part by the community (management committees and heads of families) for production purposes in agriculture, animal husbandry, handicrafts and traditional medicine, and play a particularly important role in satisfying their daily food, energy and medical needs, and more specifically by the women and young people

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[27] Bourgoutières is a type of fodder critical to the livestock production system in northern Mali.

[28] Djoros is a land owner or manager

[29] Tithes is a system of giving one-tenth of a person’s income or production for the name of God to be used for designated purposes.

[30] Zakat is an Islamic finance term referring to the obligation that an individual has to donate a certain proportion of weath each year to charitable causes

[31] **Source**: Report, Gender: of the "UNDP-GEF Project ID 10687 - Climate security and sustainable management of natural resources in the central regions of Mali for the consolidation of peace" (Project Preparation Grant (PPG)).

who supply the families with firewood and other forest products. Thanks to forest by-products, based on herbs and leaves, women also carry out income-generating activities (IGAs).

The water reservoirs/water are managed by the water masters, who are the bozos, and the ponds by the villagers.

Facilities: In rural areas, less than 20% of women have access to equipment. According to the report National Gender Profile of the Agriculture and Rural Development Sectors - Mali, access to equipment has a considerable impact on productivity and agricultural production. Given the relatively high cost of access, equipment is generally the property of the farm. Women's low incomes and limited access to agricultural credit mean that fewer of them own agricultural equipment and therefore have limited access to it. Of the women farmers who own equipment, 43% have ploughs, 35% donkey hoes and 10% carts. The other types of equipment are owned exclusively by men. Whatever the type of equipment, women's farms are always under-equipped compared with men's farms. The equipment on women's farms consists mainly of ploughs. Women's lack of equipment is a significant factor in their productivity. On the farm, they can only have access to the equipment when work has been completed on the other plots, which often leads to delays in the cropping calendar.

4.3. Access to and control of profits

In terms of access to and control over benefits, the analysis reveals that only men have access to benefits and income and decide who can have access to them. Women have control over their own income (market gardening, fattening, processing of harvested produce), but they have to inform their husbands beforehand. The use of resources is decided by the heads of household according to the results of the assessment.

4.4. Access to information, services and technologies related to natural resource management

In the villages, it is usually the village chief who is the first to be informed, then he informs the heads of the families, who in turn inform the members of their families, including the women. Information is also shared through town criers[32]. Men are also informed through village meetings, national and local radio stations, awareness-raising missions and cooperatives. Technical services (water and forestry, animal husbandry),talks in grins[33] and "word of mouth" are also sources of information for men. Some women get information related to natural resource management from radio stations (in rural household that posses radios) and meetings. In the commune administrative centres, men, women, girls and boys have access to information via meetings, radio, television, traditional communicators, posters in town halls or decentralised technical departments, social networks (WhatsApp groups) and sometimes training courses. In the livestock sector, the head of the family is informed by the shepherd in the event of illness and by the veterinary service for vaccinations.

The access to technologies and services helps both men and women to improve farming methods and techniques, increase productivity and incomes, improve living standards and raise the social and educational standards of rural life. Despite the equity efforts of support by partners (national and international NGOs), women have little access to technologies and services. The key constraints form women are: (i) land tenure issues; (ii) modes of access to land that do not encourage women farmers to invest; (iii) illiteracy among women, which limits their access to technologies and services; (iv) low access to financial resources, and v) customs and traditions

[32] Persons employed to make public announcement in the small village towns.

33 A type of non-verbal communication

that make them dependent on the men in the family. Women have very limited access to credit, especially in agriculture and the major obstacle is their inability to offer collateral[34].

Women have access to the telephone. However, they do not control it. It is the husband who buys the telephone. In the household, it is always the man who controls the device (telephone, TV or radio)[35]. In terms of time availability (daily, monthly, annually), the analysis shows that women invest more time than men, whatever the season. So men are more available than women. Measures to help women must take this aspect into account and include measures to lighten women's workloads and raise men's awareness of the need to take on certain activities for women.

4.5. Gender and value chains

Gender equality is an essential dimension of inclusive agricultural and economic growth, food and nutrition security and resilient food systems. Women play an important role in agriculture and in most value chains. Yet their work often remains invisible or insufficiently supported. Gender inequalities limit women's development and have a negative impact on the economic performance of value chains.

In Mali, women are present to varying degrees throughout the agricultural value chain, participating in the processing and marketing of farm produce. In the agricultural sector, women are present throughout the agricultural chain, but they do not have the necessary means to increase their production and productivity (lack of access to and control and ownership of land and means of production, etc.). They are generally involved in small-scale livestock farming and all the activities involved in processing and marketing agricultural products. Their role as producers in their own right is not always valued or accounted for. Women are mainly involved in growing food crops and market gardening for the family's subsistence. The challenge of food security cannot be met without counting on their contribution to food production or seeking to improve their access to land.

Also, despite the non-discriminatory texts, in practice, the influence of men dominates in terms of leadership. Decisions are generally taken by men in public and women are only consulted when necessary.The results of the gender and value chain analysis are shown in Table 3 below:

Table 3. Results of the gender and value chain analysis

Theme	Analysis results
Opportunities for women and young people in value chains from production to marketing	In the regions of Mopti, Gao, Timbuktu, Segou and Koulikoro, as far as agricultural and forestry production is concerned, there is great agro-ecological diversity, reflected in agricultural production systems ranging from cereal-based systems to horticultural products and NTFPs, from the south to the north, via the centre. There is also considerable potential for agricultural land and a young population that is predominantly agricultural. Specifically for cereals, there are : - A high level of consumption of agricultural products, particularly rice, millet, sorghum, etc. - Existence of approved input suppliers at local level. - There is a large amount of irrigable and arable land in each of these regions, including the Office du Niger zone in Ségou, the rice offices in Mopti and Ségou, and the Baguinéda irrigated perimeter office (Koulikoro), etc. - Agricultural development investment projects and programmes in the 5 regions.

[34] Fowowe B. 2022. Financial inclusion, gender gaps and agricultural productivity in Mali. Available here.
[35] **Source**: Regional consultations and Albarka Gender Analysis

Theme	Analysis results
	- Young people and women are the main players, particularly in the workforce. - Existence of a national development policy for agriculture. - Available from government research departments. Non-timber forest products are seasonal and are harvested almost all year round. There are balanites in Mopti, shea, baobab and wild jujube in Ségou, Koulikoro and Mopti, and zabban in Segou and Koulikoro. **For processing** : - A wide range of products processed from rice, millet, sorghum (couscous, crispy croquettes, cream) and PNFL (Zabban juice, baobob, shea butter, etc.). - Existence of trade fairs for processed products - Availability and existence of a large number of consumer by-products **For marketing**: there are local assembly markets with popular weekly fairs, wholesale and institutional markets (OPAM and WFP) and other TFPs for emergency food distributions. The introduction by TFPs of cereal exchanges in several regions in the 5 regions. These include the USAID Feed the Future scholarships and the national scholarships organised by AMASSA Afrique.
Opportunities for women and young people to benefit from technical capacity-building in agriculture and agroforestry.	With regard to farming and forestry in the 5 regions, it should be noted that women and young people benefit sporadically from technical capacity building in agriculture and agroforestry. At this level, it should be noted that there is a shortage of staff in the technical services of the State's Agriculture and Water and Forestry Departments to ensure better coverage for capacity building for women and young people on farming and agroforestry techniques. According to the DRA -Mopti's report from May to September 2017 on the progress of the agricultural campaign in the Mopti region "*The number of staff is 146 managers and agents available out of a need for 317, i.e. a coverage rate of 46%. There are not enough staff to actively carry out the State's urgent and demanding regalian tasks in the 2,083 villages in the region's 108 communes. The number of plant health control officers is 13.* It should also be noted that capacity building is generally carried out at the discontinuous pace of projects and programmes financed by TFPs in these areas. They are carried out with limited numbers of women and young people and in the traditional way for theory and practice on a demonstration plot. These capacity-building sessions last 2 to 3 days maximum. The participants in the capacity-building sessions give feedback to the remaining women and young people. It appears that the women and young people who give the feedback are not very good at passing on the information received during the training sessions.
Access for women and young people to market activities and infrastructure linked to agriculture and agroforestry	Women and young people will be given access to market activities and infrastructure related to agriculture and agroforestry through : - Encouraging and supporting women's and young people's entrepreneurship projects in all links of the various value chains - Prioritising the installation of women and young people on developed plots of land, as recommended by the Agricultural Orientation Law and the Agricultural Land Law;

Theme	Analysis results
	- Prioritising the selection and support (technical and financial) of women's and young people's entrepreneurship projects (individual or in organisations) in different links of the chains; - Ongoing lobbying of traditional authorities to encourage young people and women to set up rain-fed farming plots; - Raising awareness and advocating greater participation by women and young people in the decision-making bodies of farmers' organisations. - Facilitating their access to markets by providing support for tricycles
Constraints, opportunities and increased employment opportunities and improved employment options along agricultural value chains for women and young people	**In terms of constraints**, we can list low decision-making capacity and power; access to productive resources for women and young people, such as land, finance, agricultural inputs, equipment and materials; domestic and family responsibilities that limit time and mobility, especially for women. **Opportunities** include **:** - The over-representation of women and young people in informal employment is an asset for increasing the employment options of these groups. - The presence of women and young people in all segments (from production to marketing) is an opportunity, even if they benefit less from marketing.
Proposals for better employment of women and young people in value chains	You will need : - Facilitate their access to and control over resources such as land, financing, agricultural inputs and equipment - Promote and develop women's and young people's entrepreneurship by setting up new types of producers, processors and holders of resilient agri-businesses based on innovation (use of green energy) by women and young people specialised in their areas of activity and operating with positive economic and financial results. - Provide these young companies with better ongoing training, accompanied by experience-sharing visits, fair and appropriate access to resources and funding, and simplified, realistic business plans. - Facilitate intermediation between these young companies, input suppliers and potential buyers
Constraints and opportunities associated with the increased adoption of climate-smart agricultural and agroforestry technologies by women and young people	**The constraints** are many and varied, as a large proportion (around 90%) of young people, especially women, do not have the appropriate skills and techniques to adopt agricultural and agroforestry technologies. This is due to poor or non-existent technical training on the technologies, lack of information on the techniques and, above all, lack of access to improved seeds adapted to the climate and lack of equipment to enable them to adopt the techniques more widely. In addition to these constraints, there is a low level of literacy among farmers, particularly women (19%)[36] and insufficient knowledge of environmental issues. As far as **opportunities are concerned**, the five regions have a wide range of agro-ecological diversity, reflected in agricultural production systems ranging from basic systems for the production of cereals and horticultural products from the south to the north, via the centre of the country. It should also be

[36] **Source**: REPORT Demographic and Health Survey EDSM-VI 2019

Theme	Analysis results
	added that there are technical research and supervision services available in the 5 regions, most of which have carried out major research in the target areas, and the implementation of their recommendations could significantly improve production in the face of the climate. The efforts of the TFPs to give vulnerable people (young people and women) access to and adoption of technologies are noteworthy. We can cite the "green innovation" project funded by GIZ, and Feed the Future funded by USAID, which are all working to increase the adoption of technologies.
Steps to be taken to increase the adoption of climate-smart agricultural and agroforestry technologies by women and young people?	*These include* - Fair and secure access to land resources - an important factor in production and the application of technology - Developing mechanisms for adapting to climate change by promoting appropriate and ongoing training in the various technologies. - Providing information on climate, agriculture and agroforestry - Access to the necessary tools, in particular improved seeds - Exploiting water resources in an integrated and equitable manner by promoting and using renewable energies (solar panels, wind turbines) so that production is not solely dependent on rainfall.
Constraints and opportunities for women's and young people's access to knowledge about agricultural climate risks, CCA and CCM planning, and best practice in agriculture and agroforestry through new and existing climate-smart agroforestry institutions.	"Climate change is one of the major challenges facing the socio-economic development of all countries today. Mali is particularly at risk because of its low income, weak human capital and economic vulnerability. Various studies carried out on climate change (CC) in Mali have shown that the main climatic challenges facing the country are droughts, floods, poor rainfall, violent winds and sharp variations in temperature. These climate changes threaten first and foremost the primary sector (agriculture, livestock farming, fishing) and the exploitation of forests, key sectors of the country's economy. On the one hand, women are victims of the harmful effects of climate change, but on the other, some women's activities" [37] There are a number of **constraints,** including waves of drought, irregular rainfall and rising temperatures. Added to this is deforestation due to massive logging, soil degradation and a shortage of agricultural land, all of which have an impact on productivity and food security.
Proposal for better access for women and young people to knowledge about agricultural climate risks, CCA and CCM planning, and best practices in agriculture and agroforestry through new and existing institutions.	First of all, there needs to be training, information and awareness-raising on the effects and consequences of climate change, along with the various techniques for adapting to each risk, effect or consequence. In addition, making climate information, agriculture and agroforestry available. Conducting experience-sharing visits based on tangible scenarios with climate-smart villages using similar tools developed by research centres such as ICRISAT. Then, granting and/or guiding women and young people towards equipment and tools enabling intelligent production, such as soilless gardens.
Constraints to the participation/emergence of women in the value chain	It is well known that women often work in the least valued links in value chains, e.g. as homeworkers or in informal work more generally. Women tend to be underpaid and their (informal) jobs are more precarious. In the five rural areas, women are often not visible, even though they carry out a large proportion of agricultural activities.

[37] **Source**: FAO and ECOWAS Commission, Profil National Genre des Secteurs de l'Agriculture et du Développement Rural - Mali. Gender Country Assessment Series, Bamako, Mali Fatoumata Alidou GARÉKA MALLÉ, 2018

Theme	Analysis results
	In addition to social and traditional constraints, factual constraints, in particular access to productive resources (land, goods, agricultural services, financial services) and capacity and decision-making power (capacity, self-confidence, decision-making power), are holding back the emergence of women in the value chain.
Where are the men and women among the producers, retailers, wholesalers and processors? Who controls the marketing channels?	In the case of cereals, men are more than 90% present at all levels, unlike market garden produce, where women are more than 95% visible. When it comes to forestry products, particularly harvested products, women outnumber men. In the case of citrus fruit, women and young people are present both as labourers and in sales. Women are also in the majority in processing. As far as marketing channels are concerned, women are almost in the majority when it comes to market garden produce, while men are more involved in cereals.
Access to equipment and infrastructure for post-harvest management, processing, etc.	According to the report Profil National Genre des Secteurs de l'Agriculture et du Développement Rural - Mali, access to equipment has a considerable impact on agricultural productivity and production. Given the relatively high cost of access, equipment is generally the property of the farm. Women's low incomes and limited access to agricultural credit mean that fewer of them own agricultural equipment and therefore have limited access to it. Of the women farmers who own equipment, 43% have ploughs, 35% donkey hoes and 10% carts. The other types of equipment are owned exclusively by farmers. Whatever the type of equipment, women's farms are always under-equipped compared with men's farms. The equipment on women's farms consists mainly of ploughs. Women's lack of equipment is a significant factor in their productivity. On their farms, women can only get the equipment when work is finished on the other plots, often causing delays in the cultivation calendar. Post-harvest management: Women do not have control over family production, as it is the men who produce (in some cases), store and remove bundles of millet from the granary. When it comes to managing food for the family, women in monogamous households are much more involved in decision-making. On the other hand, in polygamous households, it is usually the first woman to join the household who is consulted, or at least informed, about the decisions the husband wants to take in terms of food security. In agricultural households, it is the men who decide how the harvests are distributed. They make donations to certain members of the family and to needy neighbors towards whom they feel a duty of solidarity. In charge of preparing meals, the women must inform the men of the daily ration required and warn them if it is insufficient. If the stock is exhausted, it is the man who decides what food to buy, even if the purchase has to come from the woman's own resources (income from the sale of ruminants or from parallel activities). She only decides if he asks her opinion.
Controls production resources, equipment and infrastructure	The means of production, equipment and infrastructure are controlled by people.
Positioning of men and women among aggregators	Men are in the majority as aggregators, unlike women and young people
Positioning women and men in production	Men account for over 90% of cereal production, unlike market garden produce, where women account for over 95%. For citrus fruit, women and young people are present as labourers but not as producers. They are rarely involved in production.

Theme	Analysis results
Access to and control over resources and income for women and young people	Under Malian customary law, women are excluded from land ownership, even though they play a major role in land use. Owning certain assets, such as land and houses, can confer a degree of financial and social autonomy, and can also have a beneficial effect on households. For women, owning property can increase their autonomy, their level of income and even their purchasing power, as well as giving them access to decision-making bodies and protecting them in the event of marital breakdown. According to FAO statistics, between 1996 and 2007, only 3.1% of Malian farms were owned by women. What's more, women who generate income do not have full access to these resources. "... if women had the same access to these resources as men, they would produce 20-30% more food...". - FAO at work 2010-2011Although dominated by the Bambara ethnic group, the population of the Ségou and Koulikoro regions is made up of different ethnic groups spread throughout the territory. These groups are traditionally characterised by a strong social hierarchy in which women, as mothers and wives/housekeepers, are assigned roles that determine their access to productive resources, their participation in household decisions or economic and social opportunities. They are generally responsible for accompanying their family or husband in the exploitation and use of resources (natural, financial, productive, etc.), which creates a disparity that limits their access to resources in the same way as men. The same applies to the benefits and income derived from these resources. However, it is important to point out that in rural areas, the community or the men temporarily hand over small plots of land to women, especially married women, to allow them to work the land "usufructuously", with the aim of strengthening the ingredients of the condiment and generating a little income, most of which is used to maintain the family. These plots loaned to the women can be taken back at any time. According to the EDSM VI 2018 report, 19.9% of women in the Koulikoro region and 13.4% in the Ségou region own land alone. Those who own land jointly, i.e. with other people, are estimated at 12.2% in Koulikoro and 8.5% in Ségou. The young people participate in the family farm and part of the income from these farms can be used to help the young people with their expenses, such as the cost of the young man's wedding or the young woman's dowry, or part of the resources could be given to the young men to set up a business or go on an adventure.
Participation of women and young people in decision-making	According to information gathered in the Ségou and Mopti regions, decisions concerning the community are generally taken by the village chief and the village councillors, all of whom are men. Family decisions are taken by husbands or the head of the household. Young people and women have a consultative role and issue opinions, even if the decision often concerns them. "Despite the non-discriminatory texts, in practice, the influence of men dominates in terms of leadership. Decisions are generally taken by men in public and women are only consulted when necessary".
Legal status of women and young people	The Malian Constitution clearly sets out the principle of equality and non-discrimination. Article 2 stipulates that "All Malians are born and remain free and equal in rights and duties. Discrimination based on social origin, colour, language, race, sex, religion or political opinion is prohibited". In addition, Mali has ratified, often without reservation, several regional and international conventions and treaties relating to the protection and promotion of women's rights.

Theme	Analysis results
	However, the content of the texts has not yet been translated into practice, as the influence of men dominates at almost every level. This is reflected in the fact that decisions are taken by men, even though women and young people have an advisory role.
Access for women and young people to information on climate change, the environment, agriculture and agroforestry	Information on climate, the environment and agriculture is disseminated and accessed through a variety of channels, including radio, TV, newspapers and the internet. Rural communities in the Ségou and Koulikoro regions also use radio and TV. Newspapers and the internet are generally accessible in urban areas in addition to radio and TV, and this is explained by the level of education, the means for smart phones with internet and also the source of energy. Information is accessible by telephone through Orange-Mali's "SENEKELA", but this is not widely used by the population. In Mali, overall[38] , women are less exposed to the media. In fact, 39% of women do not use any of the three media - newspapers, radio or television - for information, compared to 31% of men. It is worth noting that radio is the most widely used medium: 47% of women and 54% of men said they listened to the radio at least once a week. Furthermore, 2% of women and 5% of men are exposed to all three media at least once a week. EDSM VI 2018 report. In the Koulikoro and Ségou regions, 36.6% and 43.8% of women respectively do not use any media at least once a week, with a rate of 20.0% for urban areas and 46.1% for rural areas. For men in the same areas, 28.9% and 27.4% respectively use no media at least once a week, with a rate of 14.0% for urban areas and 37.4% for rural areas. In the light of these data, we can deduce that women's and young people's access to information on the climate, the environment, agriculture and agroforestry is relatively low compared with men. Moreover, young people, especially women, do not always have access to sources of information such as radio, television, newspapers or magazines, which are the main source of information on climate change and the measures to be taken. Without information on climate change or adaptation options, women become vulnerable to the impacts of climate change. The National Climate Change Adaptation Plan lists the main social groups vulnerable to climate change as women, young people, the elderly, the disabled, orphans and widows.

4.6. Power and decision-making.

In Mali, social norms and the sexual division of labour largely exclude women from public decision-making and conflict resolution. The representation[39] of women in all the Institutions saw a timid increase of 0.6 points between 2014 and 2015. In 2023, women accounted for 28.5% of MPs; women party leaders accounted for just 2%. Similarly, female farm managers have less access to farmers' organisations: 17% of them belong to a farmers' organisation compared with 32% of men.

The proportion of women councillors was 8.3% in the 2009 elections, 1.9 percentage points higher than in 2004 and 4.3 percentage points higher than in 1999.

In 2020, this rate rose sharply to 25.5%. This result is linked to Law No. 2015-052, which obliges political parties to include one-third women on proposed electoral lists. This obligation was met, and out of 79,238 candidates, there were 26,436 women, i.e. 33.4%[40] . The Conseil National de la

[38] Source: EDSM VI 2018 report
[39] **Source**: Report on the representativeness of women in positions of responsibility in the Public Administration_2015
[40] **Source**: Election report/synthesis, 2016

Transition (CNT) has twenty-eight (28) women out of one hundred and forty-seven (147) members, i.e. 28.57%. The current transitional government has only 6 women out of 29 members, i.e. 20.68%. Out of nineteen (19) governors, only one is a woman, i.e. 5.26%.

The causes of women's low representation in decision-making bodies are: (i) political will, (ii) negative social perception of women politicians, (iii) women's low political maturity, their low level of education and literacy, (iv) women's exclusion from decision-making bodies and structures that grant economic or development resources at all levels. With regard to decision-making, according to, the EDSM - VI 2018, less than a third of women (28%) participate in the decision for visits to their family or relatives, 20% participate in the decision for major household purchases and 20% for their own health caré. Only 10% participate in all three decisions and 63% declared not to participate in any of the three decisions mentioned above.

The 2.36% representation rate[41] of women in conflict management bodies (CSA, DDR, MOC, CRSS, special advisers to Governors, interim authorities) bears this out. At the level of the transitional justice process, efforts to take gender into account are visible, in particular at the level of the Truth, Justice and Reconciliation Commission, which has created a sub-committee on gender and adopted a work plan enabling the systematic integration of gender issues into all four pillars of the CVJR's work. The CVJR has also been the most representative peace mechanism in terms of women's participation, with 5 women Commissioners out of 25 (20%) until the appointment of women to the CSA in 2020. This consideration of gender is also reflected in the large number of statements made by women (50%)[42] .

According to the results of regional and national consultations, it should be noted that although older women are highly regarded, their role is generally limited to mediating family disputes, passing on social values and bringing up children. With the men's agreement, women can temporarily farm small plots of land. Among livestock farmers, women have the right to use income from the sale of milk. In both cases, they have little say in the management of public affairs and conflicts.

In the areas covered by the gender analysis, even when women are involved in exploiting and developing the land (market gardening and reforestation), they cannot own it, even though they have a great deal of knowledge about natural resources, the impact of socio-climatic changes and ideas to put forward for better management. At community level, there are many village groups, often made up of married and elderly women. However, they have no direct access to the customary chiefs in the traditional institutions where decisions are taken. Until the outbreak of communal violence, these groups were mixed, made up mainly of Dogon and Peulh women.

Despite the presence of women in the communal councils of the five regions covered by the IAAT project, the power relationship has not changed overall, and women's priorities relating to the management of natural resources or conflicts are still little taken into account in public decisions and local development plans. These priorities relate in particular to improving the status of women in terms of access to land management and ownership, which remains a highly discriminatory area for women. Gender relations have begun to change in Mopti, Ségou, Gao and Timbuktu as a result of the insecurity, accentuating women's dependence on the men in their families to protect them from violence. This situation can lead some women to encourage young men to join armed groups, for protection against other armed groups or the FDS, whom they do not trust. Where male family members are absent due to conflict or migration, women have to provide for their families. Many have become heads of household and income earners, particularly by developing economic activities. They have demonstrated their resilience by finding endogenous solutions to meet their needs. At the same time, women's real reproductive role and their contribution to economic

[41] **Source**: Data, Bulletin Statistique, La Femme et l'Enfant en chiffres au Mali, CNDIFE, 2018, updated.
[42] UNDP Prosmed project document

production and family management remain unrecognised and underestimated by the community, which accentuates their marginalisation and exclusion from conflict prevention and management. In addition to their resilience in providing for their families, a large proportion of women's occupations involve domestic tasks such as caring for members of the household, which are often not quantified in the various contributions made by women to economic activities in the communes targeted by IAAT.

4.7. Availability and use of gender indicators

To improve the profitability of the work of rural women active in key production sectors (agriculture, livestock, fisheries and the environment), PNG-Mali has included gender indicators in its action plan:

- Percentage of women with access to developed land,
- Percentage of women and young people who have benefited from funding from the National Agricultural Support Fund,
- Percentage of women who have applied modern techniques for the production, processing and sustainable conservation of agricultural, livestock and fishery products,
- Percentage of women trained and supervised in sustainable plant production and regeneration techniques,
- Productivity rate of women who have benefited from inputs in areas at high risk of natural disasters,
- Women's rate of access to financial resources and factors of production.

These indicators set out in the NGP-Mali should serve as a reference for the various stakeholders, particularly the IAAT project.

4.8. Rights and legal status.

4.8.1. How women and men are treated in customary and formal legal codes and by judicial systems

Discussions in groups and with key informants show that women and men are viewed differently by customary and formal legal codes, as well as by judicial systems. Details of these differences can be found in Table 4.

Table 4. How women and men are treated in customary and formal legal codes and by judicial systems[43]

Question	Answers	
How women and men are viewed and treated by :	Formal legal codes	- There is a difference between men and women in the legal code: - According to the marriage code, the man is the head of the family. - It's the man who chooses where to live. - The child takes his father's surname.
	Customary law	- In the legal and customary framework, women do not have the same right to land as men. Even though the law provides for 15% of the land developed to be allocated to women and. young people - Women are an inheritance. - There is a difference in access to employment, due to women's low level of education.
	Religious	Religion treats men and women differently.
	Coutumiers Inheritance	Women themselves are an inheritance.
	Religious heirs	The wife receives a third of the inheritance.
	Formal legal	The man does not benefit from the advantages of the death of his civil servant wife (death benefit and pension).
	Employment	There is a difference in access to employment, due to women's low level of education.

As can be seen from Table 5, formal legal codes treat men and women in the same way, without distinction, unlike customary codes, which treat men and women differently. As for the expiation of wrongs and legal representation, the legal provisions prevail and no distinction is made between men and women. Even though there is often discrimination against women, for example if a woman applies for a divorce, if the judge or magistrate is a man, he tends to incriminate the woman without trying to understand the reasons why the woman applied for a divorce44 . In this respect, it is also worth noting that formal legal codes straddle religious codes, so that when the woman concerned is a Muslim, for example, Islamic law prevails.

4.8.2. Assessment of documents issued by the State, such as identity cards, registration on the electoral rolls and title deeds

- **Identity cards**: in accordance with current legislation, the conditions for obtaining identity cards are identical for men and women. But the reality is quite different for women. In rural areas, in most cases, the birth certificates used to issue identity cards are held by men, and women have to fight hard to obtain them. The men equate women's need to obtain a card with women leaving the marital home. In some cases, women do not have the resources to obtain identity documents, and often it is the procedures for obtaining cards that they are unfamiliar with.
- **Voter registration**: the difficulties women have in registering to vote are due to illiteracy, socio-cultural constraints, refusal by husbands, etc...

[43] Based on discussions with those responsible for promoting the advancement of women
[44] **Source**: women from the Ministry of Justice

- **Property titles**: in rural areas, women have difficulty gaining access to property titles. The main reasons for this are lack of financial resources and customary management of property. For example, according to the household survey conducted by the Mopti Planning Department, **only 9.7% of 100 farm managers are women.**

4.9. Effects of climate change, consequences and categories of people affected

Table 5. Effects of climate change, consequences and people affected

Effects of Climate Change	Consequences	Categories of people affected
Land-related - Velocity of withdrawal of land from rightful owners - Access to land	▪ Weakening of the social fabric and lack of trust between men and women in the community ▪ Weakening of the social fabric and lack of trust between men and women in the community ▪ Exodus ▪ Banditry/crime/theft ▪ Decline in production ▪ Family abandonment ▪ Forced divorce ▪ Widowhood ▪ Female head of family after the man leaves; ▪ Family break-up ▪ Prostitution	- Communities, Families - Communities and families - Young girls and boys - Young boys - Households/families - Women - Women - Women - Women - Families - Women and girls
Linked to livestock farming	▪ Weakening of the social fabric and lack of trust between men and women in the community.	Communities and families
Access to agricultural residues	▪ Weakening of the social fabric and lack of trust between men and women in the community.	Communities and families
Fisheries resources	▪ Tension between users	Men, women, girls and boys

Table 5 above shows that women, like girls and boys, are the most affected by the effects of climate change.

4.10. The needs and interests of women and men, girls and boys

Taking specific account of gender in socio-economic development is legitimate and fully complies with the provisions of Mali's National Gender Policy (PNG Mali) drawn up in 2011. For the purposes of this analysis, the proposals made in support of those affected by gender are set out in **Table 6**.

Table 6 . Differentiated needs and interests of women, men, girls and boys[45]

Gender actors	Needs/interest
Women	- Training/awareness-raising on gender and human rights - Training/awareness-raising on climate change, natural resource management and cultural techniques - Training in the maintenance of materials and equipment granted - Financial support - Support for IGAs (small livestock/small ruminants, market gardening, petty trading, literacy, soap-making, dyeing, etc.), - Trades (cutting and sewing, dyeing, etc.), - Reforestation, - Agricultural inputs.
Men	- Small business, - Beef fattening.
Young girls	- Training in the trade (dyeing, soap-making, petty trading, etc.), - Crafts (cutting and sewing).
Young boys	- Small-scale livestock farming / fattening small and large ruminants, fodder production, - Trades (metal and wood joinery), - Agricultural equipment, - Agricultural inputs.
Vulnerable groups, in particular female heads of household, including people with disabilities	- Small business, - Small-scale breeding, - Agricultural inputs, - Means of transport.

4.11. Gender and COVID - 19

The COVID-19 pandemic, which was officially declared in Mali in March 2020, has claimed 461 victims, including 138 women in the Mopti region[46] . COVID-19 has further weakened the most vulnerable sectors of society, namely women and girls. In particular, those who sold food in markets, near offices and in schools saw their incomes fall and were trapped in their day-to-day activities, in most cases taking no appropriate measures to protect themselves. Women's daily informal activities in urban centres, even in conflict zones, have put them at even greater risk. The configuration of Malian society meant that it was they who were in constant contact with other members of the community, so they ran a greater risk of being contaminated and of transmitting the disease to the rest of their family. In times of conflict, the pandemic has only increased the vulnerability of women, who have been deprived of housing and whose means of subsistence have been undermined, and who can only suffer from the arrival of a pandemic on the scale of COVID-19.

A study commissioned by the UNFPA highlighted the fact that cohabitation could have led to acts of violence within households. Inactivity and lack of income have also been identified as factors leading to GBV in Mali. It should be noted that men's dominant position within the household, due

[45] Based on discussions during regional consultations
[46] **Source:** DRPSIAP - Mopti

to their financial situation, has been affected and has caused stress and irritation[47] . The closure of international borders and market restrictions have had a major impact on informal traders. Also, with the pandemic, women suffered more gender-based violence due to general social stress combined with increasing tensions surrounding the fact that the family lived in unusual proximity, in addition to limited access to food and basic supplies. The informal social safety nets and networks on which many women had previously depended for support were now weakened due to reduced physical mobility and social remoteness. Misinformation was easier to access than official information. People relied heavily on traditional and miracle healers and medicines, and rumours spread more quickly than official information from the relevant authorities in whom communities had no confidence. In addition, the broadcasts sharing information were generally shared at times when women were carrying out domestic work such as cooking or fetching water. Curfews and restrictive measures further penalised women.

In remote areas, they did not have access either to good information or to materials enabling them to protect themselves against disease. Women's work schedules were increased by the need to look after children who no longer went to school[48] . The pandemic also led to increased spending on health kits (masks, bleach, soap, hand-washing buckets, etc.). With the outbreak of the COVID-19 pandemic, it was also noted that: (i) access to basic social services had become problematic since the outbreak of COVID-19, particularly health services, GBV management services and education services; (ii) schools were closed in order to control the spread of COVID-19. As a result, the 2019-2020 school year, which was already disrupted by teachers' strikes, came close to being a blank year; (iii) Access to healthcare services was the most problematic aspect of the pandemic. Unprepared for the pandemic, like other public sectors, the health sector offered little security to users and health workers alike. The quality of the delivery of healthcare services during the pandemic period by professionals in the sector suggested that they were at the same level of psychological preparation as the general public. Fear of infection and the lack of specific protective measures for health workers, to name but a few, meant that users did not see health services as a 1er recourse; (iv) Access to drinking water was quite problematic during the pandemic period, as women and girls had to travel to obtain supplies. This might seem insignificant in the big cities, where the source of drinking water was available in the family's backyard, but in rural areas, they had to walk several metres, if not kilometres, to obtain drinking water. And these water points represented potential sources of contamination, given that masks were not worn at all, and other distancing measures were not respected.

The needs and interests of women and girls (and people with disabilities) are neither represented nor specifically taken into account in the planning and response to the pandemic. In terms of governance, women and girls were consulted very little, either on the intervention plans or on the interventions relating to COVID-19, in particular on the economic, health, sanitation and hygiene measures aimed at reducing the risk of gender-based violence (GBV). As far as the level of information is concerned, many people, especially women, knew that there were health measures in place, but did not have precise and clear information about the regulations on confinement and health. They were not aware of the care facilities, let alone the care protocol. The telephone and local radio stations were effective means of communication for reaching the most vulnerable women and girls, but e-mail was not within their reach. A number of actions were taken, including the involvement of women in community dialogue via social networks and WhatsApp groups, and the creation of **income-generating activities** (IGAs) to empower them. As a result, women have been able to assess the scale of COVID-19 and become more resilient to this pandemic.

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[47] United Nations, UNFPA: A gender perspective, protecting sexual and reproductive health and rights and promoting gender equality, Technical Briefing, March 2020 https://plateforme-elsa.org/wp-content/uploads/2020/04/French.COVID-19_A_Gender_Lens_Guidance_Note_edits_clean_file_0.pdf

[48] My own observations

4.12. Violence against women, combating female circumcision

Analysis of the EDSM-V data shows that the Malian population is lax and insensitive when it comes to domestic violence. There is still no specific law on violence against women (domestic violence, sexual harassment, forced and/or early marriages, excision). The rationalisation of these acts of violence and the social pressure on the victims explain the widespread impunity enjoyed by the perpetrators. Excision is almost universally practised in Mali, affecting 91% of women aged between 15 and 49. In terms of combating the practice, the National Programme to Combat Excision (PNLE) is an important achievement. However, like some neighbouring countries, Mali has not yet passed legislation banning female genital mutilation.

The scale of gender-based violence (GBV) has been exacerbated by the political, institutional and security crises that began in 2012, the persistence of inter-community conflicts, and the atrocities of armed groups and violent extremism. According to the MICS 2015, female genital mutilation is the most widespread form of GBV in Mali (83% of women aged between 15 and 49) and 48.8% of girls were married before the age of 18. GBVIMS recorded 4,617 incidents in 2019, 4,411 incidents were recorded from January to September 2020. According to the Regional Directorates for the Promotion of Women, Children and the Family, in Ségou, 126 cases of GBV were recorded in 2022, in Koulikoro 3,419 cases including Dioila and Nara, and in Timbuktu 88 cases during the same period.

These figures only partially reflect the real situation of GBV in the country, given the weakness of the judicial system and impunity. Women have very little access to justice services (4%), unlike men (9%), and family pressure often prevents them from speaking out about the violence they have suffered. A draft special law drawn up and validated in 2017 by a national multi-sectoral and multi-party committee under the leadership of the MPFEF is still in the process of being adopted internally at executive level.

4.13. Conflict prevention and management mechanisms

As far as women's involvement in conflict resolution is concerned, conflict resolution and prevention issues in Mali are still managed almost exclusively by men in the communities and are therefore highly discriminatory for women, even beyond any religious or cultural considerations, despite the fact that they have proven potential and capabilities in conflict resolution and prevention and, more generally, in peace-building. Since 2012, women have accounted for less than 10% of peace negotiators, and this figure has hardly changed since the adoption of resolution 1325 in 2000. Through participatory approaches that value women's contribution, this IAAT project will promote behavioural change towards a culture of inclusiveness in formal and informal conflict resolution and prevention mechanisms linked to the local level, as well as improved access for women to sustainable and resilient economic empowerment opportunities. In addition to the contestation of local leaders, opinion leaders and religious leaders, the exclusion and discrimination of women and ignorance of their role in revitalising conflict management mechanisms exacerbates women's vulnerability. IAAT presents an opportunity to reverse this trend by relying on women who have proven skills in social mediation and by strengthening their role in family and community units in matters of social cohesion and conflict management. Past experience, particularly with peace huts in Mali, has shown that if women are mobilised and involved, this can have a positive impact on conflict prevention and resolution at community level, especially as they themselves can play an active role in triggering these conflicts. The translation of Law 052 of December 2015 and its dissemination in the IAAT project areas should also be considered.

In all the areas covered by the gender analysis, prevention and management mechanisms were found to exist[49] :

Legislative provisions

- **The State judicial system**: it operates according to legal rules, it is the law, and is not very active in prevention.
- **Land Commissions (CoFo)**: these are an emanation of the State (cf. Agricultural Land Law - LFA, N°2017-001 of 11 April 2017); they are set up at the level of all communes. **Village CoFo are not set up but village chiefs are members of communal CoFo**. The communal CoFos are not yet operating satisfactorily, because they are not yet well known at village level (need for information and awareness-raising), and at communal level because they are not responsible for their operation (need for awareness-raising and responsibility for their operation). Women and young people are poorly represented in these CoFos. But with the support of certain development partners, this trend is changing.
- Decree No. 2017-0367/PM-RM of 28 April 2017, as amended, establishing the national reconciliation support mission (MARN) within the ministry responsible for national reconciliation. Article 3 of this decree provides for the establishment of regional reconciliation support teams in all regions with the following missions: to inform and raise awareness among the population about the peace process; to identify mediation and conflict management mechanisms; to prevent, mediate and manage intra- and inter-community conflicts; and to support reconciliation at local level. As of 31.12.2020, the following PRRAs are partially operational: Mopti, Ségou, Gao, Timbuktu and Koulikoro.
- Women and young people are poorly or not at all represented in the PRRAs in the regions of Ségou, Koulikoro, Mopti, Gao and Timbuktu.
- **The** RCCs are set up by the PRRAs in the municipalities. Their purpose is to prevent and resolve, as close as possible to the complex realities on the ground, sources of conflict and the dynamics of strengthening social cohesion. They should only be set up if there is not already a functioning reconciliation mechanism in the area. There is also a lack of women's participation in the communal reconciliation committees.
- **The watchdog committees**, which are usually set up by certain development actors, are not operational. Women have said that they are represented on these committees. For example, Bankass has one (1), and Teninkou has a few women.
- **Local authorities (mayors and other elected representatives)**: born out of the political reform of the State, they settle many disputes at local level. However, in some localities, the conflict resolution actions of this player are often condemned by the judiciary.
- **Village committees**: these are set up by the communities, and most of them function well in several localities. This type of organisation manages community affairs: forest surveillance committee, water point management committee, dispute management committee.

As for the successes and failures of these schemes, they are varied

The main success lies in the amicable settlement of conflicts at village level (action by the village chiefdom, with the village chief being a member of the communal CoFo). The shortcomings noted were: (i) insufficient financial and logistical resources for travel; (ii) poor understanding of the roles of the members of the watch committees, with communities wondering why the watch committees? The consequences of failure are generally the cause of social disintegration and family poverty.

[49] Regional consultations

4.14. The situation of displaced persons

With the security situation, women and girls are becoming increasingly vulnerable. Many women have become heads of household (the men have either been killed, abducted or fled) and are left to fend for themselves. The displaced are scattered in camps, in families and in houses under construction. At national level, for example, there are 412,387 displaced returnees and 120,298 internally displaced persons, 224,124 of whom are women, i.e. 54.34% of women[50] . This category of vulnerable people should be taken into account in the IAAT project.

Table 7. Situation of displaced persons in the analysis regions

N°	Regions	Total number of displaced persons - es	Number of women	Rate for women
1	Mopti	84 581	45 534	53,38%
2	Gao	61 435	35 609	57,96%
3	Timbuktu	52 153	28 599	54,83%
4	Ségou	35 295	19 682	55,76%
5	Koulikoro	4 241	2 031	47, 88%

Gao stands out from the other regions in terms of the number of displaced women, followed by Ségou and Timbuktu, as shown in table 7. As a vulnerable, marginalised group, the project will have to pay particular attention to these displaced women, especially in terms of empowering them and strengthening their resilience.

4.15. Specific and relevant knowledge, skills and experience of women as natural resource users and producers[51]

The gender analysis reveals that women have specific and relevant knowledge, skills and experience as users of natural resources and as producers.

- **As users**: they have knowledge, skills and experience in plant production and reforestation. They also have knowledge of how to raise awareness among communities, women and young people about natural resource management issues. As leaders and members of monitoring committees, they ensure the rational management of natural resources.
- **As producers**: they play a key role in the development, processing and marketing of harvested products.

4.16. Identification of stakeholders

4.16.1 Identification of existing women's groups in the project's target areas and their respective capacities, with a view to strengthening their ability to participate in climate change resilience activities[52] .

The Regional Directorates for the Promotion of Women, Children and the Family in the gender analysis regions have inventoried 102 associations/groupings in Ségou, 255 in Koulikoro, plus 251 in Mopti, 1,700 in Gao and 455 in Timbuktu. Despite the lack of material and financial resources, these associations have the capacity to mobilise women and raise their awareness of climate change. The leaders of some associations are also ambassadors for peace and can be put to good use in promoting women in value chains, climate security and the sustainable management of

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50 **Source**: DNDS, Population Monitoring Matrix Report (DTM)_ April 2022

51 Regional consultations

52 Regional consultations

natural resources. They are also involved in self-help and support for women, socio-economic development and the fight against poverty. To this end, an organisational audit of these groups and associations is more than necessary.

4.16.2. Institutions or leaders supporting women

Table 8: Institutions and leaders allied with and able to support women[53]

Institutions or leaders supporting women	Actions actually carried out	Capacity-building needs of leaders supporting women
Customary chiefs of neighbourhoods and villages	- Support and guidance for women, - Loan/grant of agricultural land, - Mobilising women.	- Human rights training. - Training on the texts, ratified conventions and commitments made by Mali in the area of women's rights.
Muslim leaders (High Islamic Council of Mali)	- Preaching on women's rights	- Knowledge of the PPG project, its objectives, results, targets and area of intervention.
Leaders of the Catholic Church	- Informing and raising community awareness of the roles and responsibilities of women in society and in the management of natural resources, - Supporting and mobilising women.	- Human rights training.
The imams	- Informing and raising community awareness of the roles and responsibilities of women in society and in the management of natural resources, - Entrepreneurial freedom for women.	- Training on climate change.
Pastors	- Informing the community and raising awareness of the role of women in society.	
Managers of local associations	- Mobilising women; - Raising women's awareness of the issue of climate change.	
The women presidents	- Mobilising women to learn about their rights and IGAs, and about the issues of natural resource management and conflict prevention and management, - Ambassadors for peace.	- Training on the challenges of climate security and natural resource management.
Elected representatives	- Raising awareness and providing information on issues such as early marriage and female circumcision.	
RECOTRADE	- Communication to strengthen social cohesion and preserve nature.	

According to Table 9 above, the coordination of neighborhood and village chiefs, the High Islamic Council, the Catholic and Protestant churches, youth leaders and traditional communicators all support the women's cause. The table above also gives details of what is being done by these various actors. The IAAT project will have to consider each category of actor as a strategic ally, and to do this, each actor will have to benefit from a reinforcement and support programme based

[53] The result of regional consultations

on the needs identified above, so that they are in a position to fully play their roles as allies and supporters.

4.17. Constraints faced by women in pursuing their practical and strategic gender interests [54]

The obstacles facing women identified by the analysis are:

- The coexistence of modern and customary law, which enshrines the hierarchy of the sexes and gender stereotypes (social roles attributed to each sex);
- Female illiteracy,
- Women's lack of access to sources of information on climate change and the measures that need to be taken,
- Low awareness among women's associations of the need to adapt to the effects of climate change,
- Women's poor knowledge of sustainable management and rational use of natural resources,
- Women's low resilience to climate change,
- The low representation of women in conflict prevention and management frameworks and mechanisms,
- Customs and traditions,
- Socio-cultural and religious constraints,
- Women's lack of self-confidence / Women's fear of commitment.

5. Recommendations for the IAAT project

The best approaches for taking action to include and empower women and girls include:

1. **Increasing access to resources needed for climate adaptation and mitigation.** The gender assessment shows that women in the project locations have less access to finance, technologies, and knowledge for better management of climate change-related risks. IAAT needs to support them to increase women's access to finance by increasing their involvement in VSLAs, linking them to financial institutions, training them on financial management, and providing knowledge on financial resources available to them. IAAT should also strongly advocate community leaders, including elected representatives, traditional chiefs, religious leaders, and administrative authorities, for greater women's access to and control of resources and benefits by bringing them into the CACs process. Gender equality must be implied while selecting beneficiaries and distributing project resources such as CSA technologies (e.g. solar pumps, biodigesters, improved seeds/breeds, etc.) and agroforestry inputs (e.g. planning materials and equipment).
2. **Creating space for women:** gender assessment shows that women have low decision-making roles in choosing production inputs and marketing activities compared to men. In addition, women's involvement in entrepreneurship is very low. IAAT's capacity-building activities can strengthen women's capabilities by increasing their access to information (both market and climate information), and building entrepreneurship skills for the selected businesses (such as value addition and post-harvest management, market gardens, agro-forestry products, etc.). Creating space for women in business environments across the

[54] Results of group work during national and regional consultations

agriculture and agroforestry value chain can improve their decision-making ability at the household and community levels.

3. **Linking with the private sector:** our assessment indicates that men are more engaged with the private sector service providers such as agriculture product collectors and aggregators and input suppliers (e.g. seed, feed, animal health services, and tools/equipment). Equal access and engagement by men and women with the private sector service provider can increase their participation in the existing market and new value chains. IAAT's should facilitate linking women with the private sector through integration into the value chains as key actors.
4. **Access to extension services:** Women's access to extension services as well as their involvement in extension service provision is limited. IAAT can develop a gender-sensitive CSA curriculum to integrate into the existing agriculture extension system (for both public and private). The gender-sensitive CSA curriculum can focus on gender rôle in input use, production and marketing in agriculture and agroforestry sectors. Currently, women's involvement in agriculture extension services is very low in Mali due to their low education rate and socio-cultural barriers as discussed in this gender assessment report. In the training and capacity building of extension service providers, women should be included as much as possible. This will help IAAT to reach more women farmers through women extension officers in Mali's cultural setting where women are less engaged with men extension officers.
5. **Empowering women in early warnings and climate action:** In Mali, women are placed at greater risk through a lack of timely and relevant information about climate risks and a lack of equal access to information services and technologies. Moreover, women's voices are often absent in designing and provisioning information services and technologies. Increasing women's access to early warning systems and climate information services can empower them for informed decision-making in climate action. IAAT can increase women's access to existing early warning systems and climate information services, and build their capacity to use this information for climate change adaptation and mitigation decisions at their farms and communities.

6. Gender action plan

GAP Goal:
Mobilization and utilization of project resources effectively and efficiently to tackle the current normative gender barriers, and biases, and contribute towards improved gender equality

Impact Statement:
Enhanced climate resilience of both women and men in farming households with improved capacity for climate change adaptation and mitigation

Project Activities	**Indicator & Targets** Please note: T = Target, F = Females, M = Males	**Timeline**	**Who Is Responsible**	**Responsibilities of GESI Advisor for SEAH**	**Cost (GCF)**
Output 1.1 Improved technical capacity and inclusiveness of extension services in climate-smart agricultural and agroforestry production					
Activity 1.1.1 Build technical capacity and reach of extension services on CSA and agroforestry techniques	GESI Action: GESI considerations are incorporated into the implementation of technical capacity building of staff working in agriculture extension services During the kick-off and inception of the project, project staff and stakeholders to be trained in gender and disability inclusion to be able to communicate our approach and objectives effectively to project participants. Indicator: *% of women agriculture extension staff participate in technical capacity-building activities* Target: 500 extension staff including at least 50% women Gender outcome: Enhanced women extension staff and their capacity to reach a large number of women	Year 1 to Year 5 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Work with local government or authorities to sensitize community members on SEAH safeguarding Conduct SEAH awareness-raising within the communities Include SEAH awareness in traveling and training events	US$ 82.153 The data collection of this indicator does not generate any additional cost as this outcome indicator will be measured during an already budgeted baseline, midline, and endline assessments[55]

[55] The GESI Lead will train data collection team and will support with analysis of the data to ensure accuracy in measuring women's representation.

	farmers in CSA capacity building and implementation				
Output 1.2: Increased use of climate-resilient practices in the production of CSA crops, livestock, and agroforestry by smallholder farmers					
Activity 1.2.1 Build awareness, capacity, community interest and field-level adoption of CSA techniques and agroforestry	GESI Action: Conduct climate adaptation and opportunity assessment in communities which will consider gender-specific climate needs of women, girls, boys, men, people with disabilities, and Indigenous people Ensure training materials mainstream gender. Trainings scheduled around the availability of women and marginalized groups[56]. Indicator: *% of women farmers, cooperative members, GAP-RU members, and CAC members trained* Target: T: 172,680, M: 86,340 F: 86,340 172,340 farmers trained, 50% of them women. 1,310 cooperative members, 50% of whom are women, trained by cooperative managers (a subset of the total) 19 emergency plans taking into account the needs and priorities of women and young people, drawn up and shared with the GAP-RU at the municipal level. 704 GAP-RU members trained in roles and responsibilities, 50% of whom are women.	Years 1 to 3 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Execute extensive community consultation to understand the social issues that may further reinforce SEAH and identify measures to address the participation barriers. Conduct SEAH awareness-raising within the communities	US$ 1,411,504

[56] As part of our operations, SCI is committed to ensuring that all venues are accessible to people with disabilities to the best of our ability, taking into consideration the limitations within our target areas. We understand that poor infrastructure remains a challenge in rural areas and the circles where we operate. Therefore, our team will discuss possible adaptations and appropriate measures to provide all necessary support to ensure increased participation. This commitment is embedded in our country's strategy, which will include the implementation of the IAAT project activities.

	627 CACs established and open to women and young people in project communes not covered by Albarka (with 1 CAC per village established) – 50% women involvement in CAC development Gender outcome: Adoption of CSA technologies and practices by large number of women farmers				
Output 1.3: Increased land area under agroforestry					
Activity 1.3.1 Develop land-use mapping at the regional level	GESI Action: Generates gender dis-aggregated land use map e.g. land ownership by gender. Indicator: *% of men and women with landownership* Target: N/A Gender outcome: Use of gender-disaggregated land use maps to reach a large number of women farmers	Year 2 to Year 3 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	NA	US$ 78,659
Activity 1.3.2 Plant agroforestry trees on community and state-owned lands	GESI Action: GESI considerations are incorporated in agroforestry need assessment Ensure training materials mainstream gender. Scheduling of training sessions will consider time availability, workload and accessibility for women and marginalized groups. Indicator: *% of women involved in agroforestry training and plantation* Target:	Year 2 to Year 5 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Provide SEAH training to project stakeholders and communities.	US$ 495,268

	Ensure 50% of women's participation in agroforestry plantation Ensure 50% of women's participation in agroforestry-related training. Gender outcome: A large number of women benefited from the improved agroforestry system i.e. increased women's access to fodder, grass, firewood, and non-timber forest products.				
Activity 2.1.1 Support the creation of inclusive private sector value chains for key CSA/agroforestry crops and technologies	GESI Action: GESI considerations are incorporated into the selected of women and girls supporting CSA/agroforestry crops and technologies Training modules must include gender-sensitive elements, including discussions on control of resources, with recommendations on how women, including young women, can increase their ownership and revenues by involving in the selected value chains. Indicator: *% of women/girls trained/supported by the IAAT project* Target: 22,778 of which 50% were women and young people. Gender outcome: Increased participation of women in CSA and agroforestry product value chains and enhanced their income	Year 1 to Year 4 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Execute extensive community consultation to understand the social issues that may further reinforce SEAH and identify measures to address the participation barriers. Provide SEAH training to project stakeholders and communities.	US$ 149,633
Output 2.2: Smallholder farming communities, especially youth and women, can more easily overcome barriers to entrepreneurship in CSA and agribusiness.					

Activity 2.2.1 Support local financial institutions to increase access to finance for smallholder farmers, especially women and youth, for CSA/agroforestry investments	GESI Action: GESI considerations are incorporated to support local financial institutions and farmers, for example, women-led VSLAs Training modules must include gender-sensitive elements to encourage participation of women-led local financial institutions Indicator: *% of women-led local financial institutions participated in the training and capacity building program* Target: T: 18,000 F: 15,300 M: 2,700 Gender outcome: Increased women's access to financial resources and investment capacity for climate change adaptation	Year 2 to Year 4 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Conduct SEAH awareness-raising within the communities and VSLAs	US$ 176,901
Activity 2.2.2 Enhance capabilities and connectivity of youth and women CSA/agribusiness entrepreneurs	GESI Action: GESI considerations are incorporated into enhancing capacity and connectivity plans for CSA/agribusiness entrepreneurship Training modules must include gender-sensitive elements to encourage the participation of women and girls' capacity building trainings Trainings scheduled around the availability of women and girls Indicator: *% of women/girls participated in capacity-building training,* *% of women linked to local financial institutions,*	Year 2 to Year 4 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Conduct SEAH awareness-raising within the communities and value chain actors	US$ 116,865

	% of women linked to private sector service provider Target: 3,750 women entrepreneurs selected and trained in basic business skills, such as drawing up business plans and market research, including loan applications. 5,250 youth and women entrepreneurs (50% women) linked to local financial institutions and private sector service providers. Gender outcome: Increased women's participation in agribusiness and entrepreneurship				
Output 2.3: Increased adoption of climate-smart agriculture technologies by smallholder farmers					
Activity 2.3.1 Install and support the productive use of biodigester systems and solar irrigation systems amongst smallholder farmers	GESI Action: GESI considerations are incorporated in the need and suitability assessment for biodigester and solar irrigation systems in the project locations Training modules must include gender-sensitive elements to encourage the active participation of women and girls, empowering them to adopt and benefit from biodigester and solar irrigation technologies Indicator: *% of women and girls participated in training events* *% of women benefited from the installation of solar and biodigester systems* Target:	Year 2 to Year 5 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Provide SEAH training to project stakeholders and communities	US$ 435,475

	1,000 solar irrigation systems installed and in operation (25% by female-headed households) 5000 biogas systems installed and in operation (25% by female-headed households). 1,000 groups of small farmers and 5,000 livestock breeders trained in the use of biodigesters and solar irrigation systems (50% women farmers) Gender outcome: Increased women's access to energy for irrigation and household use.				
Component 3: Increasing Institutional Capacity and Knowledge					
Output 3.1: Increased institutional capacity in climate change adaptation and mitigation planning and best practices to address agriculture-related climate risks.					
Activity 3.1.1 Strengthen institutional capacity in localized climate change adaptation and mitigation planning	GESI Action: GESI considerations are incorporated into the implementation of capacity strengthening activities, training materials mainstream gender aspects, lead on the assessment of institutional capacities to adopt gender-responsive actions Indicator: *% of women participation in institutional capacity building for localized climate change adaptation and mitigation planning* Target: CACs operationalize and support Community Action Plans in 48 communes, ensuring 50% women's participation.	Year 1 to Year 5 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Include SEAH awareness in traveling and training events	US$ 147,326

	58 PDESCs updated with climate change adaptation and mitigation related to CSA and agroforestry. Training on PDESCs preparation will include 50% women's participation. Gender outcome: Increased women's participation in climate change adaptation and mitigation planning				
Activity 3.1.2 Technical capacity building of national climate funding institutions for disbursements management	GESI Consideration: Capacity building for government officials on climate funding steps and tools will include key modules on the differential impacts of climate change on marginalized groups (especially women and people with disabilities) Provide targeted training to women, especially young women, in self-confidence, and communication skills to act as leaders in the national fora. Indicator: *% of women and girls participation in the technical capacity building trainings on climate finance*. Target: Workshop participation and training on Mali Climate Fund opportunities and efficiencies, ensure 50% of women's participation. Gender outcome: Increased women/girls decision making capacity in climate funding.	Year 1 to Year 5 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	SEAH Action: Include SEAH awareness in traveling and training events	US$ 53,202

Output 3.2: Enhanced knowledge sharing and coordination of best practices in CSA and agroforestry across stakeholders.					
Activity 3.2.1 Enhanced convening and contribution to national databases for CSA and Agroforestry	GESI Action: Generate gender-disaggregated information on CSA and agroforestry database. Gender outcome (long-term): Increased representation of women evidenced through use of gender-disaggregated information on CSA and agroforestry planning and funding	Year 1 to Year 5 of the project	Lead: Technical Advisor- GESI Support: MERL Advisor Technical advisor- CC Technical advisor – agriculture Technical advisor – forestry	NA	US$ 25,295

Note: % of target number of women participants/beneficiaries is based on the current Albarka project and consultation with key stakeholders in the project locations. This is the minimum level of target subjected to change after the baseline survey. IAAT prioritizes increasing the proportion of women and youth participation/beneficiaries across the project interventions. * Risk mitigation measures for Sexual Exploitation, Abuse, and Harassment (SEAH) are included based on GCF's SEAH Risk Assessment Tool (decision B.BM-2021/18).

7. Gender Assessment in Project Implementation

IAAT will engage in further consultations with community members across all 12 circles during the project's initial planning phase. These consultations and in-depth gender assessment aim to ensure that the interests and needs of women, men, girls, and boys are fully integrated during the selection of the 48 communes. By applying an intersectionality analysis, the assessment will examine how different forms of marginalization—such as gender, education level, head of household status, marital status, disabilities, age, and other social factors—intersect and impact vulnerability. This approach will facilitate the identification and selection of IAAT beneficiaries and the prioritization of key interventions that address gender and climate change adaptation and mitigation needs. Additionally, the assessment will help address root causes of vulnerability, supporting more sustainable and impactful gender outcomes. This assessment will support to generate the baseline for gender integration in IAAT's activities, monitoring project outcomes during the project period and impact evaluation.

Appendices

Appendix 1: Mapping of sectors

Name of stakeholder	Type of stakeholder (partner, direct beneficiary, indirect beneficiary)	Profile of stakeholders	Consultation methodology	Consultation findings	Role in implementing the project, and/or consultation methodology and timetable	Comments
AEDD: gender unit	Partner	Technical partner	interview	Institutional support for the project Ensure gender mainstreaming and monitoring of the project	Ensure that gender is taken into account	
Ministry of Justice	Partner	Partner	Interview	Under formal legal codes, men and women are treated in the same way, no distinction is made, unlike customary codes which treat men and women differently. As for the expiation of wrongs and legal representation, it is the legal provisions that prevail, no distinction is made between men and women. Even though there is often discrimination against women, for example if a woman applies for a divorce, if the judge or magistrate is a man, he tends to incriminate the woman without trying to understand the reasons that led the woman to apply for a divorce[57] . In this respect, it is also worth noting that formal legal codes straddle religious codes, so that when the woman concerned is a Muslim, for example, Islamic law prevails.	Advice on formal texts	
Regional Directorates for the Promotion of Women, Children and the Family	Partner	Technical partner		Low involvement of women and young girls and boys in NR management mechanisms and frameworks	Support for gender mainstreaming in the IAAT project	All partnerships will be developed

[57] **Source**: Ministry of Justice

			Interview	Confinement of women to reproductive and community roles and earlier than reproductive roles Women's access to land (small area) and limited control.		through agreements and or partnership or collaboration agreements.
Regional Water and Forestry Directorates	Partner	Technical partner	Interview	There are promising niches in the production of improved stoves and the processing and marketing of harvested products, which are the domain of women.	Support for the implementation of RNA actions, set-asides, identification of harvested products, production, processing and marketing of forest products. Capacity-building for women, girls and boys.	
Regional Directorates of Agriculture	Partners	Technical partners	Interview	There are few support projects for women in the agricultural sector, and those that do exist are one-off projects lasting between 3 and 4 months. The presence of women in market garden production, groundnuts, fonio, cowpeas and rice. There is potential in terms of developed land and opportunities for support and capacity-building for women in cultivation and market gardening techniques. Availability of agricultural agents to support women	Support for the development of market garden sites, support for agricultural inputs and capacity-building for market gardeners.	

b.1 Customary chiefs / male community leaders	Partners and beneficiaries	Partners/ Beneficiaries Supporters and allies of women	Focus groups supplemented by individual interviews	Socio-cultural constraints limit women's access to and control over resources and benefits, and determine the social division of labour. They also constitute constraints for women Women and men are treated differently by formal and informal legal codes. Women are rare in conflict prevention and management mechanisms and frameworks. Women have knowledge and experience of NR management as users and producers. Male leaders are allies and supporters of women.	Support for community and women's mobilisation and awareness-raising	
b.2 Women community leaders / presidents of CAFOs and FENAFERs	Partners and beneficiaries	Partners/ Beneficiaries Supporters and allies of women	Individual interviews	Are committed to protecting the environment through reforestation and market gardening. Key players in the processing and marketing of harvested produce, fish and milk. They are involved in raising small ruminants, poultry and often cattle. They are involved in conflict prevention and management. Despite this, they come up against socio-cultural constraints. Involved in capacity building for women	Support for community and women's mobilisation and awareness-raising	
b.3 The Chambers of Agriculture	Partners and beneficiaries	Partners/ Beneficiaries Supporters and allies of women	Consultations	Involved in supporting and mobilising women	Advice support Support	

Appendix 2: List of interviewees[58]

N°	First name	Name	Function	Structure	Region
1	Isae	COULIBALY	Head of Division	Regional Department for the Promotion of Women, Children and the Family	Koulikoro
2	Daouda	TRAORÉ	Technical Advisor	Ministry for the Promotion of Women, Children and the Family	Bamako
3	Mamadou	SOW	Documentalist / Archivist	Regional Budget Monitoring Committee	Ségou
4	Boureima	KOMNOTOUGO	Project manager	G- FORCE/Alabarka project	Mopti
5	Almahadi	Ag AKERATANE	Program Manager	Tassagh/Alabarka project	Gao
6	Goumar	ABOUBACRINE	Program Manager	ADICOM/ Albarka Project	Timbuktu
7	Mariama	TANGARA	Gender Officer	AEDD	Bamako

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